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Abstract

The real estate sector in Nepal is among the rapidly developing yet the least digitalised sectors in the country. Although there is an escalating need in the residential and commercial property purchasing and selling, the property buying, selling, and renting process in Nepal is still very dependent on offline modes, personal contacts, and mediator agents. Such a fragmented system can pose buyers and sellers to high risks such as fraud, inflated prices, forged documents and transparency. It is evident that there is an urgent need to have a trusted centralised digital platform that can help to bridge this gap.

The report is the record of the entire development of Open House Nepal, the full-stack web-based real estate marketplace that was developed based on the following technologies: React.js on the front-end, Node.js on the back end, and SQL Lite as the database. The portal is structured in such a way that property owners and agents can post listed verified properties, whereas the buyers and renters will be able to browse, search, and engage with listings through a secure and user-friendly interface. The main characteristics of Open house Nepal are role-based user authentication, property verification system by the administration, review and rating system, interactive comment system, 360-degree view of the property with the Pannellum library, advanced search and filtering, real time notification system that is provided through email and in-app notifications.

The process of developing the project was based on the Agile Scrum approach, which enabled the iterative development and constant improvement throughout ten planned sprints. Extensive design documentation in the form of Entity Relationship Diagrams (ERDs), Use Case Diagrams, Data Flow Diagrams, System Architecture Diagrams and UI wireframes supported the development. A pre-survey of 100 people proved that such a platform is necessary as people identified the following essential pain points: the lack of transparency, threat of fraud, and the unavailability of digital property tools. The results of post-survey feedback provided after user testing were strong satisfaction with the usability and features of the platform.

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Table of Abbreviations

Abbreviation	Fullform
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DOM	Document Object Model
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
npm	Node Package Manager
ORM	Object-Relational Mapping
REST	Representative State Transfer
SQL	Structured Query Language
UI/UX	User Interface / User Experience
OHN	Open House Nepal

Table 1 Table of Abbreviations

CHAPTER 1: INTRODUCTION

1.1 Project Description

Open House Nepal (OHN) is a web-based portal, which is centralized, and is specifically created to solve the problems of the Nepalese property market (Kumar, 2021; Shrestha, 2022). It is an online platform that can bring together property owners, sellers, buyers, and renters to conduct property transactions in a secure, transparent and effective way. (Pant, 2020; Adhikari, 2021).



Figure 1 Open House Nepal

It has been built with a current technology stack. React.js is used to provide the frontend user interface, providing a responsive and interactive experience on all devices. All the server-side logic and REST API endpoints are handled with Node.js, and the relational database to store property listings, user information, reviews, and record of transactions is SQL lite. 360° virtual tour, I utilized the Pannellum framework to create an immersive, high-performance panoramic experience.

Central to Open House Nepal is the three pillars of transparency, trust and accessibility. This transparency comes with confirmed property listings, standardised property information, as well as a system of public reviews. The administration level document

checks and role-based access control are used to build trust. An easy-to-use design based on a responsive design that supports mobile, tablet and desktop devices, and improved search and filter features that allow visitors to find their desired information within a short time makes the site easy to navigate. (Nielsen, 2012; Garrett, 2011)

The Agile Scrum methodology informed the development of this platform, enabling it to be developed and tested in terms of an individual module through a series of sprints. This method enabled feedback integration at periodic stages and priority readjustment, depending on results of testing. (Beck et al., 2001; Highsmith, 2009)

1.2 Current Scenario

Nepal real estate has experienced a tremendous growth in the last 10 years due to urbanisation, internal migration and the rising purchasing power of the middle class. (World Bank, 2022; Central Bureau of Statistics, 2021) There has been a demand boom in residential property and commercial space in cities such as Kathmandu, Lalitpur, Bhaktapur and Pokhara. Although this is increasing, the actual manner of property dealings in Nepal has not changed much and is still highly dependent on the traditional and offline means (Adhikari, 2021).

Majority of the individuals seeking to purchase or rent property in Nepal continue relying on word of mouth, personal contacts or visiting physical broker offices to obtain listing. No single trusted site allows one to collect verifiable properties in the entire country and offer them in an easily searchable and easily filterable format. Even limited available online platforms like Lalpurja Nepal, Gharbazar.com, and EProperty Nepal have the simplest listing features available but none of them have the essential functions of document verification, user feedback, and virtual tours that are interactive (Adhikari, 2022).

This absence of a contemporary digital infrastructure poses severe issues. Customers often receive scams when they are sold fake documents of ownership of land or when sellers are selling property that they have no legal right to sell. As per the statistics of the Nepal Police Department (2024), the number of fraudulent cases in property has been

steadily rising over the past few years, with a total of more than 2,500 cases reported within the past five years. The sellers, however, find it hard to access many people and are usually compelled to transact with numerous intermediaries, increasing the costs of transactions (OECD, 2020).

Technologically, there has been a significant improvement in smartphone penetration and internet connectivity in Nepal. The Nepal Telecommunications Authority indicates that mobile internet users in Nepal reached over 20 million in 2023, which means that there is an existing and eager user base of a digital real estate platform in case a user-friendly and reliable form was made available (NTA, 2023).

1.3 Problem Domain and Project as Solution.

According to the secondary research and the main pre-survey survey carried out on 100 participants the following are the main issues in the Nepalese real estate market presently (Creswell, 2014; Saunders, Lewis and Thornhill, 2019)

- Lack of transparency: The prices of the property are usually overvalued or reported inaccurately without a standardised, publicly available pricing database. There is no means of comparing prices across listings to buyers because it requires contacting more than one broker (Adhikari, 2021).
- Document fraud risk: Documents of property ownership, such as land titles (lalpurja) and citizenship certificates are readily forged. This is a repeat issue due to the lack of any verification system within the current systems (Nepal Police, 2024).
- Lack of user reviews: There is no way to leave a review or rating of the property or seller, so fraudulent sellers or false information cannot be held accountable (Filieri, 2015; Deloitte, 2022).
- Extensive reliance on offline operations: Starting with property search all the way to signing a contract, almost all operations demand face-to-face attendance, which is time-consuming and unavailable to buyers located outside Nepal or in other cities (World Bank, 2022).

- None of the virtual viewing features: Buyers are supposed to visit the site several times before they settle on a property, this is costly and inefficient especially to the diaspora buyers or those who would want to buy properties in other cities.

All these issues are dealt with by Open House Nepal (Porter, 2008). The site also brings in administrative validation of property documents which means that only authenticated listings will be visible to the public. The comment and review system brings about some form of accountability and gives the opportunity to the real buyers to share their experiences. The 360 virtual tour capabilities minimize the use of physical location visits. The search and filtering tools are advanced to enable one to find properties that fit certain criteria. Real-time notifications make sure that the users are informed about new listings and modifications of previous ones. All these attributes combine to make Open House Nepal a holistic and effective resolution to the systematic issues in the Nepal real estate market (Porter, 2008).

1.4 Aim and Objectives

1.4.1 Aim

This project is designed to create and build a secure, user-friendly and transparent web-based real estate marketplace in Nepal that makes transaction of property simpler to buyers, sellers, renters and agents and solve the major problems of trust, transparency and accessibility of digital markets (Porter, 2008).

1.4.2 Objectives

The project had the following specific objectives:

- To establish a safe user authentication framework, role-based access control of buyers, sellers and administrators.
- To develop and install a property listing module that will enable sellers to add, update, and maintain listings by uploading photos and listing property details.

- To establish an administrative system of checks to verify property ownership documents and accept or reject listing prior to public visibility.
- To have the ability to have a review and rating system that enables approved users to post reviews about properties, including administrative capabilities.
- To create a system of comments and interaction that allows buyers to pose questions and interact with sellers in real-time on pages of the listing.
- To add a 360-degree virtual property tour feature with Pannellum JavaScript library.
- To introduce a real-time notification system, which will notify the users through email and in-app notifications about new listings, listing updates, and change of the verification status.
- To create high-level search and filtering features enabling users to search based on location, price, property type and size.

1.5 Structure of the Report.

1.5.1 Background

The background chapter gives a careful study of the end users of the platform, the technical terms and technologies that the project will utilize, three comparable projects that are currently in operation in the Nepalese market and a comparative study that will help identify the gaps that the Open House Nepal will fill (Porter, 2008; Yin, 2018).

1.5.2 Development

The chapter on development takes us through the whole technical development process. It discusses the methodologies that have been considered and why Agile Scrum has been chosen (Schwaber and Sutherland, 2020). It contains both the pre-survey and post-survey results of the survey. The requirement analysis section captures both the functional and non-functional requirements. All major diagrams such as Use Case Diagram, ERD, DFD, and system architecture diagram are shown in the design section.

Lastly, the implementation section records every sprint together with development screen shots and sprint retrospectives.

1.5.3 Testing and Analysis

The chapter of testing and analysis explains the test plan and gives the results of unit testing and system testing. The test objective, steps to be followed, expected result and the actual result are all given in each test case. The chapter ends with a critical analysis, which thinks on what was best, what failed to achieve as anticipated and what could be done to improve.

1.5.4 Conclusion

In the conclusion chapter, the legal, social, and ethical aspects of the platform are discussed. It enumerates the strengths and weaknesses of the existing system and details the intended future work and functionality of later stages of development.

CHAPTER 2: BACKGROUND

2.1 About the End Users

Open House Nepal is built in such a way that there are four types of users, each having various needs, objectives, and access to the system (Pressman and Maxim, 2020; Sommerville, 2016).

Buyers and renters are the largest and the first group. These are individuals or organisations interested in purchasing a property either as a residential, commercial or investment property. This group in Nepal usually comprises working class in urban centres, families planning to relocate, business owners seeking commercial space, and Nepal diaspora planning to invest in property in home country. The buyers desire to easily search and see the properties, read the detailed and precise information, see the photos and virtual tours and connect with the sellers promptly. They would also like to read the reviews of other buyers to make some informed decisions (Filieri, 2015).

Sellers and property agents will be the second group. They are people or registered agencies that would like to advertise properties on the site to sell or rent them. This comprises homeowners, real estate firms and standalone brokers in Nepal. The sellers require an easy and effective method of listing, sending pictures and other supporting documents, control their listings, and answer questions of the prospective purchasers (Garrett, 2011).

The third one is administrators. Administrators are to ensure that the platform is not compromised. They check property documents provided by sellers, approve or reject listings, moderate reviews and comments that break community guidelines and handle user accounts. The administrator is a very important position that ensures the integrity and credibility of the site (OECD, 2020).

The fourth group is visitors or unregistered users. These are individuals who will visit the site but not open an account. They can search and see property listings but cannot contact sellers, leave reviews or save favourite properties. This segment is significant as it signifies the potential future registered users who can turn into those who are convinced of the value of the platform by being able to experience it themselves (Kotler and Keller, 2016).

2.2 Understanding the Solution

This section contains easy to understand explanations of the major technical ideas and technologies employed in the construction of Open House Nepal, to ensure that the reader has a general understanding of what has been produced and how it functions regardless of whether they have a strong technical background.

2.2.1 React.js

React.js is a free, open-source JavaScript library that was created by Facebook to create user interfaces. Put simply, it enables developers to develop web pages that dynamically update without the necessity of reloading the whole page. In an illustration, as a user uses a filter on the property search page, the search results update immediately without a browser refreshing. React does this with the help of a mechanism known as the Virtual DOM that efficiently computes the smallest possible changes that can be used to modify the page.

React.js was employed in Open House Nepal to create all the pages that users can interact with, including the login screen, and the pages that list the properties (Garrett, 2011).

2.2.2 Node.js and Express.js

Node.js is a JavaScript runtime environment, which enables JavaScript code to execute on the server (backend) as opposed to only in the browser. Express.js is a lightweight platform, which is built on top of the Node.js platform enabling it to simplify the creation of web servers and API endpoints. They all combine to do the logic behind the scenes of Open House Nepal, when the frontend (the part handling requests) makes a request (like to get all the property listings), they respond to that request through their processing (a query to the database), and send the answer to the request (a list of the properties in JSON format). The fact that Node.js has a non-blocking, event-driven design is one of the factors why it was selected in this project, as it implies that it can serve numerous parallel requests without going offline, and it is an ideal choice in a marketplace platform where numerous users can be browsing simultaneously (Richards, 2015).

2.2.3 SQLite

The relational database management system that was adopted to handle the data layer of the application was SQLite. It stores data in a well-organized table where there are well-established relations among them. In the Open House Nepal site, SQLite is used to store all the necessary data, such as user accounts, property listing, image data (in JSON format), reviews, comments, and notifications (Pressman and Maxim, 2020).

The relational character of SQLite is relevant because it allows querying the interrelated data effectively. The system can access related data such as property details, images, reviews and seller profiles in one query using the Sequelize ORM, which allows the system to execute the more complicated joins found in SQL. SQLite was selected instead of NoSQL databases such as MongoDB, as the data in the app is highly structured and is based on relationships.

The major benefits of SQLite are that it is serverless, file-based, which makes it light and simple to operate in development and testing. Although it is simple, it provides complete compliance of the ACID, which is necessary to uphold integrity in data, particularly the information that is sensitive to users and transactions. The system is further made database-agnostic, so that it can be easily migrated to more scalable databases such as PostgreSQL or MySQL in the future without any major alterations to the underlying application code (Pressman and Maxim, 2020).

2.2.4 REST API

A REST (Representational State Transfer) API is a standardised mechanism by which various components of a software system communicate with one another over the internet through API using the HTTP protocol. In Open House Nepal the React.js frontend and the Node.js backend are in touch with each other via a REST API. A user writes the login form, and the frontend makes a POST request to the API endpoint on the backend. The backend checks the credentials, creates a JWT (JSON Web Token) and returns it to the frontend. This token is stored in the frontend, and all further requests would include it to confirm that the user has been logged in. This decoupling of frontend and backend with a REST API ensures that the system is more modular and easier to maintain and may be extended in the future to a mobile application (Pressman and Maxim, 2020).

2.2.5 360-Degree Virtual Tour (Pannellum)

The 360-degree virtual tour feature was installed with the Pannellum JavaScript library. This library enables web page developers to integrate interactive 360-degree panorama images into web pages. Sellers can post 360-degree photos that have been made with a compatible camera or smartphone, and buyers can then access the virtual tour on the listing page to explore the property as though they were in it. The users can zoom in, have a pan and see the property in any direction. This is a very useful feature to buyers who cannot visit the property and be physically present as in the case of those living in a different city or even abroad.

2.2.6 Role-Based Access Control

Role-Based Access Control is an access control tool that is used to control the accessibility to sections of a system depending on the role of a user. Open House Nepal has four roles, Visitor, Buyer, Seller, and Admin. Every role is assigned some permissions. As an example, property listing can only be created by a user with the Seller role, listing can be approved or rejected by an Admin user, and only a logged-in Buyer can leave a review. This will allow every user to have access to only perform actions that are within their role and safeguard the platform and those who use it against unauthorised actions (NIST, 2020).

2.3 Similar Projects

2.3.1 Lalpurja Nepal

Lalpurja Nepal is among the more established real estate platforms in Nepal, functioning as an online marketplace, as well as a brokerage service. It provides property sale, rent, and lease listing under various categories such as residential, commercial and land. Besides mere listing, Lalpurja is also offering other services like property valuation, buyer representation and contractual support. The site is quite established within the Nepalese real estate industry and has established itself as a service-based agency and not a listing portal alone.

Figure 2 Lalpurja Nepal

But in terms of technology, Lalpurja is constrained. It does not provide any kind of online document authentication, i.e. listing can be supported or not supported with original ownership documents. No system of user review or rating, therefore buyers cannot judge the credibility of a seller by previous experience. There are no virtual tours, and this implies that all interested buyers must visit the property in person. The value of the platform highly relies on human brokers and, therefore, the platform is not a self-service digital marketplace but rather it is vulnerable to the same kind of inefficiencies as the traditional brokerage (Porter, 2008).

2.3.2 Gharbazar.com

Another popular property listing portal in Nepal is called Gharbazar.com where people and agencies post listings of properties on sale, rent or lease. The site has a clean design and has varying listing plans, such as premium and featured that provide sellers with more exposure. It encompasses residential, commercial, and land in the Nepal big cities. Regardless of its sensible user base, Gharbazar has serious gaps in comparison with what is required of a modern real estate platform.



Figure 3 Gharbazar.com

There is no document verification available, so there is no guarantee that a listing is a

legitimate property. The system has no review or comment system and therefore, the buyers are unable to share their experiences and ask questions in public. There is no virtual tour, and this limits the platform to use by remote buyers. The search and filter option is not advanced as other more mature real estate sites across borders. Like Lalpurja, Gharbazar is more of a classified advertisements board of property than it is a real integrated online marketplace (Porter, 2008).

2.3.3 EProperty Nepal

EProperty Nepal is a listing-based real estate portal that gathers the properties on the major cities of Nepal such as Kathmandu, Lalitpur, Bhaktapur and Pokhara. It enables real estate shoppers to search the listing and see some basic information. The platform is more geographically covered than others and navigable. EProperty Nepal, however, has most of the same limitations as its competitors.



Figure 4 EProperty Nepal

It does not authenticate property documents, lacks a system of user reviews, does not provide virtual tours and has limited search capabilities. The platform lacks any interactive aspect, lacks a means of buyers and sellers interacting directly with each other through the platform, and there is no mechanism to notify users of new listings that would meet their criteria. EProperty Nepal in short is a high-end non-interactive listing board with minimal interactivity or trust-building capabilities (Porter, 2008).

2.4 Comparisons

The following comparison table evaluates the four platforms across the key features relevant to a modern real estate marketplace. Note that 'Yes' indicates the feature is fully available, 'Partial' indicates limited availability, and 'No' indicates the feature is absent.

Feature	Lalpurja	Gharbazar	EProperty	Open House Nepal
Verified Listings	Partial	No	No	Yes
Document Verification	Manual	No	No	Yes (Admin)
User Review System	No	No	No	Yes
Comment / Interaction	Limited	Limited	No	Yes (Full)
360° Virtual Tour	No	No	No	Yes
Advanced Search & Filter	Basic	Moderate	Moderate	Advanced
Real-Time Notifications	No	No	No	Yes
Role-Based Access	No	No	No	Yes
Mobile Responsive	Partial	Yes	Partial	Yes
Trust / Transparency Level	Medium	Low	Low	High

Table 2 Comparisons

As the comparison table clearly shows, Open House Nepal offers the most comprehensive set of features among all four platforms. The document verification system, combined with user reviews and 360-degree virtual tours, creates a level of trust and interactivity that none of the existing Nepalese real estate platforms currently provide. This positions Open House Nepal as not just an incremental improvement but a fundamentally different kind of platform that addresses the root causes of mistrust and inefficiency in Nepal's property market.

CHAPTER 3: DEVELOPMENT

3.1 Considered Methodologies

3.1.1 Waterfall Model

Waterfall model is an old and simplest software development methodology. It is a highly sequential process where one step must be finished before the next step is initiated: requirements gathering, system design, implementation, testing, deployment, and maintenance. What is so appealing about the Waterfall model is that it is simple and the entire project is planned at the start, hence it is easy to estimate cost and time. Nevertheless, the Waterfall model was not applicable to this project due to several significant reasons (Sommerville, 2016).

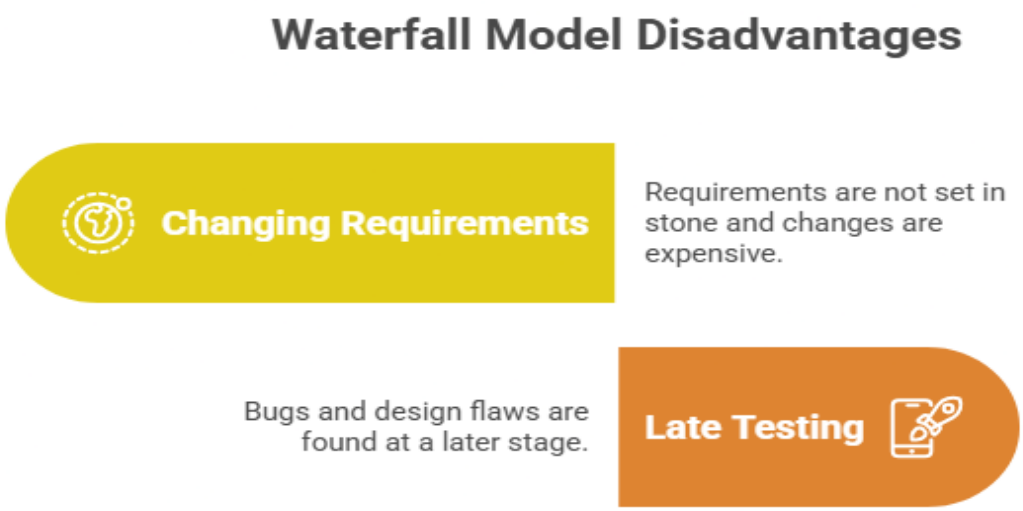


Figure 5 Waterfall Model

To begin with, the requirements of a real estate platform are not as set in stone. The requirements changed and new features were introduced as the survey responses were received and the supervisor gave feedback. Any change in requirements after design is done is exceedingly expensive and time consuming in a Waterfall approach. Second, in Waterfall the testing process is only done when the development is finished, and the bugs and design flaws are found at a later stage. In the case of a complex system such as Open House Nepal, where there are numerous interconnected modules, a core architectural issue being identified towards the end of the project may have been disastrous (Beck et al., 2001).

3.1.2 Prototype Model

The Prototype model entails the development of a working prototype of the system at the beginning of the development process and then through the feedback of the users or stakeholders, the prototype is refined and improved. This design process level of iteration can prove extremely handy in specifying requirements and early detection of usability issues.

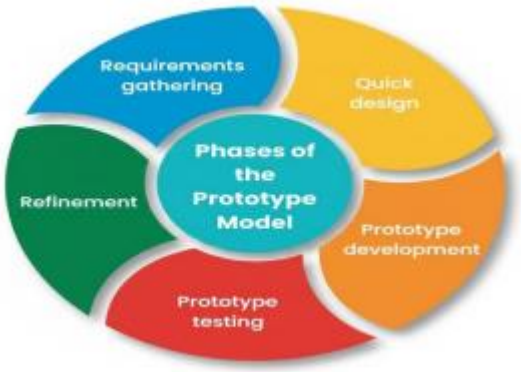


Figure 6 Prototype Model

Although it has certain benefits, prototyping is not the most suitable approach to use in this project. Prototyping and re prototype is a time-consuming activity especially when there is only one developer and a limited amount of time. Also, prototypes are likely to simplify complex backend functionality, including document verification processes, which

might provide a false impression of the end system functionality (Sommerville, 2016). The project had very clear modules and a clear set of objectives so the time spent in prototype iterations would have been less time to work on the actual development (Pressman and Maxim, 2020).

3.1.3 Spiral Model

The Spiral model is a risk-based approach that incorporates the repetitive development with a risk evaluation at every cycle. It is ideal to use in projects that are big, complex, and risky and the risks must be always controlled during development. Whereas the focus on risk management in the Spiral model is attractive in a platform, which handles sensitive property information, it was not selected in this project (Sommerville, 2016).



Figure 7 Spiral Model

The Spiral model mandates formal risk assessment on each iteration, resource-intensive, and necessitates a bigger team with special risk management skills. Being a single developer with a restricted four-month time frame, overhead costs of formal risk analysis per cycle would have been counterproductive and would have cut the time to do the actual development (Pressman and Maxim, 2020).

3.2 Selected Methodology

3.2.1 Agile Scrum Selected Methodology

Upon reflection of the above options, Agile Scrum has been chosen as the main methodology of development of Open House Nepal. Scrum is a methodology based on the wider Agile philosophy that structures the development process into time-constrained sprints known as sprints, usually two- to four-week sprints. Sprinted increments of the product are delivered at the end of each sprint in a form that is a working and testable increment of the product (Pressman and Maxim, 2020).

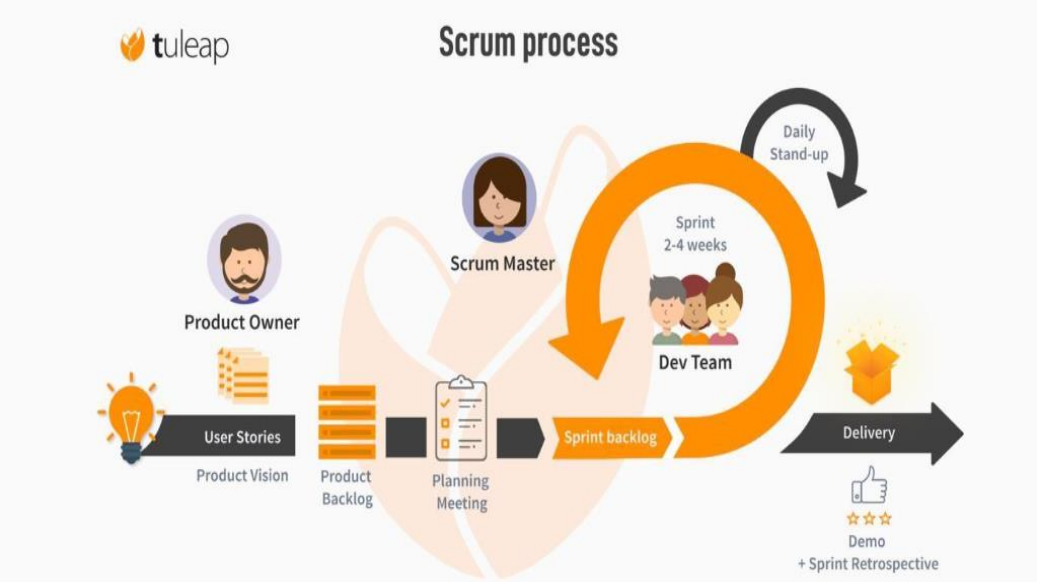


Figure 8 Agile Scrum

Scrum was selected due to several reasons which were directly related to the nature and constraints of this project. First, it fits well with projects that have changing requirements, which was evident here considering that new insights were added to the project due to survey feedback and supervisor guidance as the development progressed. Second, the sprint-based format of Scrum gave a natural method of structuring the development of the ten modules of Open House Nepal, each sprint centred around one or two modules.

Third, the retrospective at the conclusion of every sprint was a welcome change to look back and improve, and, as a one-person developer, I had to learn and evolve continuously. Fourth, the focus of Scrum on producing working software at the conclusion of each sprint meant that there was a continuous product that could be shown and tried and the risk of realizing that something was wrong was minimized (Pressman and Maxim, 2020).

Some of the classic ceremonies of Scrum were modified as a solo developer: the daily standup was substituted with a short daily review log, the Product Owner role was taken up by the project supervisor, and the Scrum Master role was assumed by the developer. Irrespective of these changes, the fundamental tenets of iterative development, constant feedback and sprint delivery were all upheld during the project (Schwaber and Sutherland, 2020).

3.3 Phases of Methodology

Sprint 1: Project Planning and Research

The initial sprint was dedicated to the knowledge of the project scope, preliminary requirements and the establishment of the development environment. An in-depth literature review was performed to know the actual situation in the real estate market of Nepal and the voids in the current digital platforms. The technology stack was finalised, and the development environment was set up with VS Code, Node.js, React.js, and SQL lite. The risk register was developed, and the possible technical, operational, and project risks plus mitigation strategies were identified (Beck et al., 2001).

Sprint 2: System Design and Foundation

All-important design artefacts were developed in the second sprint. These involved the Entity Relationship Diagram (ERD), the Use Case Diagram, the Data Flow Diagram, the System Architecture Diagram and the original wireframes of the UI. The GitHub repository was configured, the project structure was established, and the scaffolding of the basic

project of both the frontend (React app) and the backend (Node.js/Express app) was done. The database schema was developed and made in SQL lite (Elmasri and Navathe, 2016).

Sprint 3: User Authentication Module

Sprint Three was concerned with deploying a secure User Authentication and Authorization Module. This covered the creation of the registration interface, the login portal and a full forget password recovery workflow. The system is based on the stateless session management with the help of JSON Web Tokens (JWT), which provides secure communication between the frontend and the backend (JSON Web Token (JWT); NIST, 2020).

The use of Role-Based Access Control (RBAC) to control user access was introduced, which has four roles: Visitor, Buyer, Seller, and Admin. To increase the level of data security, user passwords are encrypted with the help of the bcrypt hashing algorithm and then stored in the database. Specific authentication middleware was also created to secure sensitive routes, and apply role-based access control, making sure that only authorized users (Sellers and Admins) can access actions, such as creating a listing (Sommerville, 2016).

Also, the registration and the login steps were extensively tested with different edge cases to test reliability and easy user experience before they moved to the next development sprint (Beck et al., 2001).

Sprint 4: Property Listing Module

Sprint Four was working on Property Listing and Management Module. The module allows sellers to compile detailed property listings by listing details like title, price, size and detailed description. To improve the accuracy of the location, Google Maps integration was added, which also enables users to indicate exact addresses of the property. Also, Node.js Multer middleware was utilized to handle multi-file uploads with an option of users uploading property images and legal documents effectively (Node.js; Pressman and Maxim, 2020).

Another important feature that was added in this sprint is Automated Image Optimization, which is implemented based on Sharp library. This capability operates the uploaded photos in the background and resizes as well as compresses them, which enhance the performance of the application and reducing the time of loading the page. In addition, a 360 Virtual Tour was also provided where customers can interactively view the property in 360-degree viewer (PwC, 2021).

Status Management was created to take care of the entire property listing cycle. This enables the property owners to change the listing status to Sold, Under Rent or Archived. It also developed a special dashboard called My Listings, which has tab-based filtering to make it easy to manage listing. Also, a verification system was added to the back end by adding an is verified flag, allowing administrators to view and approve listings before they are publicly exposed NIST, 2020).

Sprint 5: Review Listing Module

The review and rating module was created in the fifth sprint. This will enable the users to leave a comment after using the properties and agents. Whenever a user buys a property, he/she can rate Open House Nepal agent on his/her performance such as the helpfulness, responsiveness and professionalism they showed during the process. In the same way, when renting a property, users are allowed to rate it using the 1-5 stars scale and provide written feedback about their experience, e.g, the property condition, facilities (e.g., AC), and general experience. The system has back-end validation to make sure that only verified users having conducted a transaction could place reviews. Also, it has an admin moderation feature, enabling the administrators to delete reviews that contravene the rules of the community (Sommerville, 2016).

Sprint 6: Comment and Booking Module

The comment and interaction system was developed during sprint six. This functionality enables people to interact on the pages that present the property listing directly and leave comments, ask the seller questions, and answer other users. The comment management system (backend) is CRUD (Create, Read, Update, Delete) to effectively manage interactions with users (Pressman and Maxim, 2020).

Moreover, a reservation system was also added, which allows the customers to use the button of Book to book a tour of the property. After a booking request is done, the email notification is received by both the seller and the admin. The seller is then allowed to respond to the booking via the site by accepting, offering an alternative time or rejecting the request. The admin is informed by email too and can mediate between the buyer and the seller, when necessary, assists in arranging the property tour and facilitating the deal finalization (OECD, 2020).

Sprint 7: 360-Degree Property Viewer

Pannellum was used to implement the 360-degree virtual tour module in sprint seven. As a part of listing, sellers can post 360-degree panoramic images. The buyers are then able to interact with the tour on the listing page, zooming in and panning, and going to full-screen mode. This sprint also introduced image optimization to make sure that panoramic large images do not slow down the page load time (Pannellum; PwC, 2021).

Sprint 8: Notifications Module

The notification system was created in the eighth sprint. The use of in-app notification icon was introduced to inform the user about key activities on the platform. Whenever a new property is listed, a notification is sent out to all users to inform them of the new listing. Also, a notification is sent to a user, when a user lists a property; he/she is informed that his/her property has been listed. On the same note, a user also receives a notification about his or her booking when she or he books a property. The use of email notifications is primarily related to verification. Once a property has been booked, the admin is sent an email with detailed information on the booking including the citizenships details that the user has provided. This will enable the admin to check the booking and make sure that the user is genuine and then go ahead (NIST, 2020).

Sprint 9: Testing, Integration and Optimisation

Sprint nine was purely on testing and bug fixing. Unit-testing of all the modules was done, and system level testing was done to ensure that the modules were functioning properly as a unit. This sprint also had performance optimisations, such as optimisation of

database queries, image compression, and splitting the frontend code to lower initial load times (Sommerville, 2016).

Sprint 10: Documentation and Submission

The last sprint was dedicated to the final project documentation, such as this final report, the technical documentation, user manual, and Software Requirements. The code was finally put into GitHub, and the project was ready to submit (Pressman and Maxim, 2020).

3.4 Survey Results

3.4.1 Pre-Survey Results

The pre-survey was made to learn the real-life complaints of users in the Nepal real estate market at the outset of the project, and it is to confirm the necessity of the suggested platform. The questionnaire was sent online and 100 people who participated in the questionnaires were varied and included students, working professionals, self-employed individuals, buyers, sellers and investors.

Some of the important results of the pre-survey were as follows. Most of the respondents (98% of the respondents) affirmed that they had looked at property online in Nepal indicating high demand of online property tools. Risk of fraud was chosen as the largest problem they encounter in the current property market by 62 per cent of the respondents, by far the most frequent issue mentioned by them. The greatest challenge cited by 19 percent is offline communication, 12 percent are lack of transparency and 7 percent are no user reviews or feedback.

When questioned on the feature they considered most valuable in an online property site, 74 percent of them picked the option of verified property listing as the most preferred feature. 14% picked review listing system, 7% picked comment or testimonial section and very few 5% picked secure payment system. This greatly confirmed the choice to focus on administrative verification as the main feature of this platform. When asked about reviews, 90 percent of the respondents gave the rating of reviews as very important or

important in selecting a property. 88 percent of the respondents believed that only verified users should leave reviews and this was a direct influence in the decision to limit the review system to verified buyers.

On the question about virtual tours, 72 percent strongly agreed and 29 percent agreed that a 360-degree viewer will assist them to know a property without visiting it physically. Three-quarters (74 percent) and a quarter (26 percent) strongly and agreed respectively that they would rather use a site that provides virtual tours. These findings validated the usefulness of the 360-degree tour feature and the move to invest time in incorporating the Pannellum.

Link to pre survey

3.4.2 Post-Survey Results

The post-survey followed the creation of the Open House Nepal platform to assess the user satisfaction, usability and effectiveness of the features implemented. The questionnaire was distributed online and got more than 100 replies among the students, working people and general users who did test the system.

The following are some of the major findings of the post-survey. The general user experience was very positive with over 90 percent of the survey showing that their experience was excellent with a small percentage of respondents rating their experience as good or average. This means that the system has been able to provide high-quality user experience.

About design and layout, most users (approximately 8590) rated the site 5 out of 5, some rated the site 4 and very few rated the site 3. That indicates that the interface is attractive and is organized. In the same way, navigation was listed as a strong point with more than 90 percent of users citing that the site is “Very Easy” to navigate and only a few citing as Easy or Neutral. In response to the question of having found the information they sought, about 85-90% of the respondents answered in the affirmative, yes, completely and a small percentage answered partially. This implies that the site can be useful in providing appropriate and convenient property data. The system features were also highly rated.

The 360-degree house view option has also been deemed as Important and implemented successfully since most of the users (approximately 90 percent or more) have rated it as Very Useful. Similarly, the map and location feature also received good feedback and majority of the users said they were Very Satisfied, with some few saying they were Satisfied. When it came to expectations, most respondents (approximately 90-95) claimed that the site was exceeding expectations, with the other users claiming that the site was meeting expectations. This shows that the system has not only met but, in most cases, exceeded the expectations of the users. The rates of retention and recommendations by the users were also very high. Virtually all the respondents said they would use the site again, with very few of them choosing the option of Maybe. On the same note, more than 95% of the users indicated that they would refer others to the platform, which showed high trust and satisfaction.

Overall, the post-survey results clearly show that the Open House Nepal platform has been successfully implemented, meeting user needs and validating the decisions made during the design and development phases. The most important elements, including the 360° virtual tour and the integration of maps, were especially welcome, which proved the results of the pre-survey and proved the usefulness of the system to overcome the real-life user issues.

Link to post-survey

3.5 Requirement Analysis

3.5.1 Functional Requirements

Functional requirements are the requirements that outline what the system should do. Following the survey, literature review, and supervisor recommendations, the following functional requirements of Open House Nepal were identified:

- The system should enable users to create accounts using email address and password, and log-in and out safely. In addition, the system should be able to have four roles: Visitor, Buyer, Seller, and Admin. Sellers should be able to create, modify and delete property listings (adding photos and supporting documents).
- Admin users should have the possibility to look through, accept, or deny property listings and uploaded documents. Only the approved listings should be shown to buyers and visitors.
- Authenticated customers should have the ability to post, update and cancel reviews of properties.
- The community should have the ability of the admin users to moderate the reviews and delete those that do not meet the community standards.
- Registered users should have the capability of posting, editing and removing comments on listing pages.
- Sellers should also have the capability to post 360-degree panorama images to integrate virtual tours
- The users should be able to filter through the properties based on location, price bracket, type of property and size.
- The system should give out real-time in-app notifications of listing changes, verification, and new reviews or comments.

3.5.2 Non-Functional Requirements

Non-functional requirements explain the behaviour of the system regarding performance, security, usability and scalability:

- Performance: Page load times are not supposed to take more than 3 seconds when there is normal load.
- Security: Before storing user passwords, it is necessary to encrypt them with bcrypt. JWT tokens must be used for session management. All API routes must be protected by authentication middleware.
- Usability: The interface should be mobile, tablet, and desktop friendly. Non-technical users should be able to use the system without training.
- Reliability: Uptime of the system should be 99%. Periodic database backup should be put in place.
- Scalability: The system architecture should accommodate future expansion of features and user base without necessarily re-designing the architecture.

3.6 Design

3.6.1 The Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) will be used as the hierarchical map of the whole Open House Nepal project to break down an intricate real estate platform into achievable and rational stages. The WBS allows considering all the technical needs, starting with the initial architecture design, to the final security optimization, by dividing the top-level goal into five main pillars: Project Planning, System Design, Development, Testing, and Documentation. Such systematic method enables the allocation of resources more efficiently and gives a clear picture of the lifecycle of the project, so that the development will be flawless in terms of matching the original project objectives (Sommerville, 2016).

During the Development stage (Section 3), you have been able to deploy a sprint-based approach to developing the core modules of the platform. This will involve the creation of a secure User Authentication system (Sprint 1), a powerful Property Listing module (Sprint 2) that will support the necessary document uploads (Lal Purja, Citizenship) and the interactive features (Reviews, Comments and 360-degree Property Viewer) (Sprints 3-5). Finally, the Testing and Optimization phase (Section 4) test the whole system with intense unit, integration, and security testing to ensure that the product is not only rich in features but also performant and secure to real-life real estate transactions (IEEE, 2014).

The Work Breakdown Structure (WBS) is in appendix 8.4.1

3.6.2 Gantt Chart

A Gantt Chart is a graphical project management tool that depicts a project schedule within a given time frame. It presents a list of tasks on the vertical axis and time intervals on the horizontal axis in the form of horizontal bars, which indicate the date of start, the duration of the task and the date of end of the task respectively. This enables project managers to observe the progress of activities that overlap, the critical path of the project, and all dependencies, including the need to have the backend API completed prior to beginning the frontend UI, are handled effectively to meet the ultimate deadline (Pressman and Maxim, 2020).

In the case of the Open House Nepal project, the Gantt chart illustrates a step-by-step, systematic implementation of the development lifecycle. It points out the successful transition between Requirements and Design phases (Project Planning and System Foundation) and a high-intensity Sprints-based Development phase. As the chart indicates, as the backend was being developed, parallel frontend modules such as the Property Listing and 360-degree Viewer were being developed, to ensure maximum time efficiency. Lastly, the chart also demonstrates the last phases of the intensive Testing and Bug Fixing, then the last phase of Documentation and Submission, which clearly demonstrates that the project was completed within the deadline and as per the initial plan (PMI, 2021). Gantt Chart is in appendix 8.4.2

3.6.3 Use Case Diagram

The Use Case Diagram illustrates the interactions between the different types of users and the system's functionalities. The actions available to Visitor/Registered Users include user login/register (checking credentials and checking the user database), view image gallery and use the 360-degree tour option (Sommerville, 2016).

Property Owner/Sellers can post property listings (including posting property details, authenticating property data and storing property data), and participate in the review and comment system. Admins can control users (verify and see users), system reports (property and user reports, daily activity reports), and control the reviews and comments. Payment Gateway is an outside party that communicates with the payment module (Laudon and Traver, 2021).

Use Case Diagram in appendix 8.4.3

3.6.4 Entity Relationship Diagram (ERD)

The ERD establishes the structure of data of the application. These are the fundamental entities and their associations. User entity contains all the user account data such as name, email, password hash, phone, user type and verification status. The Property entity contains all the listing data and is many-to-one to the Property Owner (Pressman and Maxim, 2020).

Image entity is where the photo URLs are stored and it is a many-to-one relationship with Property. Document entity is the metadata of a document and is many-to-one related to Property. Review entity contains the ratings and review text and has many-to-one relationships with Property and User. The Comment object contains the text of comments and is many-to-one related to Property and User (Garrett, 2011).

Notification entity contains notification messages and is many-to-one related to User. Search Log is a database that tracks search queries of users. The Favourite entity contains the properties of individual users that are saved (Pressman and Maxim, 2020).

Entity Relationship Diagram (ERD) in appendix 8.4.4

3.6.5 Data Flow Diagram (DFD)

DF Level 0 (Context Diagram) presents Open House Nepal as a single system, which takes inputs of Buyers (search queries, registration data, reviews and comments), Sellers (property listing and documents), and Admins (verification decisions and moderation actions). The system results are property search results to Buyers, listing management feedback to Sellers and system reports to Admins (Sommerville, 2016).

The DFD Level 1 breaks the system down into the following processes: User Authentication, Property Listing Management, Document Verification, Search and Filter, Review and Comment Management and Notification Management. The following data stores are the User Database, Property Database, Document Store, Review Database, Comment Database, and Notification Queue (Elmasri and Navathe, 2016).

Data Flow Diagram (DFD) in appendix 8.4.5

3.6.6 Flowchart

A flowchart is a diagrammatic illustration of a system or a process which involves the use of symbols, arrows, and shapes to illustrate the order in which steps are taken and where decisions are made. It assists in breaking down complicated processes through a clear demonstration of how tasks are undertaken at the beginning and the end and therefore, it is easier to comprehend, examine and convey system activities (Pressman and Maxim, 2020; Sommerville, 2016).

The provided flowchart is a visualization of the working process of the Open House Nepal platform, where the user access is the starting point and is then followed by registration or login, and then role identification (Buyer, Seller, or Admin). All roles have certain functions to execute, including browsing properties, listing, uploading documents, and listing verification or approval. The system will guarantee an organized and safe procedure of property transactions with the verification of the admins being a major aspect in ensuring trust and authenticity (Dennis, Wixom and Roth, 2019; Sommerville, 2016).

Flowchart Diagram in appendix 8.4.6

3.6.7 System Architecture Diagram

The open house Nepal system architecture is based on client-server architecture with distinct frontend, backend and database layers. The frontend is a single-page application written in React.js and is executed in the browser of the user. It talks to the backend by making RESTful API calls over HTTPS. The backend is an Express.js/Node.js application which does all business logic, authentication and any database operations (Sommerville, 2016).

SQL lite database is also hosted on the same server with the backend. Outside services such as email (through Nodemailer/SMTP) and map services are included in the backend. Another layer in the system is a document storage layer that contains property photos and uploaded documents (Pressman and Maxim, 2020).

System Architecture Diagram in Appendix 8.4.7

3.6.8 UI

The layout, structure and user flow of the Open House Nepal platform were planned to use UI in Figma before development. The design is clean and has a uniform teal colour scheme (#4DB6AC) throughout the pages.

The Landing Page has a big hero section consisting of a background image and a search bar at the center where one can search properties. At the bottom of it, one can find a section of recent property listings that are represented in a grid format with the property cards containing images, price, and basic information. Other sections like location highlights and informational content are also provided and finally a footer.

The Property Listing Page has several properties in grid format with a side filter panel. The filtering feature enables users to narrow down the search results by factors like price, location, and type of property. All property cards have a picture, price, and some small information with a possibility to get more details (Figma; Nielsen, 2012).

The Property Detail Page shows comprehensive details of a chosen property. It features a huge gallery of images on the top, then property description, features and price. Section of agent/seller information and other property related information such as floor plans are also included. The design is organized in such a way that it provides the user with a total view of the property within a page (Sommerville, 2016).

UI in appendix 8.4.8

3.7 Implementation

3.7.1: 360° Virtual Tour

The 360° panorama viewer in action.

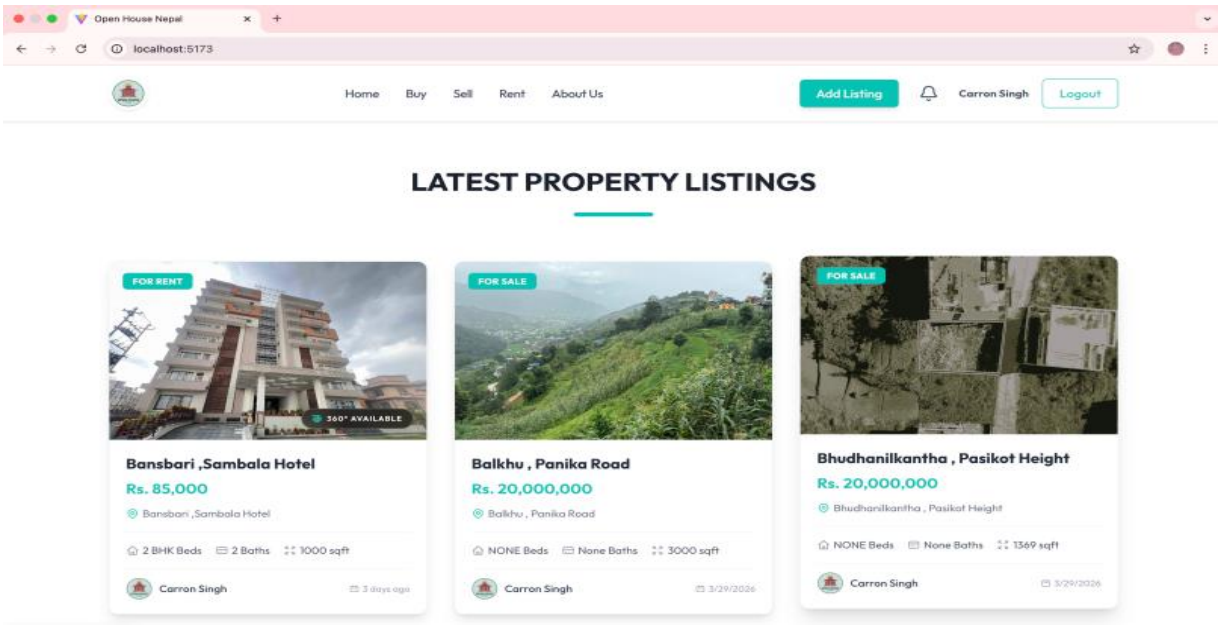


Figure 9 360° Virtual Tour

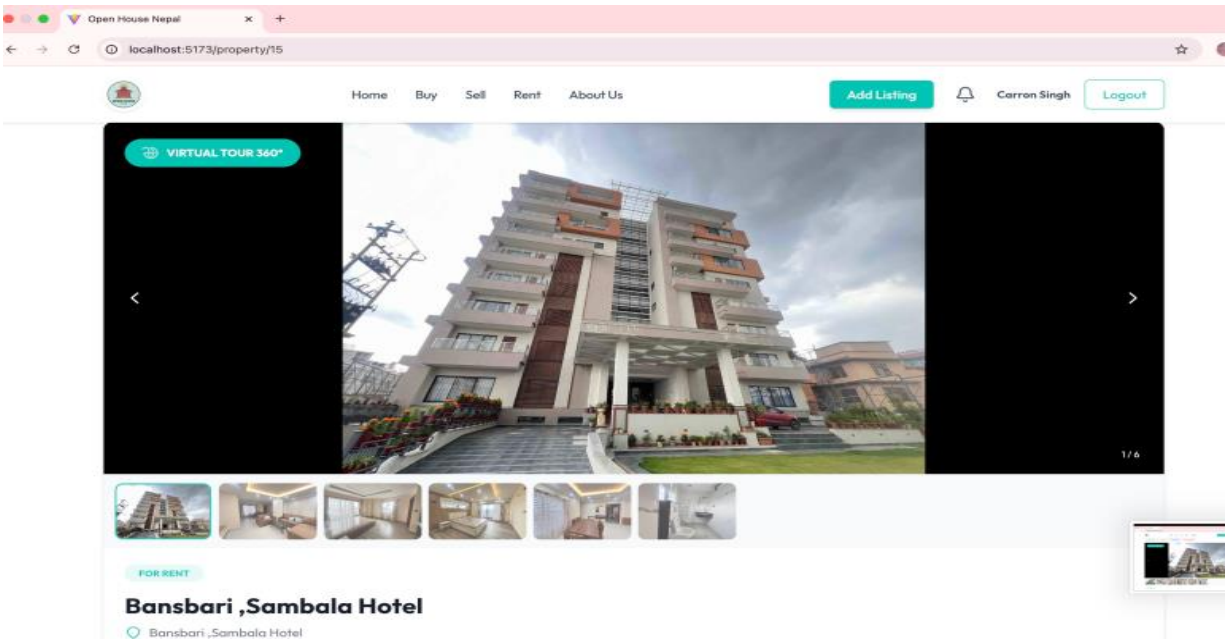


Figure 10 360° Virtual Tour

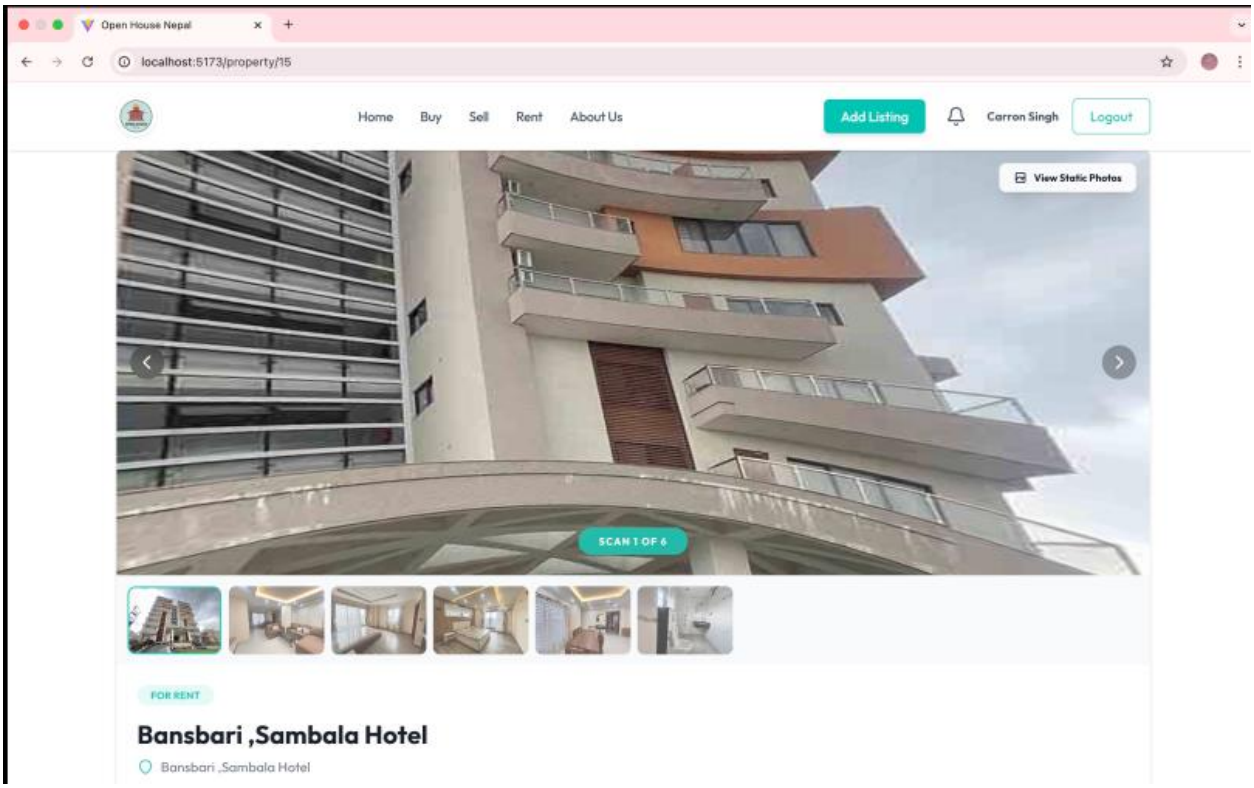


Figure 11 360° Virtual Tour

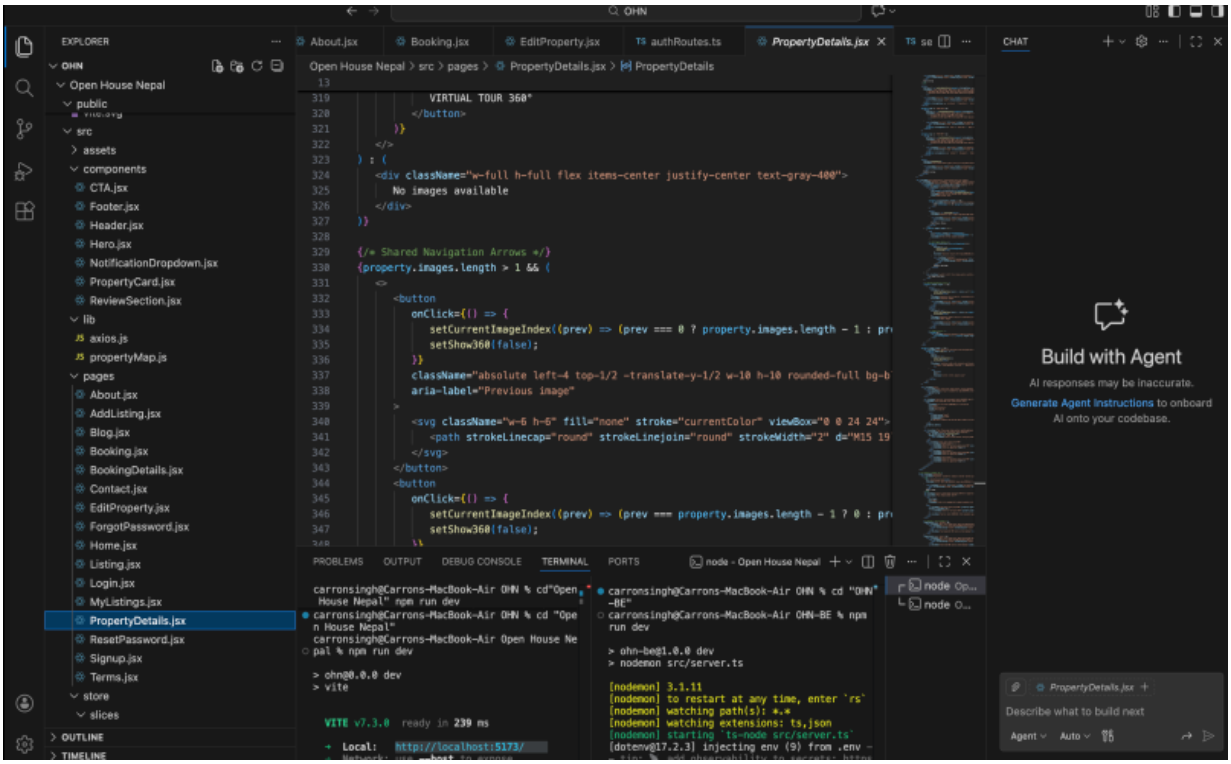


Figure 12 360° Virtual Tour Code

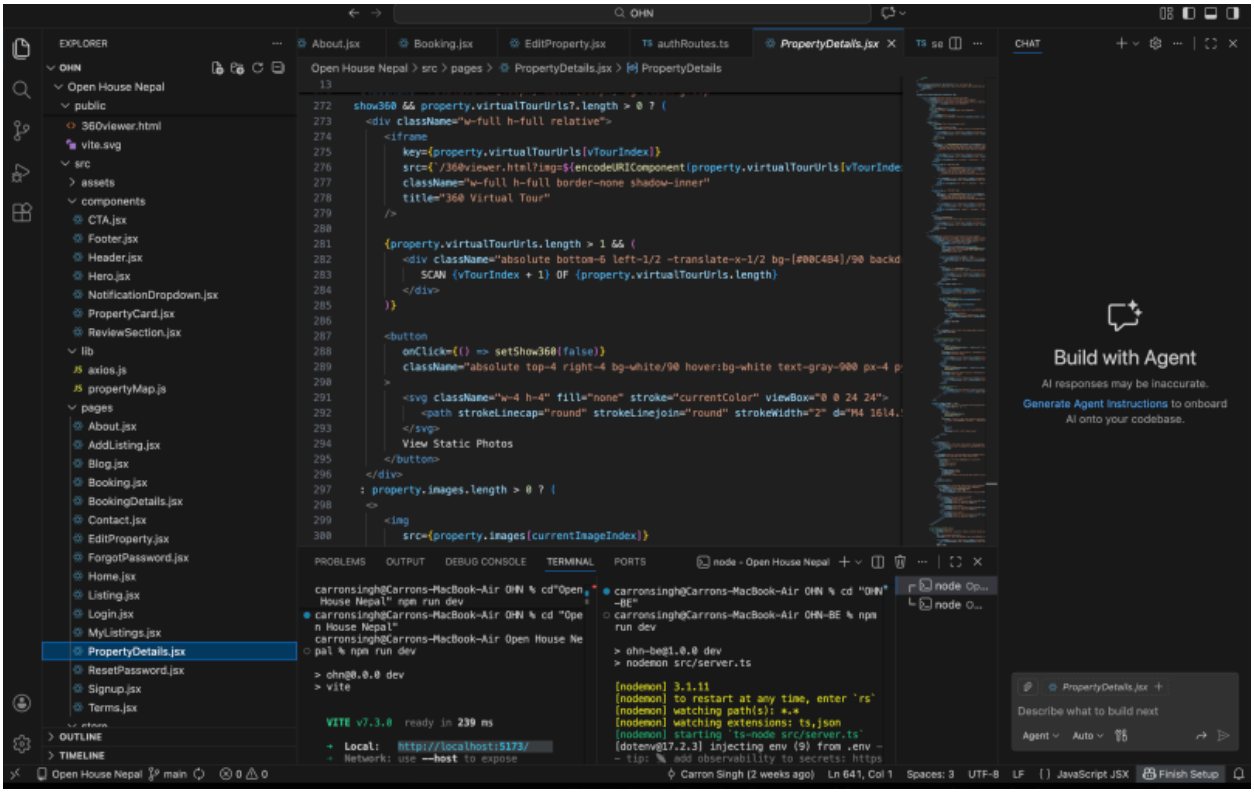


Figure 13 360° Virtual Tour Code

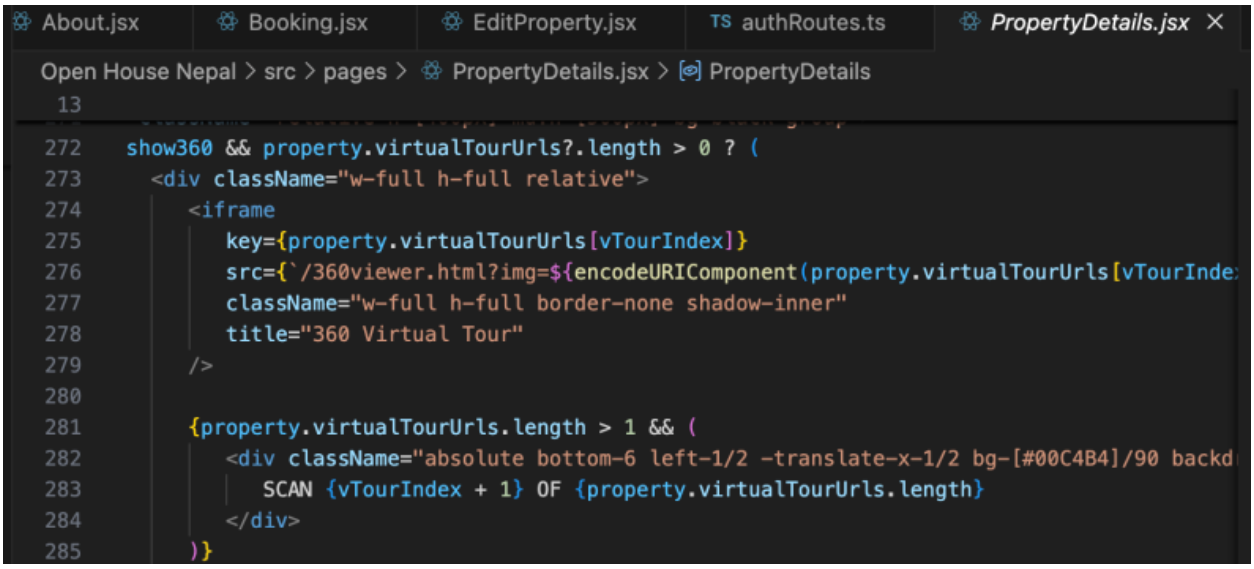


Figure 14 360° Virtual Tour Code

Feature name: Immersive Virtual Property Exploration.

To make the viewing of the property more interactive, I added a 360-degree Virtual Tour option that effectively enables potential buyers and renters to see the inside of a property in all its details, without the necessity of physically visiting the location. "Technically, it will be an isolated iframe with a custom-built viewer component that renders high-resolution panoramic images and allows the user to have a realistic experience of the property.

3.7.2: Interactive Map Integration

The Google Map on the Property Details page showing the property location.

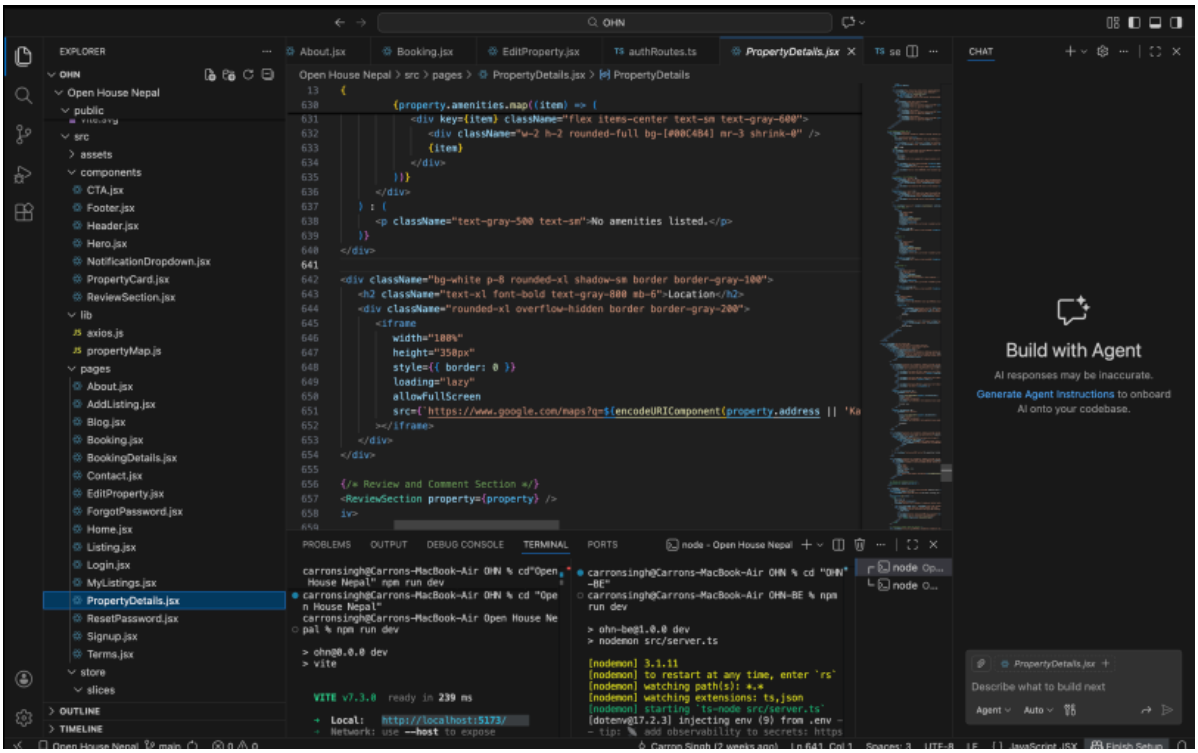


Figure 15 Interactive Map Integration Code

```
<div className="bg-white p-8 rounded-xl shadow-sm border border-gray-100">
    <h2 className="text-xl font-bold text-gray-800 mb-6">Location</h2>
    <div className="rounded-xl overflow-hidden border border-gray-200">
        <iframe
            width="100%"
            height="350px"
            style={{ border: 0 }}
            loading="lazy"
            allowFullScreen
            src={`https://www.google.com/maps?q=${encodeURIComponent(property.address ||
        ></iframe>
    </div>
</div>
```

Figure 16 Interactive Map Integration Code

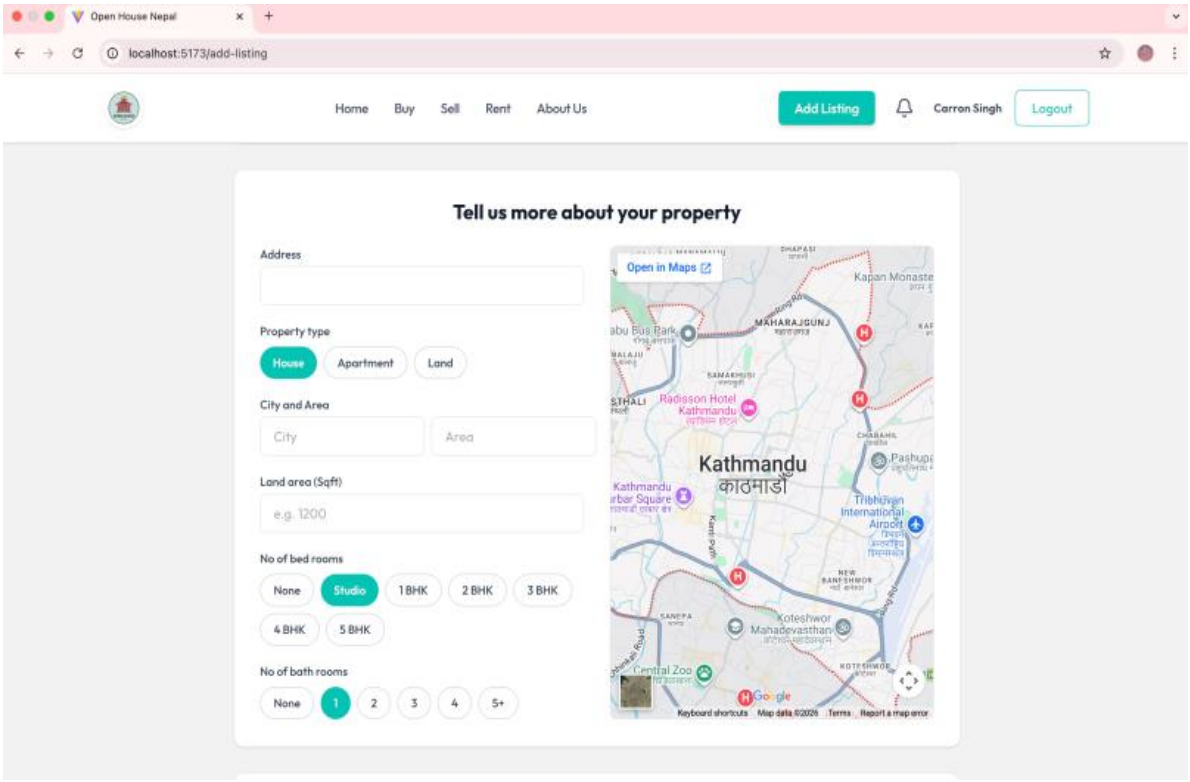


Figure 17 Interactive Map Integration

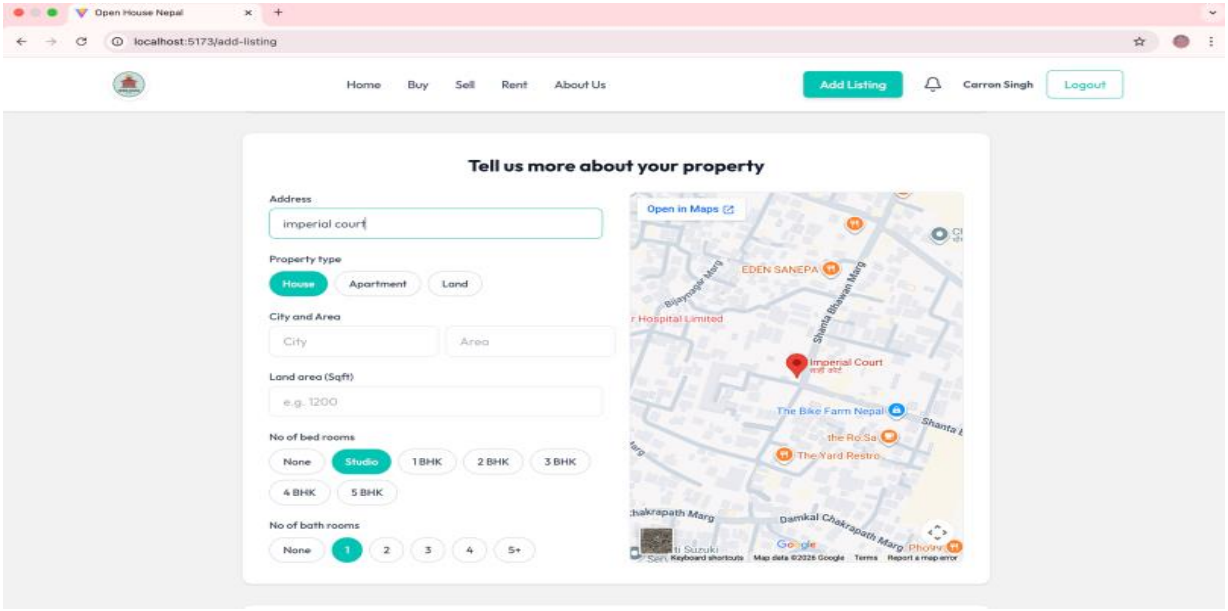


Figure 18 Interactive Map Integration

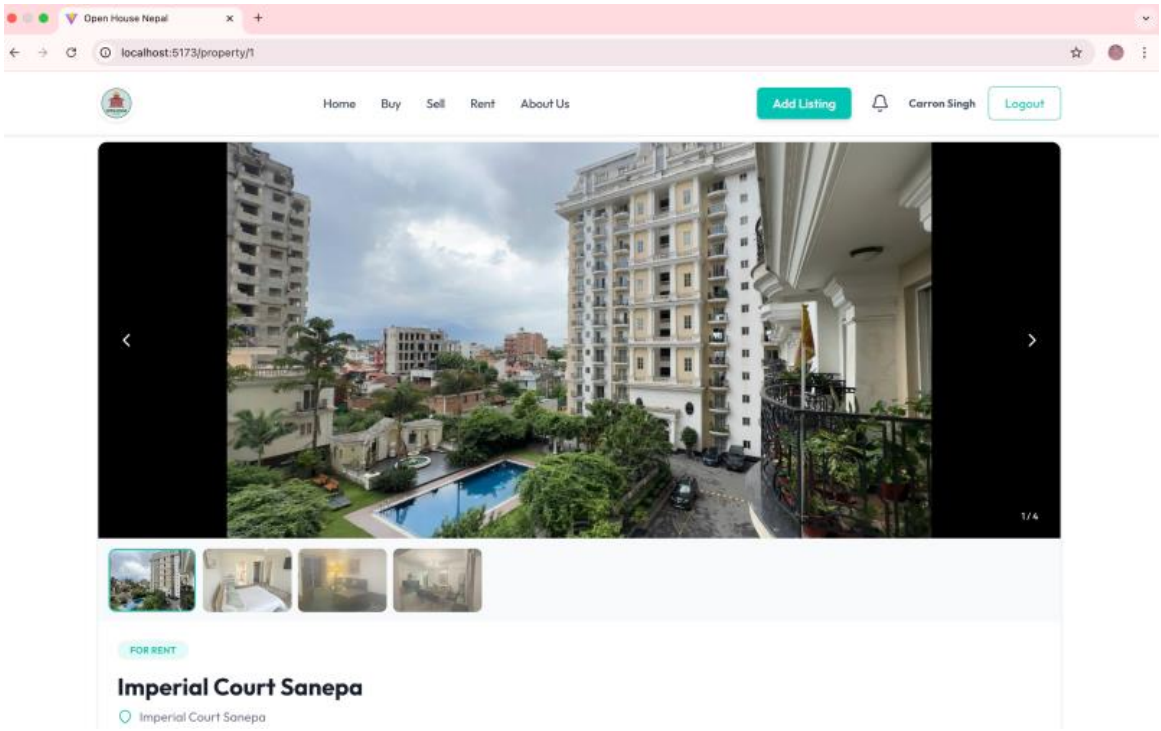


Figure 19 Imperial Court Sanepa

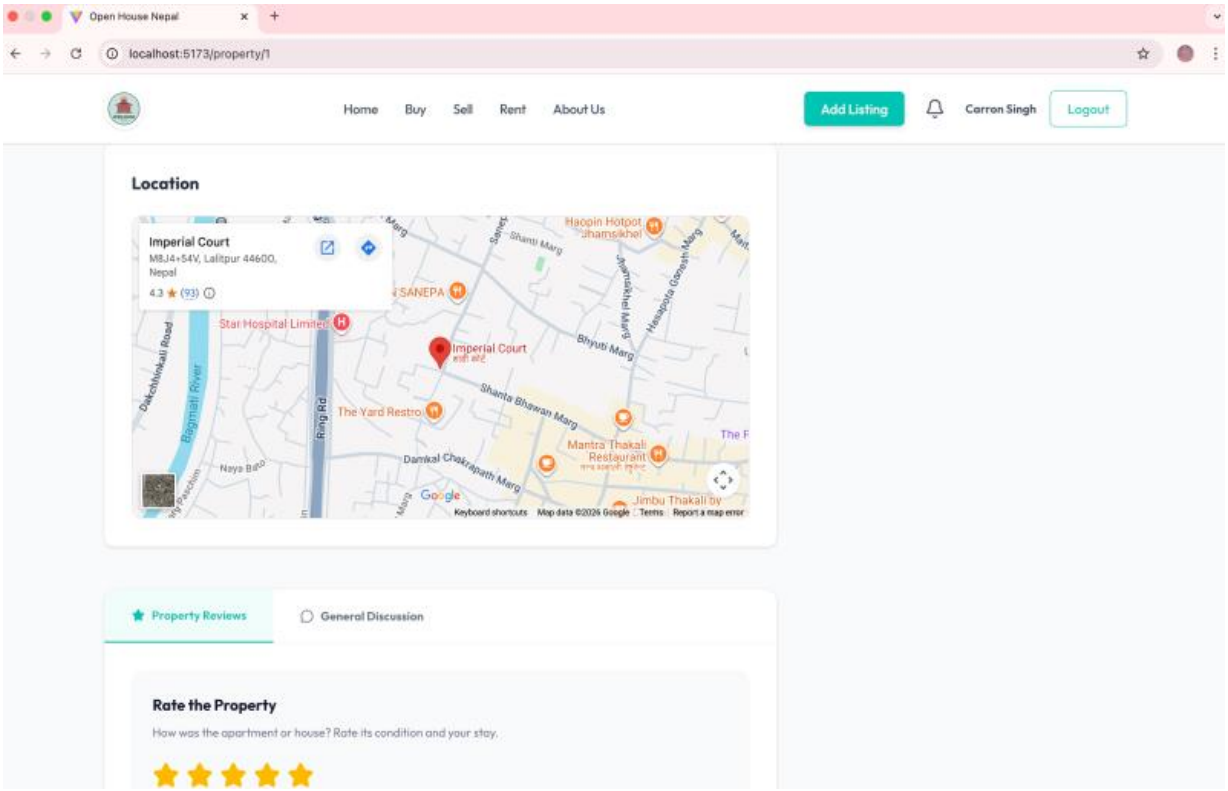


Figure 20 Imperial Court Sanepa

Feature name: Geospatial Data Integration via Google Maps API.

The platform is equipped with an interactive map system, based on the Google Maps API. On each listing, the system automatically geocodes the address provided and displays an accurate location marker on a dynamic map interface. This type of geospatial search experience is more contextual compared to a search using pure text addresses.

3.7.3:Reviews & Ratings

The Reviews section at the bottom of a property page with star ratings.

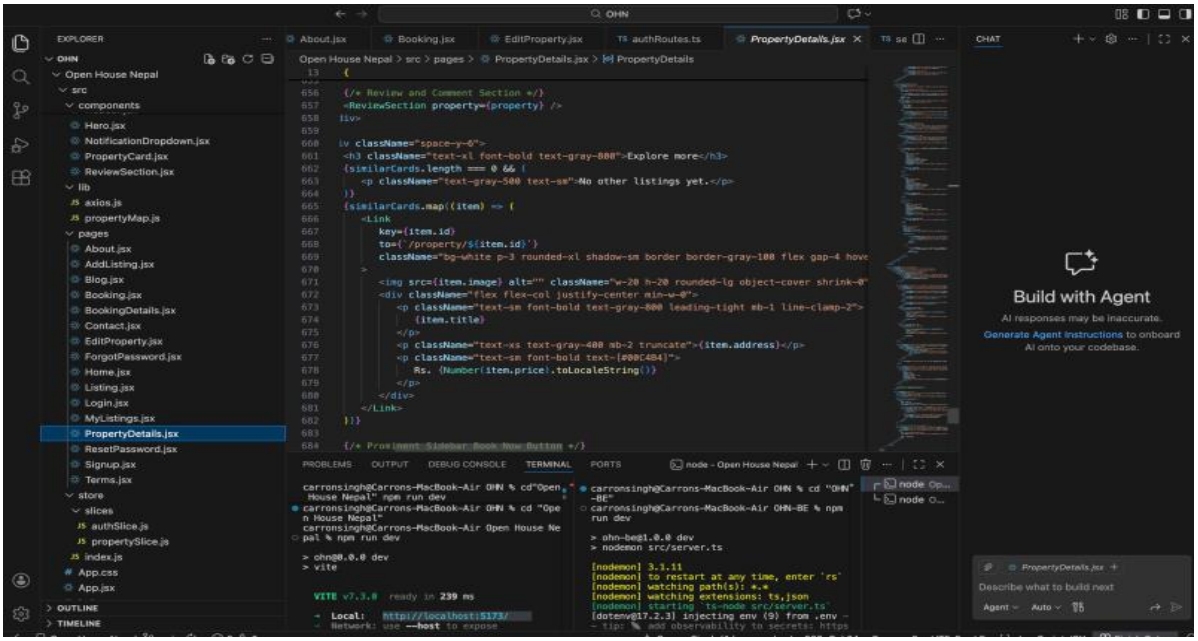


Figure 21 Reviews & Ratings Code

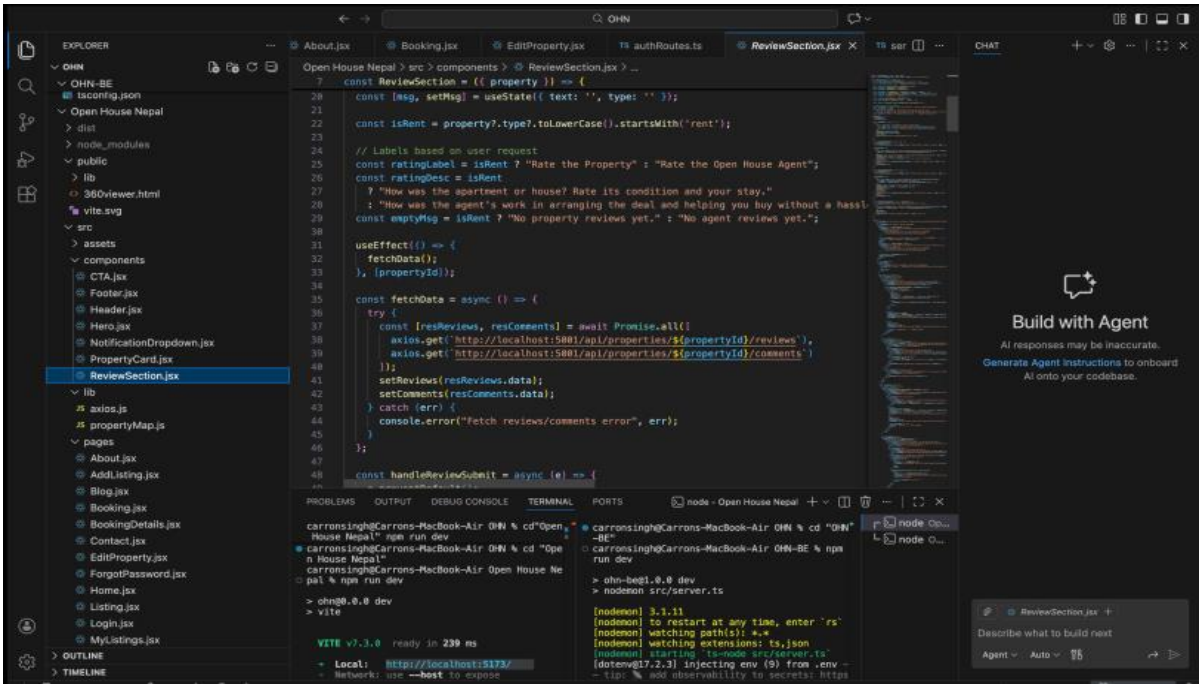


Figure 22 Reviews & Ratings Code

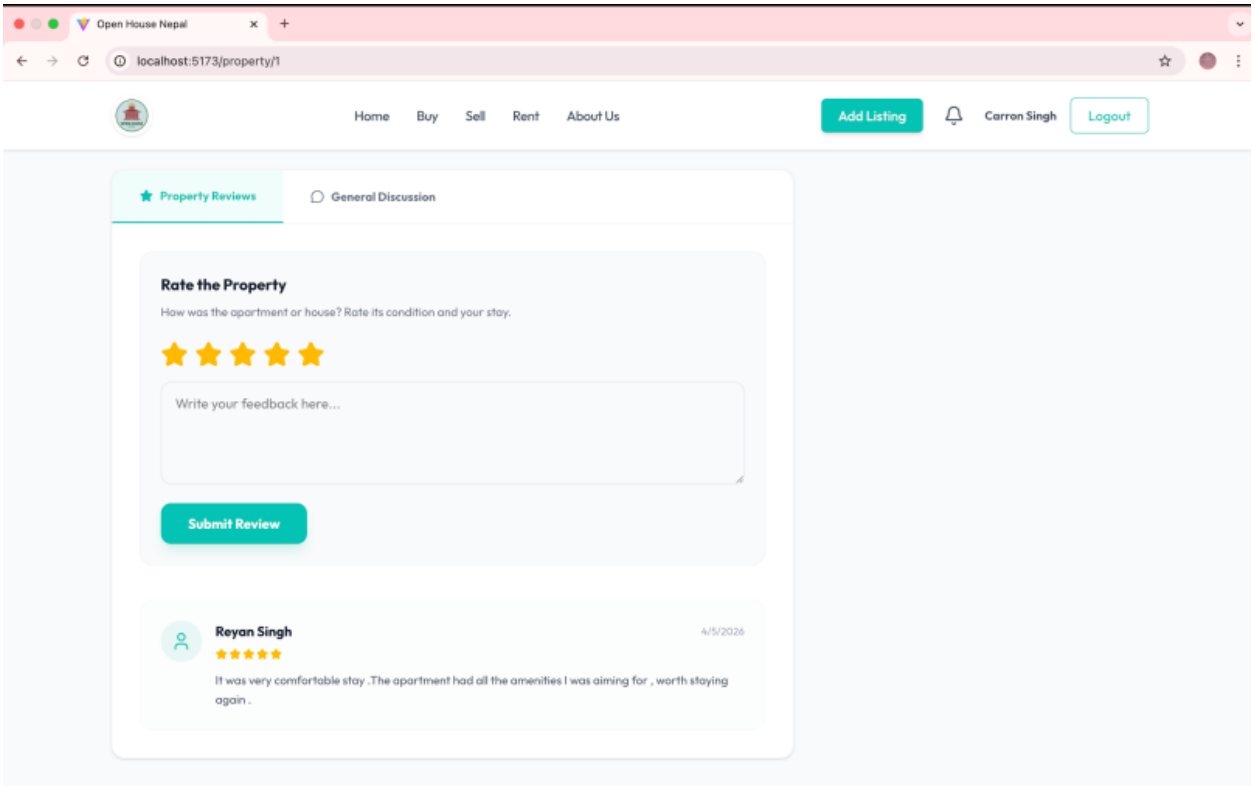


Figure 23 Reviews & Ratings

Feature name: User Feedback and Trust System.

They introduced a strong User Feedback and Trust System that allowed building credibility in the marketplace, wherein verified users can leave a badge of an average rating, and all the reviews can be stored and recovered efficiently with relational tagging.

3.7.4: Property Viewing Booking Feature

The Book a Viewing form or the Booking page.

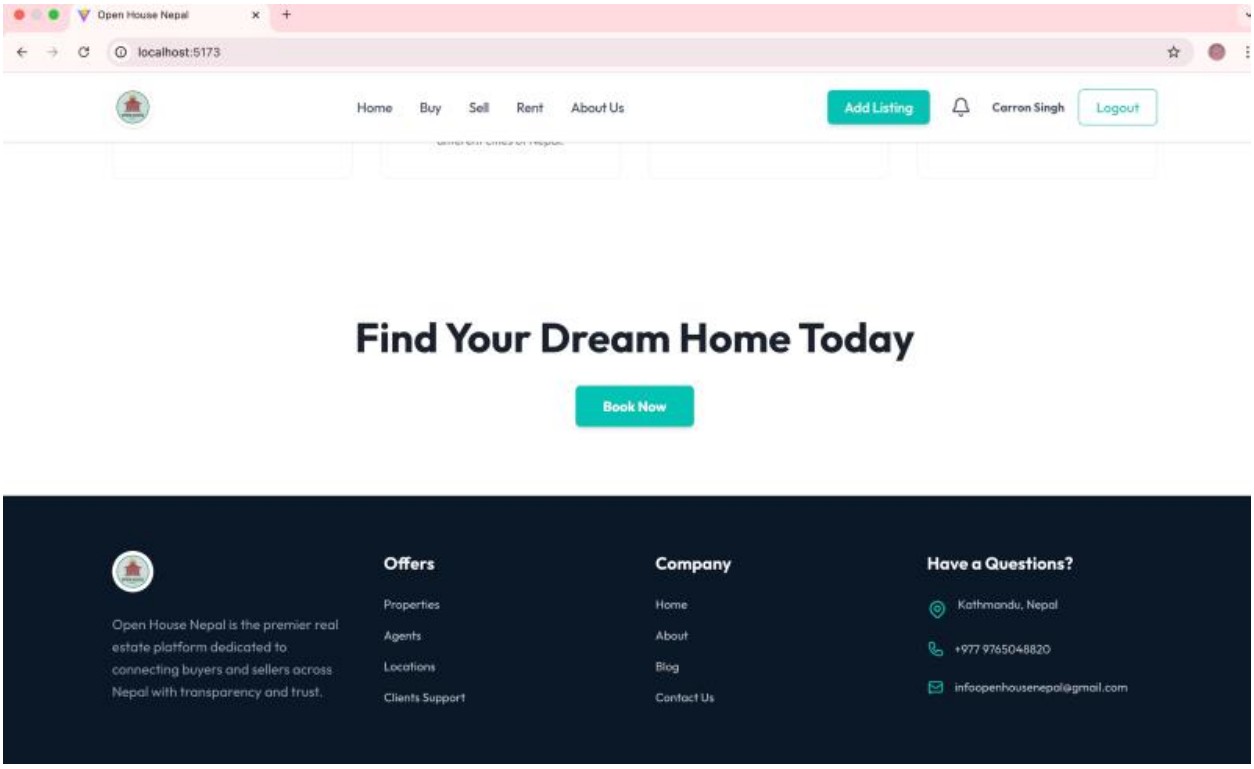


Figure 24 Property Viewing Booking Feature

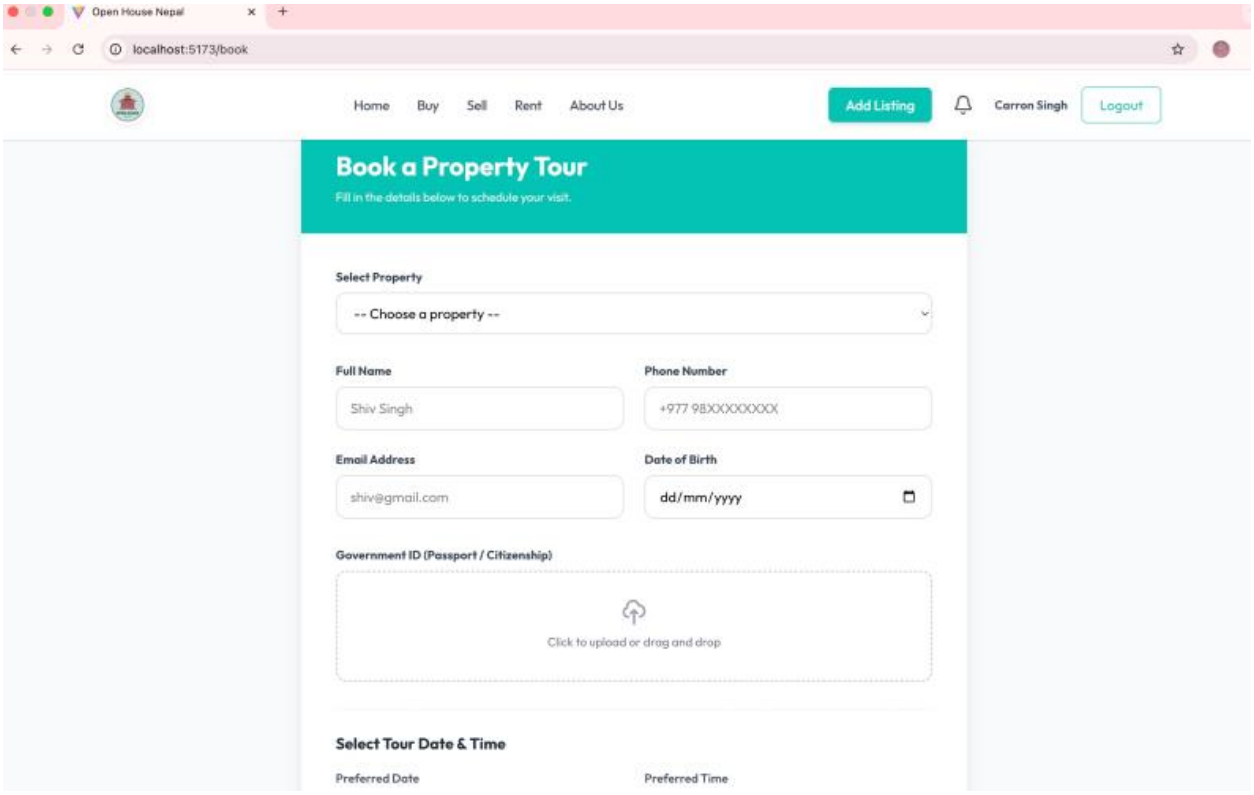


Figure 25 Property Viewing Booking Feature

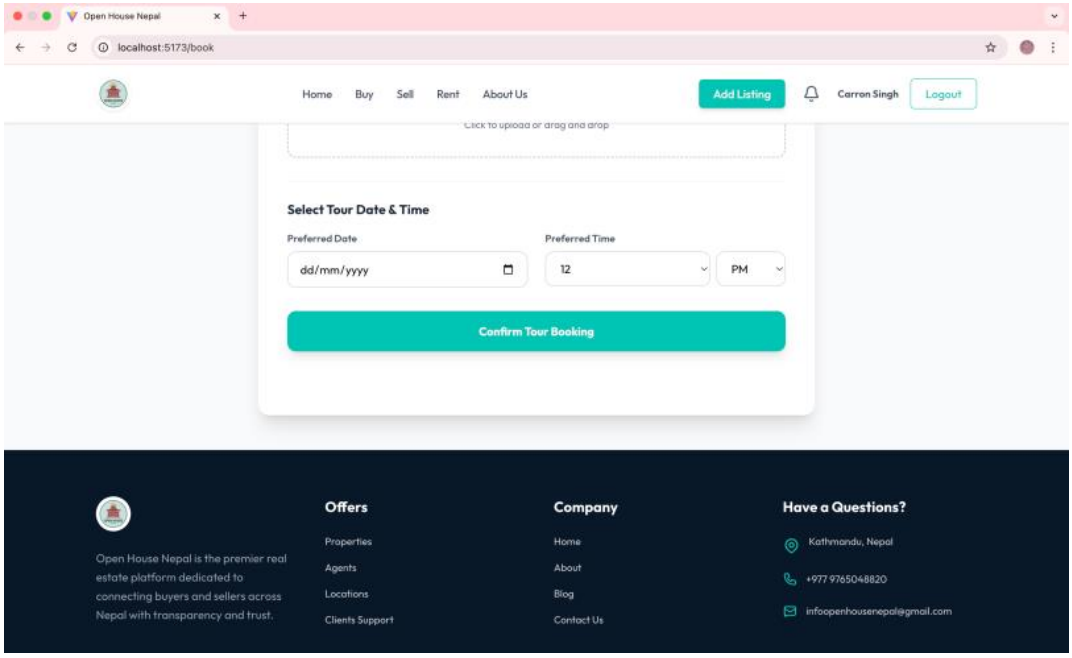


Figure 26 Property Viewing Booking Feature

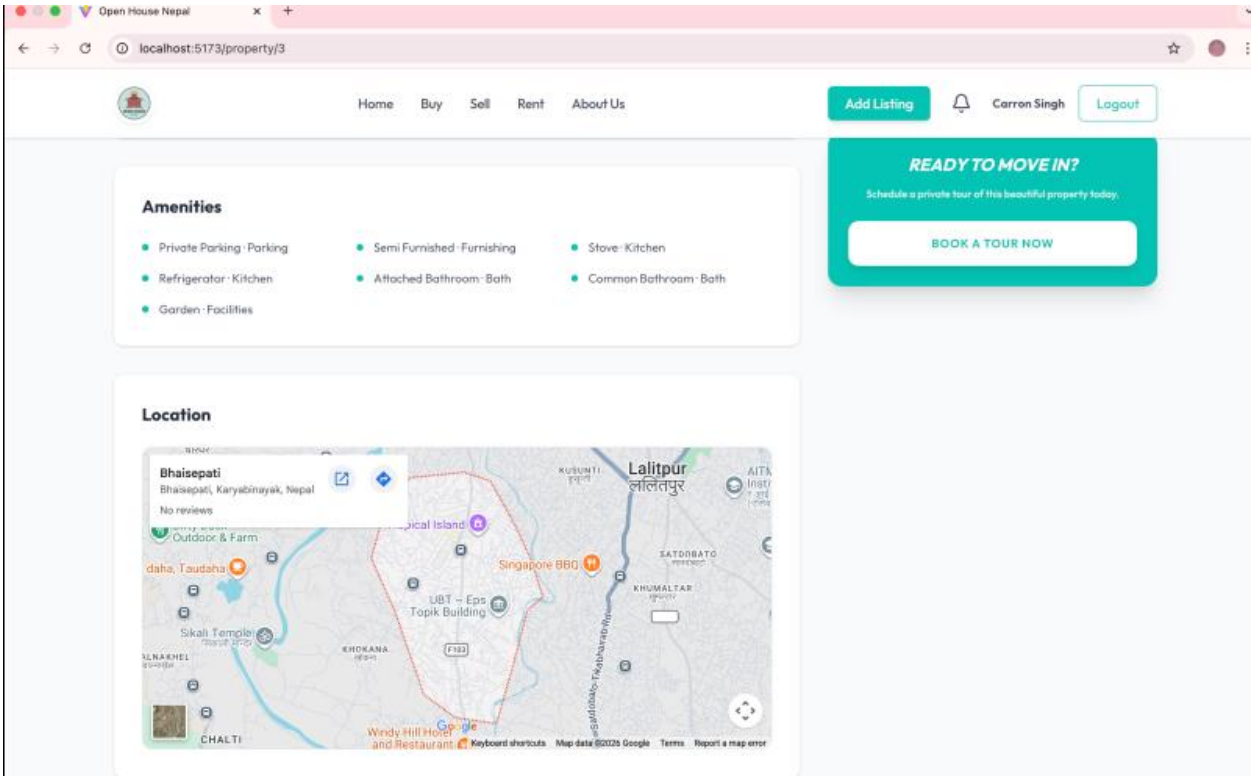


Figure 27 Property Viewing Booking Feature

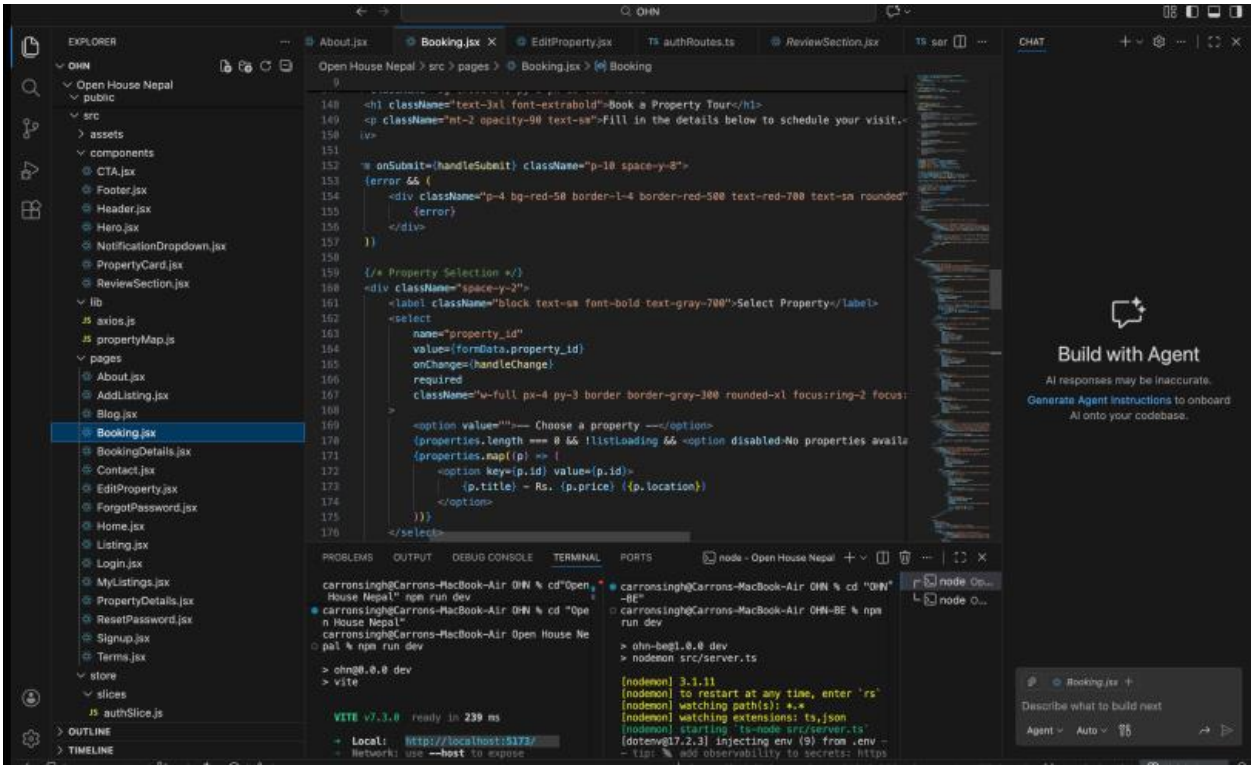


Figure 28 Property Viewing Appointment Management Code

Feature name: Property Viewing Appointment Management.

The Property Viewing Appointment Management module is a module simplifying site visit scheduling, enabling interested users to complete a special booking form on the listing page, specifying their desired date and time, with Multer (verified on the backend) verifying the information.

3.7.5: Real-time Notifications

The Notification bell icon open, showing recent alerts.

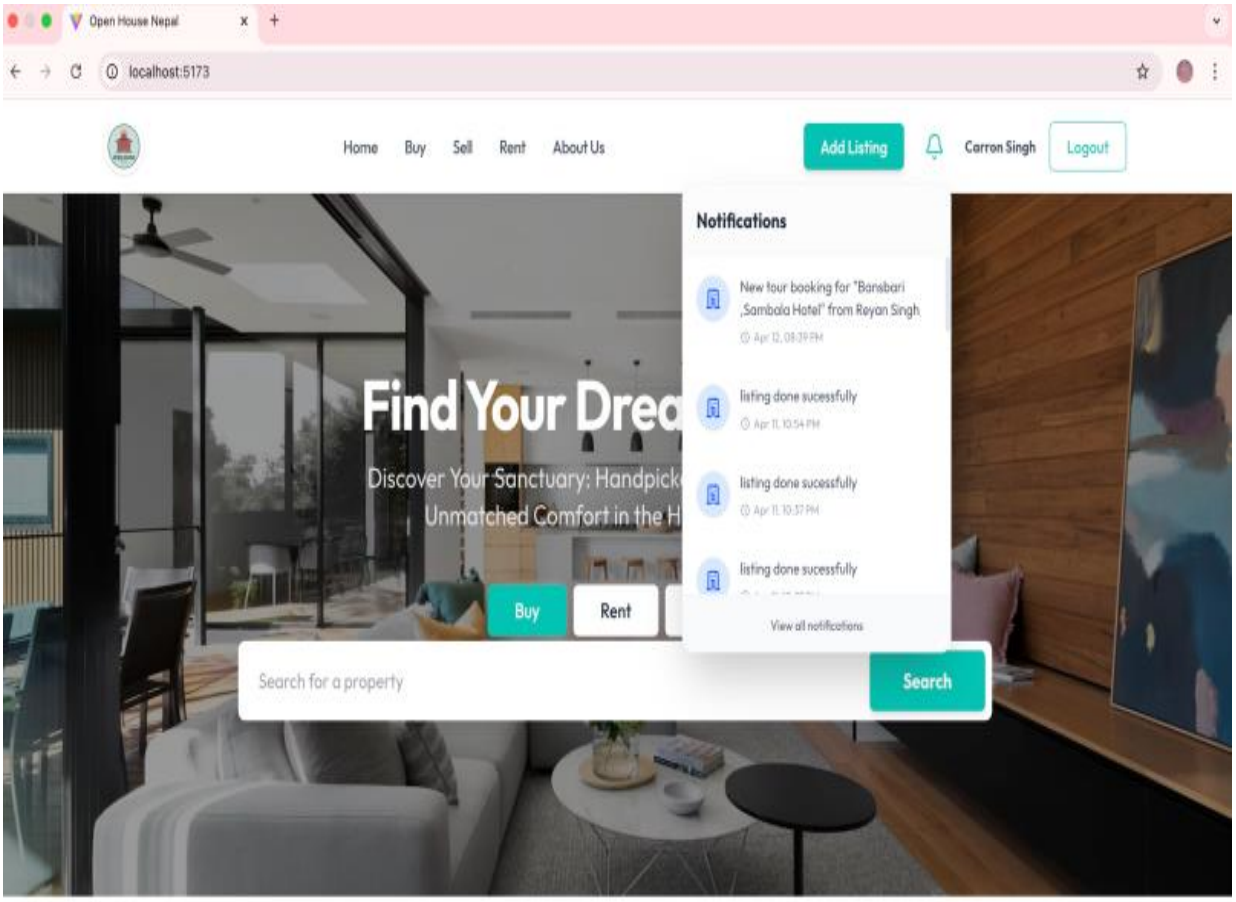


Figure 29 Real-time Notifications

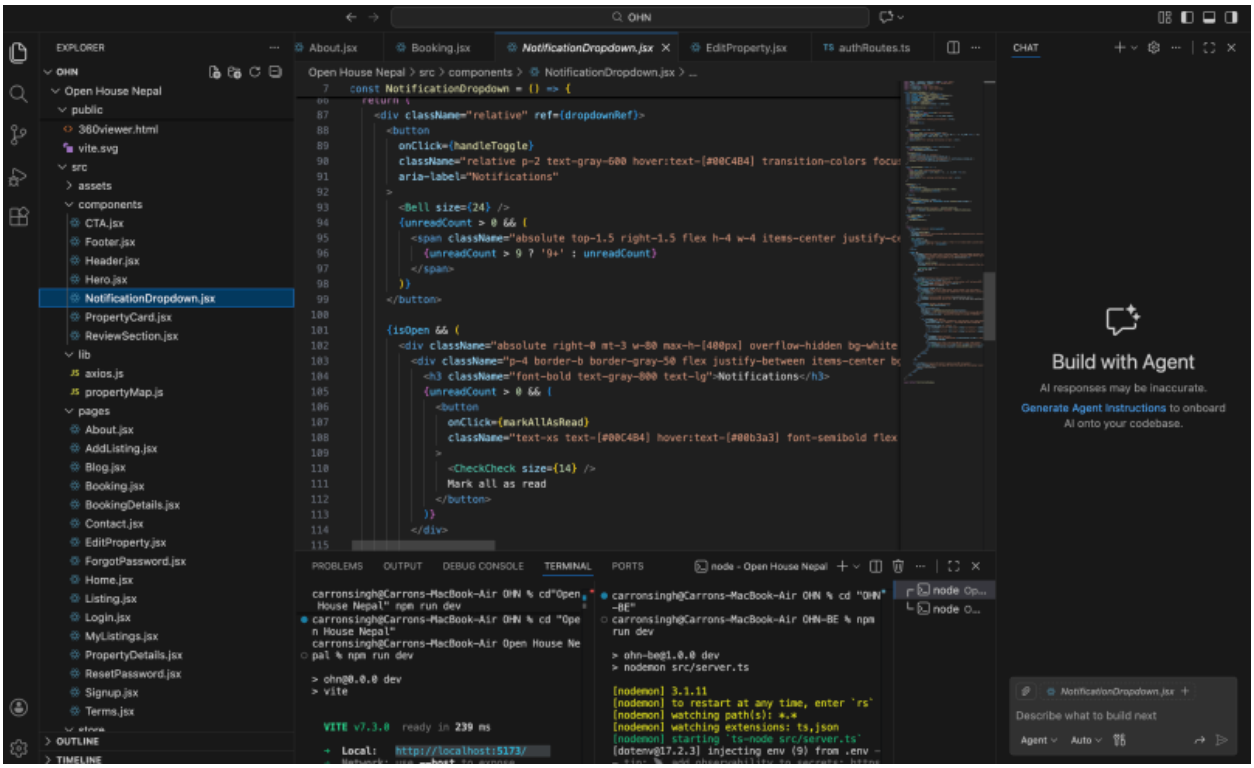


Figure 30 Real-time User Activity Alerts Code

Feature name: Real-time User Activity Alerts.

To maintain user interest and stay continually updated, I created a system of Real-time User Activity Alerts. This system uses a notification architecture based on polling that checks the backend to identify new events every 30 seconds and, as a result, users can see relevant alerts; e.g., a red badge with unread messages on the notification bell immediately when a new inquiry is posted on their property or a booking request has been approved.

CHAPTER 4: TESTING AND ANALYSIS

4.1 UNIT TESTING, TEST PLAN

Unit testing is a software testing methodology that involves evaluating individual system units or components separately to make sure they carry out their intended tasks as planned. Unit testing in this project mostly concentrates on controller functionality and backend API endpoints. Before being integrated with the frontend, each feature such as user registration, authentication, and data retrieval is tested separately. This procedure guarantees the accuracy and dependability of the system's primary business functions and aids in the early detection of logic problems.

Test No.	Test Name	What it Tests
Test 1	User Registration API Test	Validates that the system correctly hashes passwords and saves new user records in the database.
Test 2	User Login API Test	Verifies that the system authenticates credentials and generates a valid JWT security token.
Test 3	Property Listing Retrieval Test	Ensures the API can successfully fetch and return an array of all properties stored in the database.
Test 4	Single Property Detail Retrieval	Tests if a specific property can be retrieved using its unique ID for the "Details" view.
Test 5	Create Property Listing API	Validates Multipart/Form-Data handling for both property text details and file uploads (Lalpurja/ID).
Test 6	Property Information Update	Ensures that authorized users can modify specific fields (like price) of an existing property record.
Test 7	Delete Property API Test	Verifies that the system removes a record from the database and checks for proper user authorization.

Table 3 UNIT TESTING

4.1.1 Test 1: User Registration API Test

4.1.1.1 Authentication Module - User Registration

This test verifies that the system correctly accepts a new user's information (Name, Email, Password, Role) and successfully saves it into the database, returning a 201 Created status. Test Type: Unit Testing / API Functional Testing.

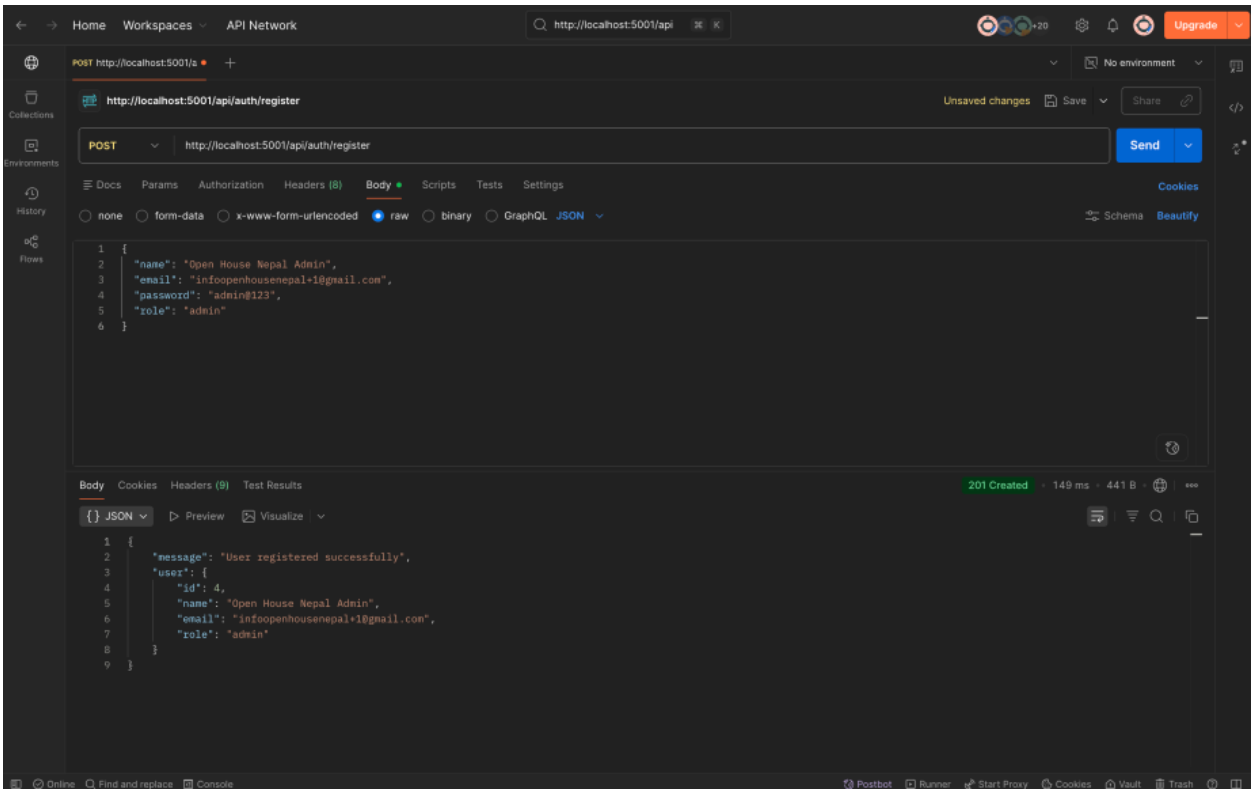


Figure 31 User Registration API Test

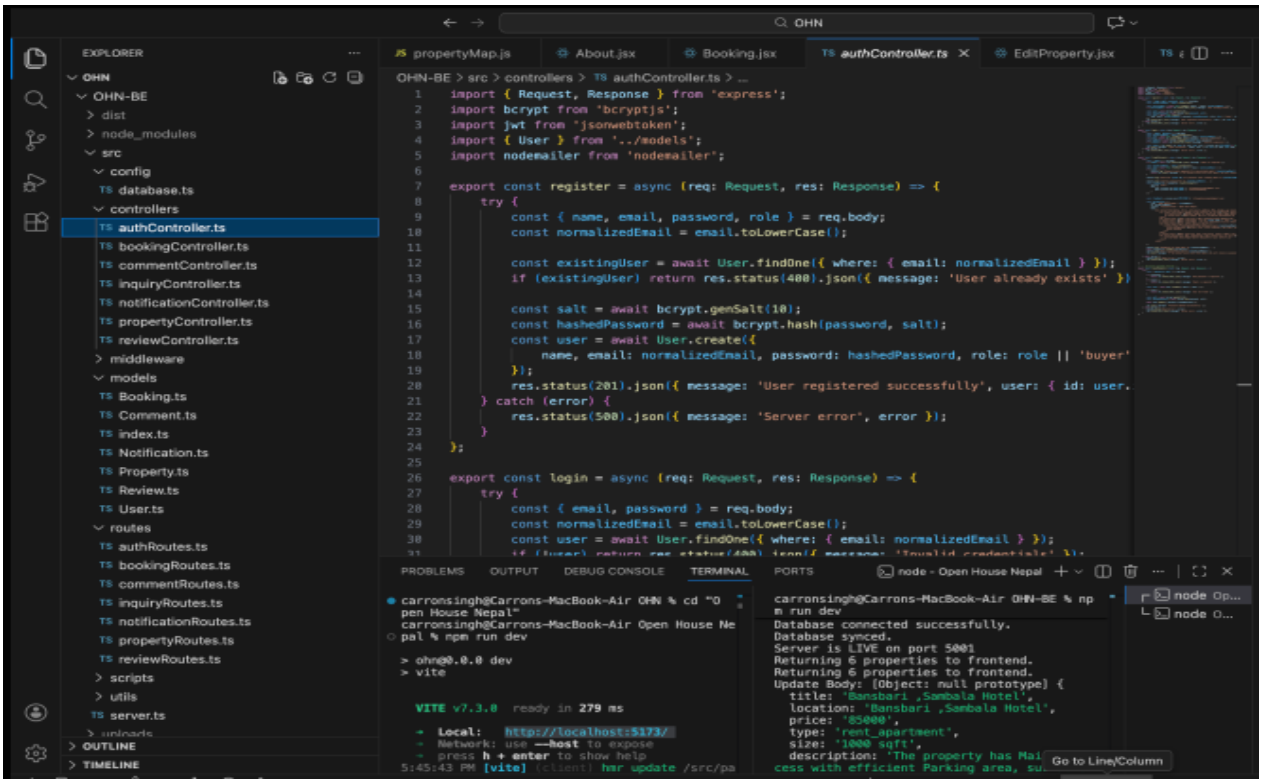
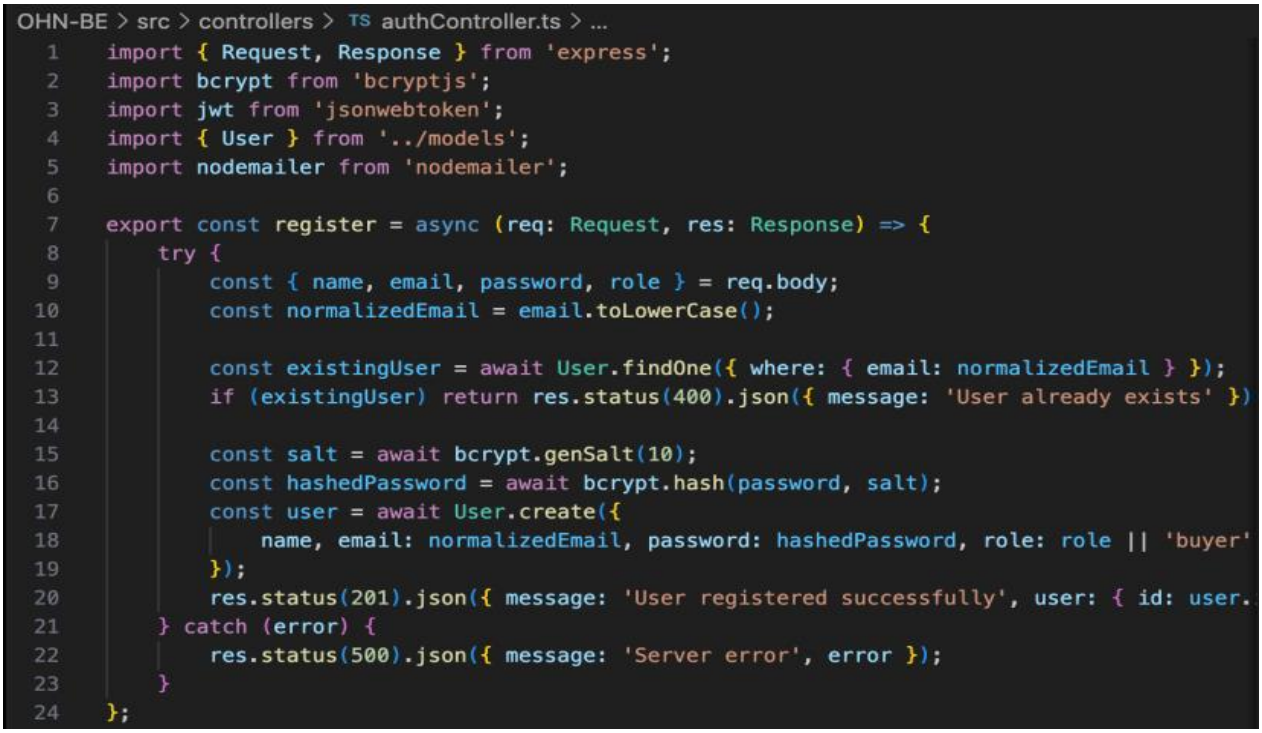


Figure 32 User Registration API Test



```
OHN-BE > src > controllers > TS authController.ts > ...
  1   import { Request, Response } from 'express';
  2   import bcrypt from 'bcryptjs';
  3   import jwt from 'jsonwebtoken';
  4   import { User } from '../models';
  5   import nodemailer from 'nodemailer';
  6
  7   export const register = async (req: Request, res: Response) => {
  8       try {
  9           const { name, email, password, role } = req.body;
 10           const normalizedEmail = email.toLowerCase();
 11
 12           const existingUser = await User.findOne({ where: { email: normalizedEmail } });
 13           if (existingUser) return res.status(400).json({ message: 'User already exists' })
 14
 15           const salt = await bcrypt.genSalt(10);
 16           const hashedPassword = await bcrypt.hash(password, salt);
 17           const user = await User.create({
 18               name, email: normalizedEmail, password: hashedPassword, role: role || 'buyer'
 19           });
 20           res.status(201).json({ message: 'User registered successfully', user: { id: user.
 21       } catch (error) {
 22           res.status(500).json({ message: 'Server error', error });
 23       }
 24   };
```

Figure 33 User Registration API Test

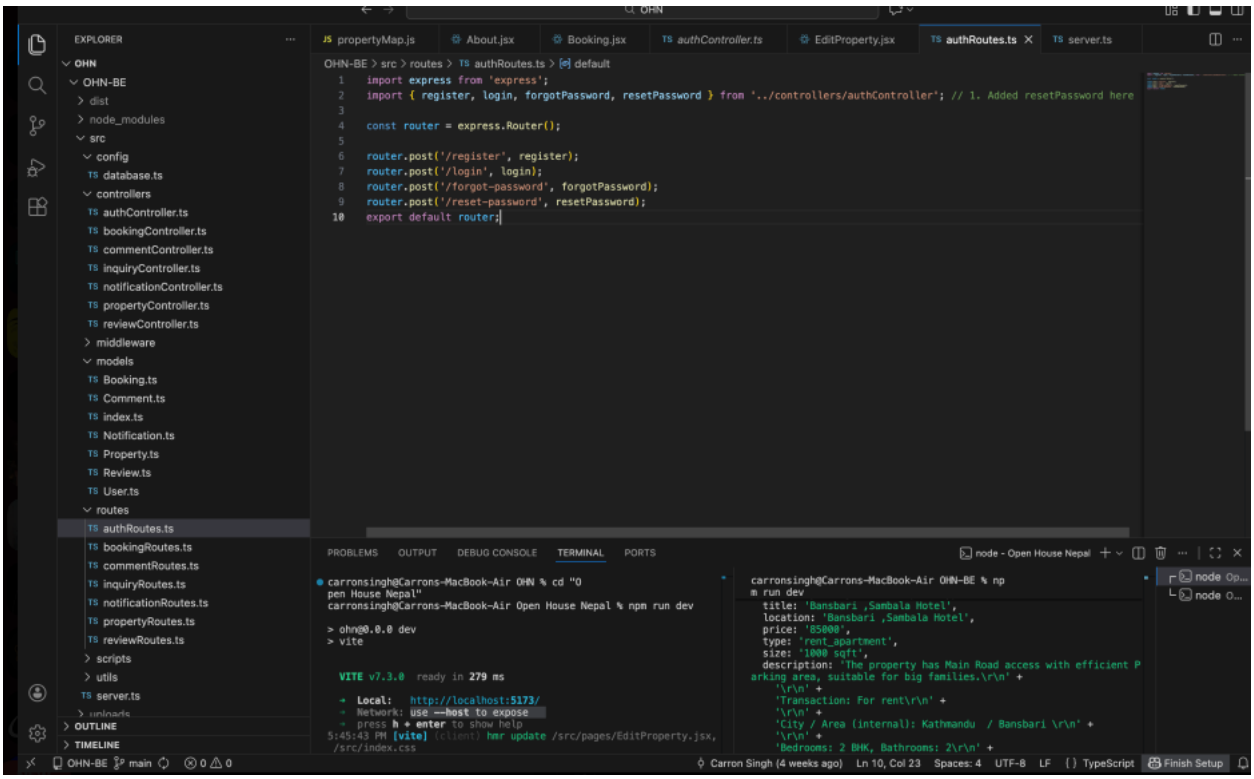


Figure 34 User Registration API Test

4.1.2 Test 2: User Login API Test

4.1.2.1 Authentication Module - User Login

This test verifies that the system correctly authenticates a user with valid credentials and generates a secure JWT Token for session management. Expected Result: 200 OK status and a valid token string in the response.

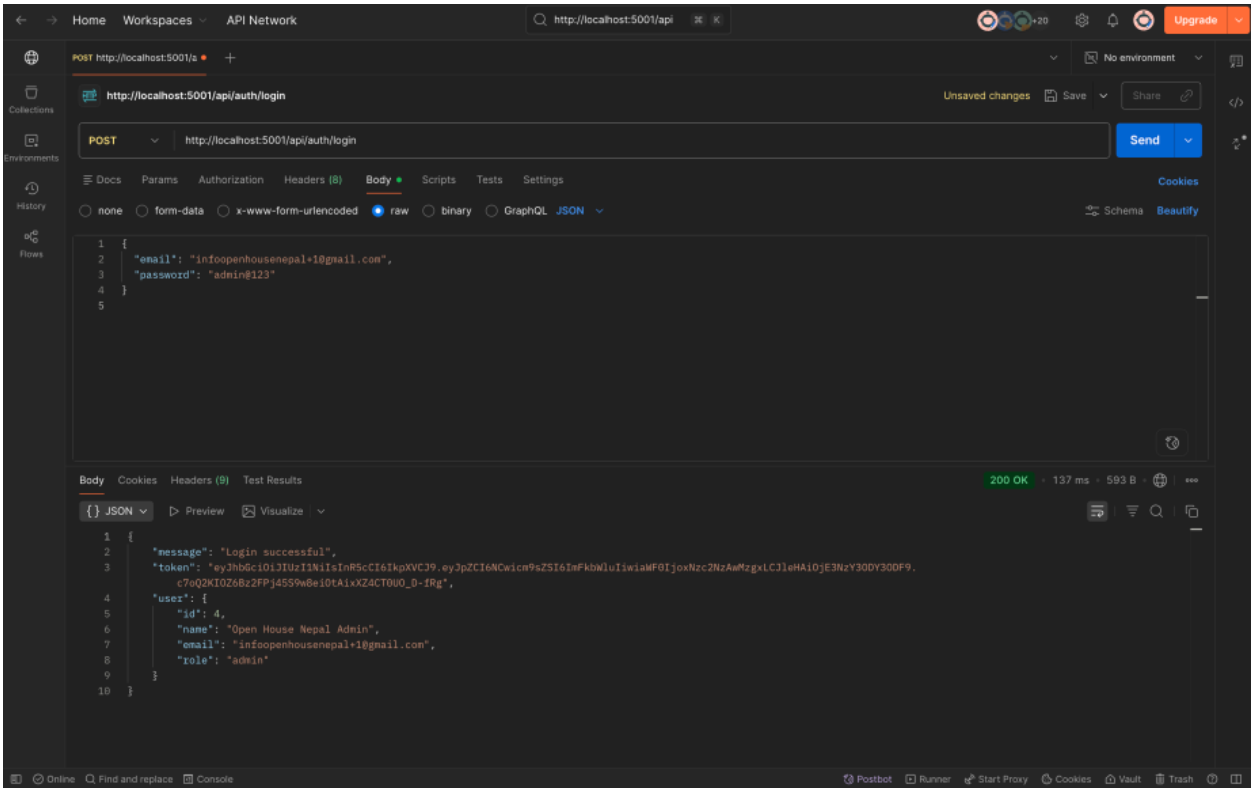


Figure 35 User Login API Test

4.1.2.2 Business Logic Implementation for User Authentication (authController.ts)

This code handles the validation of user credentials and the generation of secure JWT tokens. For the Route Screenshot:

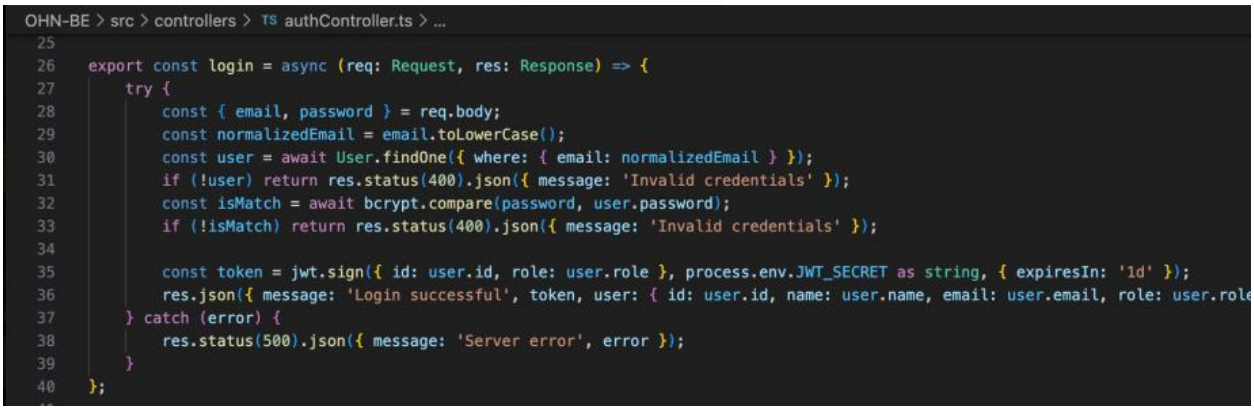


Figure 36 User Login API Test

4.1.2.3 API Endpoints Definition for Authentication Module (authRoutes.ts)

This file maps the incoming HTTP POST requests to their respective controller functions.

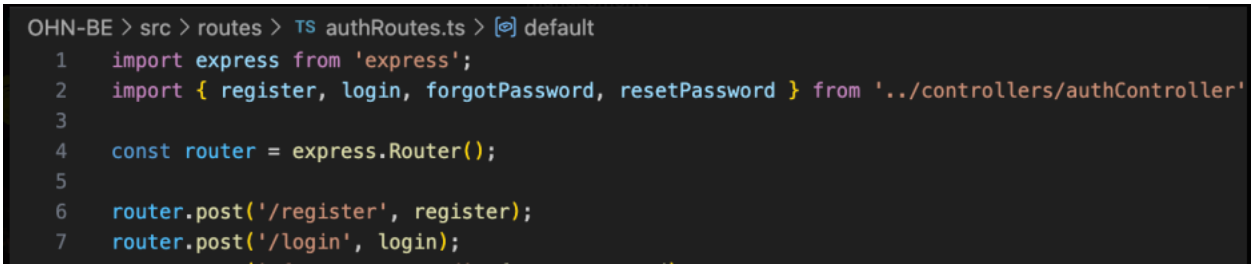


Figure 37 User Login API Test

4.1.3 Test 3: Property Listing Retrieval Test

4.1.3.1 Property Module - Data Retrieval

This test verifies that the system can correctly fetch all property listings from the database. It ensures that the database connection and the data mapping logic are working as expected. Expected Result: 200 OK status and a JSON array containing all property objects.

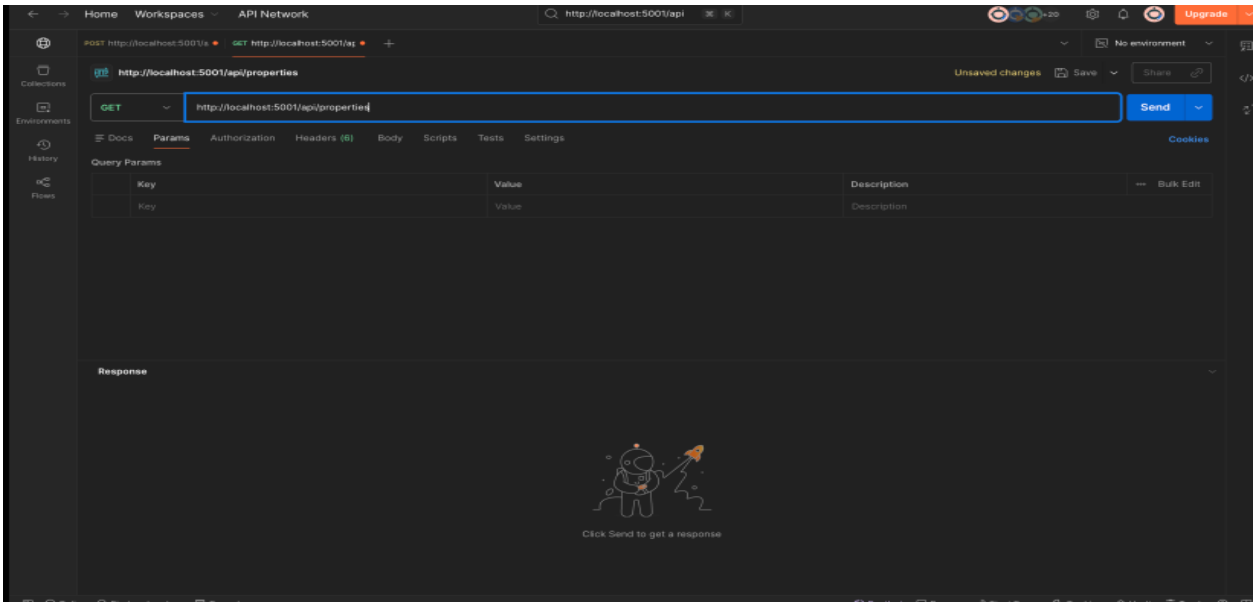


Figure 38 Property Listing Retrieval Test

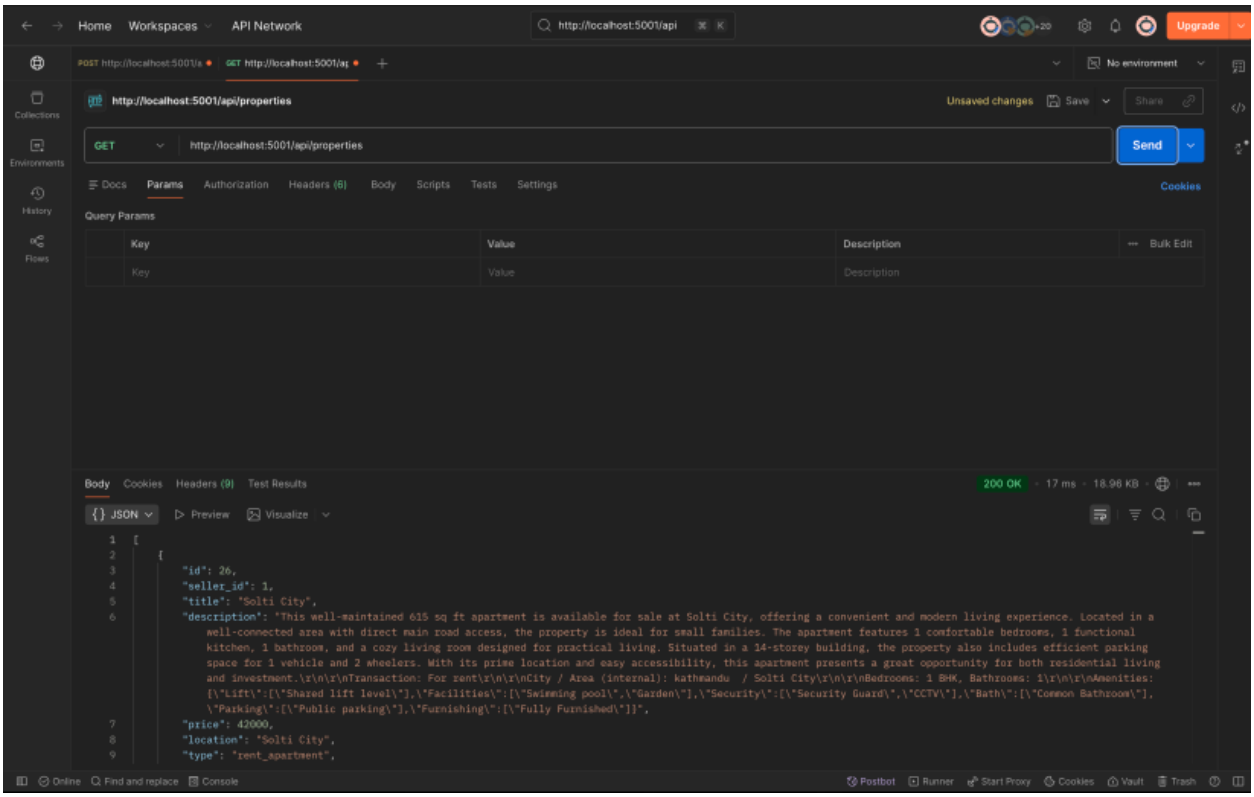


Figure 39 Property Listing Retrieval Test

4.1.3.2 The Logic (Code): File: src/controllers/propertyController.ts

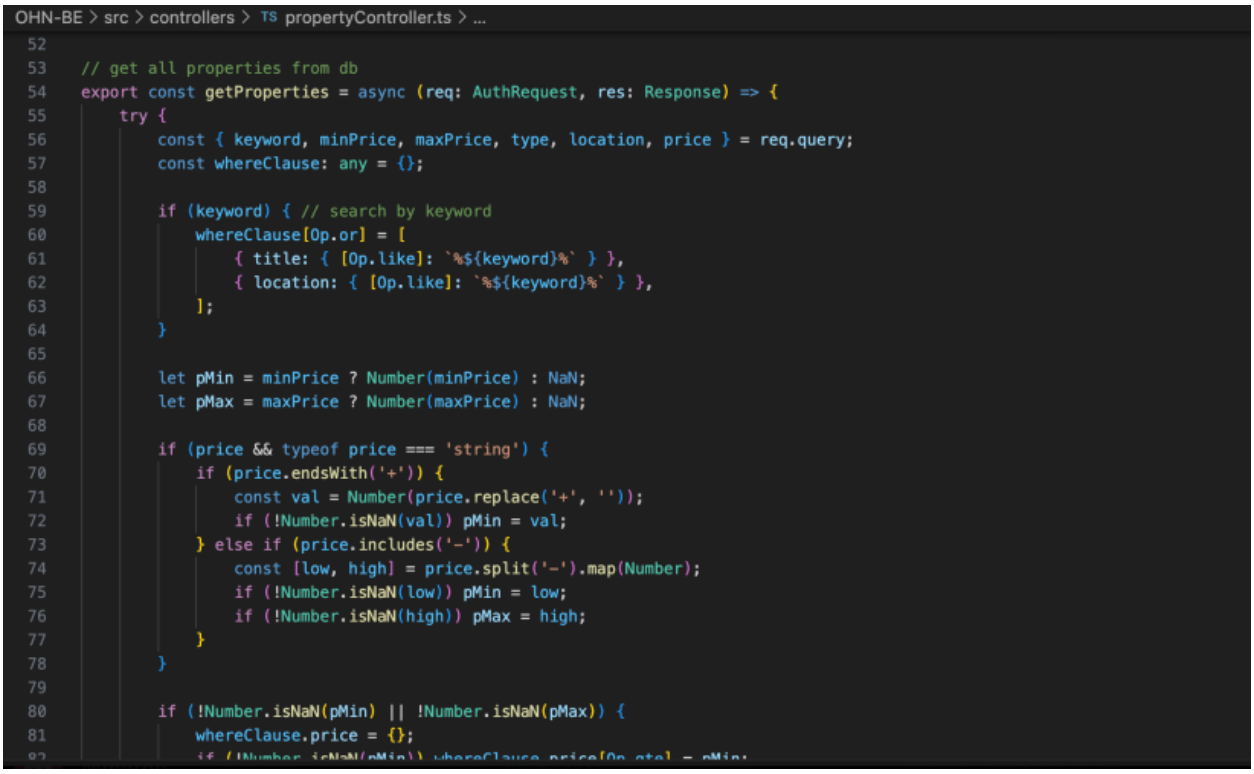


Figure 40 Property Listing Retrieval Test

4.1.3.3 Logic for Fetching All Property Listings (propertyController.ts)

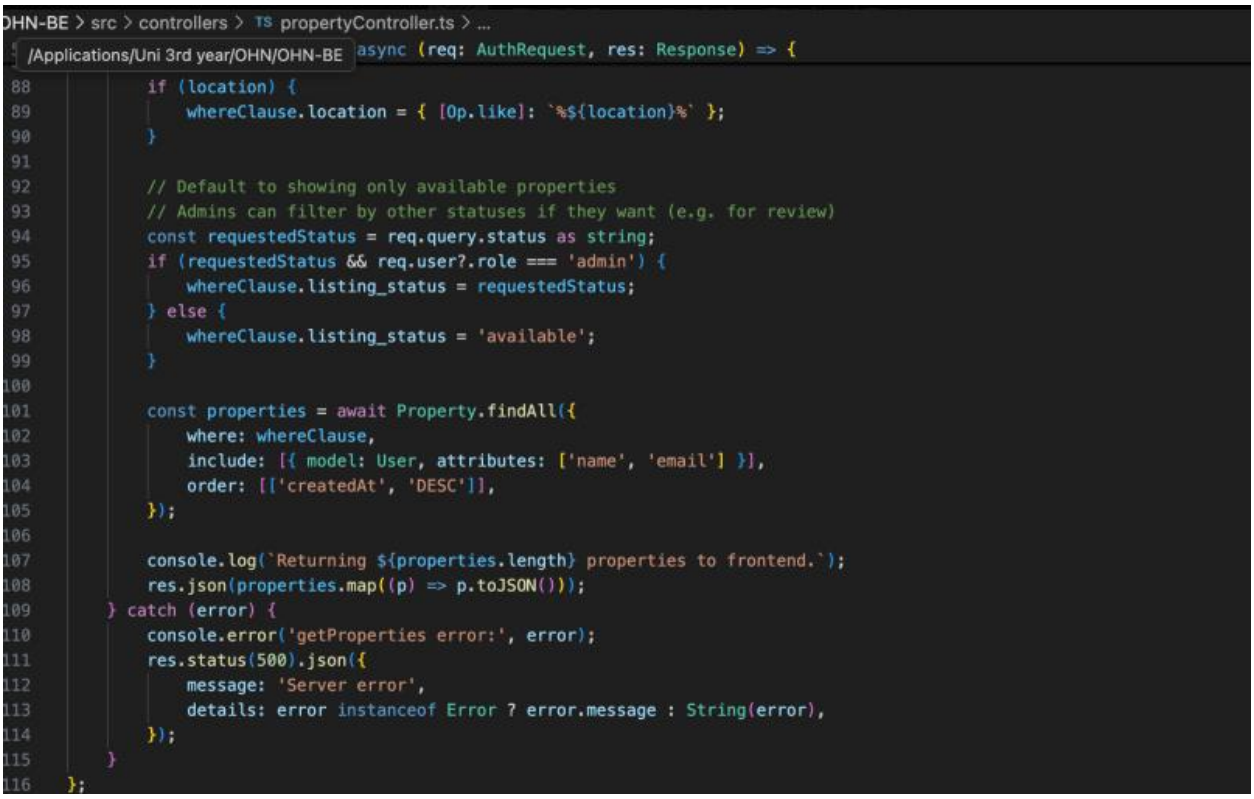


Figure 41 Property Listing Retrieval Test

4.1.3.4 The Route (Code): File: src/routes/propertyRoutes.ts

Endpoint Definition for Property Module (propertyRoutes.ts)

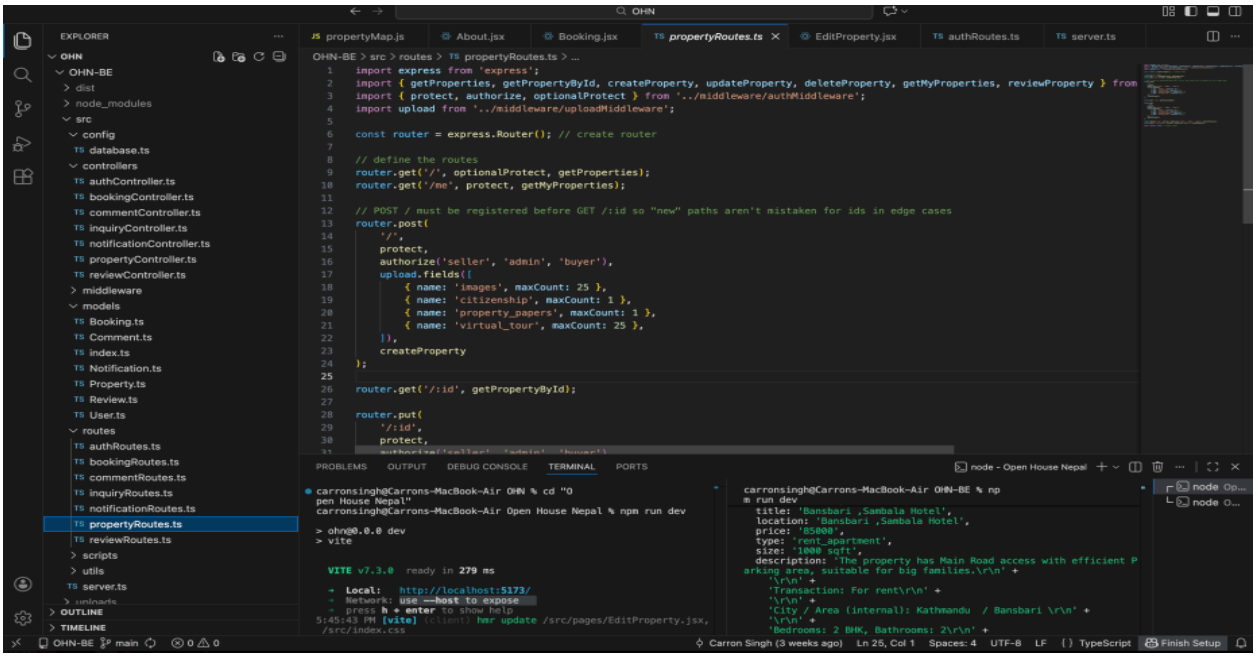


Figure 42 Property Listing Retrieval Test

4.1.4 Test 4: Individual Property Detail Retrieval API Test

4.1.4.1 Property Module - Single Data Retrieval

This test verifies that the specialized data for a unique property is correctly retrieved from the database using its ID. This is essential for the "Property Details" view of the application.

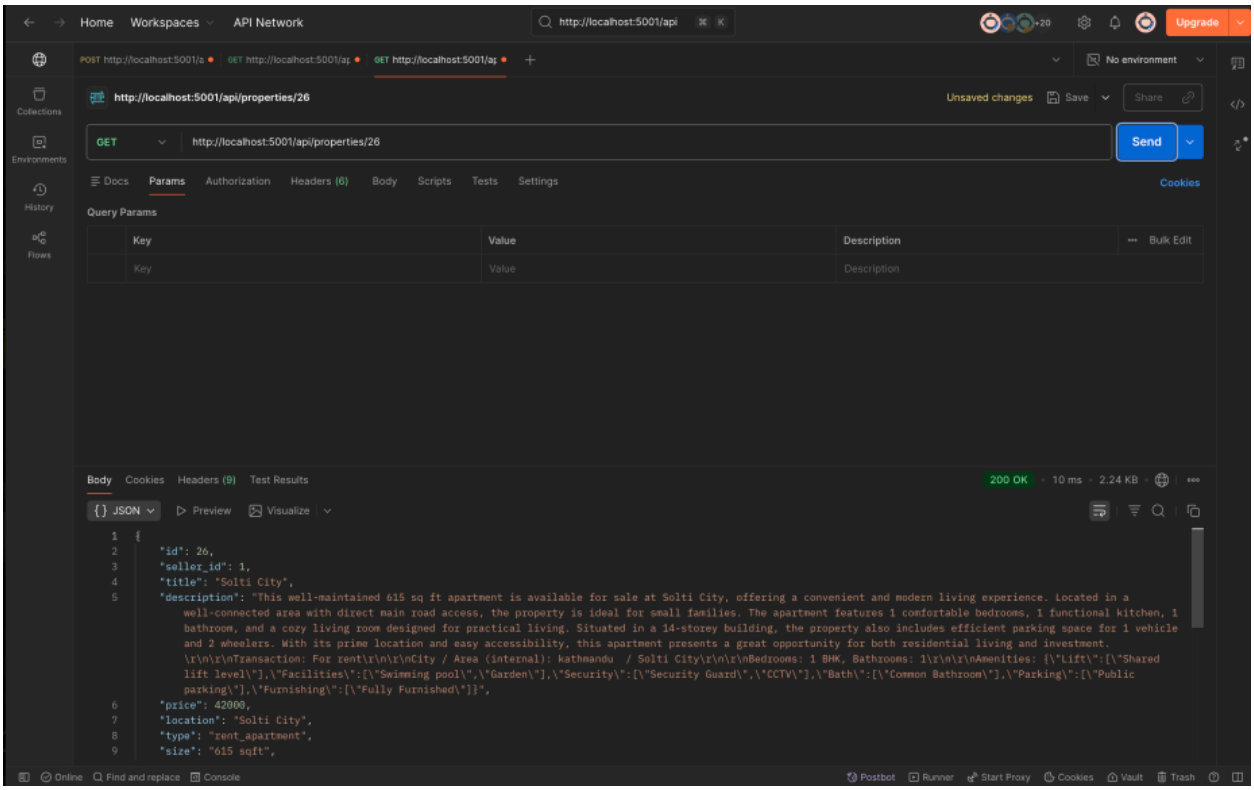


Figure 43 Property Module - Single Data Retrieval

4.1.4.2 Business Logic for Individual Property Lookup (propertyController.ts)



Figure 44 propertyController.ts

4.1.4.3 Dynamic API Route for Property Details (propertyRoutes.ts)



Figure 45 propertyRoutes.ts

4.1.5 Test 5: Create Property Listing API Test

4.1.5.1 Property Module - Secure Creation & Multipart Data Handling

This test validates the entire workflow for listing a new property. It confirms that the system can securely process JWT Tokens, handle Multipart/Form-Data for text and files, and correctly store images and verification documents (Lalpurja and Citizenship).

Expected Result: 201 Created status, successful backend file processing, and a new database entry set to "pending" for admin approval.

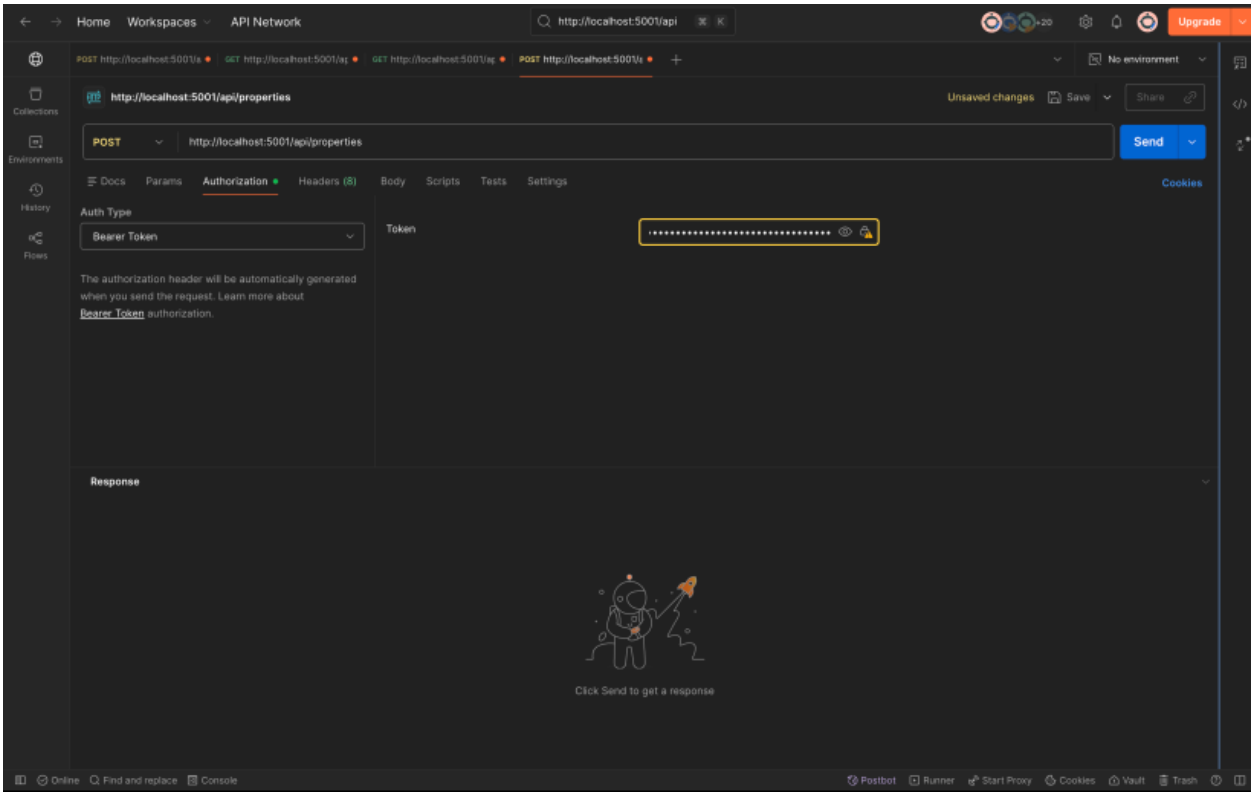


Figure 46 Secure Creation & Multipart Data Handling

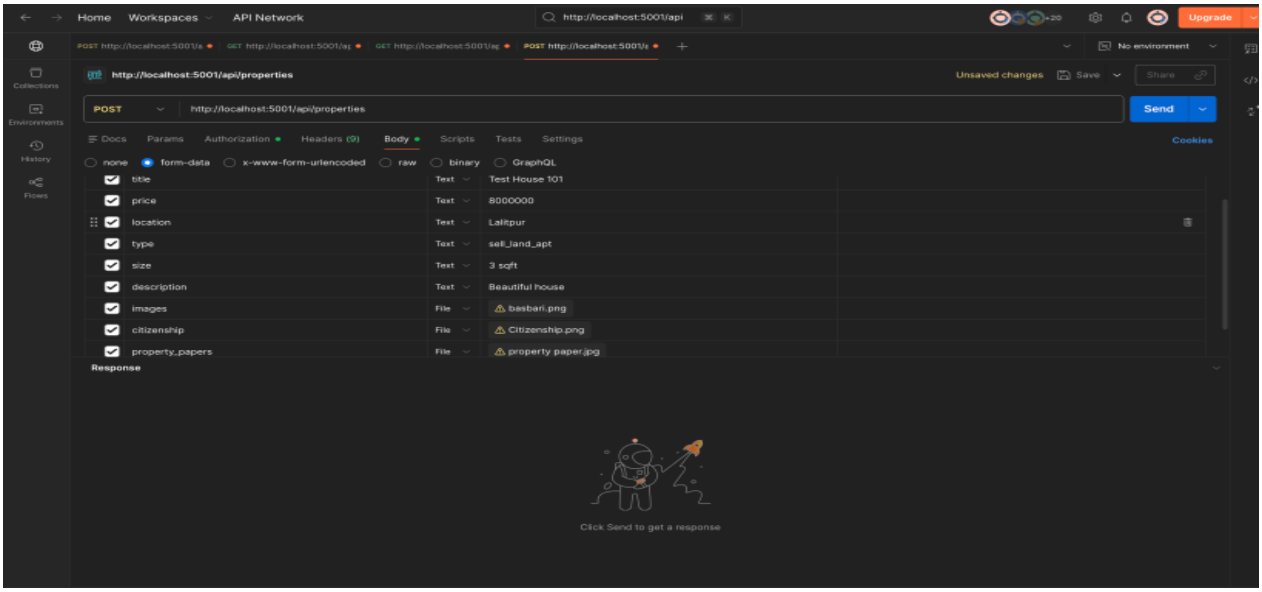


Figure 47 Secure Creation & Multipart Data Handling

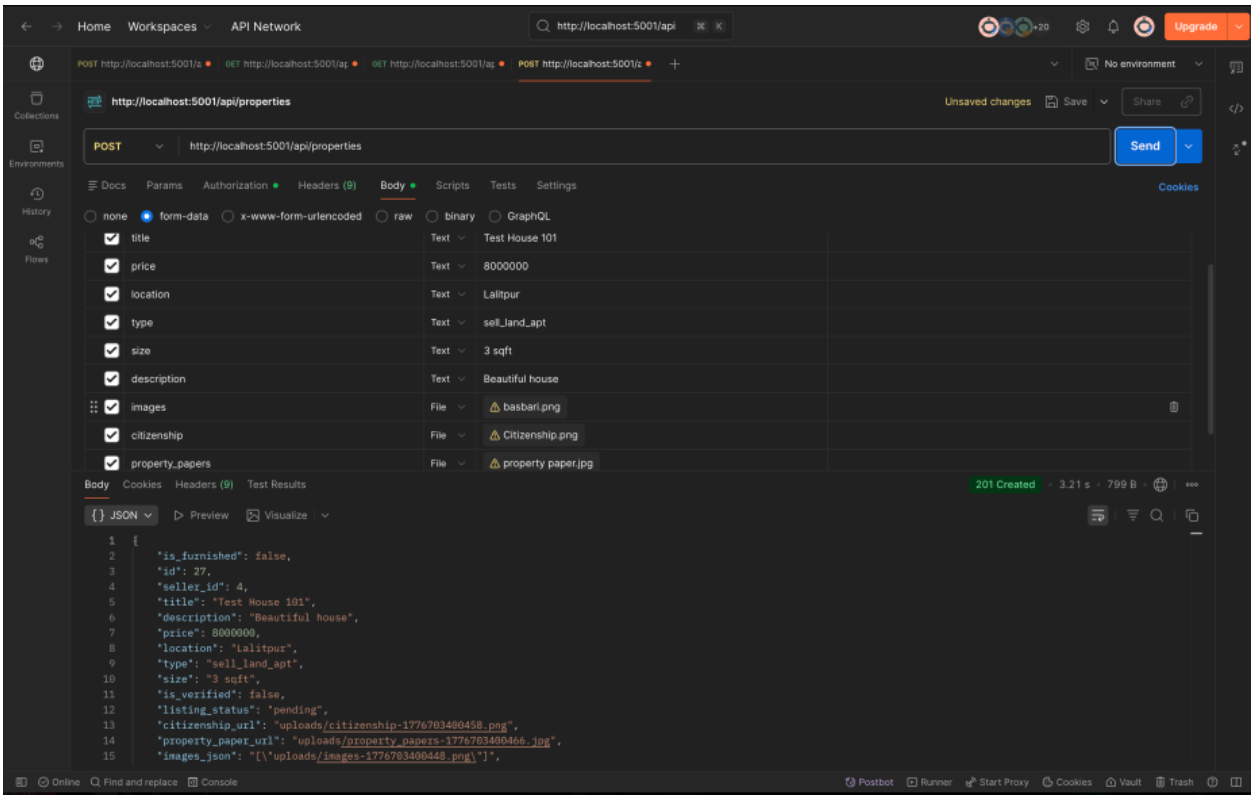


Figure 48 Secure Creation & Multipart Data Handling

4.1.5.2 Business Logic for Property Creation & File Metadata Storage (propertyController.ts)

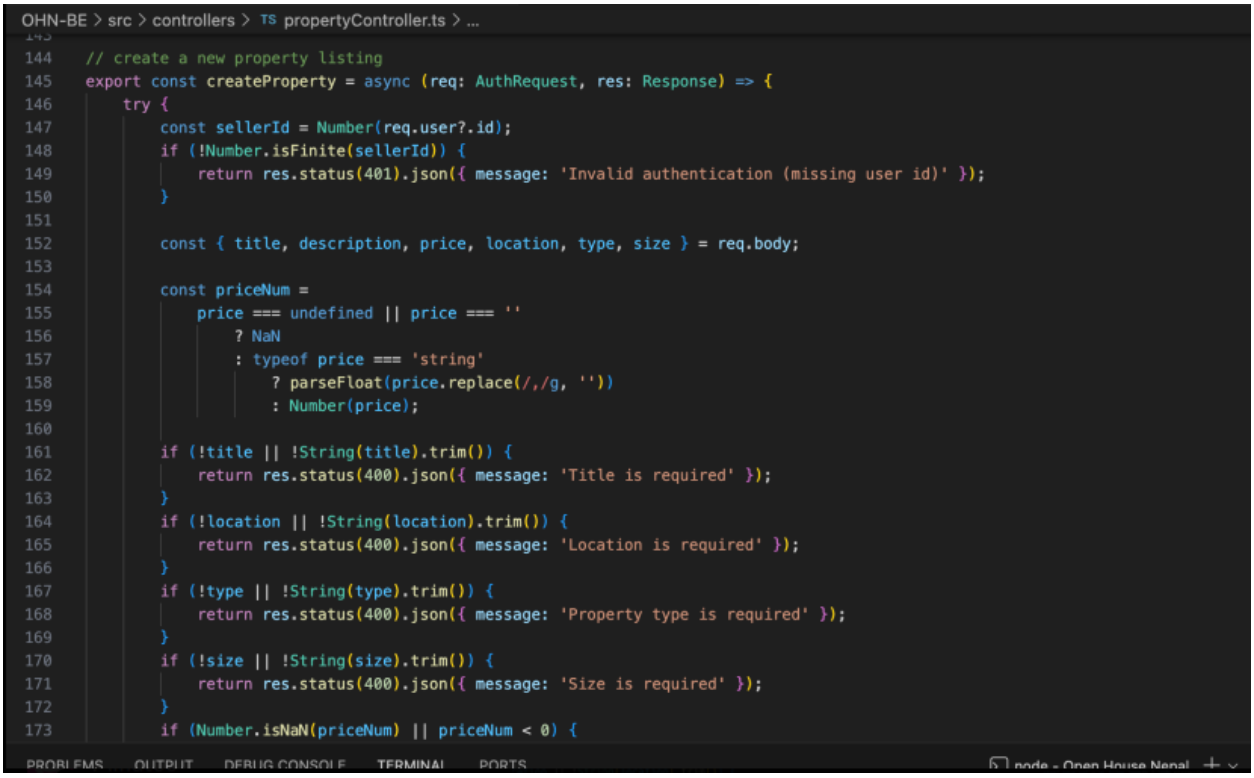
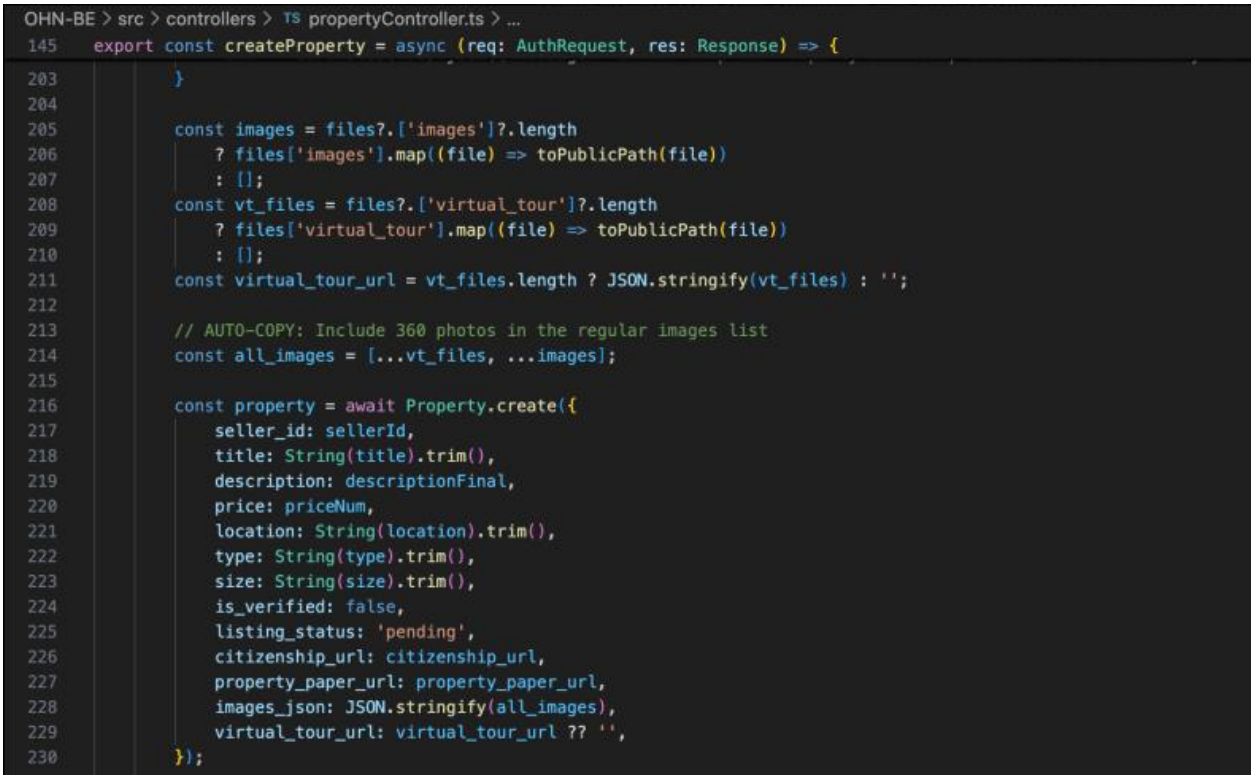


Figure 49 Property Creation & File Metadata Storage



```
OHN-BE > src > controllers > TS propertyController.ts > ...
145  export const createProperty = async (req: AuthRequest, res: Response) => {
177      const descText =
178          description !== undefined && description !== null ? String(description).trim() : '';
179      const descriptionFinal = descText.length > 0 ? descText : '—';
180
181      const seller = await User.findByPk(sellerId);
182      if (!seller) {
183          return res.status(401).json({
184              message: 'Your session is no longer valid',
185              details: 'Please log out and sign in again, then try creating the listing.',
186          });
187      }
188
189      // req.files is handled by multer
190      const files = req.files as { [fieldname: string]: Express.Multer.File[] } | undefined;
191
192      const toPublicPath = (file: Express.Multer.File) => {
193          const p = file.path.replace(/\\/g, '/');
194          const idx = p.indexOf('uploads/');
195          return idx >= 0 ? p.slice(idx) : `uploads/${file.filename}`;
196      };
197
198      const citizenship_url = files?.['citizenship']?.[0] ? toPublicPath(files['citizenship'][0]) : null;
199      const property_paper_url = files?.['property_papers']?.[0] ? toPublicPath(files['property_papers'][0]) : null;
200
201      if (!citizenship_url || !property_paper_url) {
202          return res.status(400).json({ message: 'Citizenship and Property ownership documents are mandatory for verification
203      }
```

Figure 50 Property Creation & File Metadata Storage



```
OHN-BE > src > controllers > TS propertyController.ts > ...
145  export const createProperty = async (req: AuthRequest, res: Response) => {
203      }
204
205      const images = files?.['images']?.length
206          ? files['images'].map((file) => toPublicPath(file))
207          : [];
208      const vt_files = files?.['virtual_tour']?.length
209          ? files['virtual_tour'].map((file) => toPublicPath(file))
210          : [];
211      const virtual_tour_url = vt_files.length ? JSON.stringify(vt_files) : '';
212
213      // AUTO-COPY: Include 360 photos in the regular images list
214      const all_images = [...vt_files, ...images];
215
216      const property = await Property.create({
217          seller_id: sellerId,
218          title: String(title).trim(),
219          description: descriptionFinal,
220          price: priceNum,
221          location: String(location).trim(),
222          type: String(type).trim(),
223          size: String(size).trim(),
224          is_verified: false,
225          listing_status: 'pending',
226          citizenship_url: citizenship_url,
227          property_paper_url: property_paper_url,
228          images_json: JSON.stringify(all_images),
229          virtual_tour_url: virtual_tour_url ?? '',
230      });
```

Figure 51 Property Creation & File Metadata Storage

4.1.5.3 Secure POST Endpoint for Listing Submission (propertyRoutes.ts)



Figure 52 Secure POST Endpoint for Listing Submission

4.1.6 Test 6: Property Information Update

4.1.6.1 Property Module - Existing Data Modification

This test ensures that the system allows authorized owners to update their property details (e.g., changing the price). It validates that the backend correctly identifies the property by ID and updates only the requested fields in the database.

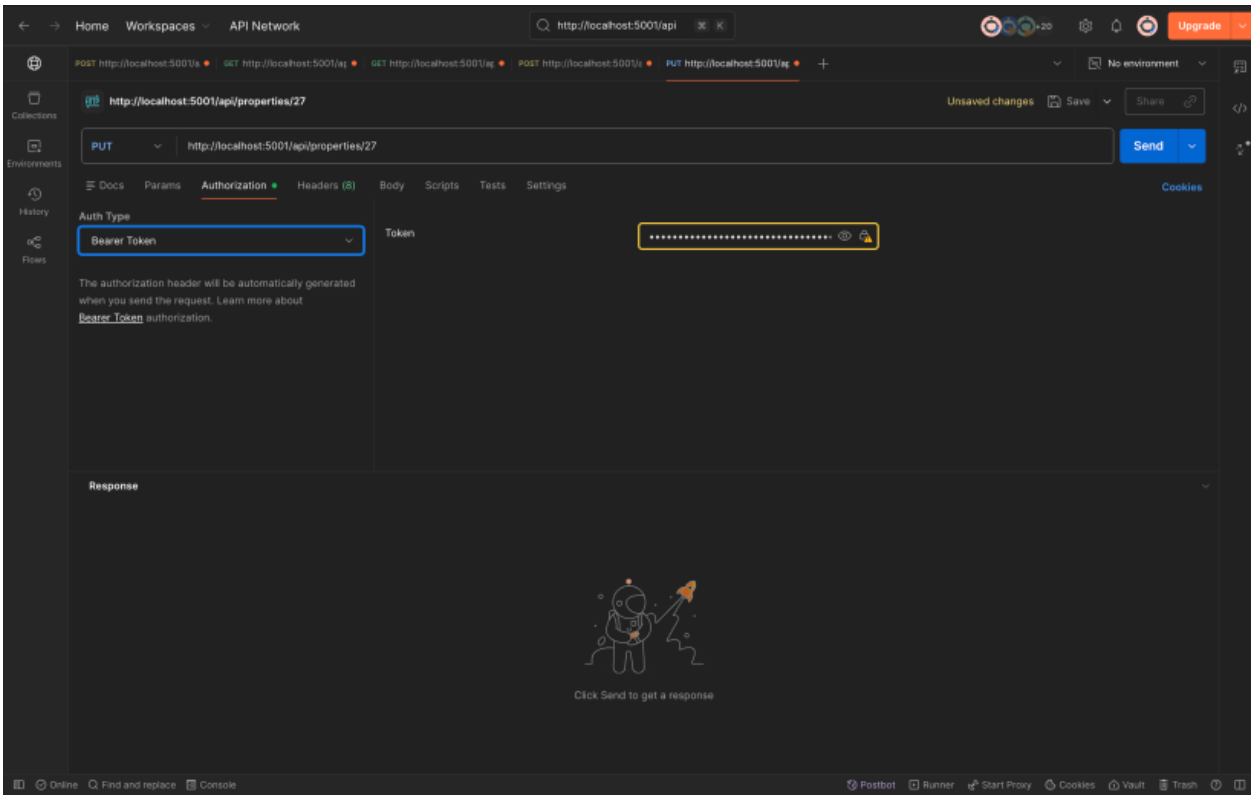


Figure 53 Existing Data Modification

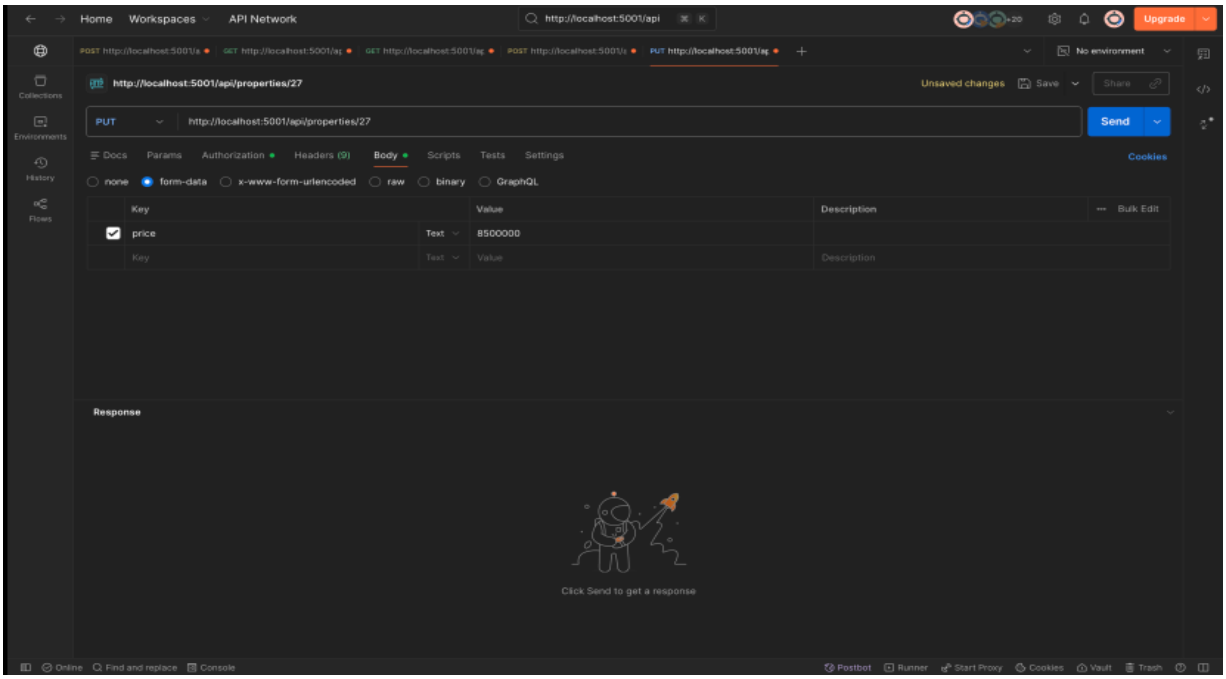


Figure 54 Existing Data Modification

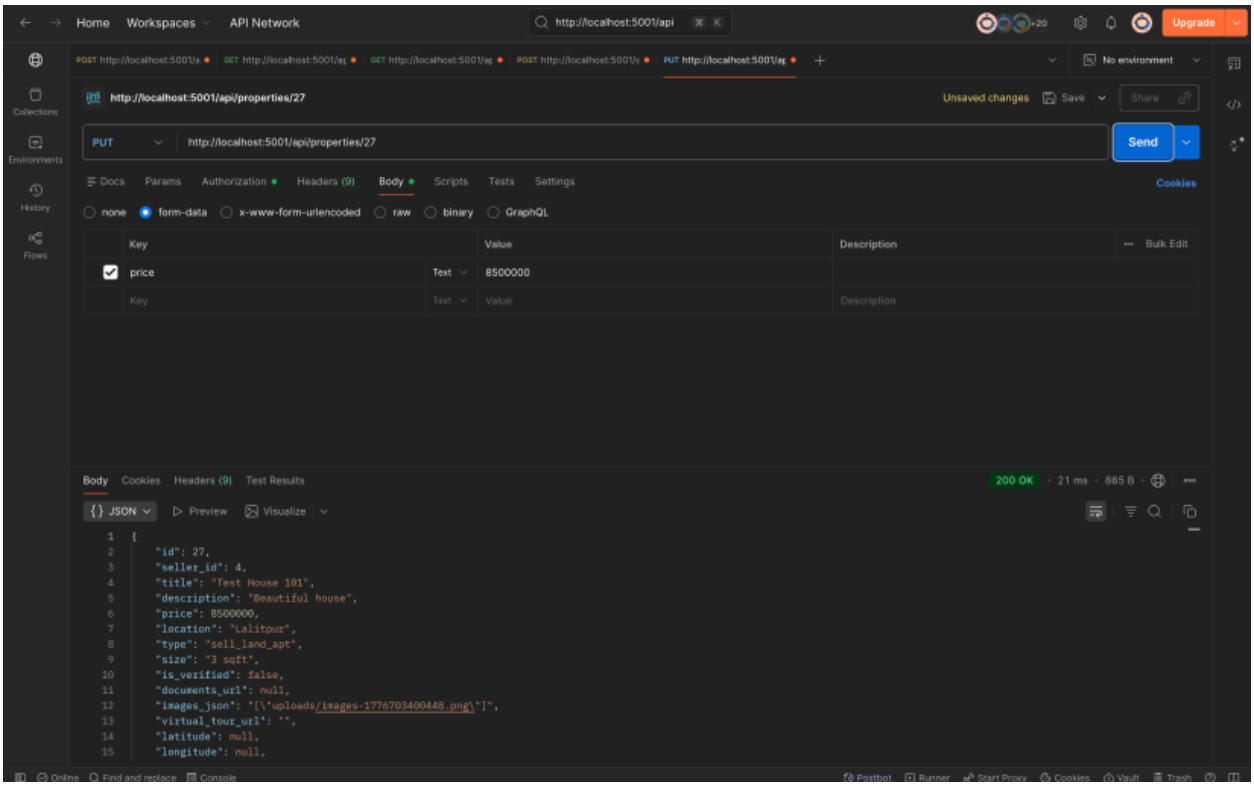


Figure 55 Existing Data Modification

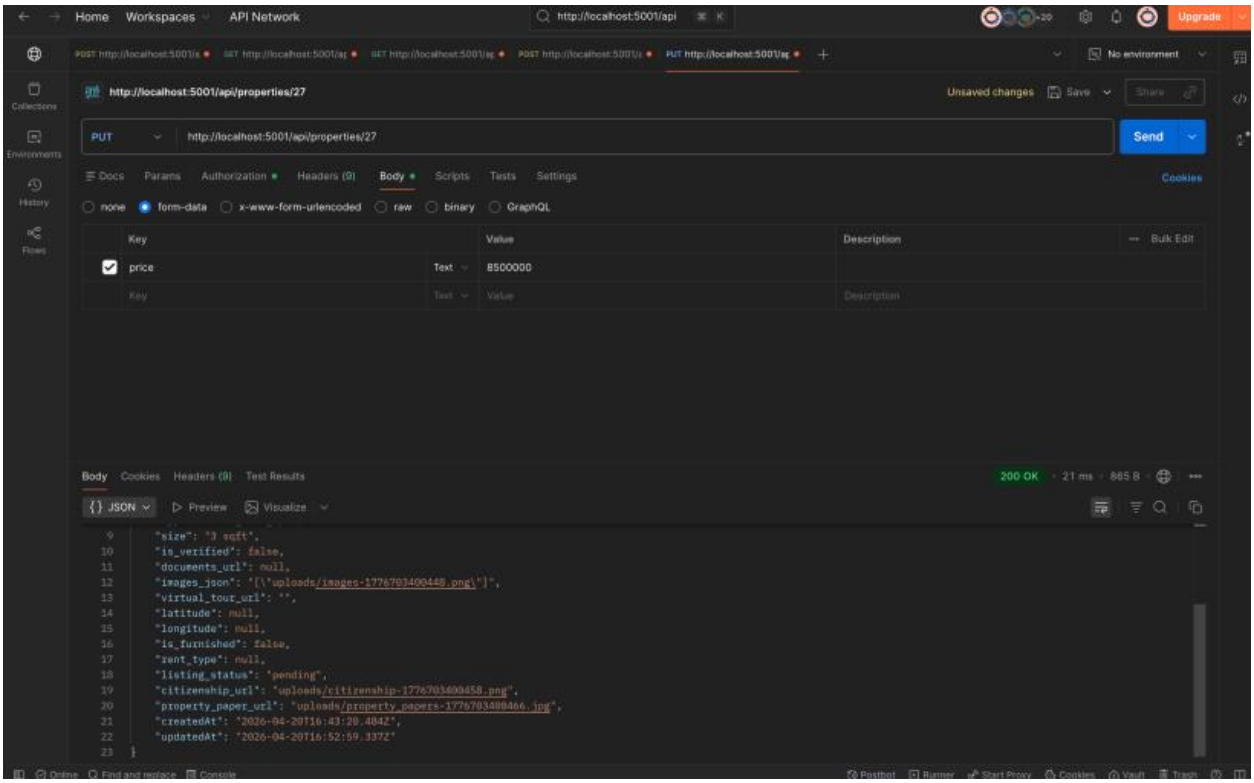


Figure 56 Existing Data Modification

4.1.6.2 Logic for Authenticated Property Information Update (propertyController.ts)

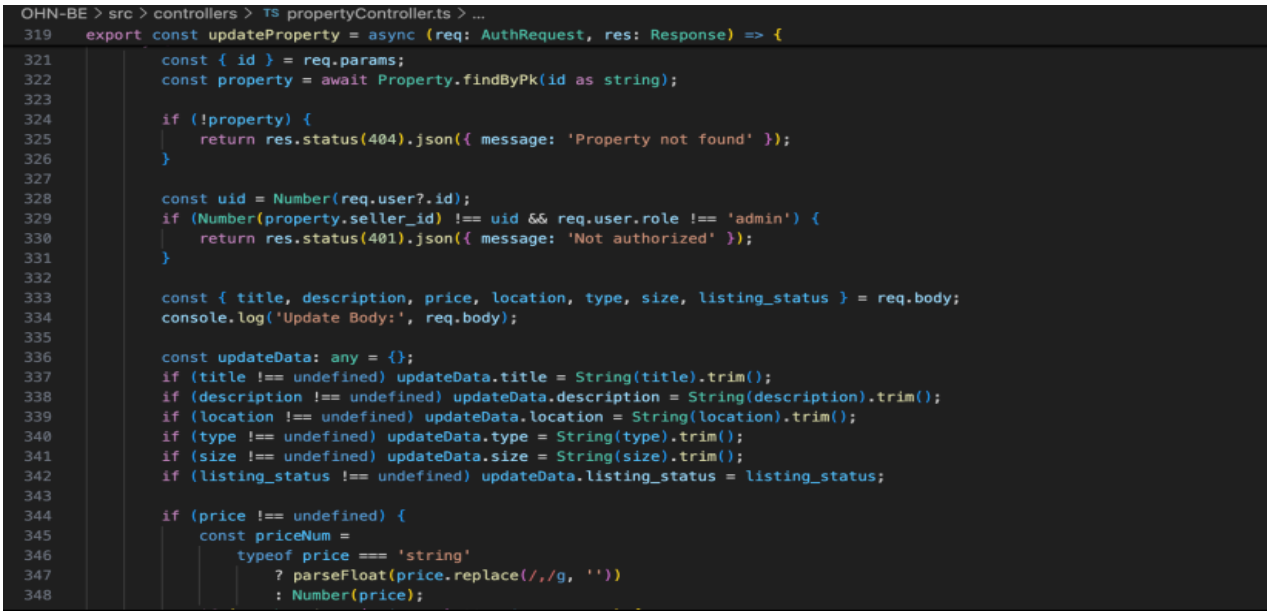


Figure 57 Authenticated Property Information Update



Figure 58 Authenticated Property Information Update

4.1.6.3 Secure PUT Endpoint for Property Modification (propertyRoutes.ts)

OHN-BE > src > routes > TS propertyRoutes.ts > ...

```
27
28    router.put(
29        '/:id',
30        protect,
31        authorize('seller', 'admin', 'buyer'),
32        upload.fields([
33            { name: 'images', maxCount: 25 },
34            { name: 'citizenship', maxCount: 1 },
35            { name: 'property_papers', maxCount: 1 },
36            { name: 'virtual_tour', maxCount: 25 },
37        ]),
38        updateProperty
39    );
```

Figure 59 Secure PUT Endpoint for Property Modification

4.1.7 Test 7: Delete Property API Test

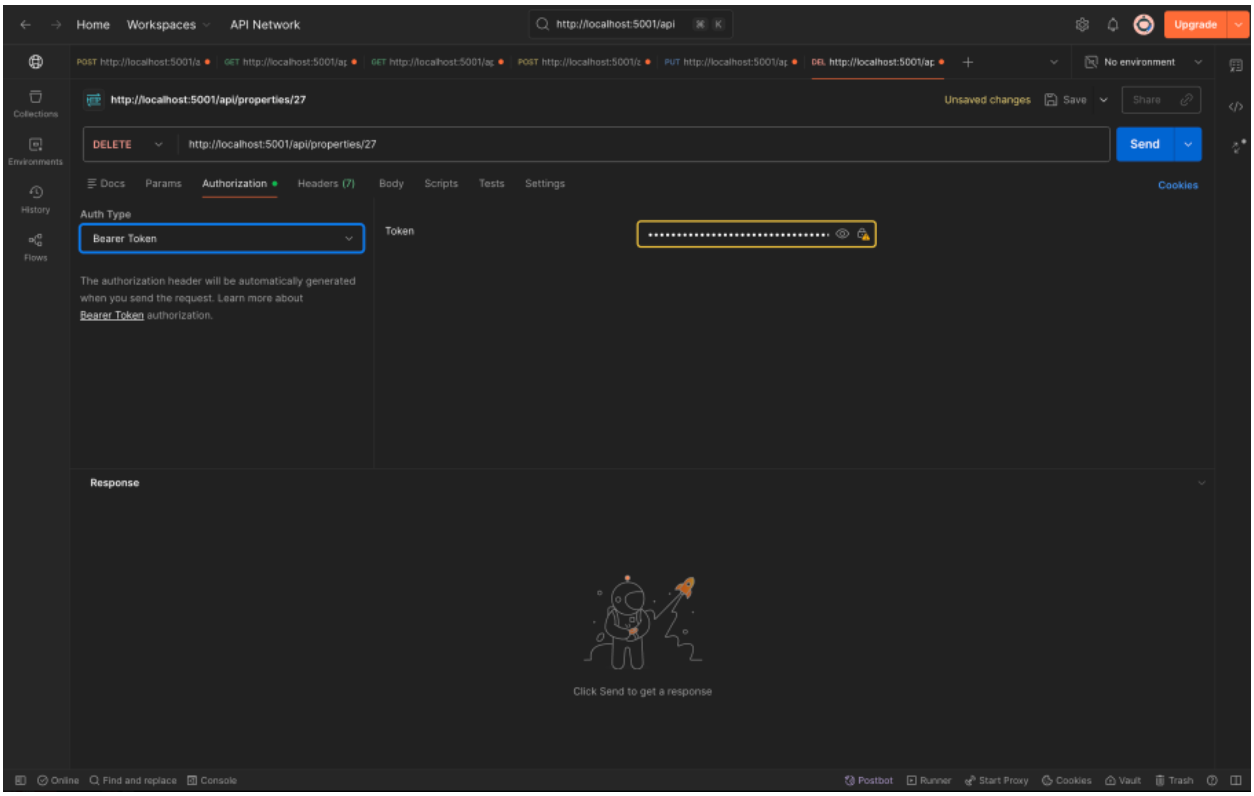


Figure 60 Delete Property API Test

This test validates the functionality for permanently removing a property listing from the database. It verifies that the system checks for proper Authorization (only the owner or an admin can delete) and successfully removes the specific property record.

4.1.7.1 Business Logic for Authorized Property Deletion (propertyController.ts)

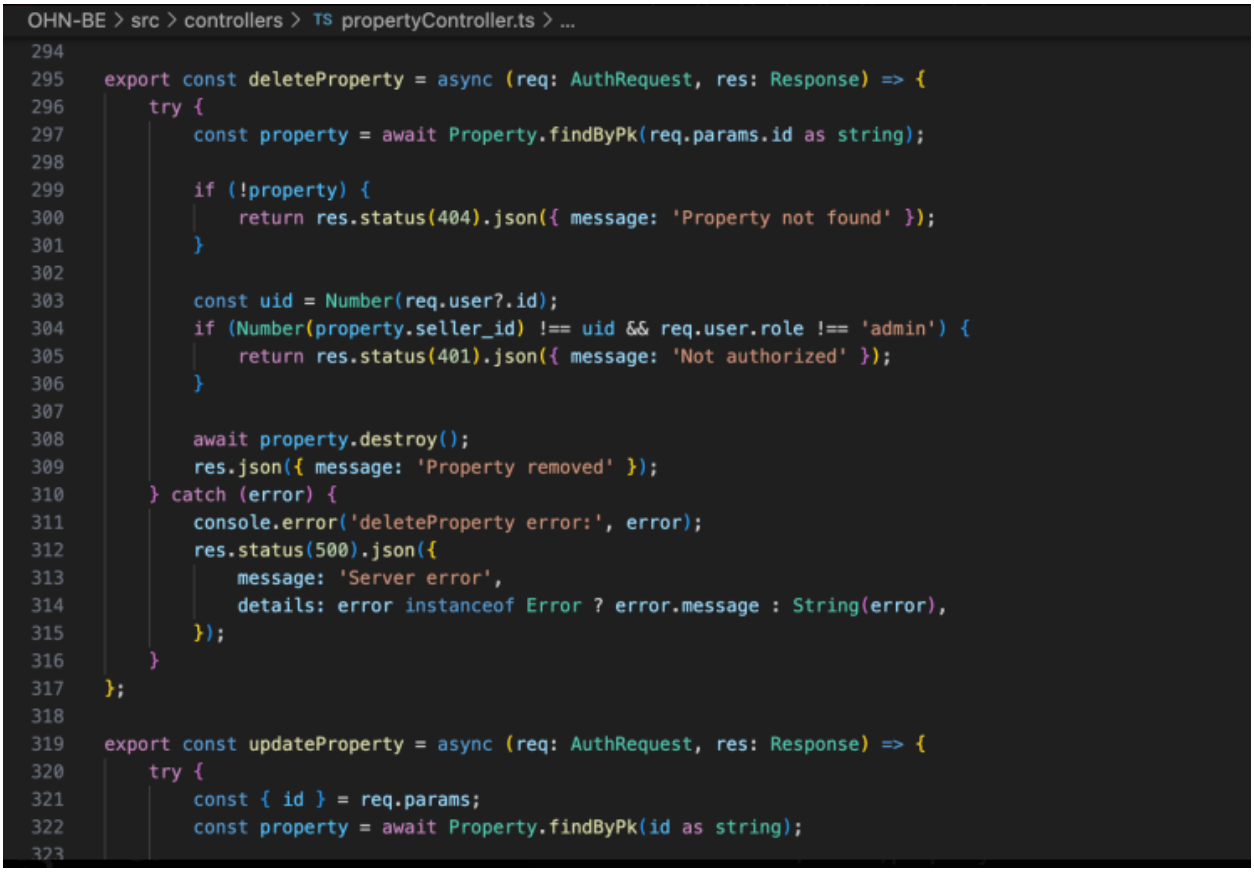


Figure 61 Business Logic for Authorized Property Deletion

4.1.7.2 Secure DELETE Endpoint for Listing Removal (propertyRoutes.ts)

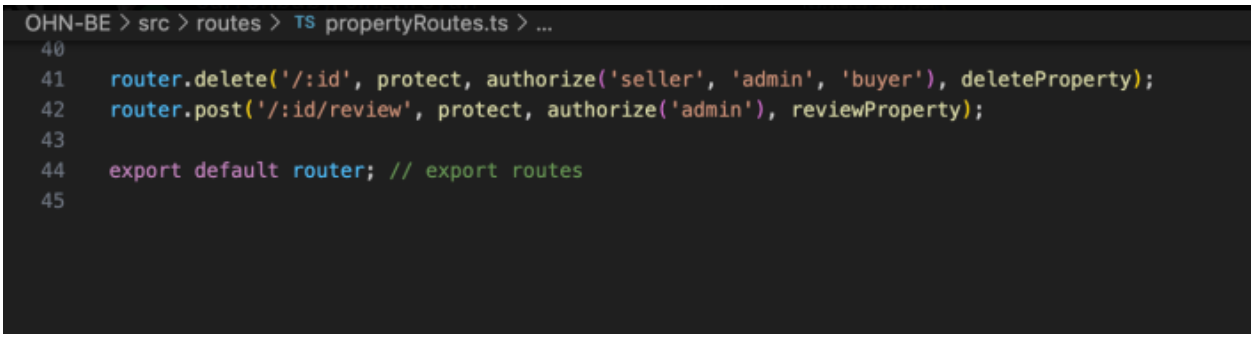


Figure 62 Secure DELETE Endpoint for Listing Removal

4.2 SYSTEM TESTING, TEST PLAN

System testing is a comprehensive testing phase where the complete, integrated software is evaluated to ensure it meets the specified functional and non-functional requirements. It evaluates the end-to-end user flows, validating that the frontend, backend, and third party services (like Google Maps & Nodemailer) work together as a unified system. These tests are performed from the perspective of an actual user to ensure a seamless experience, security, and data accuracy across the entire application interface.

Test Name	What it Tests
Signup Validation Test	Verifies that form errors appear if mandatory fields are left empty during registration.
Successful Account Registration	Confirms that a user can create an account and access the platform via the UI.
Login Security (Negative Case)	Ensures that incorrect passwords or unregistered emails are rejected by the system.
Successful Login & Navigation	Validates that a user is correctly redirected to the Dashboard after valid authentication.
Keyword / Location Search	Verifies that the search bar correctly filters results based on a specific location (e.g., Kathmandu).
Price Range & Category Filter	Ensures the sidebar filters successfully narrow down property results by budget and type.
360° Virtual Tour Module	Validates that the panoramic viewer correctly loads and allows interactive movement.
Add Listing Form & Uploads	Verifies that the full multipart form (text + citizenship + papers) submits without errors.

Admin Dashboard & Approval	Confirms that an Admin can verify documents and make a "Pending" property visible to the public.
Email Notification Verification	Ensures that automated emails (Nodemailer) are received for approval/rejection and inquiries.
Password Reset (Forgot Pass)	Tests the entire account recovery flow from email request to setting a new password.
Interactive Map Navigation	Validates that Google Maps loads the correct marker and allows zooming/panning.
Review & Comment System	Verifies that users can leave ratings and comments that appear in real-time on the page.
Property Unarchive/Delete	Tests the ability of an admin or owner to restore or permanently remove a listing via the UI.

Table 4 System Testing

4.2.1 System Test 1: Authentication System Tests

4.2.1.1 Signup Form Validation Test

Verifies that the signup form correctly identifies missing information and prevents submission until all mandatory fields are filled.

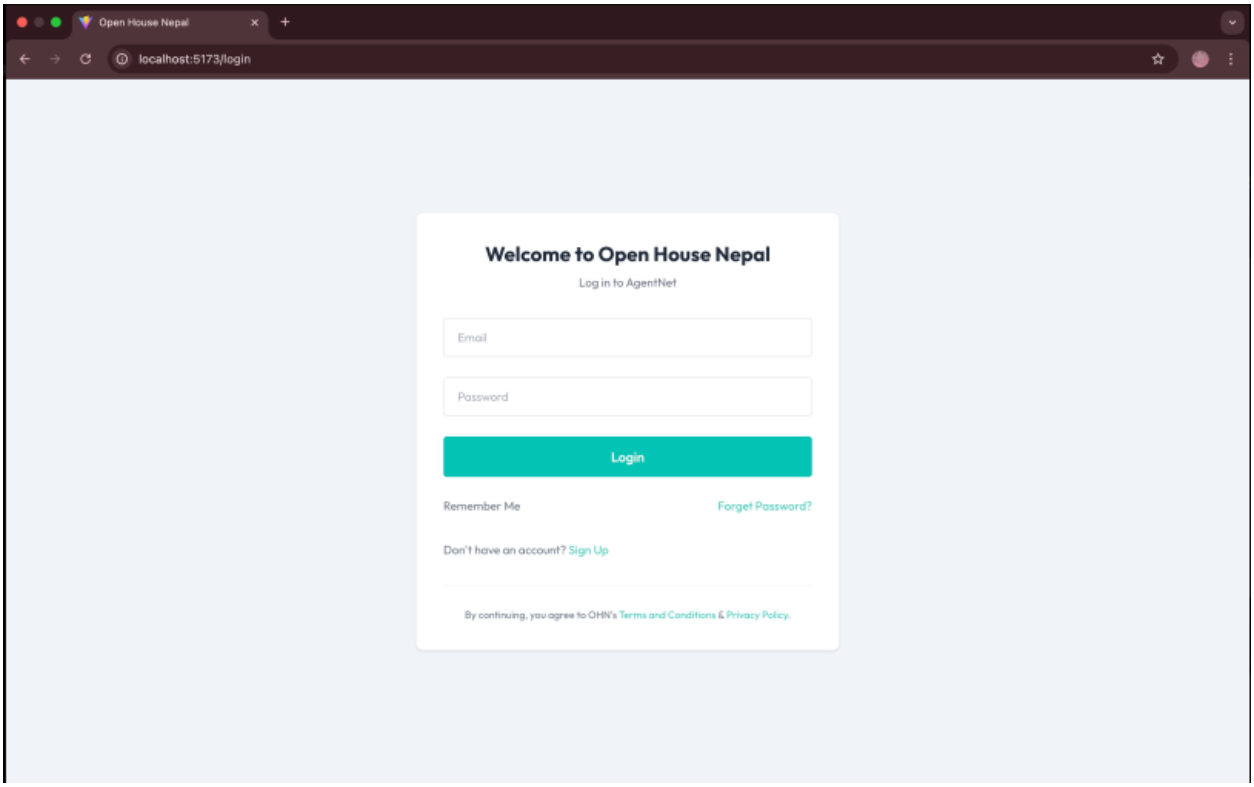


Figure 63 Signup Form Validation Test

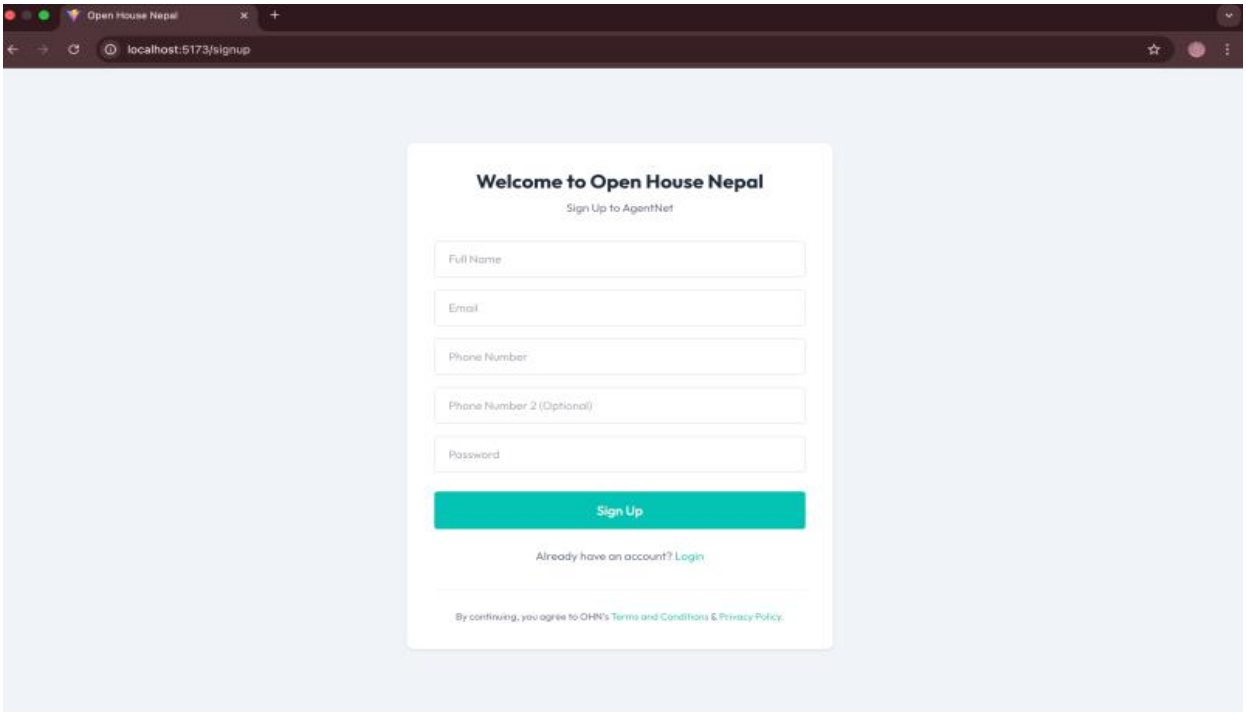


Figure 64 Signup Form Validation Test

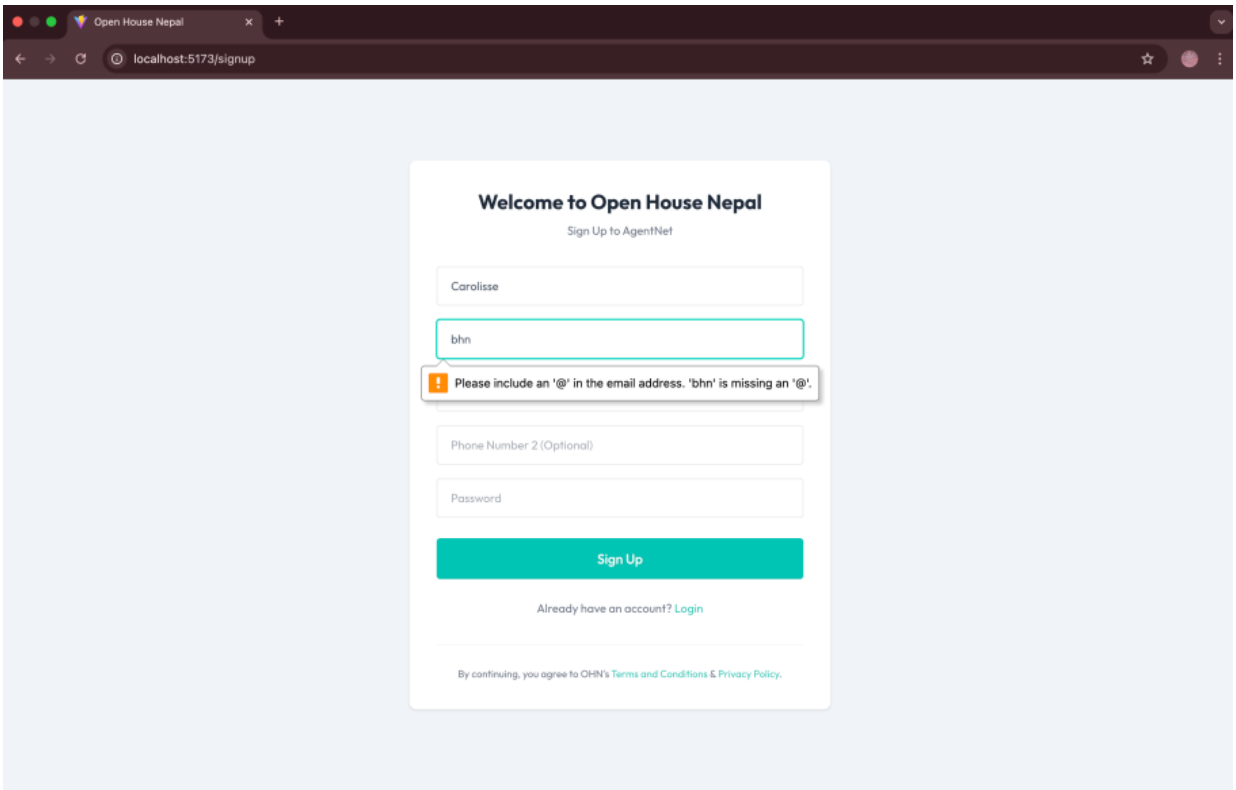


Figure 65 Signup Form Validation Test

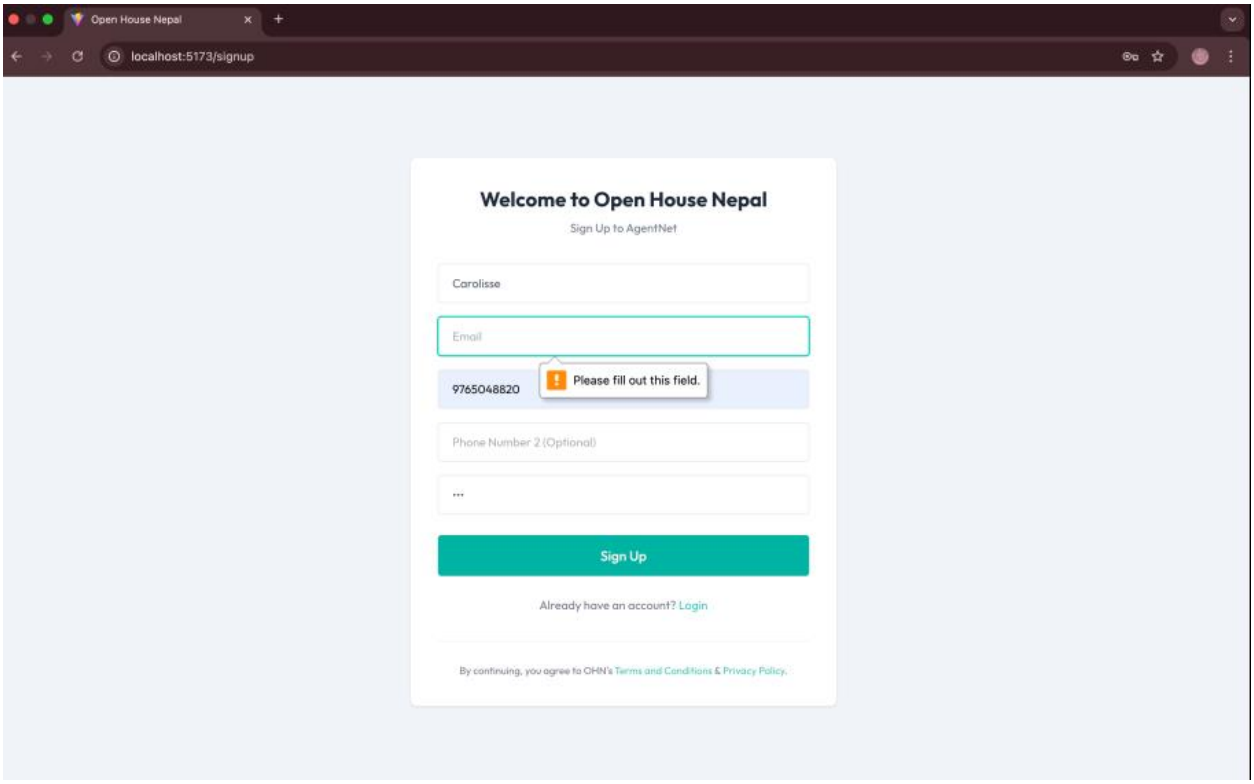


Figure 66 Signup Form Validation Test

4.2.2 System Test 2: Successful Account Registration

Confirms that a new user can successfully create an account when providing valid information

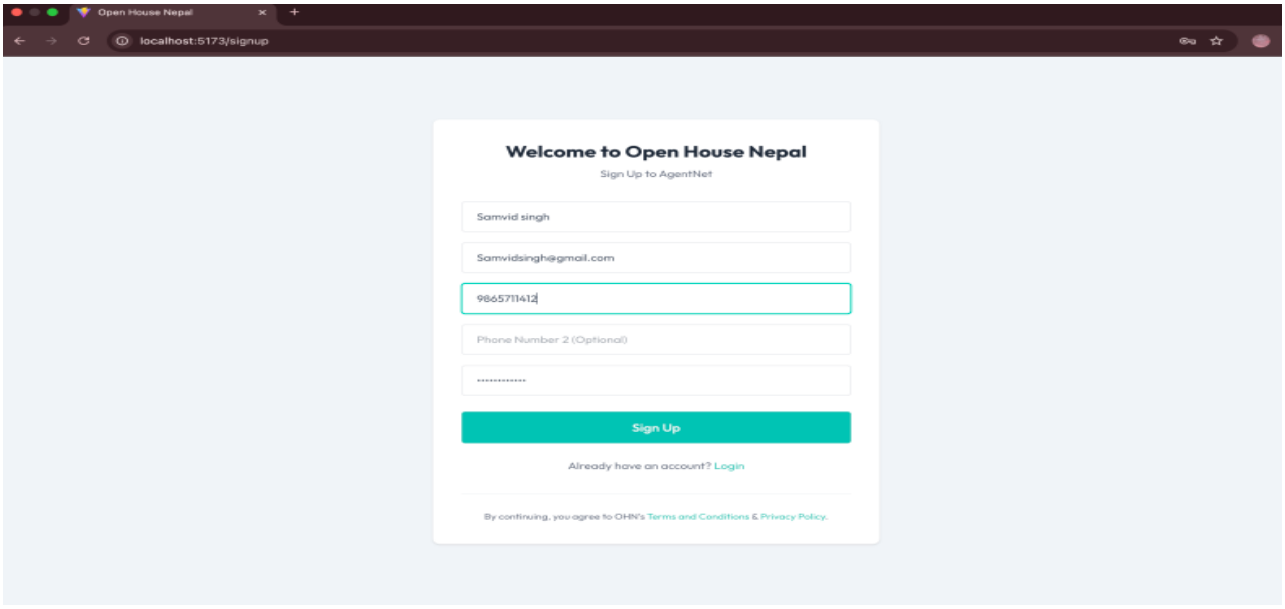


Figure 67 Successful Account Registration

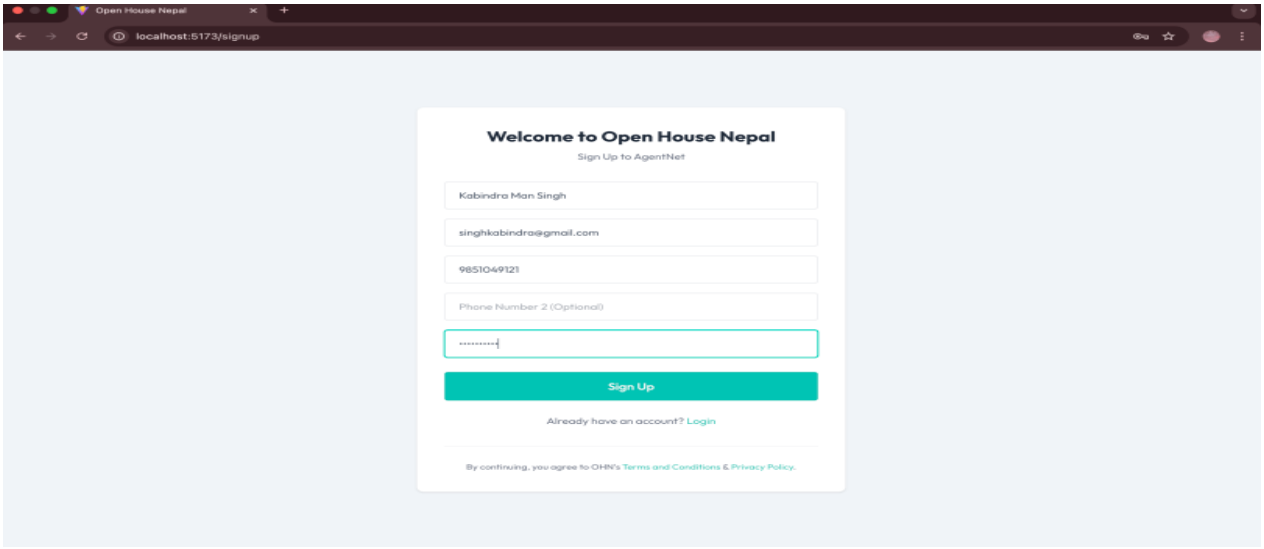


Figure 68 Successful Account Registration

4.2.3 System Test 3: Login Security Test (Negative Case)

Tests the security of the login system by ensuring it rejects incorrect passwords or unregistered email addresses.

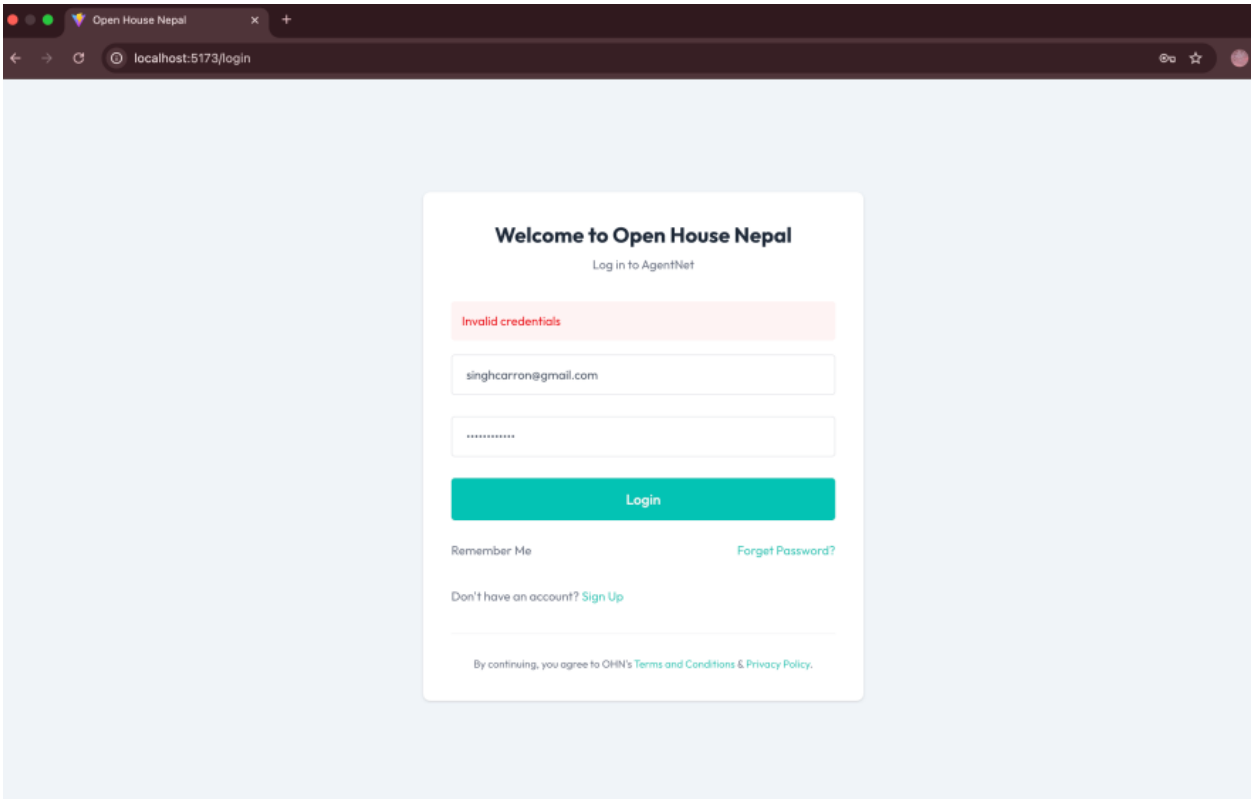


Figure 69 Login Security Test (Negative Case)

4.2.4 System Test 4: Successful Login & Navigation

Verifies that a user is correctly authenticated and redirected to the appropriate landing page (Home or Dashboard) after entering valid credentials.

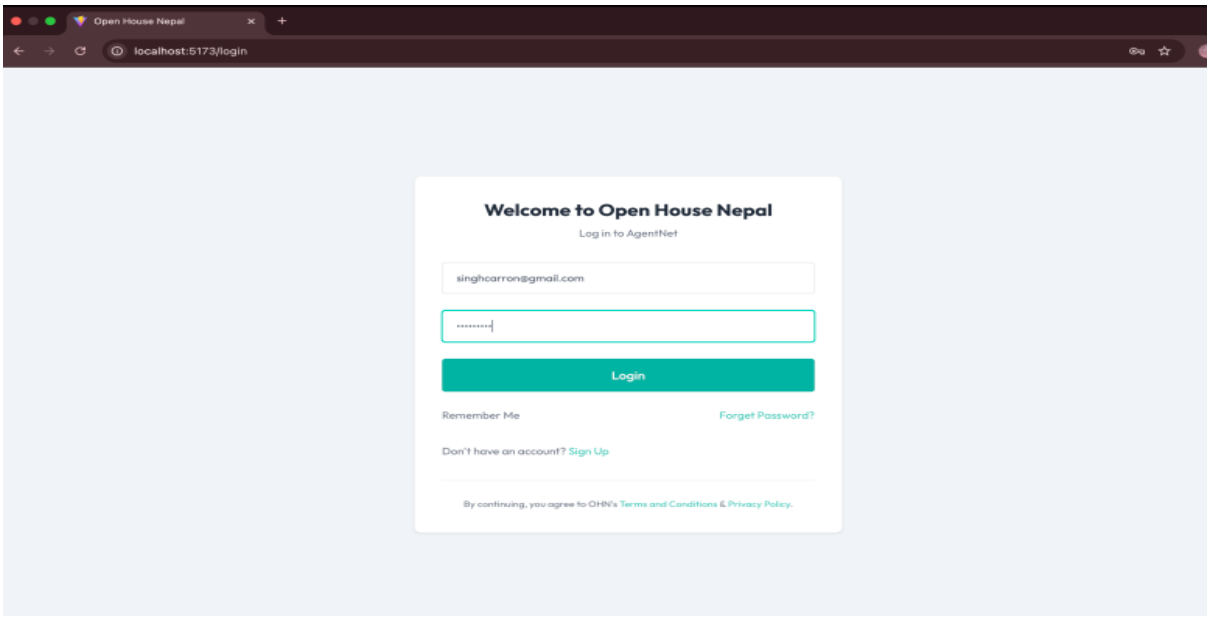


Figure 70 Successful Login & Navigation

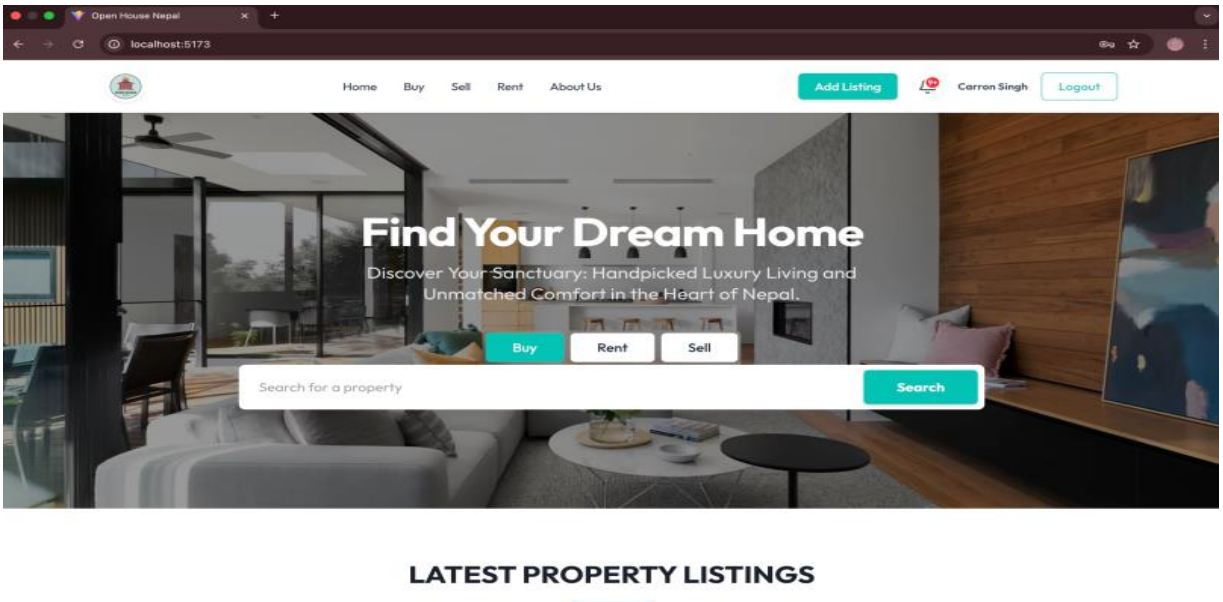


Figure 71 Successful Login

4.2.5 System Test 5: Homepage Search Interface

Ensures the main search interface is accessible and ready for user input.

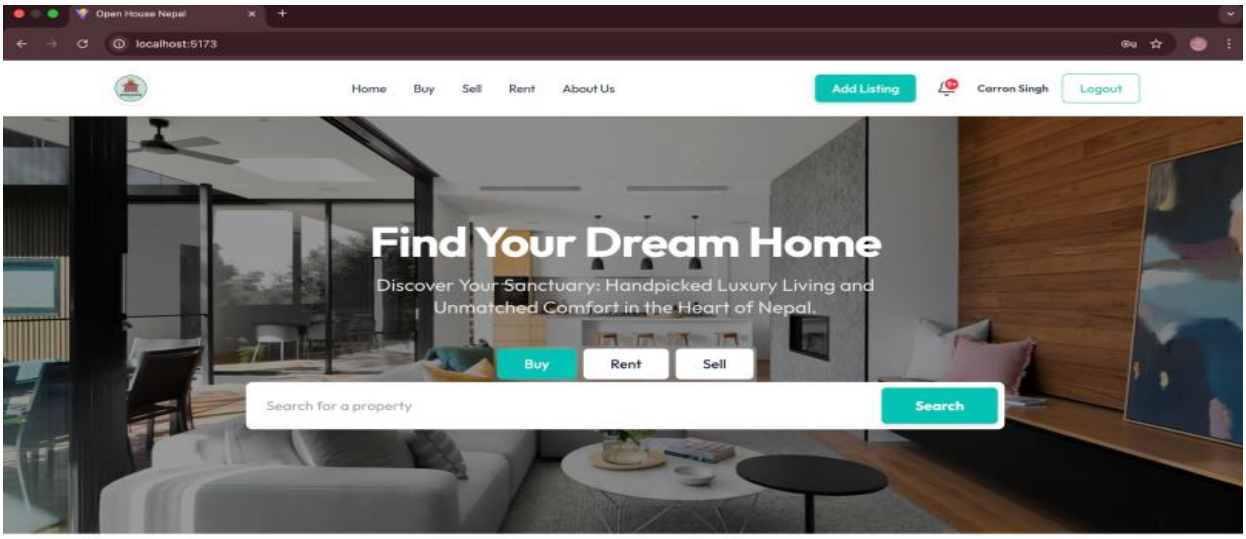


Figure 72 Homepage Interface

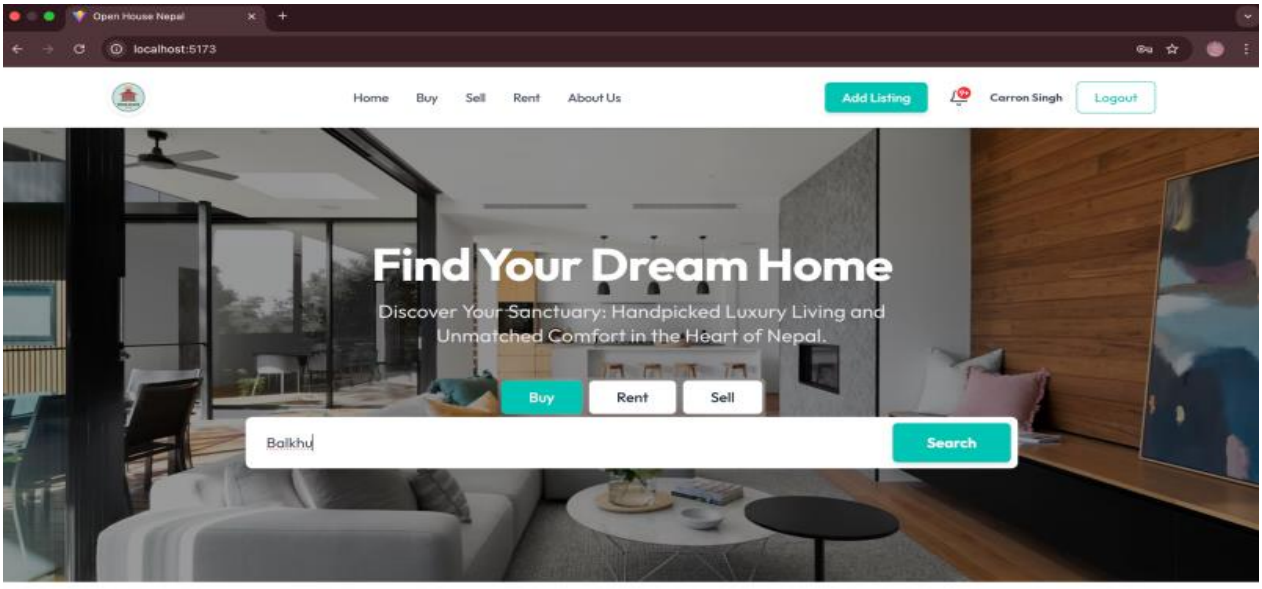


Figure 73 Search Interface

4.2.6 System Test 6: Keyword Location Search

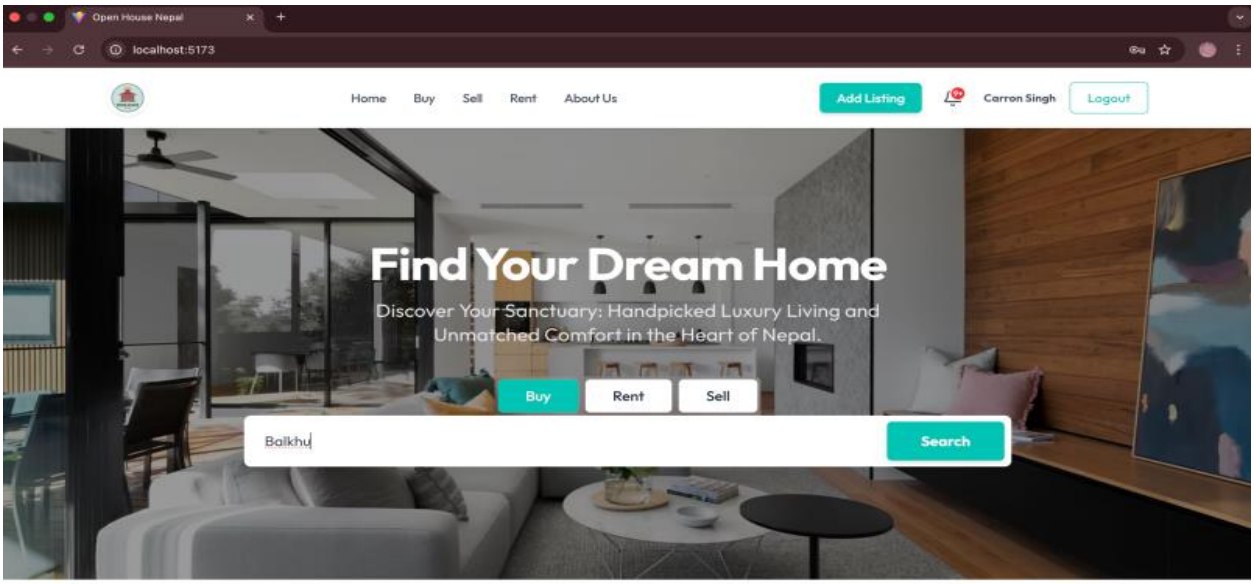


Figure 74 Keyword Search

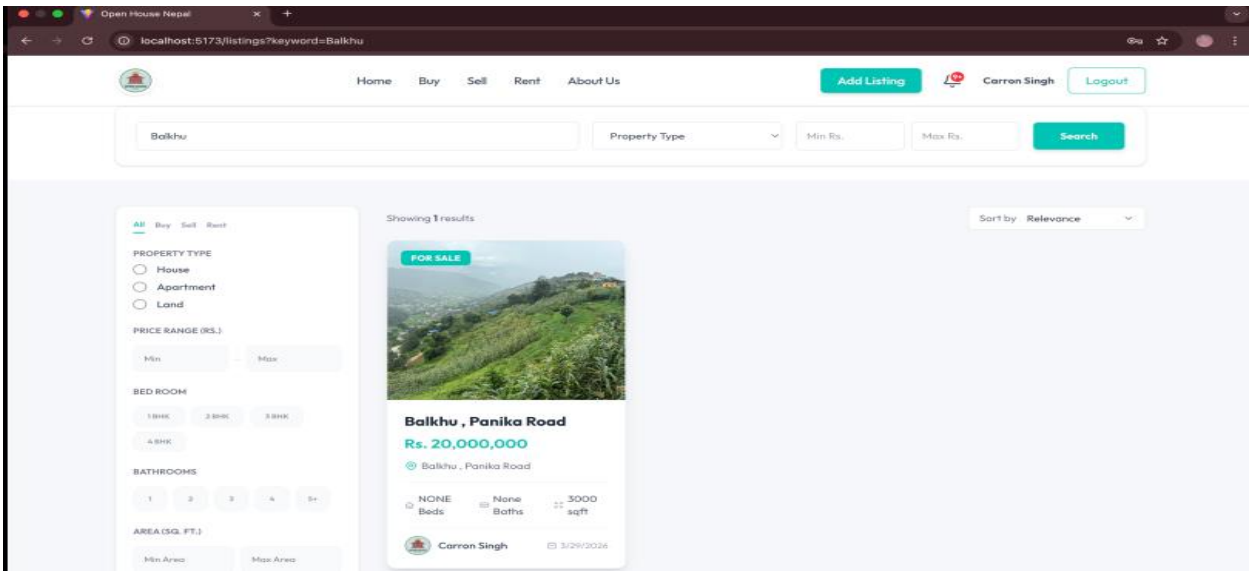


Figure 75 Search successful

4.2.7 System Test 7: Price Range Filtering

Tests the system's ability to filter properties within a specific budget range Minimum and Maximum price.

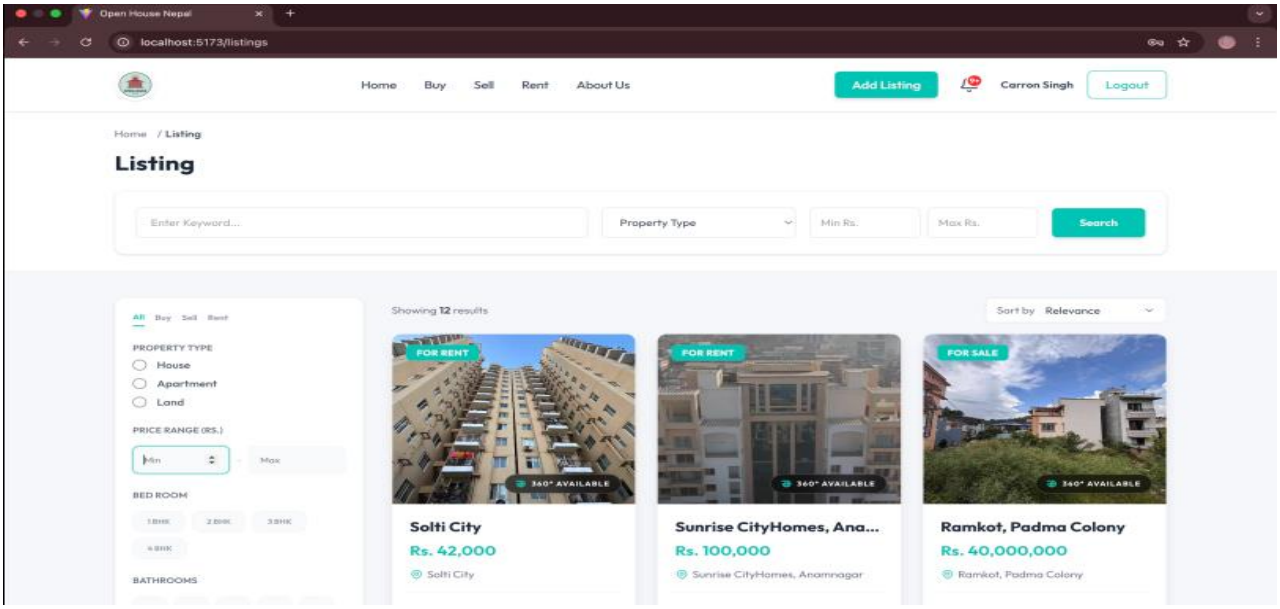


Figure 76 Price Range Filtering

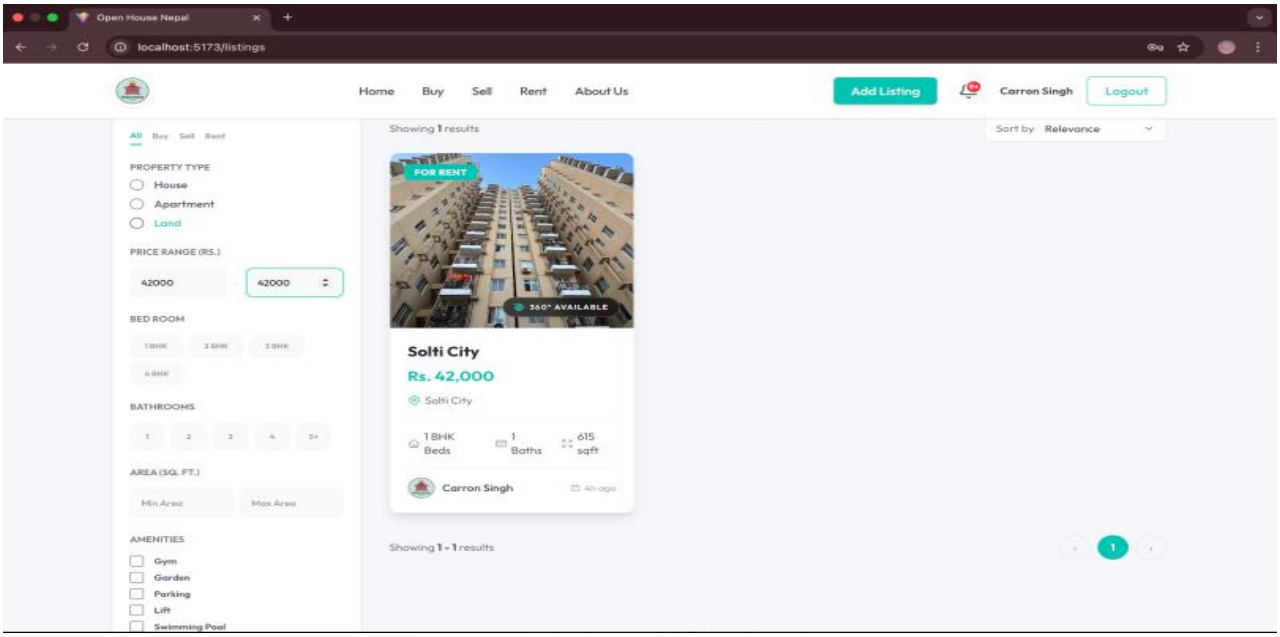


Figure 77 Price Range Filtering Successful

4.2.8 System Test 8: Property Type Filtering

Confirms that the user can filter listings by category like searching specifically for House or Apartment or Land.

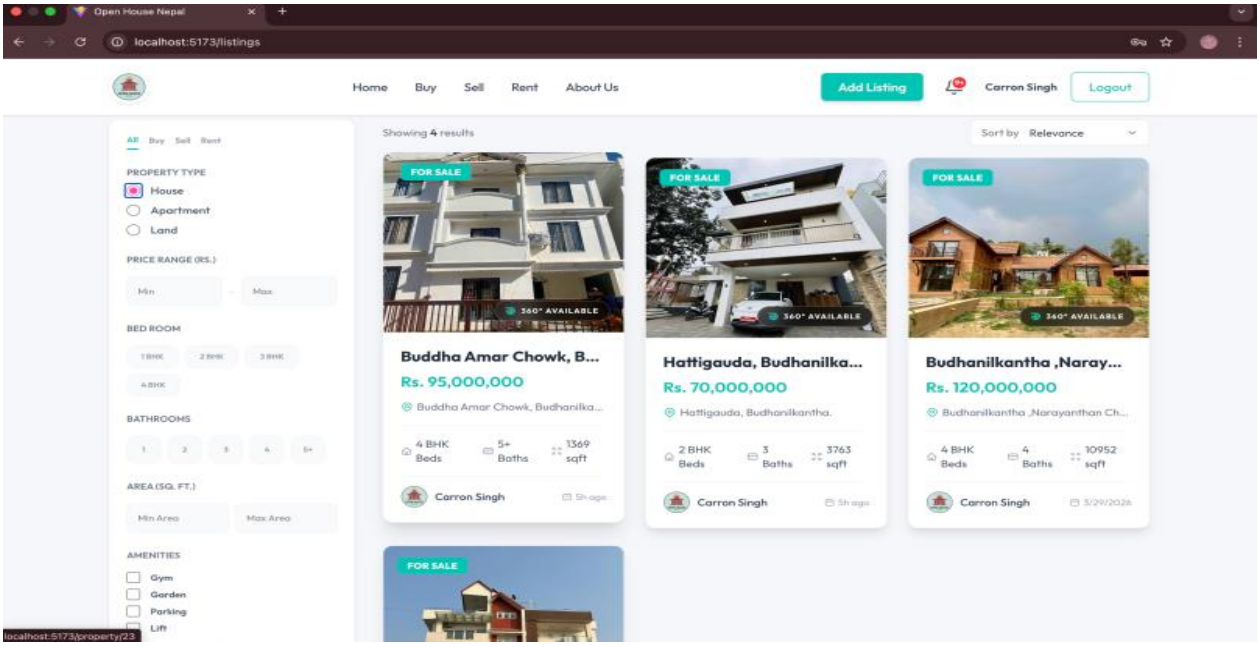


Figure 78 Property Type Filtering

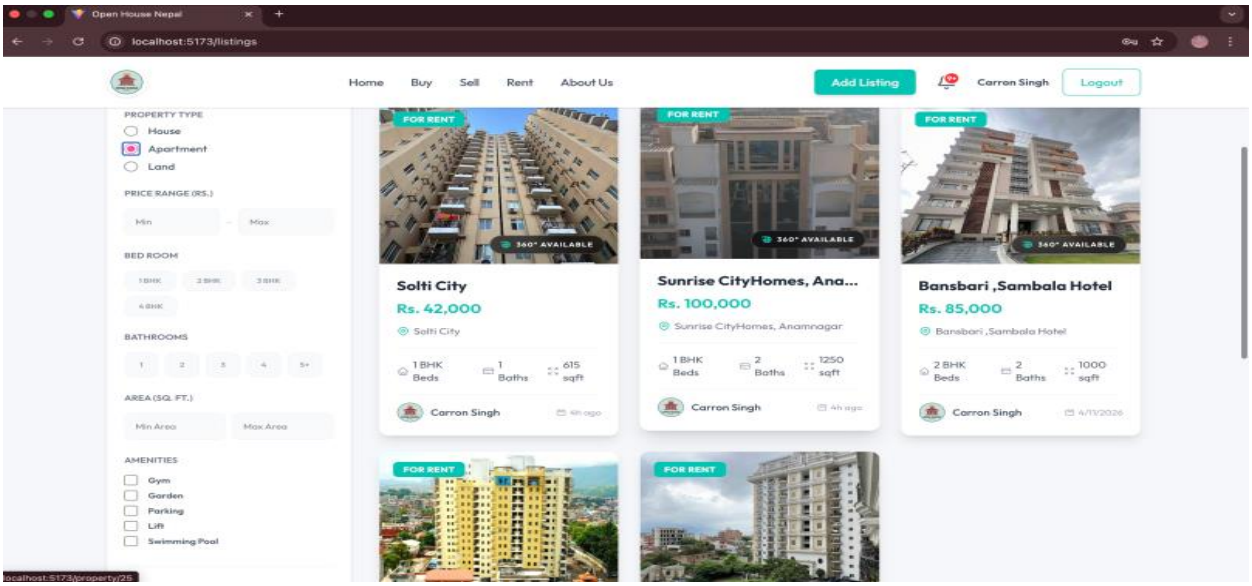


Figure 79 Property Type Filtering Successful

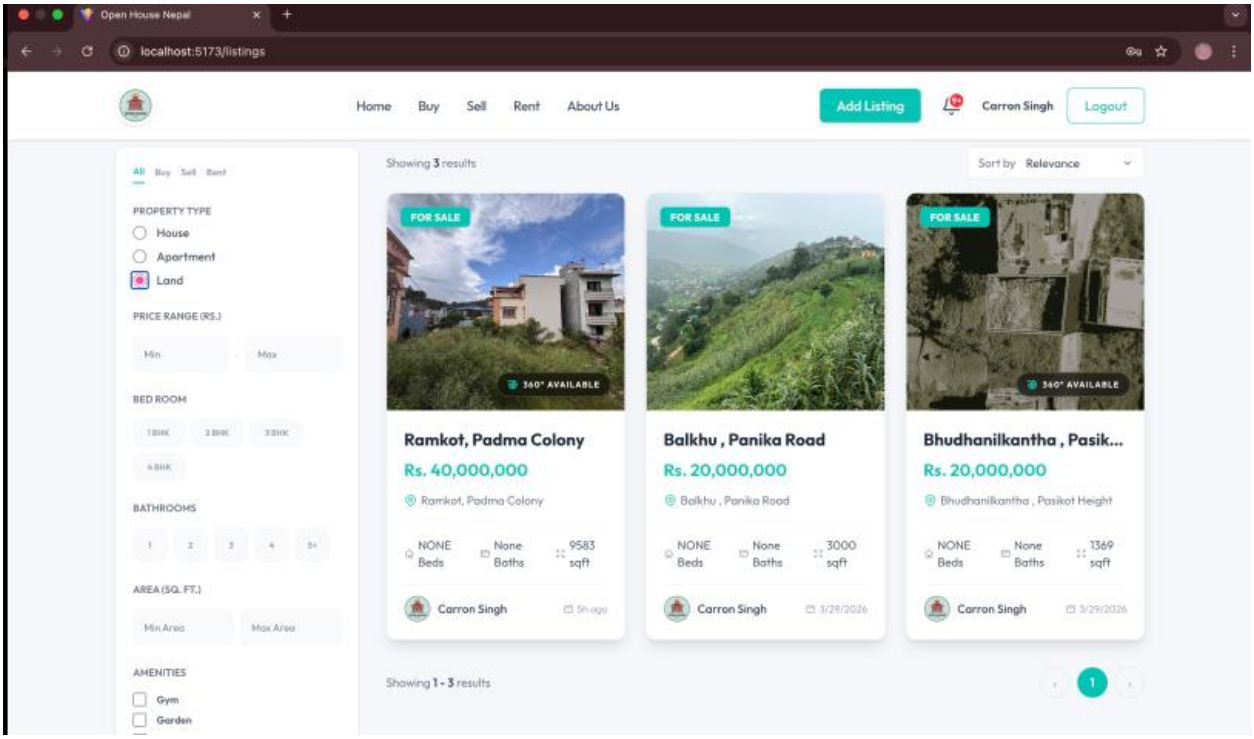


Figure 80 Property Type Filtering Successful

4.2.9 System Test 9: 360° Virtual Tour Module

Verifies that the Virtual Tour viewer loads correctly and allows the user to interact with the interactive panorama.

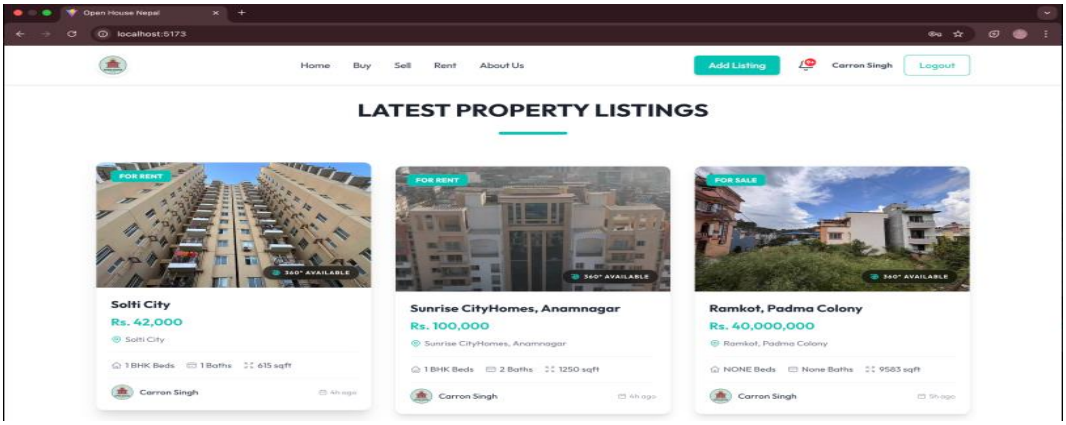


Figure 81 Property with 360° Virtual Tour Module

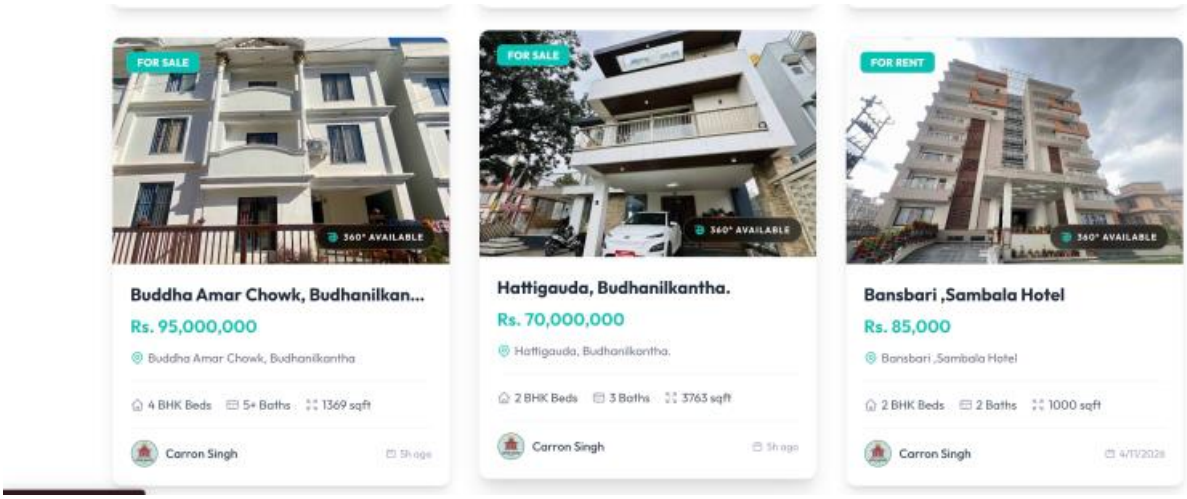


Figure 82 Property with 360° Virtual Tour Module

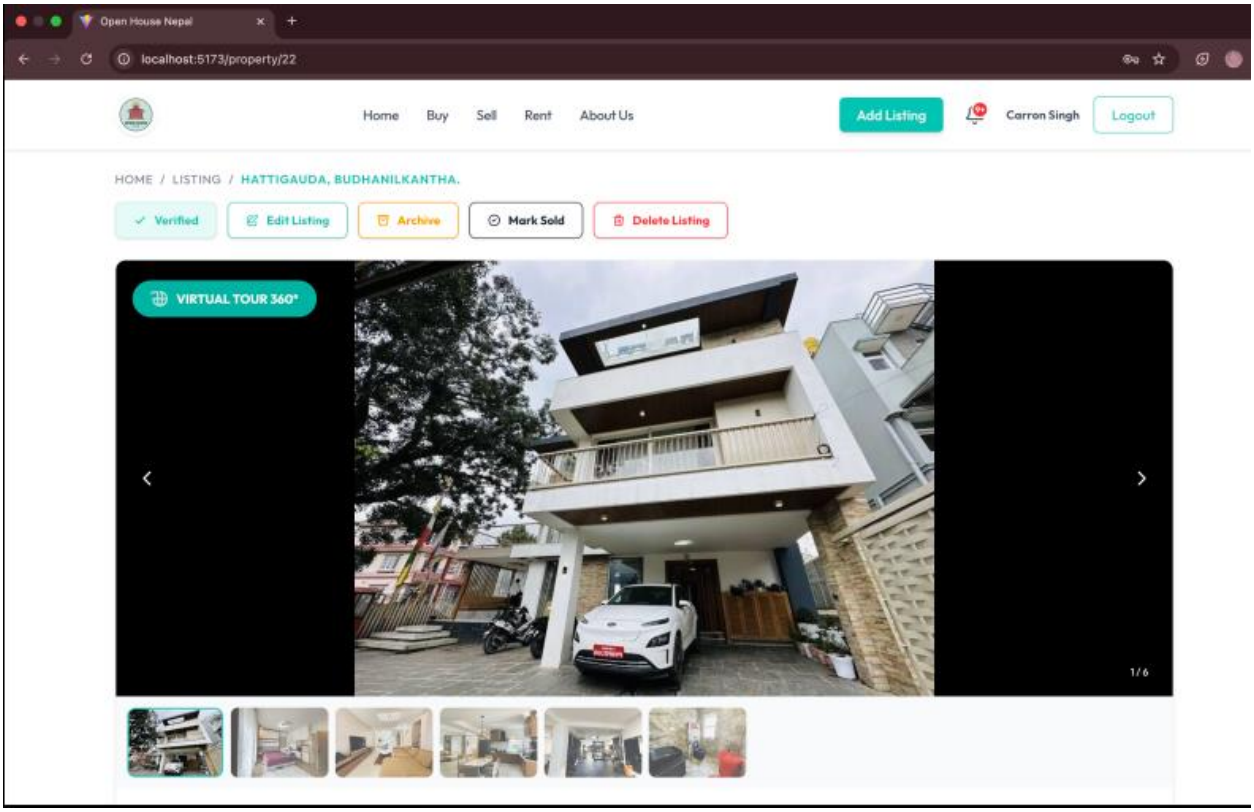


Figure 83 Select Property with 360° Virtual Tour Module

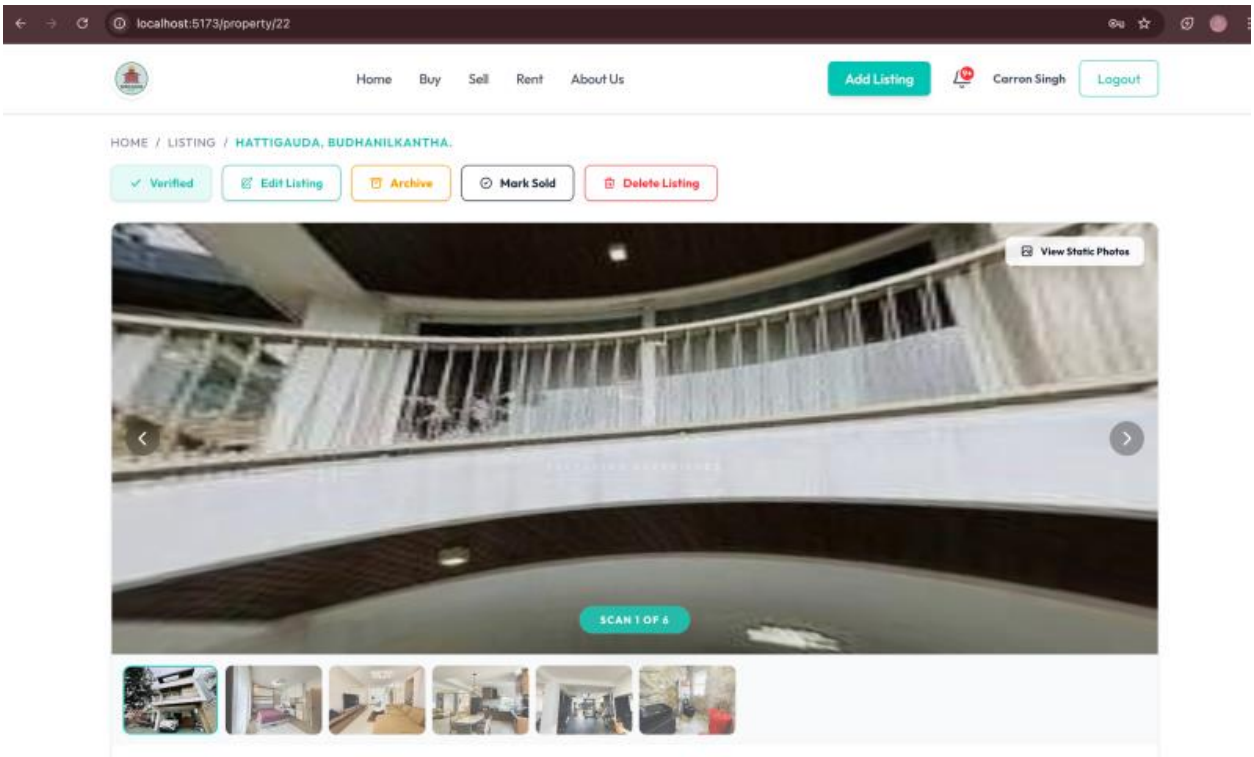


Figure 84 Property with 360° Virtual Tour Module

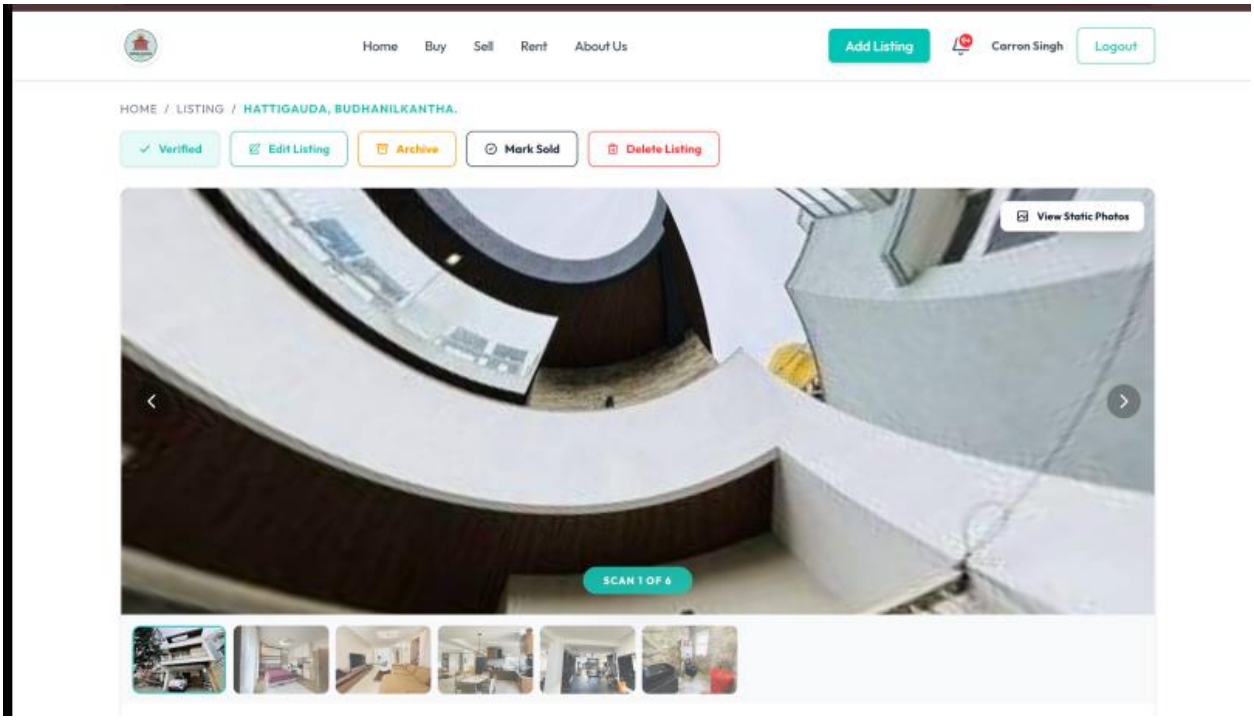


Figure 85 Property with 360° Virtual Tour Module

4.2.10 System Test 10: Add Listing Form Interface

Verifies that the “Post Property” form is fully functional and includes all required input fields such as Title, Price, Location, Description, Property Type, and Images. It ensures that sellers can properly enter all necessary property details.

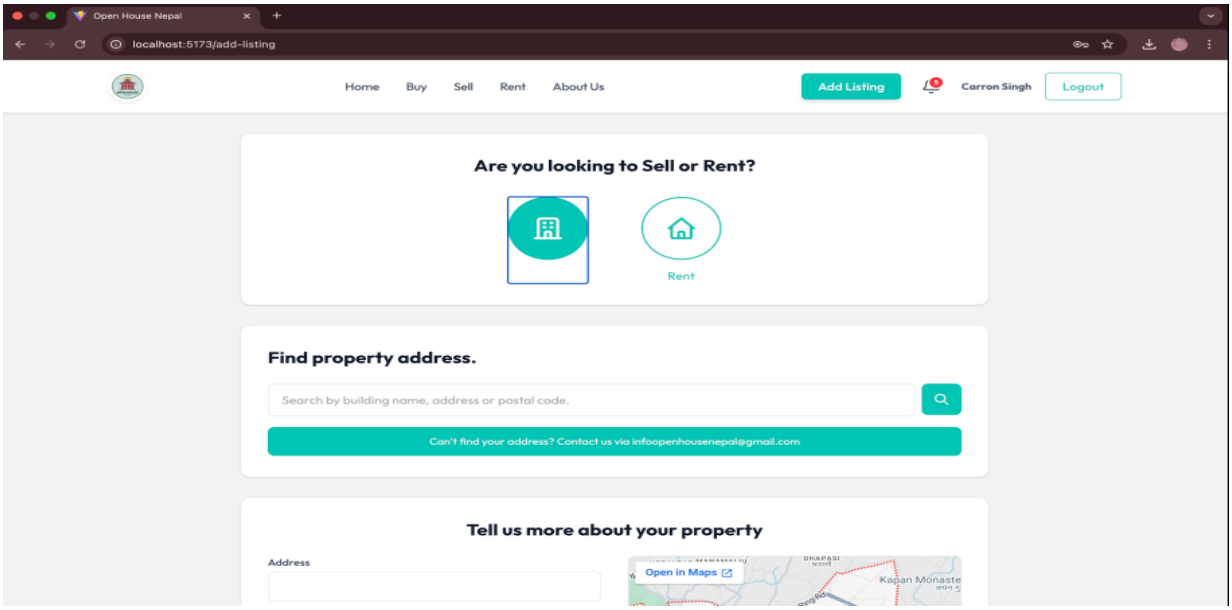


Figure 86 Add Listing

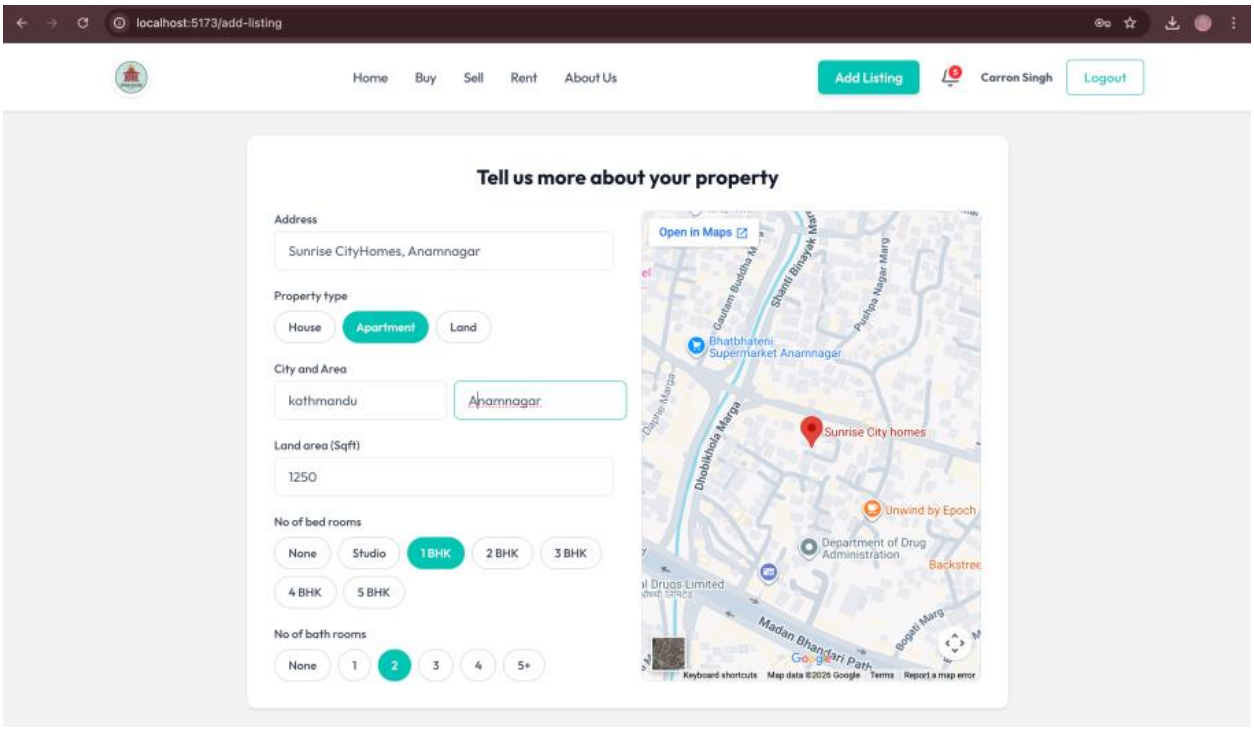


Figure 87 Add Listing Form Interface

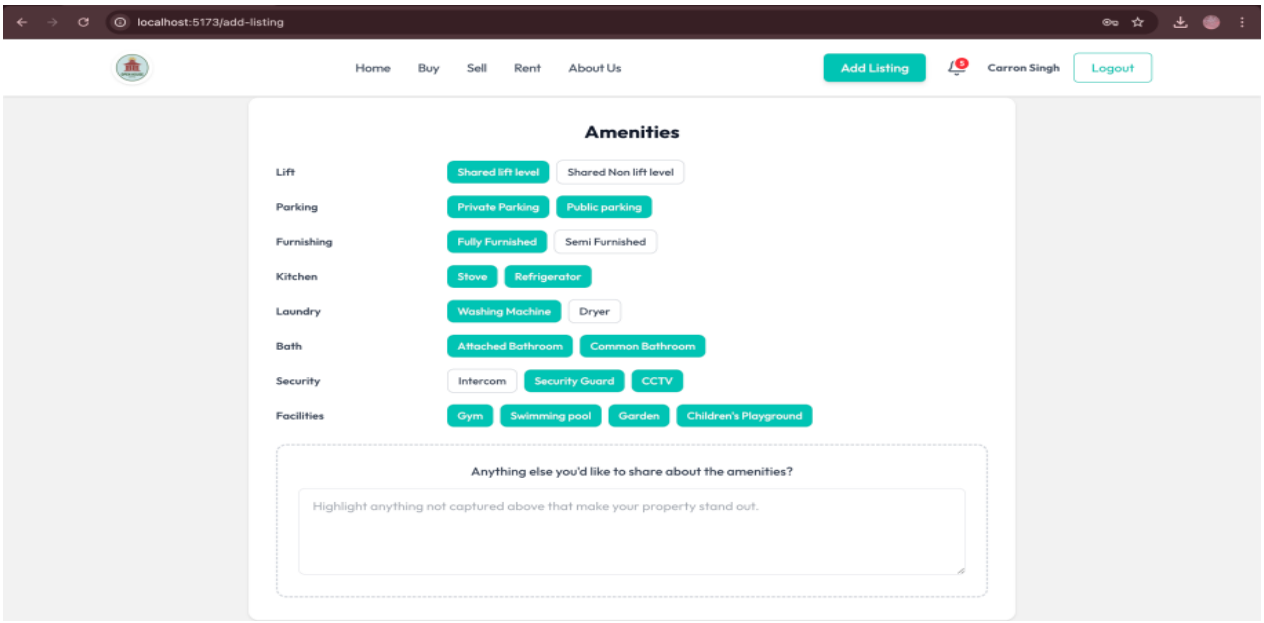


Figure 88 Add Listing Form Interface

4.2.11 System Test 11: Document Upload Interface

Tests the system's capability to handle multiple file uploads, specifically for required documents such as Citizenship and Property Ownership papers. It ensures files can be selected, displayed, and prepared for submission without errors.

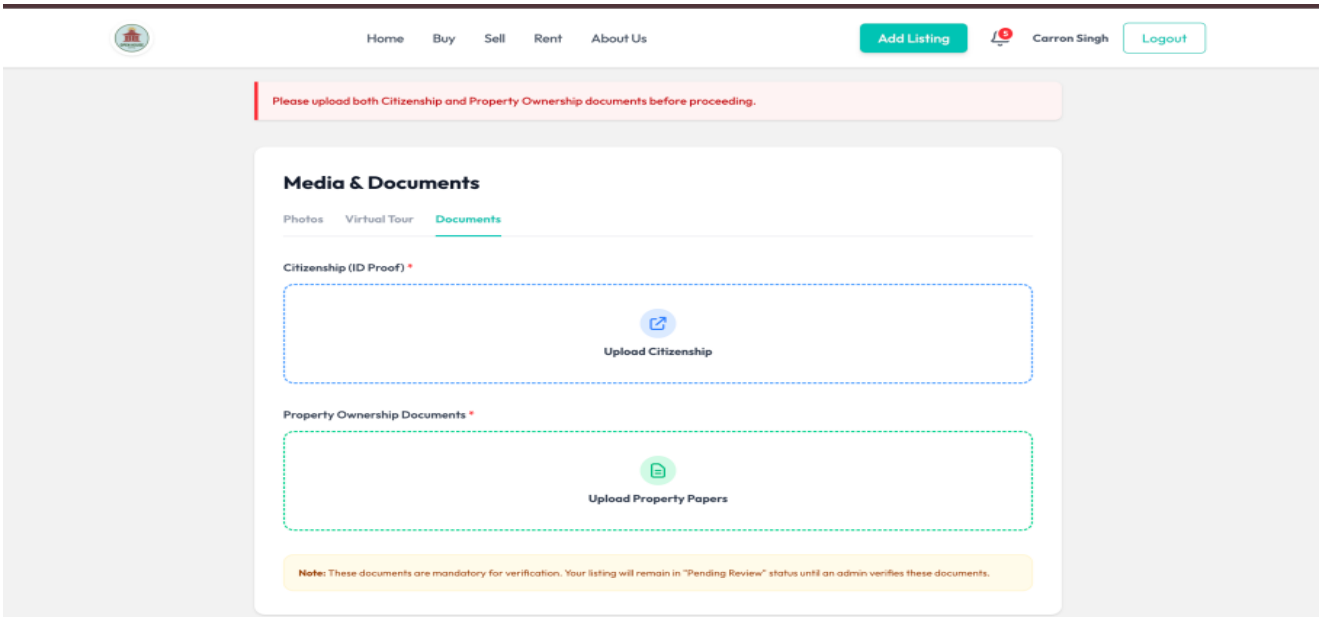


Figure 89 Document Upload Interface

4.2.12 System Test 12: Submission & Status Feedback

Verifies that after successfully submitting a property listing, the system provides clear confirmation feedback indicating that the listing is “Pending Approval” by the admin.

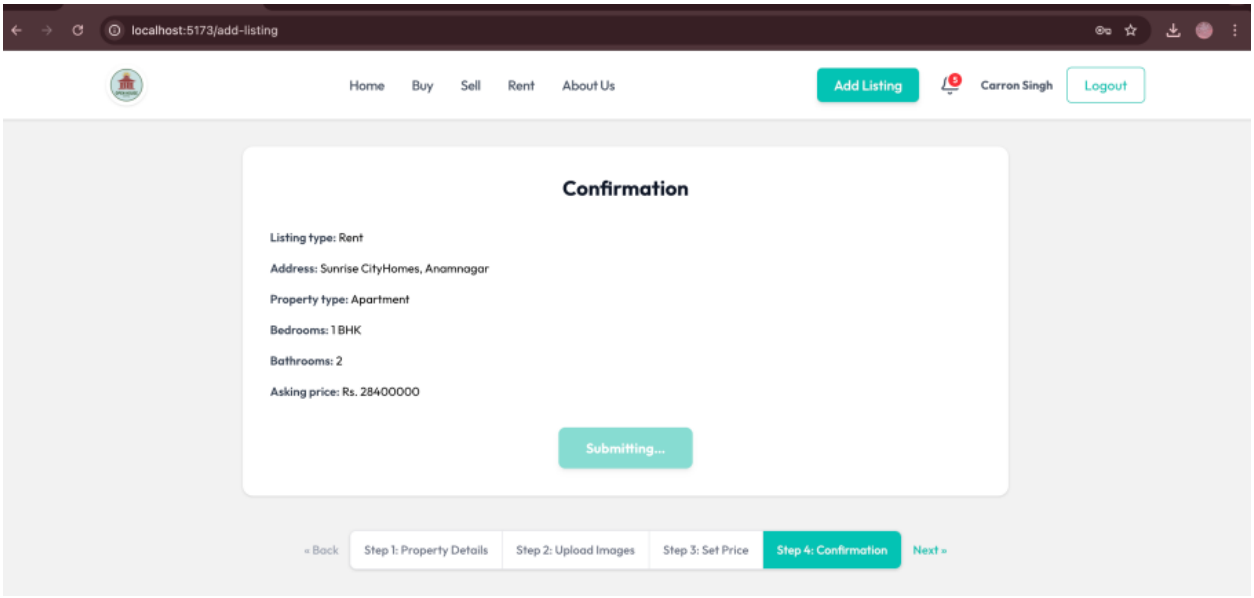


Figure 90 Submission

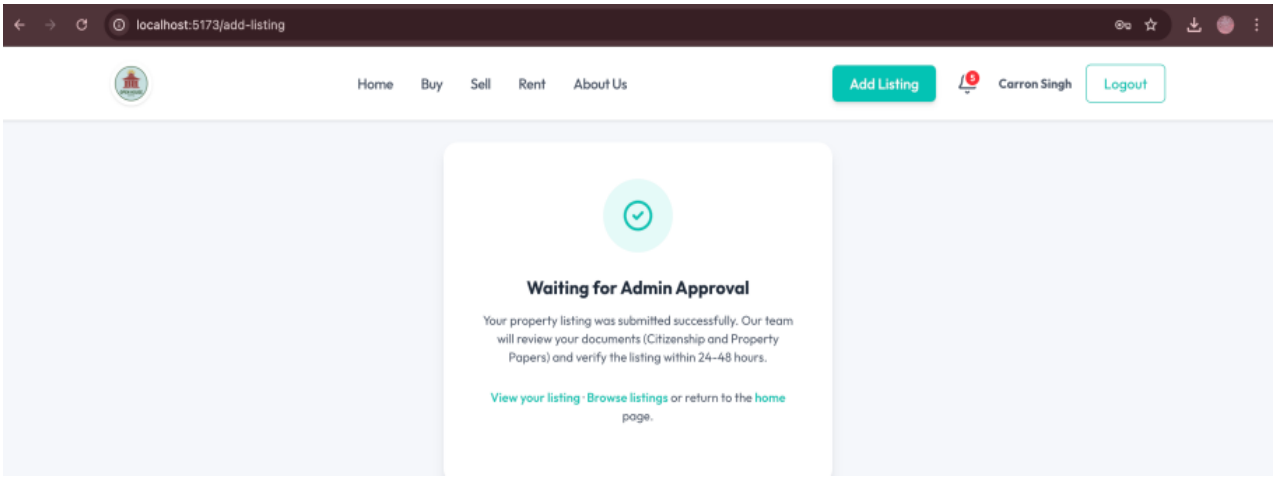


Figure 91 Status Feedback

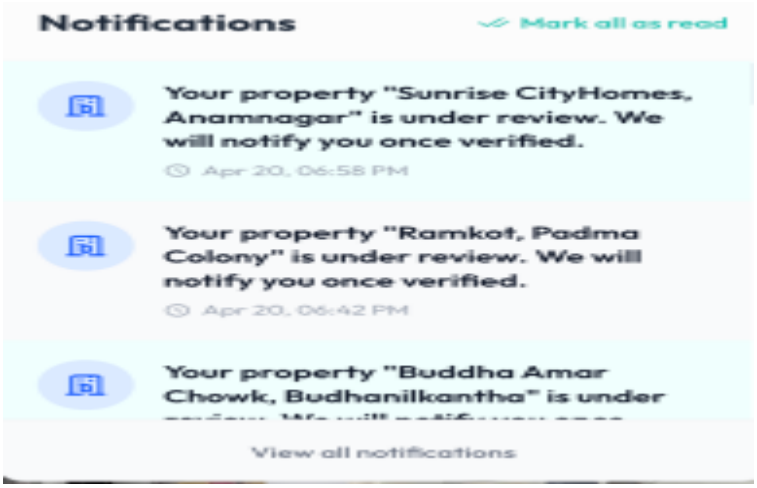


Figure 92 Under Review



Figure 93 Listed Successfully

4.2.13 System Testing 13: Seller Personal Dashboard

Verifies that the seller can access a personal dashboard where they can view, manage, and track all their submitted property listings, including their status like Pending, Approved, Rejected.

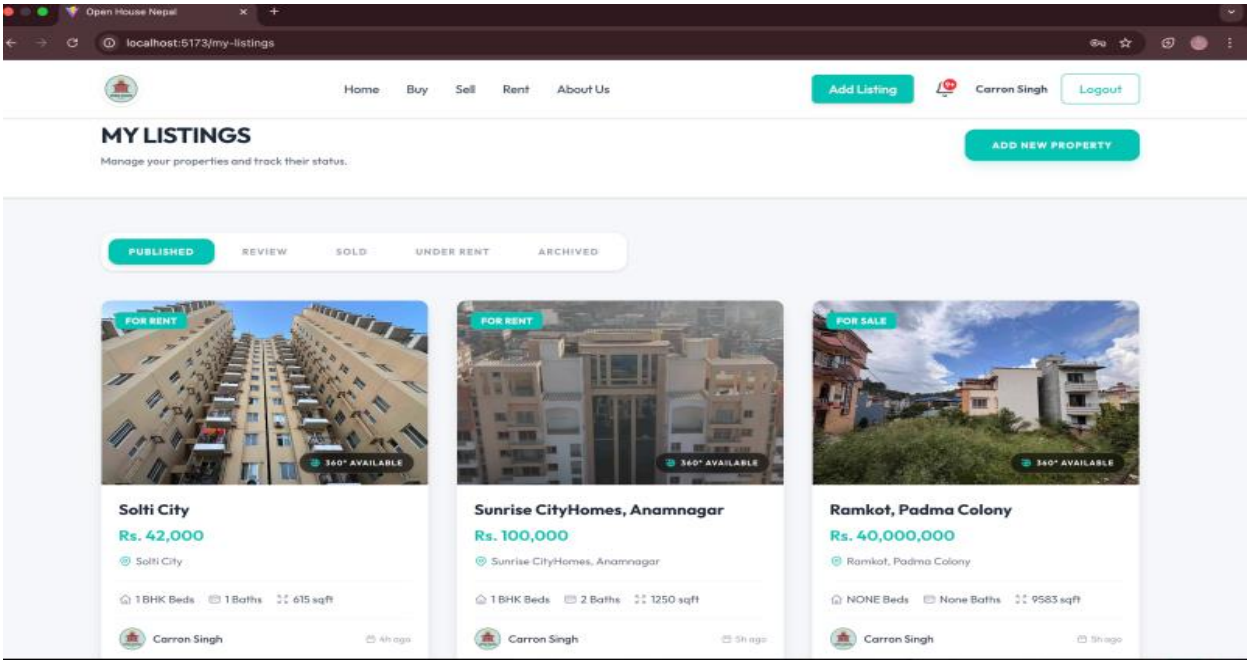


Figure 94 Seller Personal Dashboard

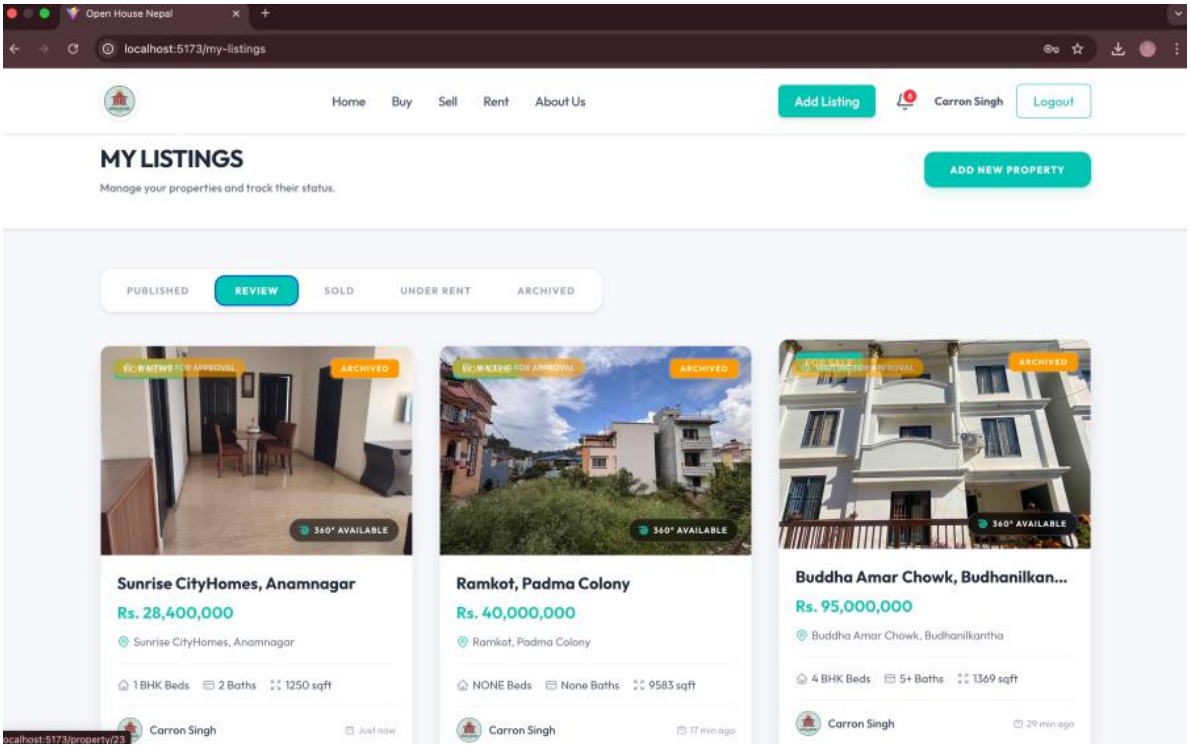


Figure 95 Seller Personal Listing

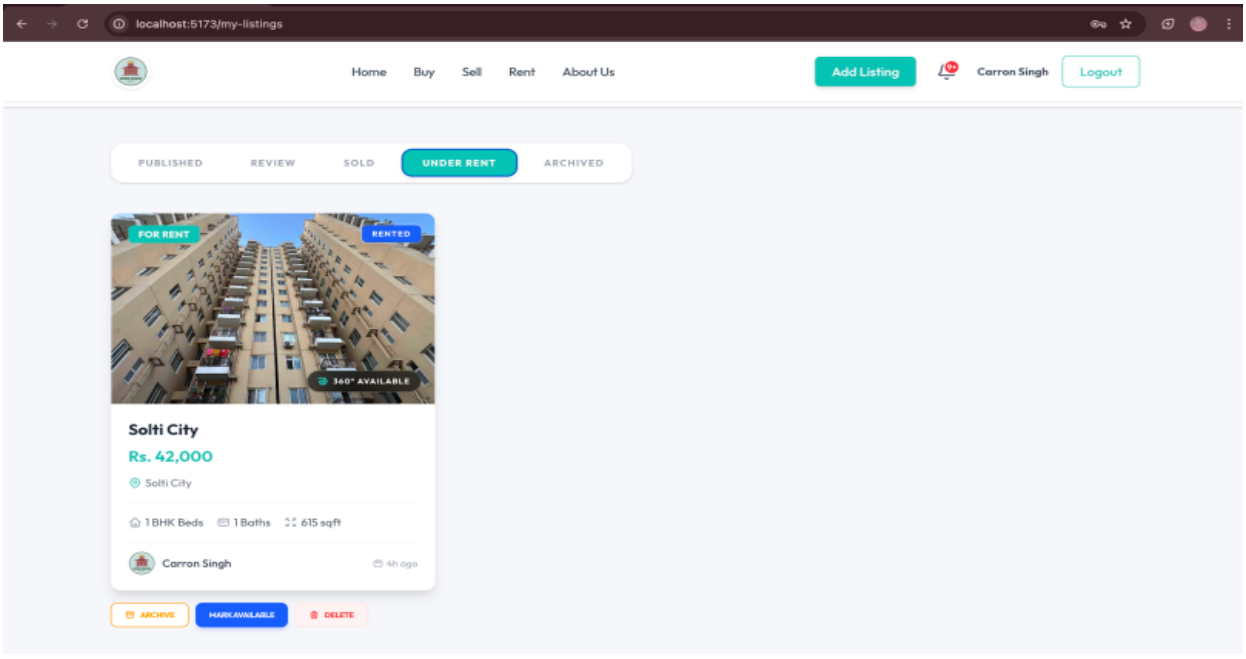


Figure 96 Seller Listing Under Rent

4.2.14 System Testing 14: Update Listing (Edit Mode)

Tests the functionality to enable sellers to modify and update their existing property listings using the user interface. It ensures that changes like price, description, images are saved and reflected correctly.

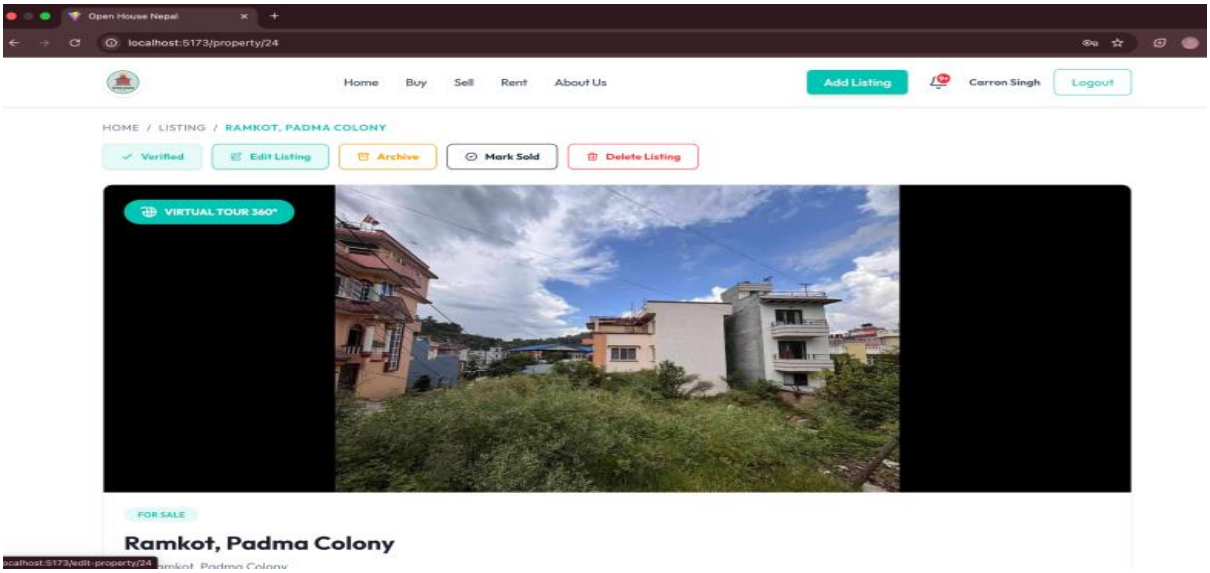


Figure 97 Update Listing (Edit Mode)

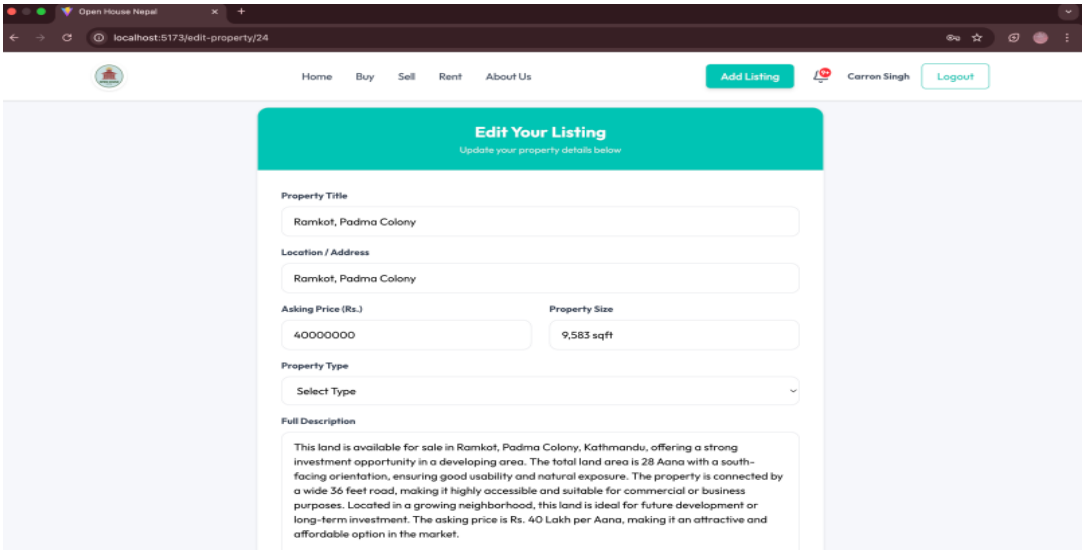


Figure 98 Update Listing (Edit Mode)

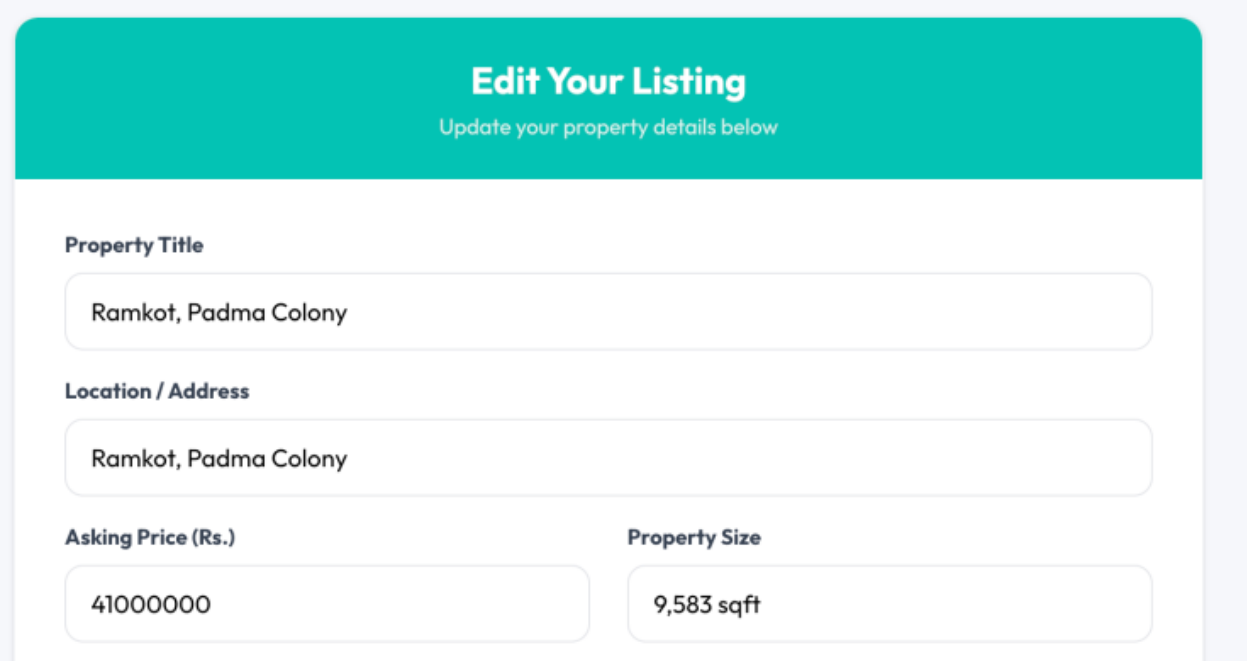


Figure 99 Update Listing (Edit Mode)

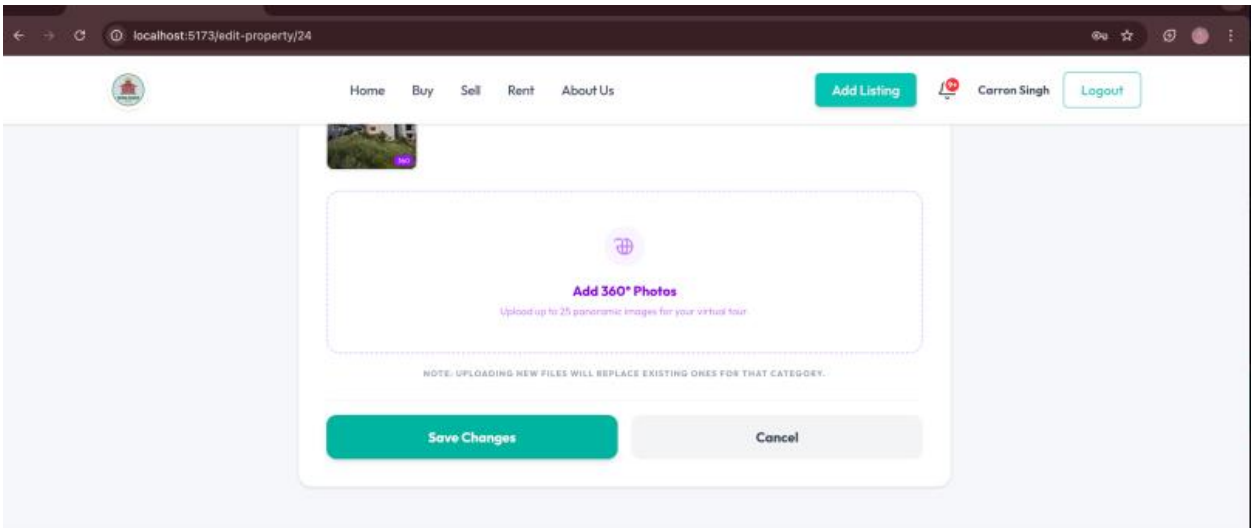


Figure 100 Update Listing (Edit Mode)

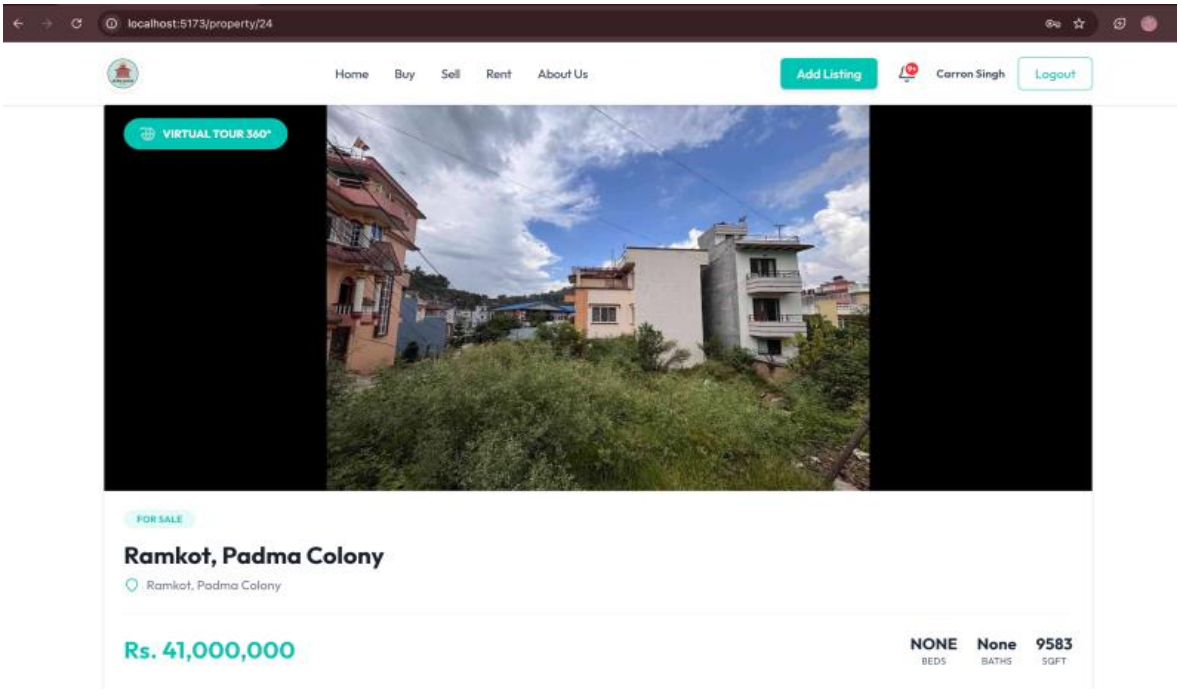


Figure 101 Listing Updated

4.2.15 System Testing 15: Password Reset Request (Forgot Password)

Tests the case where a user can request a password reset by typing his/her registered email into the system.

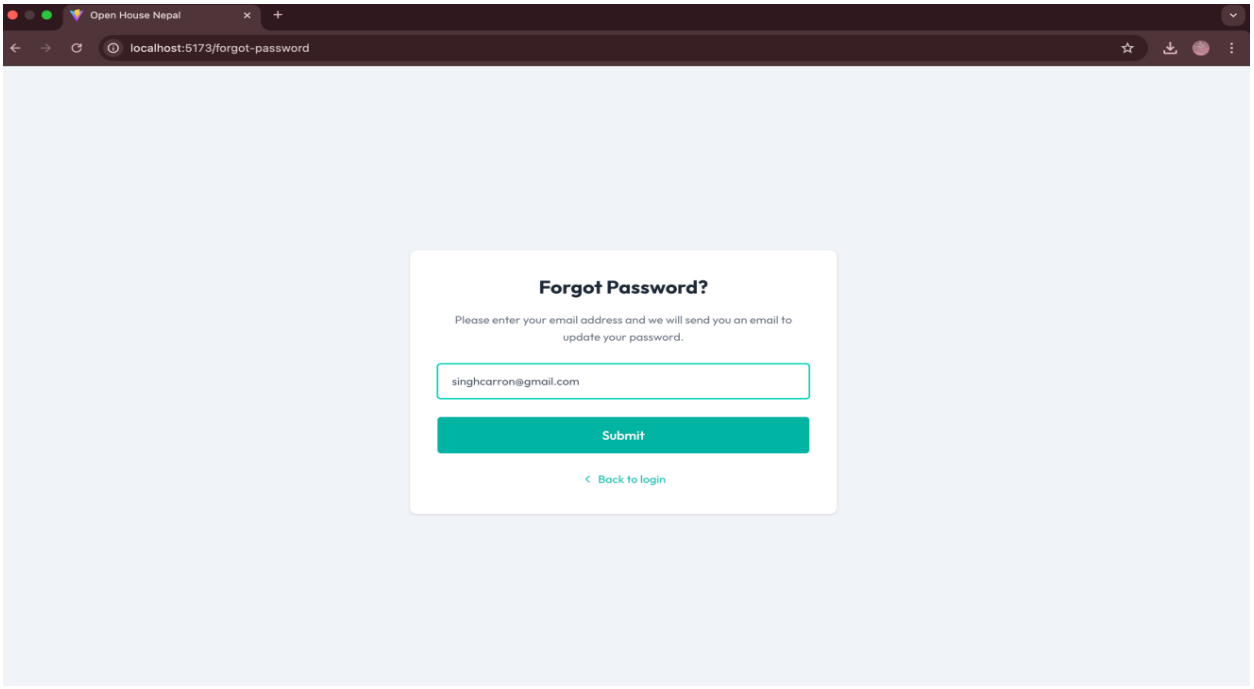


Figure 102 Password Reset Request

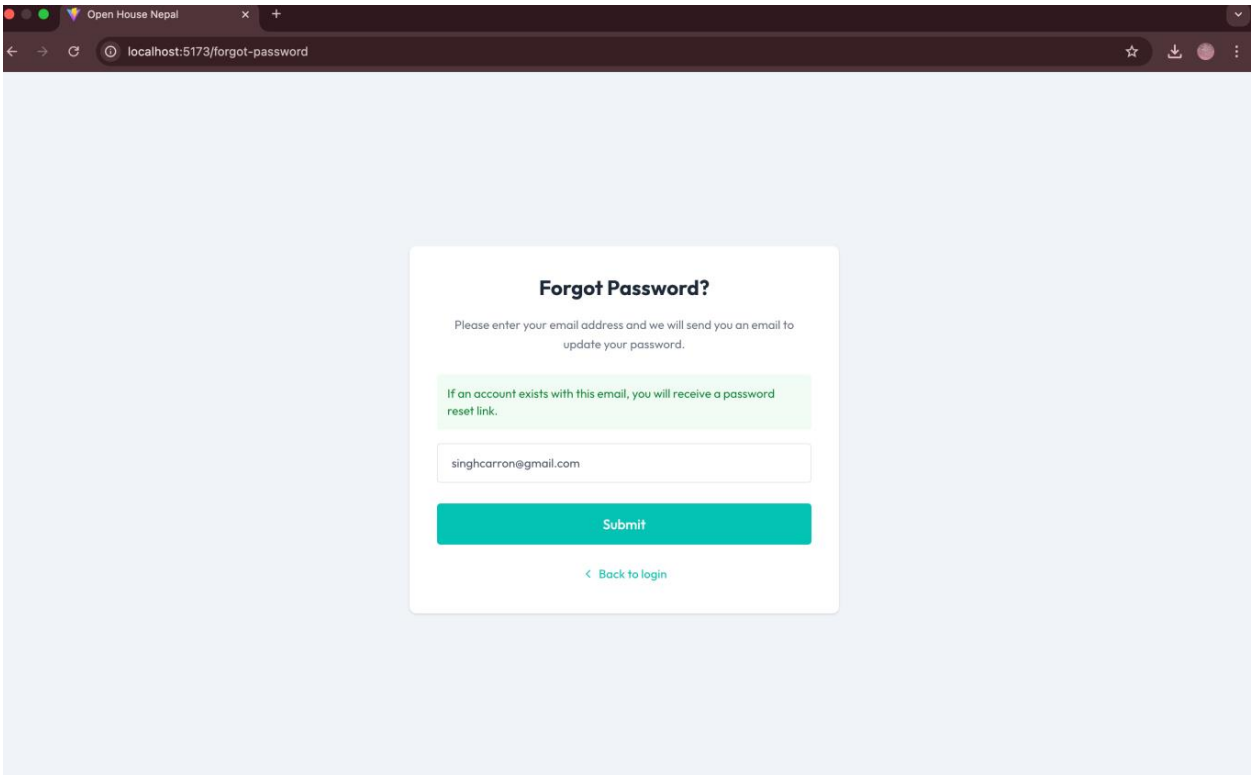


Figure 103 Password Reset Request sent

4.2.16 System Testing 16: Password Reset (Email Check-up)

Verifies that the system can send a working password reset email to the inbox of the user.

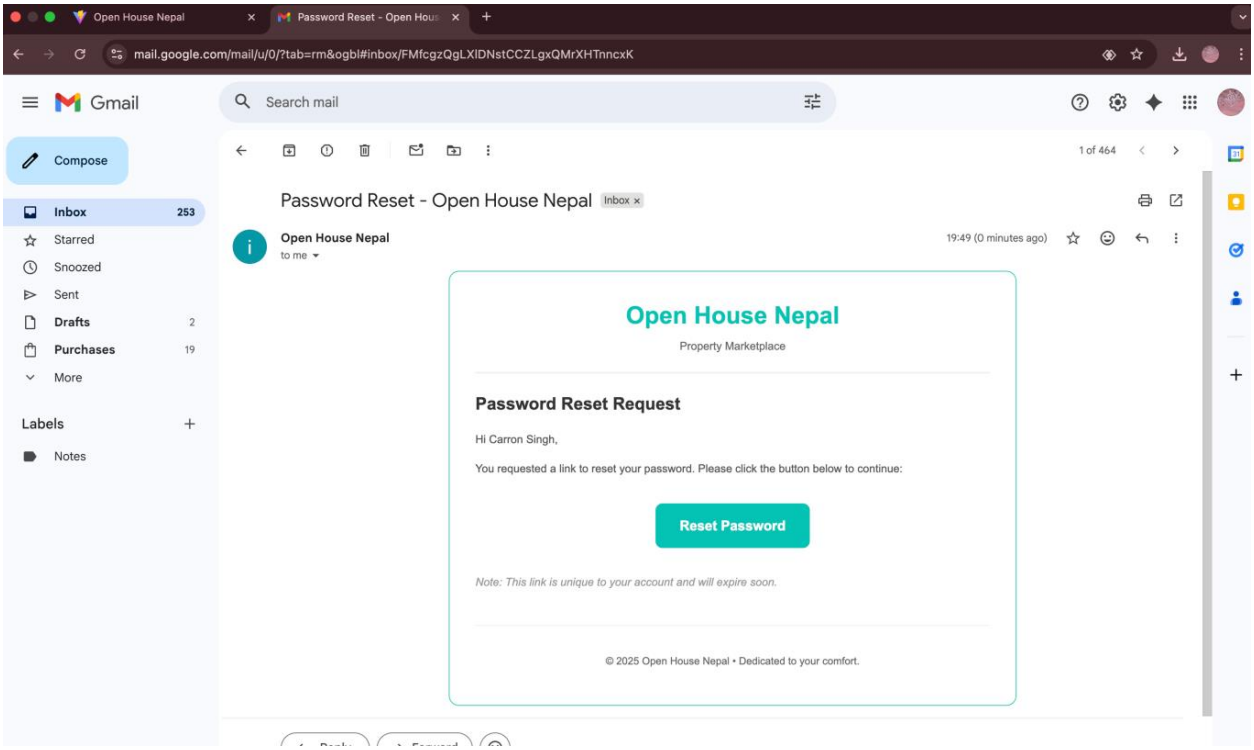


Figure 104 Password Reset Gmail link

4.2.17 System Testing 17: New Password settings

The test is performed to determine whether the user can set a new password or not via the secure reset link included in the email.

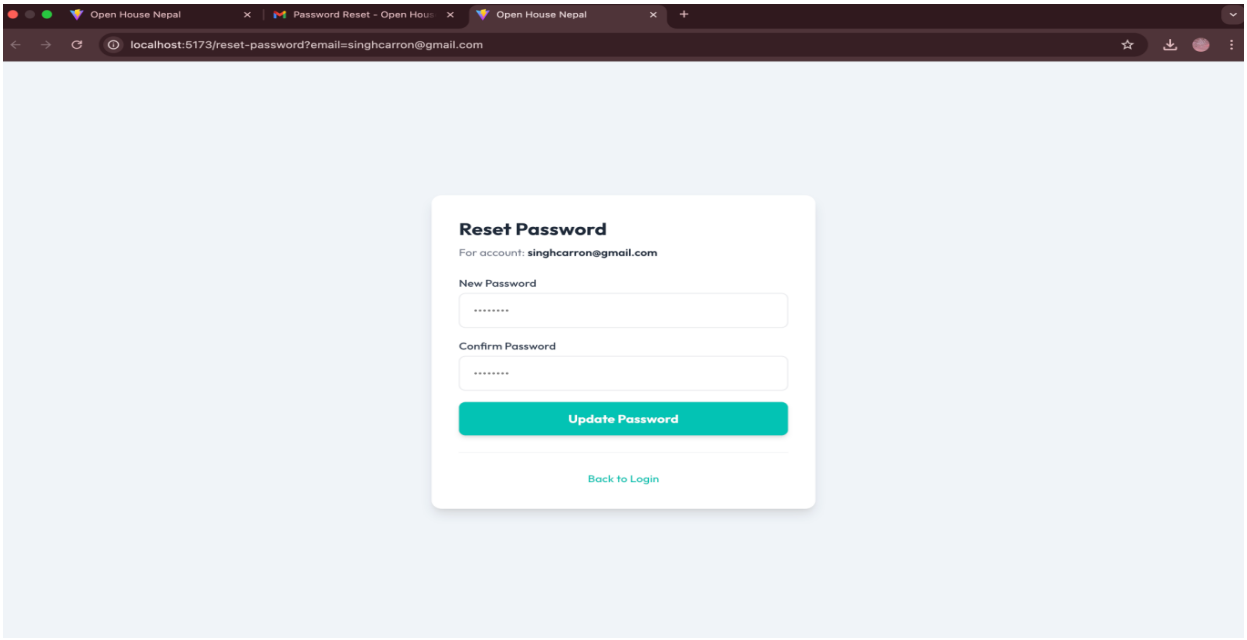


Figure 105 New Password settings

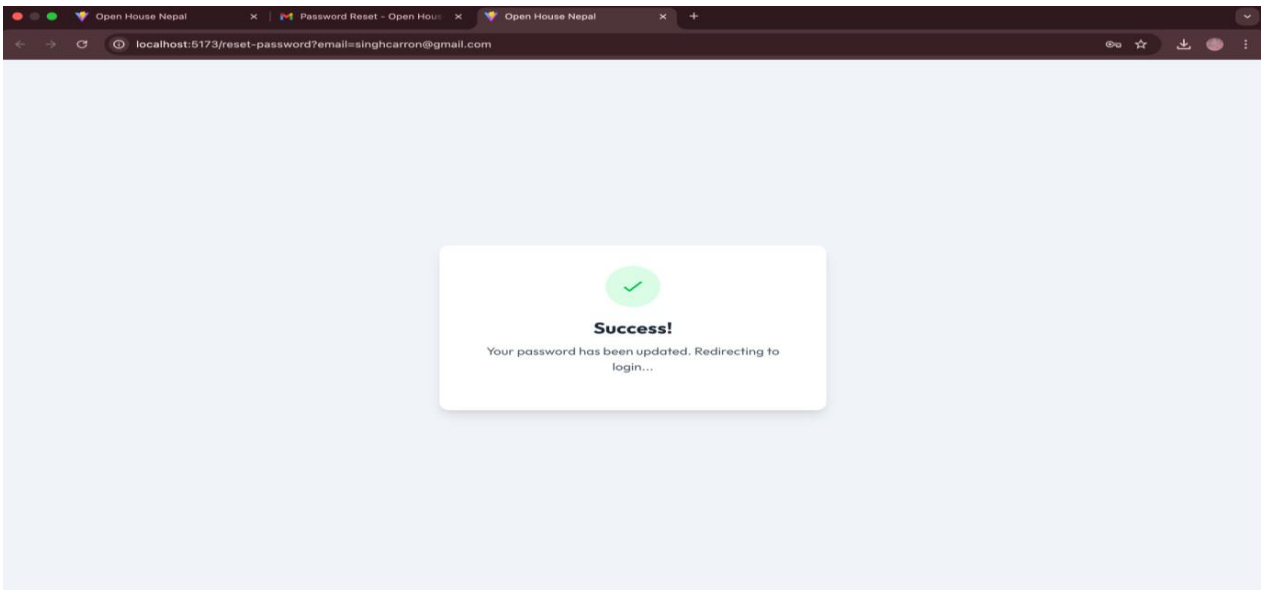


Figure 106 New Password settings Success

4.2.16 System Testing 18: Verifying a post-Reset login

Makes the old password ineffective and makes the system to allow one to log in with the newly updated password.

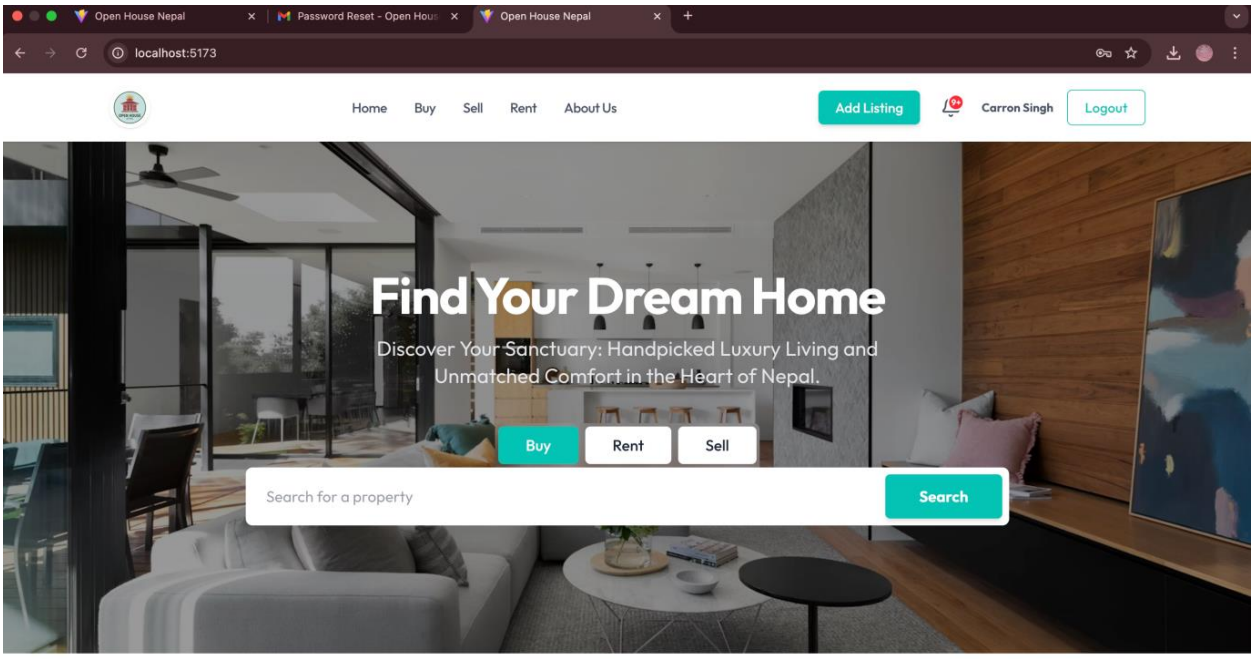


Figure 107 Verifying a post-Reset login

4.2.19 System Testing 19: Interactive Map Display

Checks that the map service (Google Maps API or Leaflet) loads properly on the Property Detail page and shows the position following the information given by the seller.

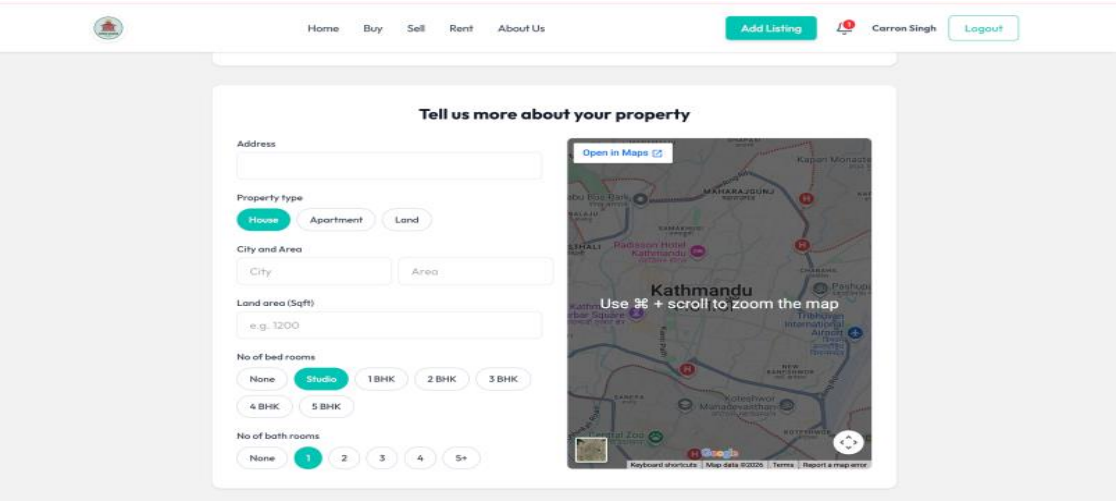


Figure 108 Interactive Map Display

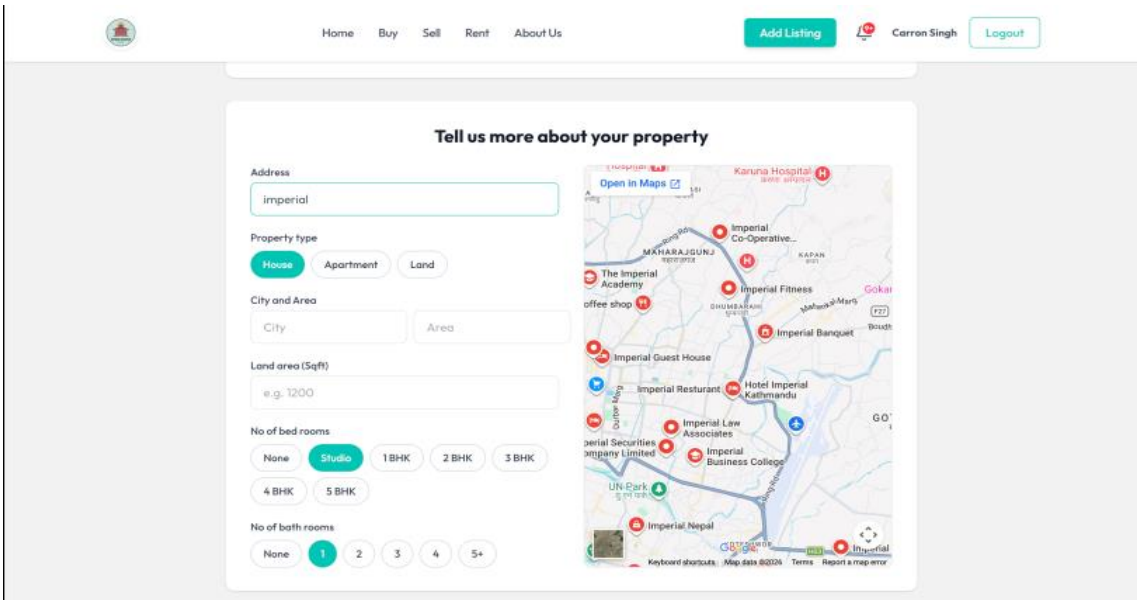


Figure 109 Interactive Map Display

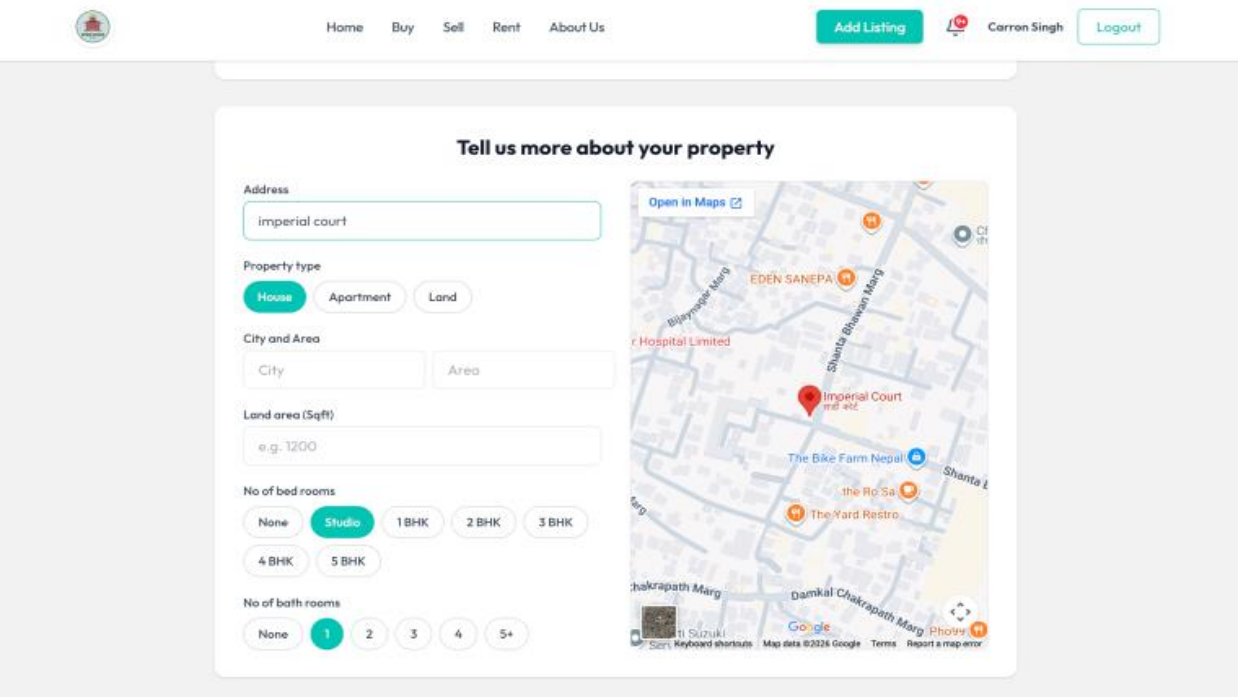


Figure 110 Interactive Map Display

4.2.20 System Testing 20: Map Navigation & Interaction

Practical Tests the interactive map can be zoomed in/out and panned around the map to discover the immediate surroundings of the property.



Figure 111 Map Navigation & Interaction

4.2.21 System Testing 21: Submission of Property Review

Confirms that an authenticated user (buyer) will be able to post a star rating (1-5) and a written review of a property listing.

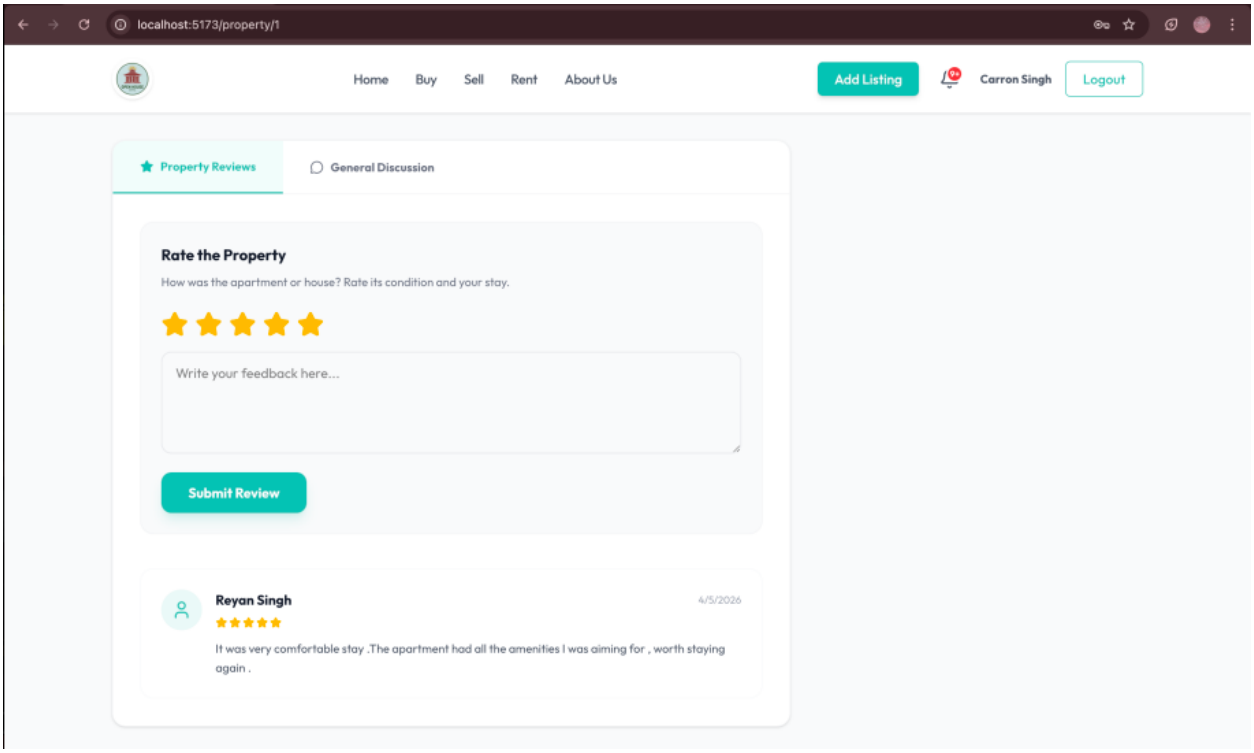


Figure 112 Submission of Property Review

4.2.22 System Testing 22: Commenting System in Real Time

Ensures that users can leave comments or questions on a property page and that the comments are displayed immediately in the comment thread without requiring a page refresh when real-time is used.

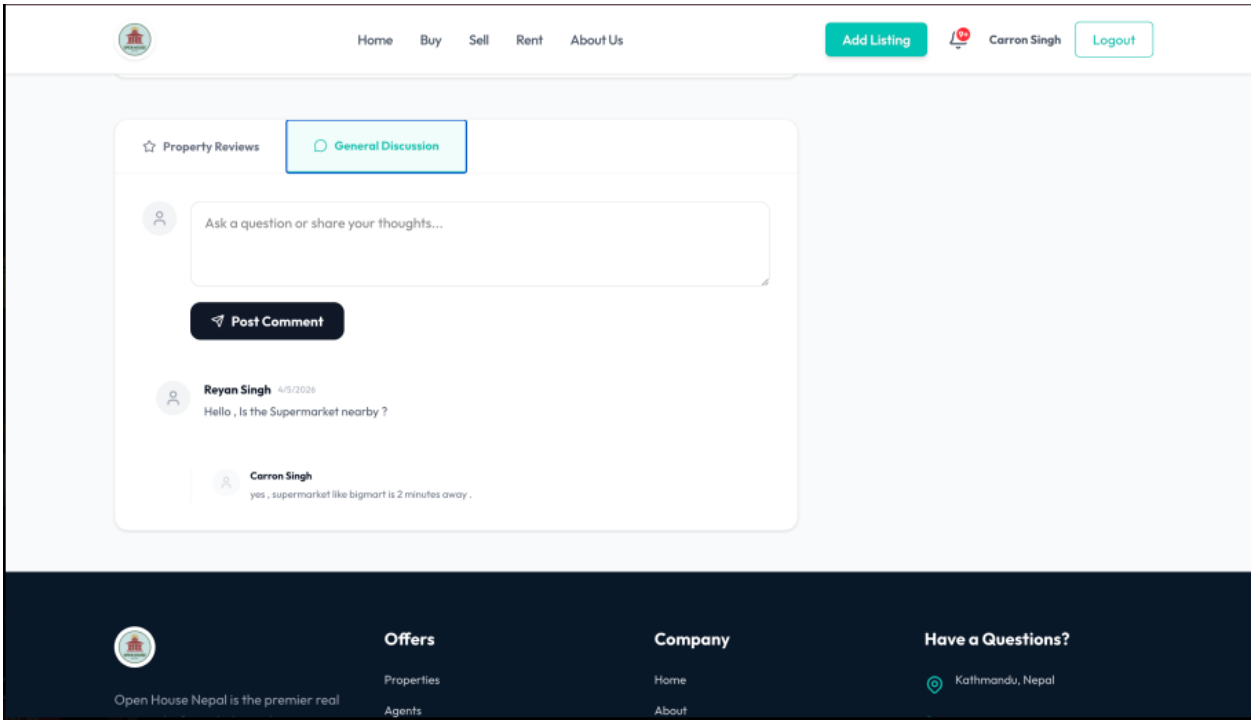


Figure 113 Commenting System in Real Time

4.2.23 System Testing 23: Amenity-Based Filtering

Checks the functionality of the advanced search filters to filter properties according to the desired amenities (e.g. this means that only properties with a Garden should be displayed).

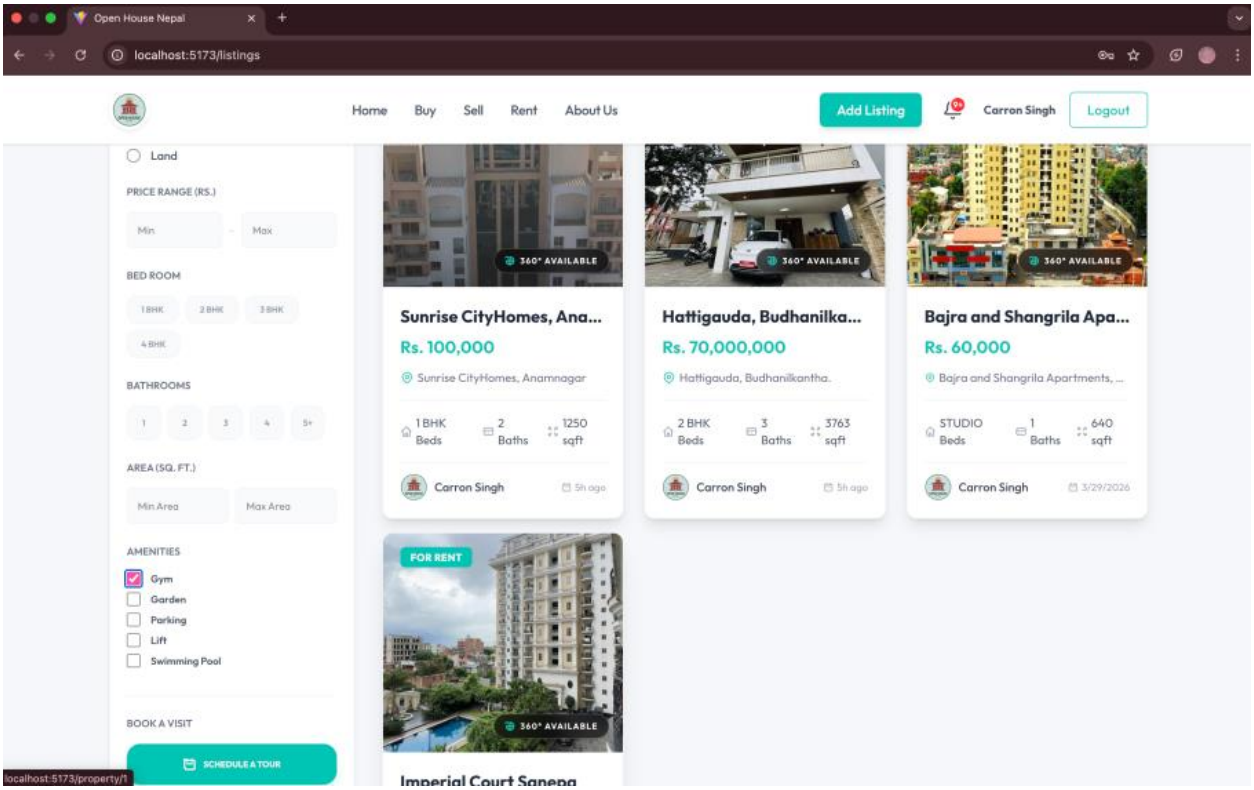


Figure 114 Amenity-Based Filtering

4.2.24 System Testing 24: Dynamic Loading of Amenities

Checks the presence of the amenities (icons and labels) chosen by the seller when creating the property in the property detail page.

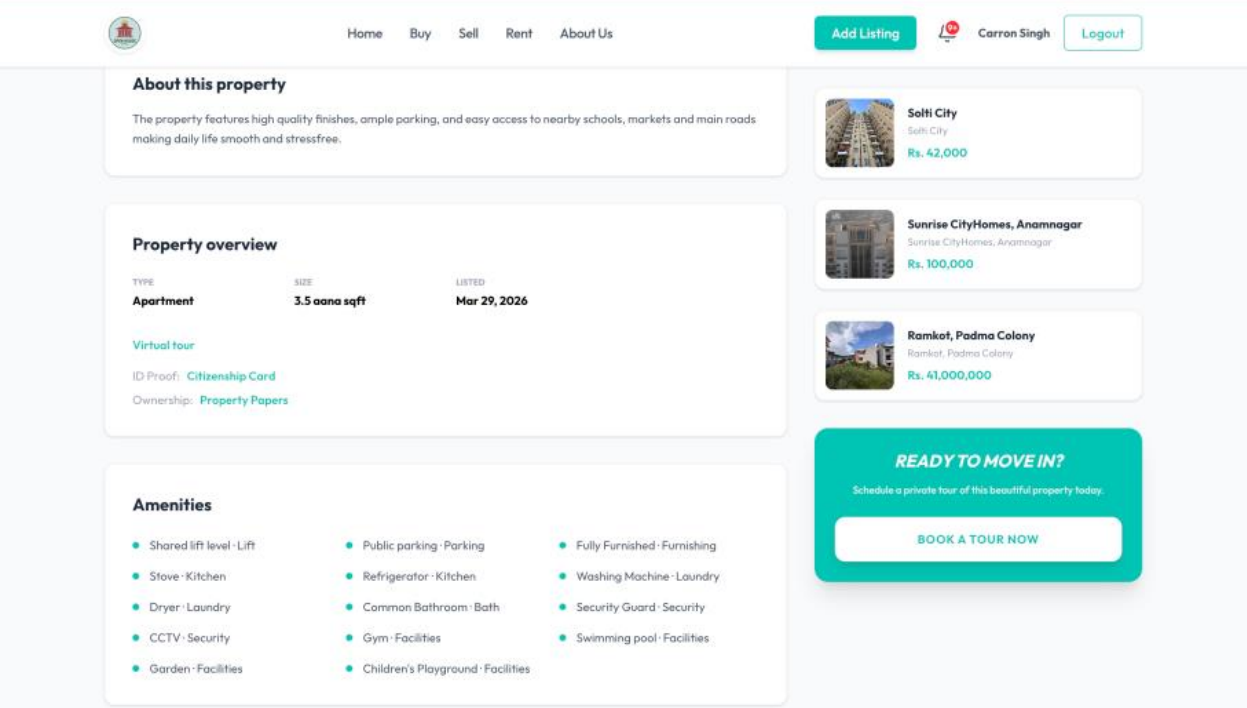


Figure 115 Dynamic Loading of Amenities

4.2.26 System Testing 26: Tour Booking System

Checks that users can book a property tour and do so successfully by choosing a date/time and processing the booking request using the system.

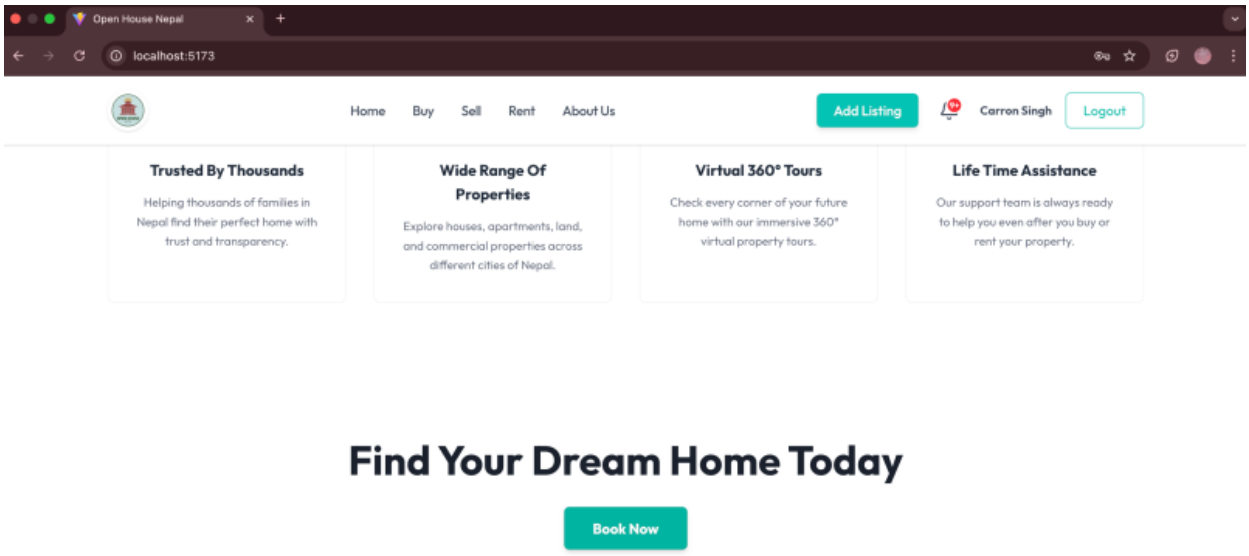


Figure 116 Tour Booking System

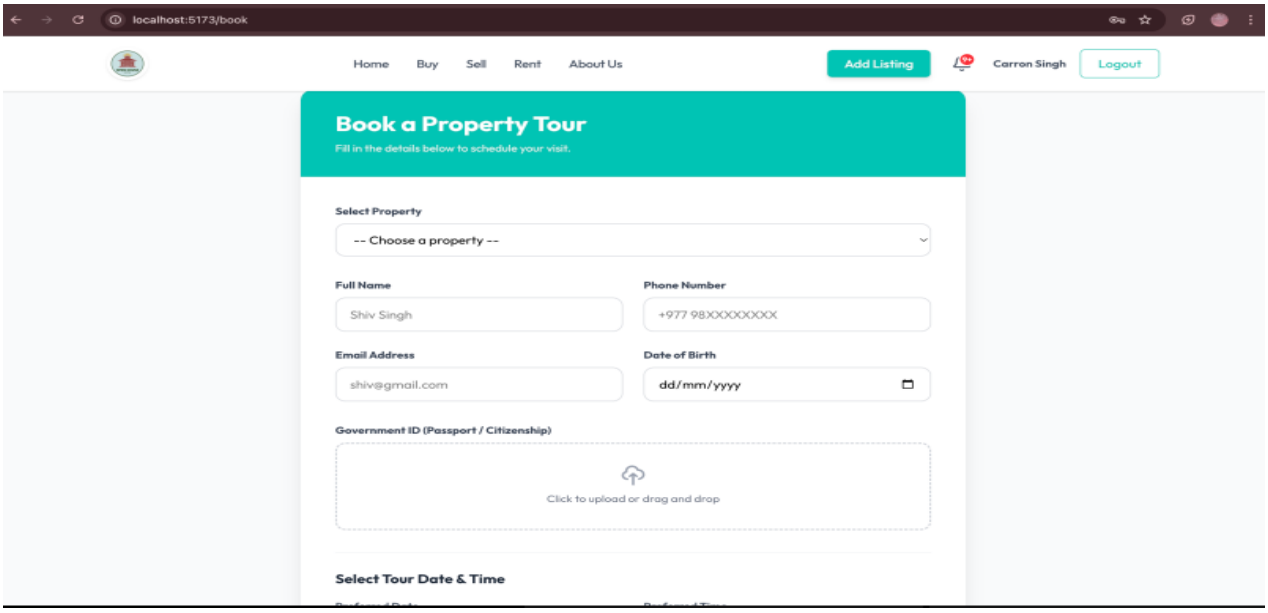


Figure 117 Tour Booking System

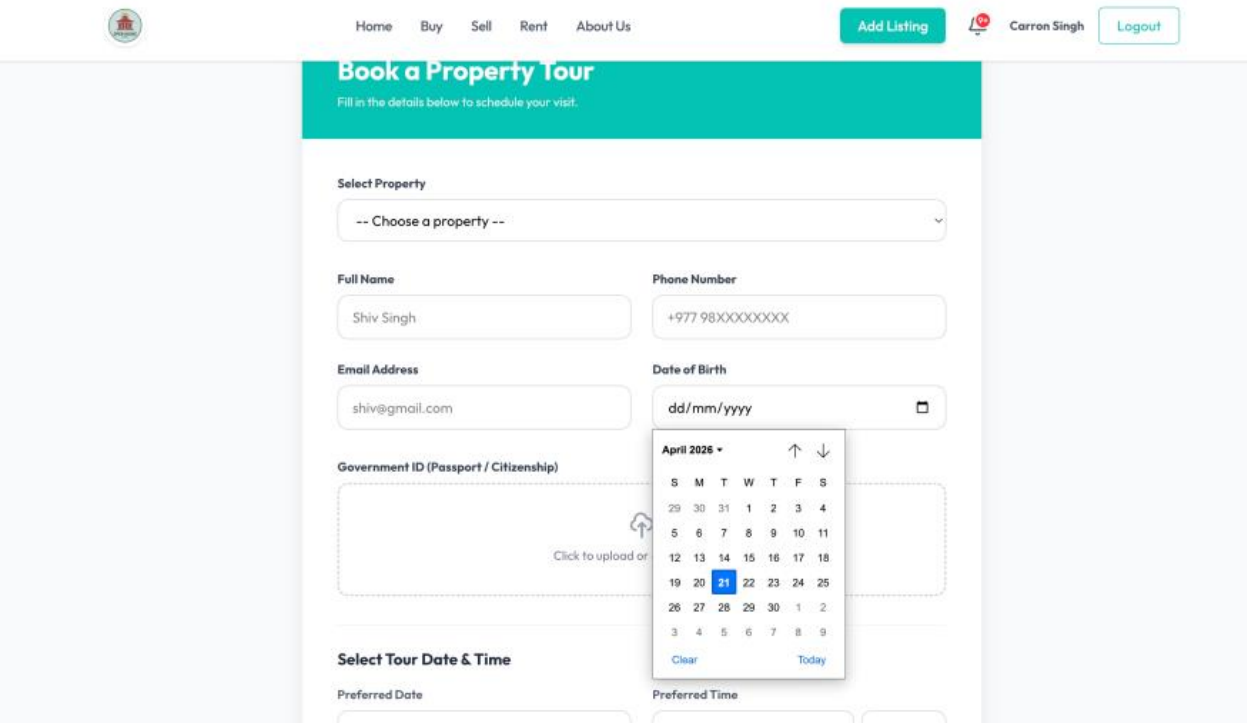


Figure 118 Tour Booking System Confirm Date

4.2.26 System Testing 26: Admin Review & Approval Flow

Verifies that the admin can access the admin dashboard to review newly submitted property listings, check uploaded documents (e.g., Citizenship and Ownership papers), and either approve or reject the listing. It also ensures that the system updates the listing status accordingly and sends an email notification to the seller informing them of the decision.

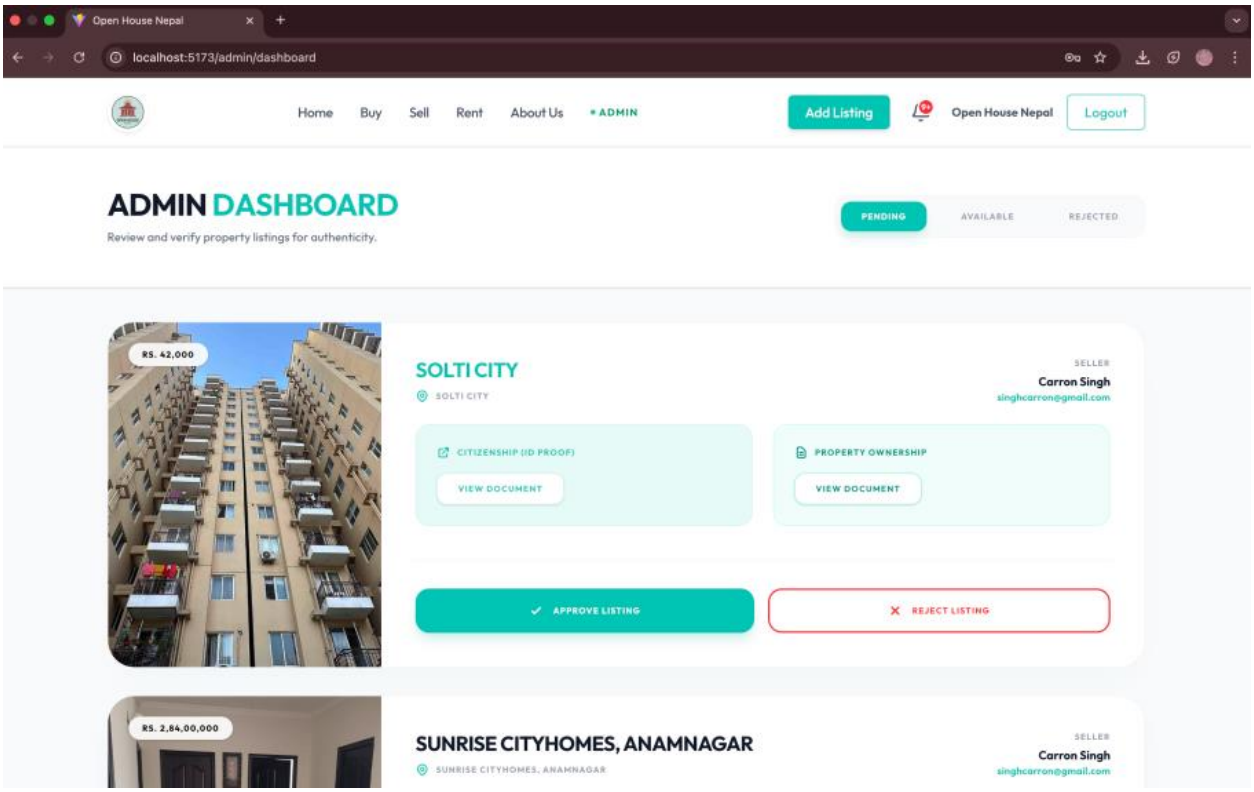


Figure 119 Admin Dashboard

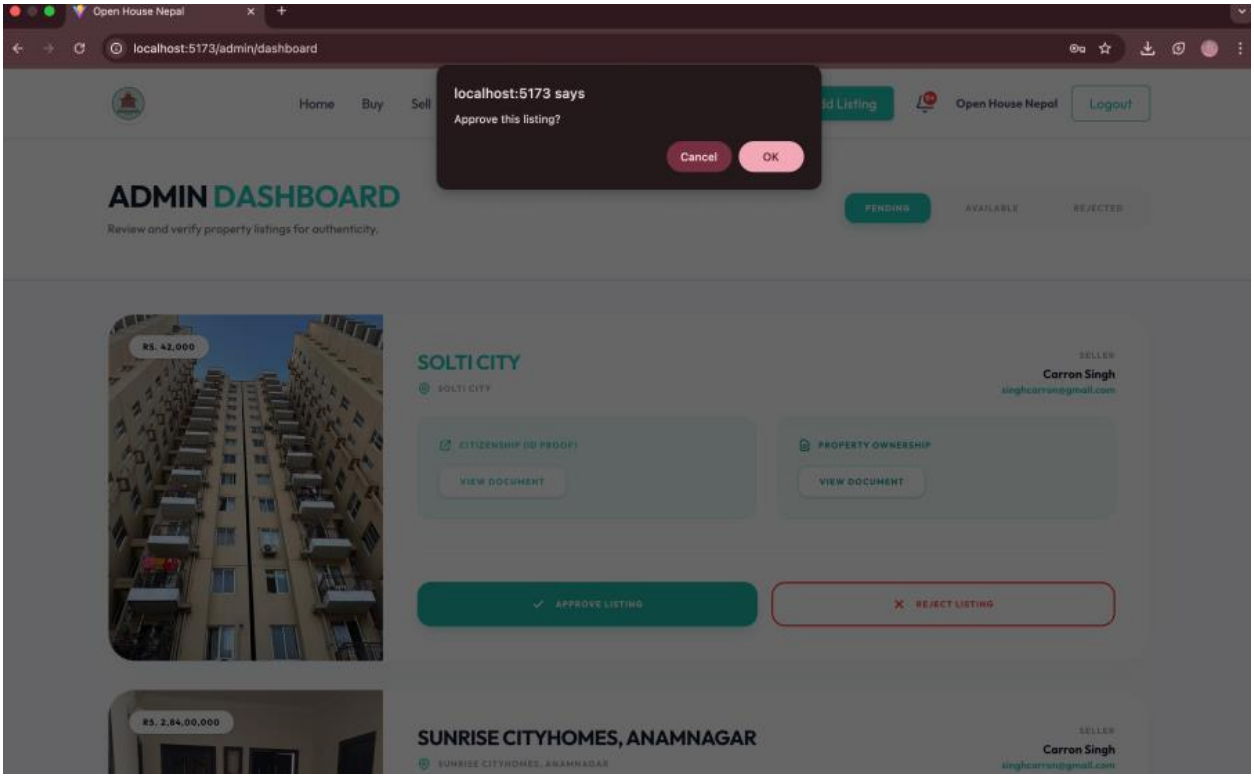


Figure 120 Admin Confirm listing

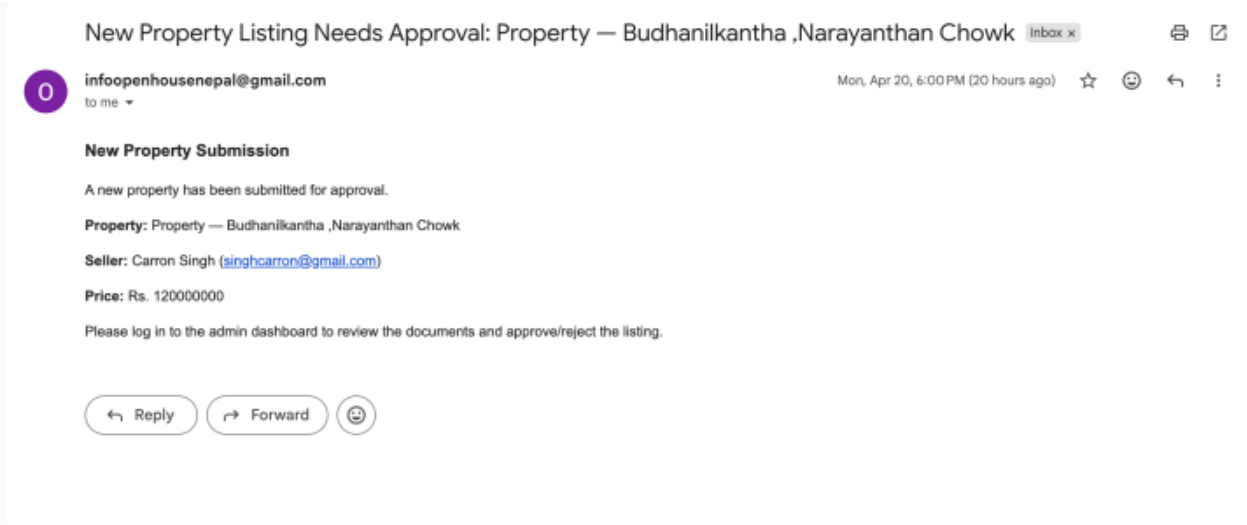


Figure 121 New property Listing Approval in Gmail

4.2.27 System Testing 27: Admin Dashboard Access

Verifies that only authorized admin users can log in and access the admin dashboard securely.

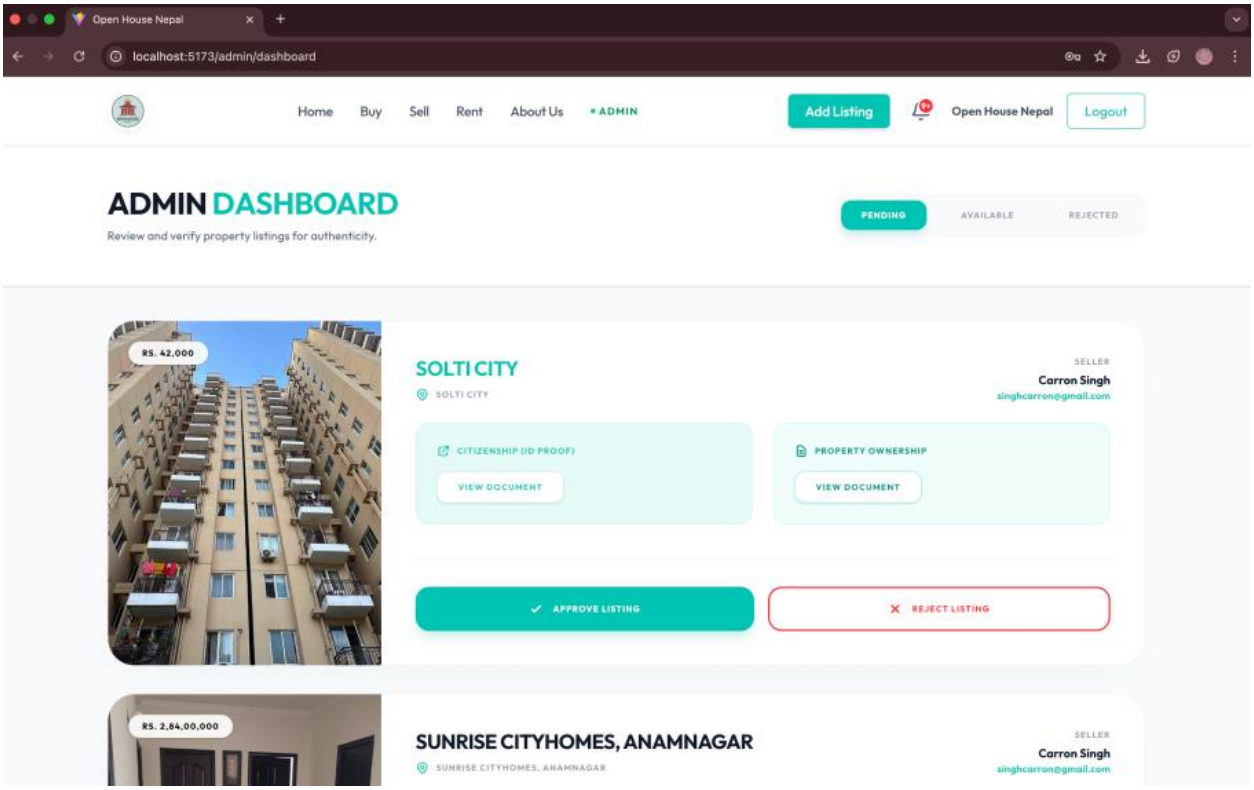


Figure 122 Admin Dashboard

4.2.28 System Testing 28: Unarchive Listing Functionality

Confirms that the restore an archived listing back to active status, making it visible again on the platform.

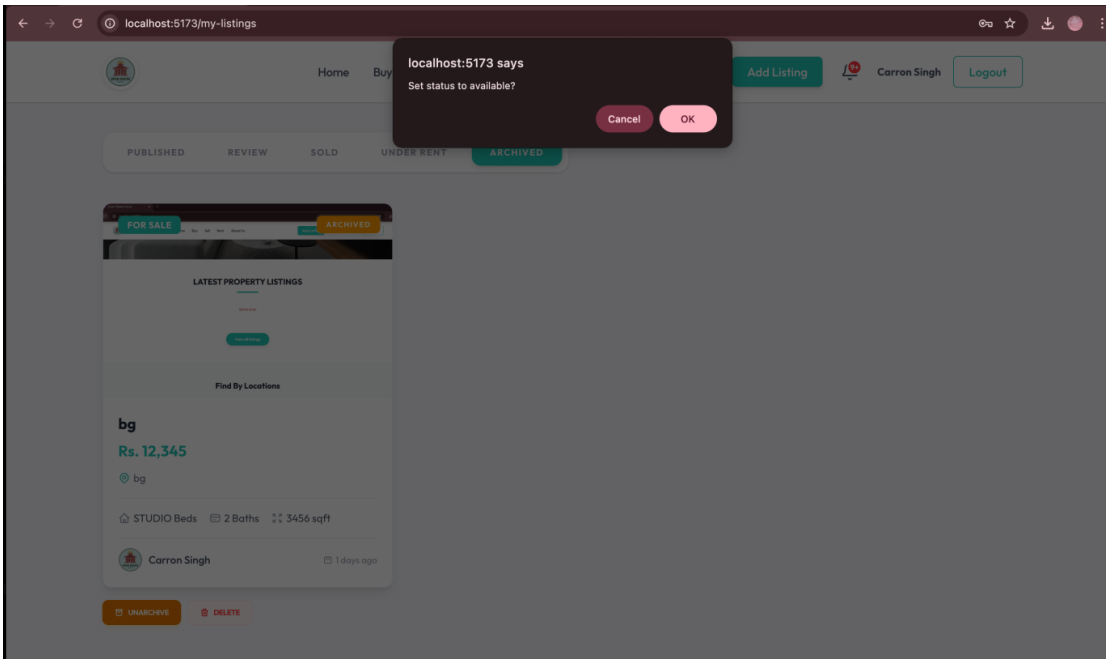
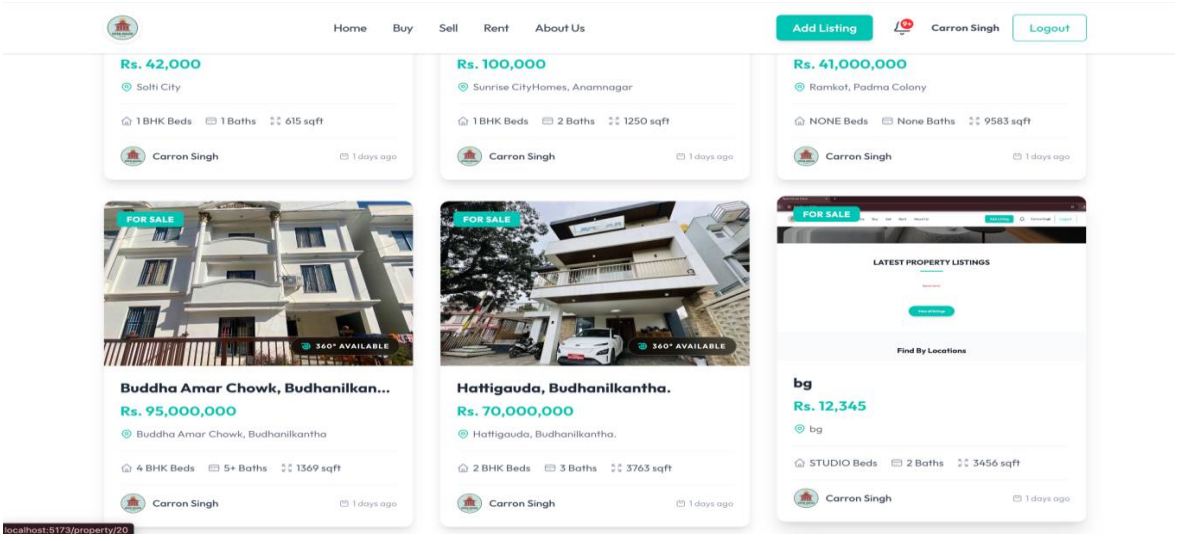


Figure 123 Unarchive Listing Functionality



4.2.29 System Testing 29: Delete Functionality

Tests that the admin can permanently delete a property listing from the system, ensuring it is removed from both the admin panel and user view.

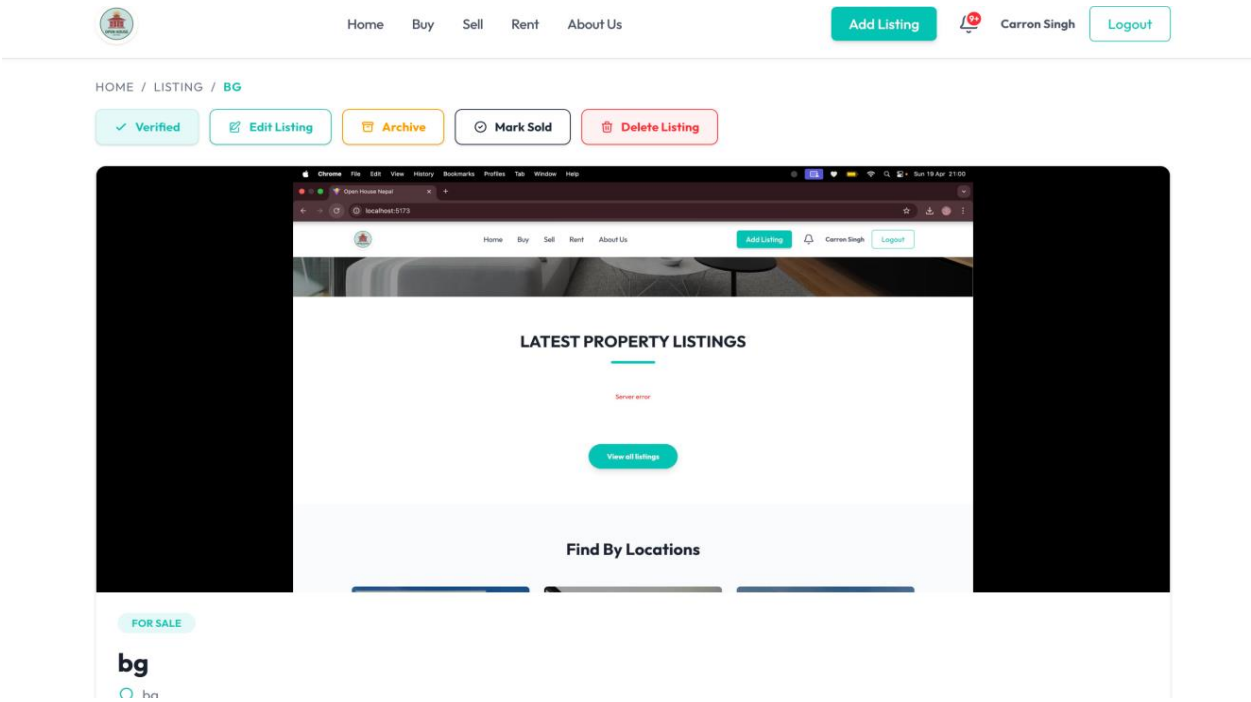


Figure 124 Delete Functionality

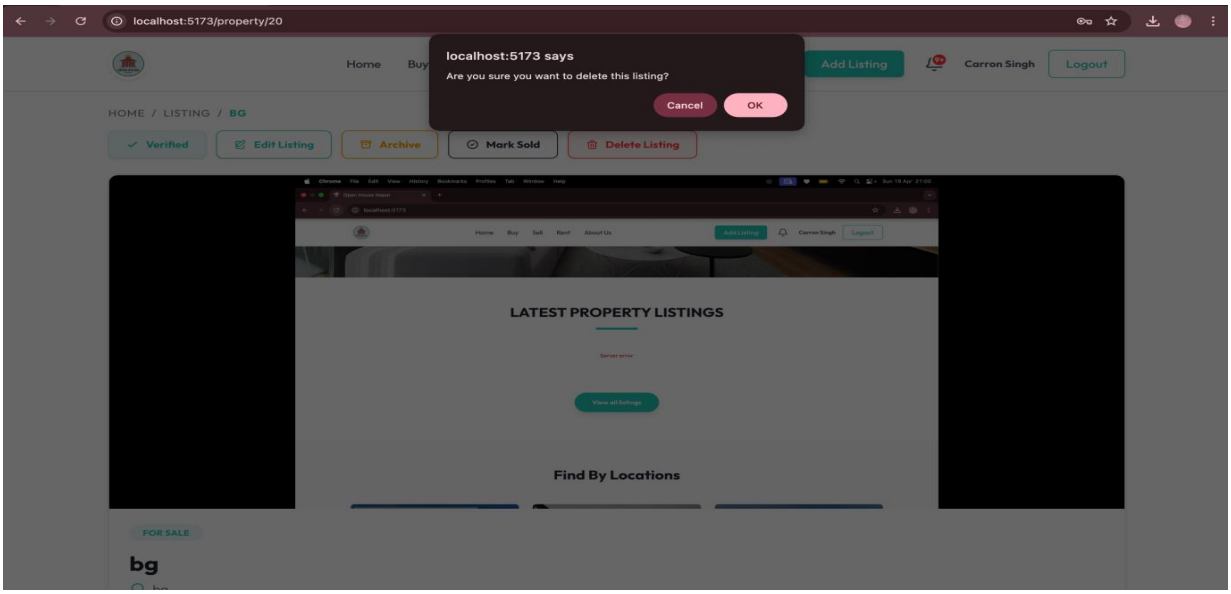


Figure 125 Confirm Delete listing

4.3 CRITICAL ANALYSIS

The testing stage of the Open House Nepal project was effective to ensure that the application fulfils its essence functionalities, especially in providing safe, trusted and interactive real estate platform. The unit testing outcomes revealed that the backend architecture is sound, and authentication and data protection mechanisms are well-performing. Use of JSON Web Tokens (JWT) and password hashing (bcrypt) can make sure that user data is safely kept, and access control is effectively implemented. This is particularly so considering the sensitivity of the data that the system deals with such as property ownership documents and information of the user. Also, multipart-form API testing proved the system can process complicated data like high-resolution images and PDF files without making the system unstable.

On a usability side, the system testing revealed the effectiveness of the platform in providing a user-friendly and captivating experience. The 360° virtual property tours and the use of Google Maps are some of the features that greatly increases user interaction by giving an insight of the realistic, spatial perception of properties. These features assist in closing the information asymmetry between the buyers and sellers, thus enhancing transparency and trust in the platform. Furthermore, the admin approval process turned out to be a key element in the integrity of the platform since the verification of all the property listing is carried out prior to its public availability. This will directly solve the problem of fraudulent listing, which was revealed as one of the major risks in the initial planning stage.

Nevertheless, the testing process also revealed some performance-related issues. In initial testing, the system had lagged when processing high-resolution images affecting overall response time. This was successfully addressed by the introduction of the Sharp image processing library, which optimizes images in the dynamic compression and subsequent storage and display. Moreover, the existing email notification system based on Nodemailer works well, but testing feedback indicates that a real-time notification system based on WebSockets could be a further enhancement to the existing system in subsequent versions to enhance the responsiveness and user interaction.

In general, the overall testing process is needed to ensure the application is stable, secure, and functionally complete. Although some small performance improvements can still be achieved, the system is in good shape to be used in the real world, and it can serve the purpose of the offering of a modern, reliable, and interactive real estate platform.

CHAPTER 5: CONCLUSION

5.1 LEGAL, SOCIAL AND ETHICAL ISSUES

Social, Ethical, and Legal Concerns relate to the legal obligations, user effect, and ethical standards that a system must adhere to. Compliance with regulations such as privacy and data protection legislation is a legal concern. Fairness, user safety, trust, and making sure the system is used sensibly and without causing harm are the main social and ethical concerns (Floridi, 2013; Mason, 1986).

5.1.1 LEGAL ISSUES

Several legal considerations are important and can be applied in the running of Open House Nepal.

5.1.1.1 Data Protection and Privacy:

Open House Nepal gathers personal data of its users which includes full names, email, phone numbers and in certain cases identity documents (Electronic Transactions Act, 2063 BS; NIST, 2020). Placing trust in the Electronic Transactions Act (2063 BS) in Nepal and new data protection regulations, any platform that gathers personal data must ensure that it is secure, retains it only as long as needed, and does not provide it to any third party without the consent of the user. Encrypted storage of passwords, use of JWT to manage sessions, and role-based access control are the solutions to this in Open House Nepal as they restrict access to sensitive user data.

5.1.1.2 Intellectual Property:

The site is written on open-source software such as React.js, node.js, express.js, and Pannellum. All this is licensed by their respective licenses (MIT, Apache 2.0). Photos and documents of the property posted by the users are still the intellectual property of the user who posted them. The terms of service (which will be prepared prior to the launch of the platform) should clearly specify the licensing agreement according to which the users will give the platform the right to show their uploaded contents (Laudon and Traver, 2021).

5.1.1.3 Consumer Protection:

The document verification feature of the platform will shield buyers against fraudulent listings. It should be pointed out, though, that administrative verification does not constitute a certain legal property, nor it can substitute the formal conveyancing of a licensed lawyer or the Department of Land Management. The site should also have clear warnings informing the customer to exercise his due diligence prior to finalizing any property deal (NIST, 2020).

5.1.2 Social Issues

5.1.2.1 Digital Divide:

Although internet penetration in Nepal is rapidly increasing, a large population, especially rural residents, do not have access to quality internet connectivity or smartphones. Open House Nepal being a web-based platform, is reliant on internet connections. It implies that the platform can initially be better used by urban population, compared to rural population, which can make the disparity in access to property market information even bigger, instead of smaller. To partly overcome this, the platform has been made to support low-bandwidth connections and run on entry-level smartphones (Pressman and Maxim, 2020).

5.1.2.2 Implication on Traditional Brokers:

The advent of a digital real estate platform has the potential to disrupt the livelihood of traditional property brokers and agents in Nepal, who depend on commissions that may be earned when brokering property deals. Although the platform opens opportunities where agents can register and post properties on the internet, it has also enabled the buyers and sellers to have direct transaction which may eliminate the need of having intermediaries. It is a fair social factor and must be openly considered (Porter, 2008).

5.1.2.3 Faith in Digital Systems:

Mistrust of online systems Nepalese users, especially older ones, are highly distrustful of online platforms, especially when it comes to high-value transactions. Trust will not come easy and will need a steady supply of what the platform promises; that of verified listings and open information. The go to market strategy of the platform will include community outreach and user education campaigns (World Bank, 2022).

5.1.3 Ethical Issues

5.1.3.1 Administrative Verification Bias:

The system of administrator verification depends on human administrators to check and verify the listings of the property. This creates the element of inconsistencies or biasness in decision-making. As an illustration, administrators may fast track listing by familiar sellers or be more critical when listing unfamiliar sellers. To counter this, a standardised checklist of verification has been prepared, and all the administrators are expected to comply with the checklist and verification decisions are recorded in the system to enable auditing (ISO 27001, 2022).

5.1.3.2 User Review Integrity:

The review and rating system is designed to promote transparency, but it could also be misused. There is the chance that sellers leave counterfeit positive reviews concerning their properties with the help of secondary accounts, or rivals leave unjust negative ones. The option to only do reviews based on verified buyers is a great protection against this, yet not a full solution. In the future, review authenticity checks should be considered, including making the reviewers prove that they visited the property in person (OECD, 2020).

5.1.3.3 Data Retention:

The platform should have clear policies regarding the duration of retention of user data like personal information, uploaded documents etc. Storing the data forever will be risky to privacy and destroying it too fast may result in the platform failing to meet its legal duties in case of a dispute. There must be a policy that balances these competing interests in terms of data retention that must be developed and made clear to users (ISO 27001, 2022.

5.2 Advantages

5.2.1 Document Verification System

The sellers must post ownership documents, and an administrator must check them to approve them. This minimizes fraudulent entries and boosts confidence. It directly caters to the requirements of 74% of the pre-survey users.

5.2.2 360-Degree Virtual Tour

Enables customers to visit homes remotely on their devices. This comes in handy particularly to users who are not within the city or even abroad besides minimizing unnecessary visits to the sites.

5.2.3 Review and Comment System

Allows customers to browse reviews by other shoppers and leave comments. This enhances transparency and promotes the honesty of sellers in their information.

5.2.4 Advanced Search and Filters

The users can filter properties based on price, location, type, and size. This is timesaving and assists users to locate relevant listings in a short period.

5.3 Limitations

5.3.1 No Specialized Mobile Application

The platform is only available as a web application. The mobile application would be a more user-friendly experience and push notifications.

5.3.2 Polling-Based Notification System

Notifications are not entirely real-time and can be inefficient with growth of the platform.

5.3.3 Manual Document Verification

Administrators check this, and this will slow down with more listings. Scalability might be enhanced by automation.

5.3.4 No Private Messaging System

The buyer/seller communication is transparent through comments, so users might not interact with the site when sensitive questions need to be asked.

5.3.5 Price Analytics Lack

The site lacks market trends or price comparisons, and it is more difficult to assess prices of properties.

5.3.6 Real-Time Notification System

The updates provide the user with updates as new property that fits their preferences is listed, so they can move swiftly in a competitive market.

5.4 Future Work

There is a clear and exciting roadmap for the future development of Open House Nepal. The next development phase has been identified as the following features and improvements as priorities.

The highest priority will be to develop a specific mobile application on iOS and Android that will be developed post-launch. A SaaS native mobile application will facilitate push notifications, greater integration with the camera, and an improved user experience on smartphones, which comprise most internet users in Nepal. The notification system will be improved and changed to the WebSocket-based architecture based on Socket.io, which will offer the real-time push notifications without the additional cost of constant polling requests.

A price forecasting system will be developed using AI to assist buyers and sellers in knowing the fair market prices. The tool will use historical data of transactions and property characteristics to provide price estimates which will be presented in listing pages and search results.

Blockchain technology will also be discussed as the means of establishing irrevocable, validating ownership records of property. The platform could enable a degree of trust and tamper-proofing that is greater than can be achieved by manual document verification by documenting ownership transfer on a blockchain.

Integration of Augmented Reality (AR) will be implemented in the future, which will enable buyers to see what a property would be like with various furniture and interior decorations using their smartphone camera, further minimising the physical visit to the site.

To supplement the current system of public commenting, a direct messaging system between buyers and sellers will be adopted, which will offer a privacy to the sensitive negotiations.

The support of multiple languages (especially Nepali Devanagari script) will be introduced to make the platform more available to a larger audience and meet the requirements of the government on digital services in Nepal.

CHAPTER 6: REFERENCES

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CHAPTER 8: APPENDIX

8.1 APPENDIX A: PRE-SURVEY

8.1.1 PRE-SURVEY FORM

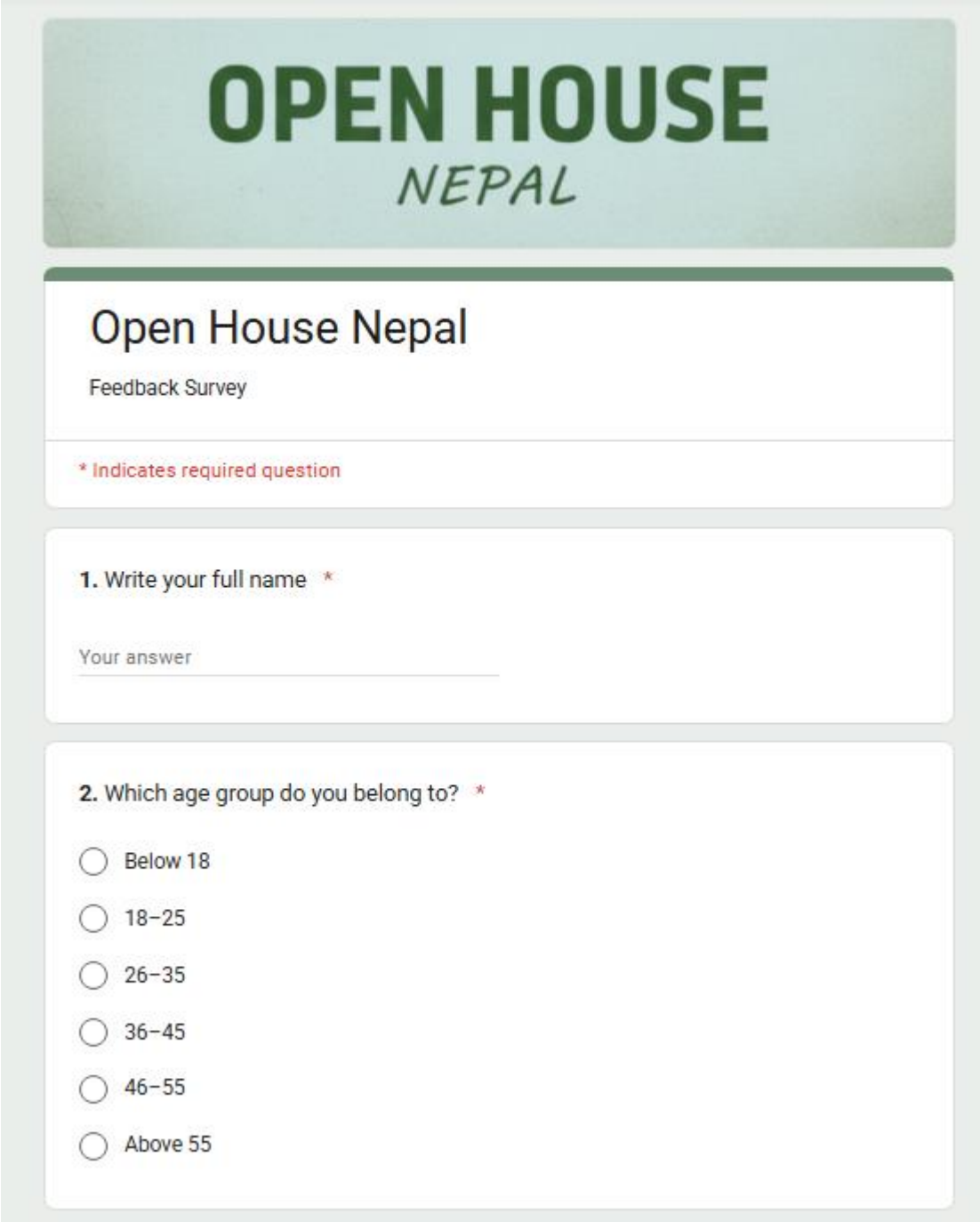


Figure 126 PRE-SURVEY FORM

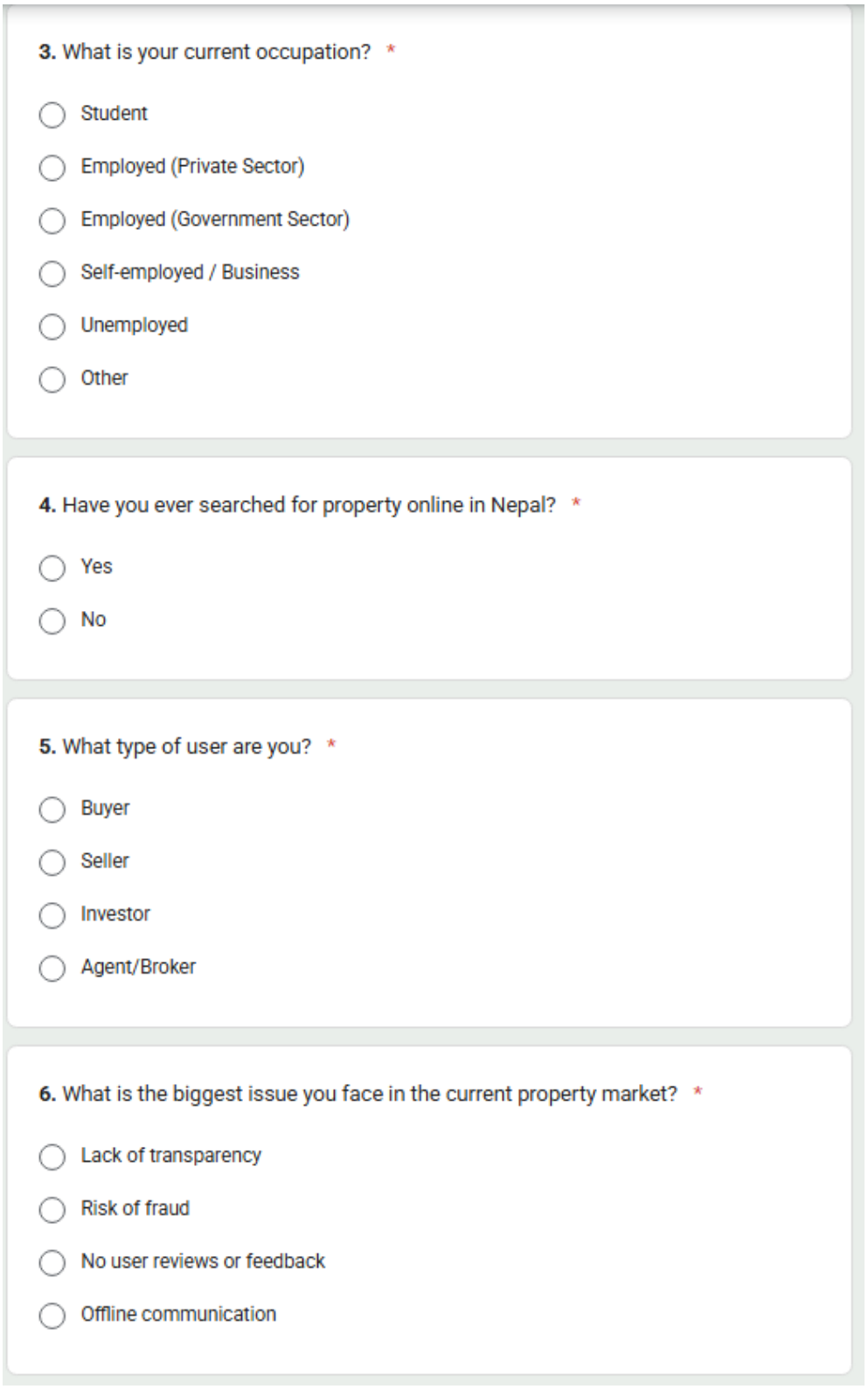


Figure 127 PRE-SURVEY FORM

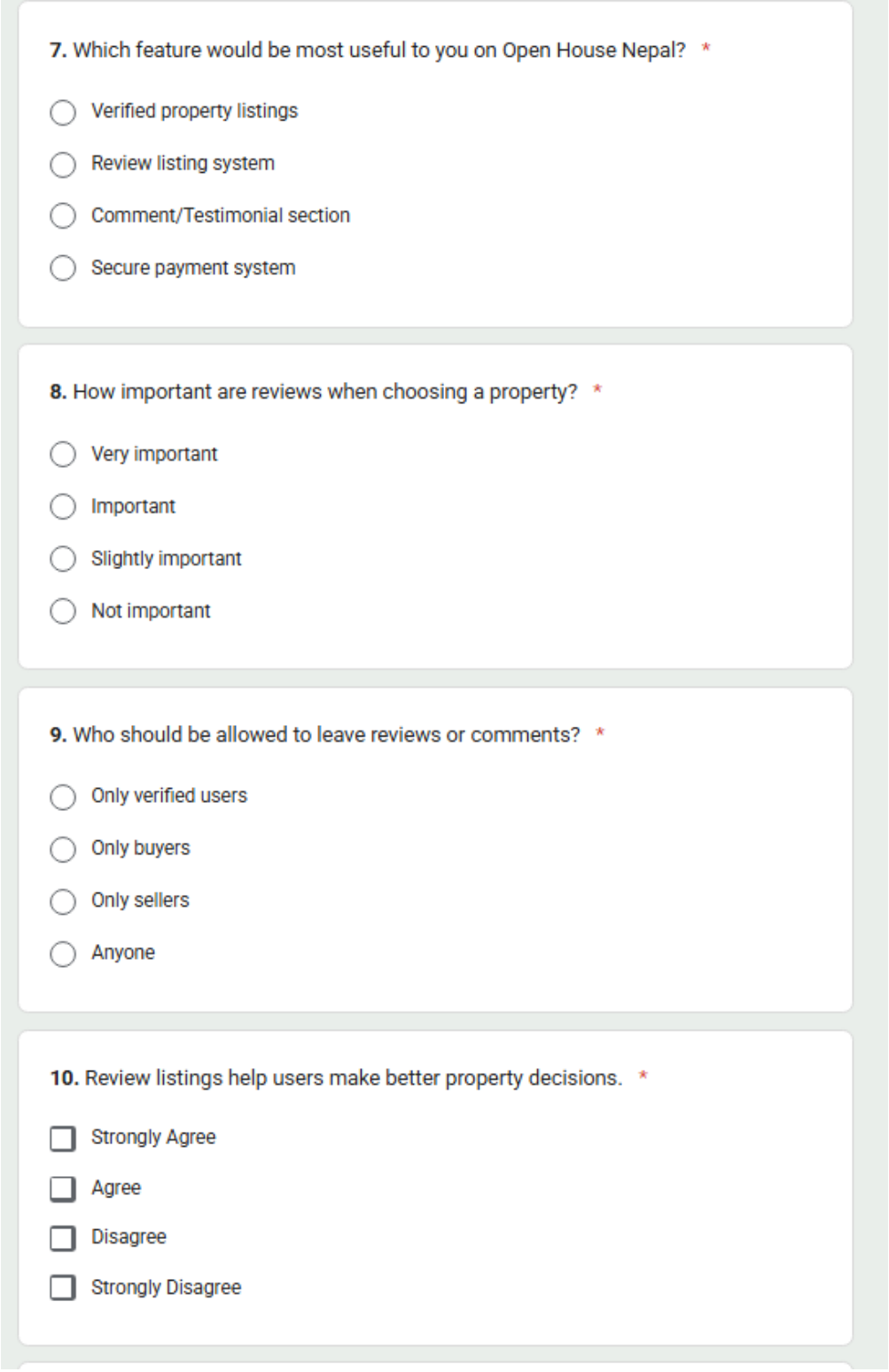


Figure 128 PRE-SURVEY FORM

11. Comment and testimonial features increase trust in property listings. *

- [ ] Strongly Agree
- [ ] Agree
- [ ] Disagree
- [ ] Strongly Disagree

12. User reviews can reduce fraud in online property platforms. *

- [ ] Strongly Agree
- [ ] Agree
- [ ] Disagree
- [ ] Strongly Disagree

13. Reading other users' experiences improves confidence in sellers. *

- [ ] Strongly Agree
- [ ] Agree
- [ ] Disagree
- [ ] Strongly Disagree

14. A 360-degree property viewer helps me understand the property layout without visiting in person. *

- [ ] Strongly Agree
- [ ] Agree
- [ ] Disagree
- [ ] Strongly Disagree

Figure 129 PRE-SURVEY FORM

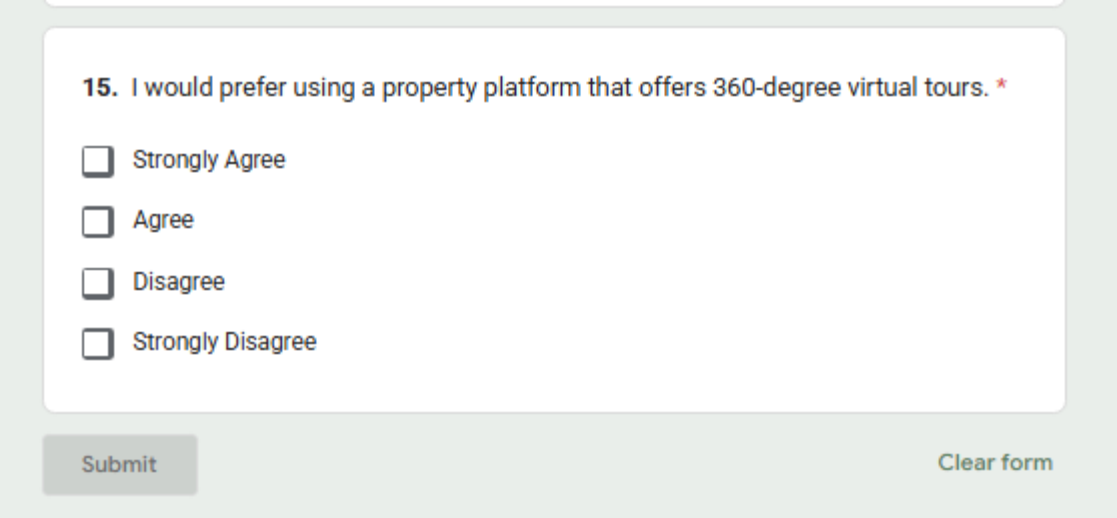


Figure 130 PRE-SURVEY FORM

8.2.2 SAMPLE OF FILLED Pre-SURVEY FORMS

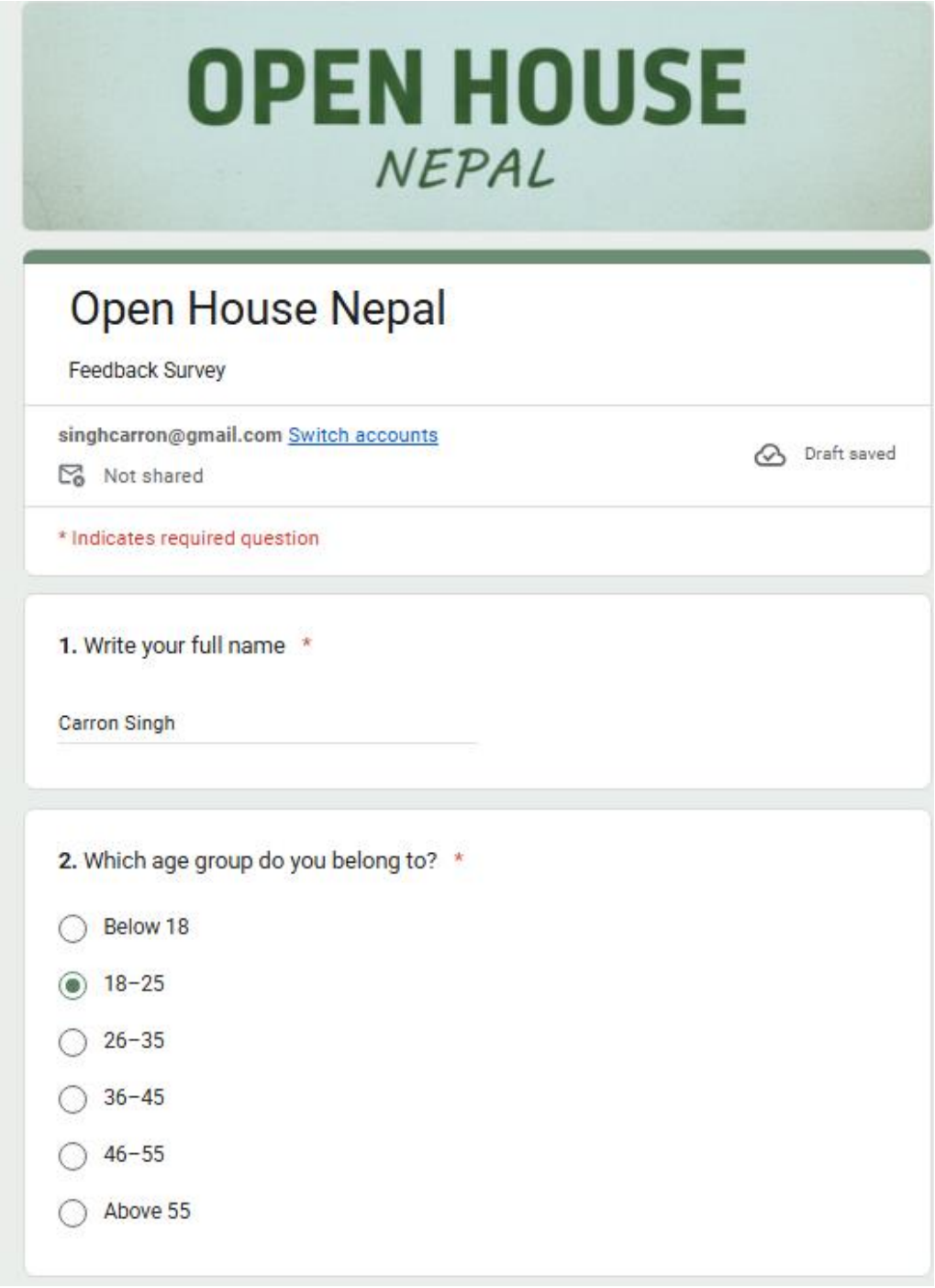
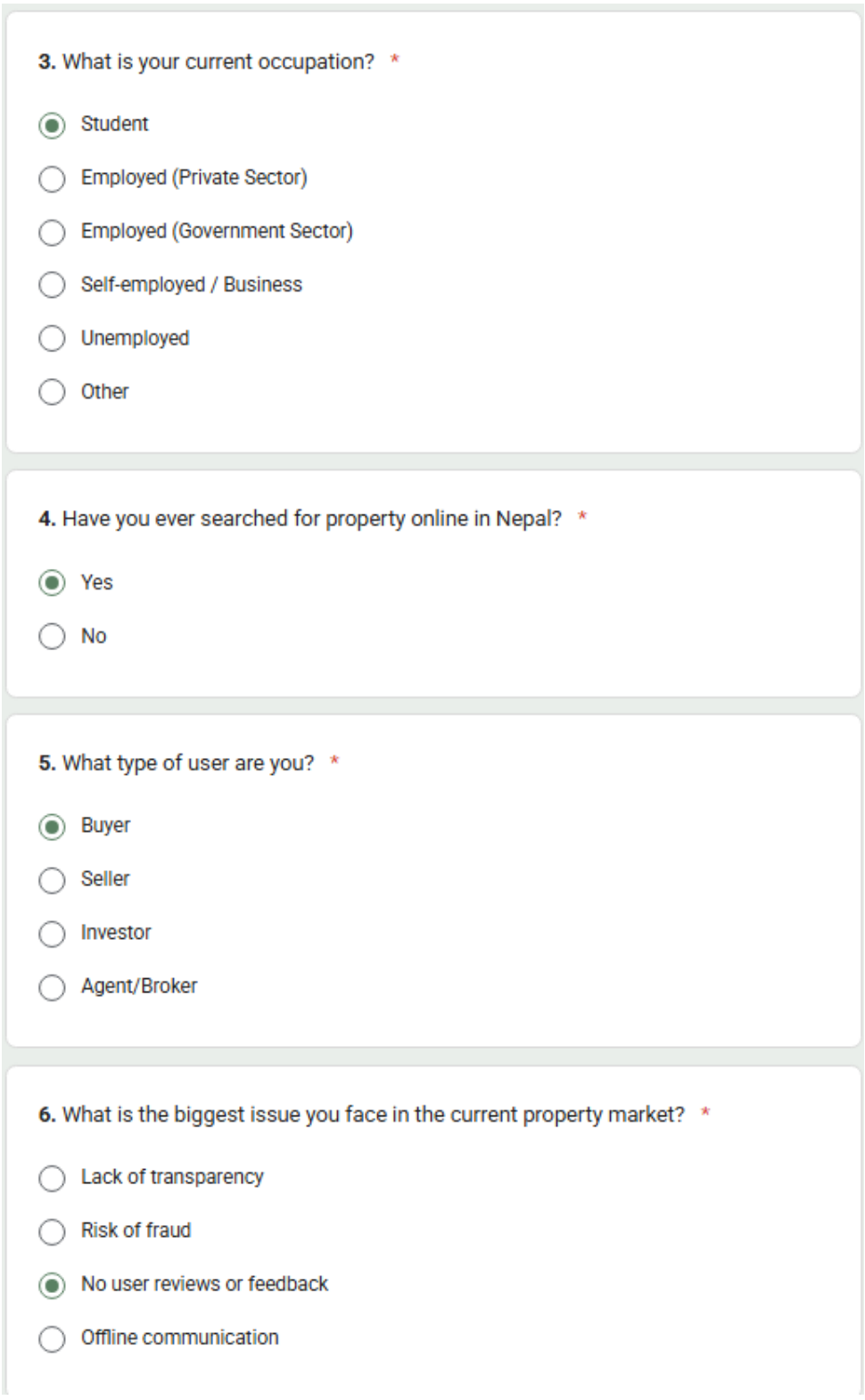


Figure 131 SAMPLE OF FILLED PRE-SURVEY FORMS



3. What is your current occupation? *

- (●) Student
- ( ) Employed (Private Sector)
- ( ) Employed (Government Sector)
- ( ) Self-employed / Business
- ( ) Unemployed
- ( ) Other

4. Have you ever searched for property online in Nepal? *

- (●) Yes
- ( ) No

5. What type of user are you? *

- (●) Buyer
- ( ) Seller
- ( ) Investor
- ( ) Agent/Broker

6. What is the biggest issue you face in the current property market? *

- ( ) Lack of transparency
- ( ) Risk of fraud
- (●) No user reviews or feedback
- ( ) Offline communication

Figure 132 SAMPLE OF FILLED PRE-SURVEY FORMS

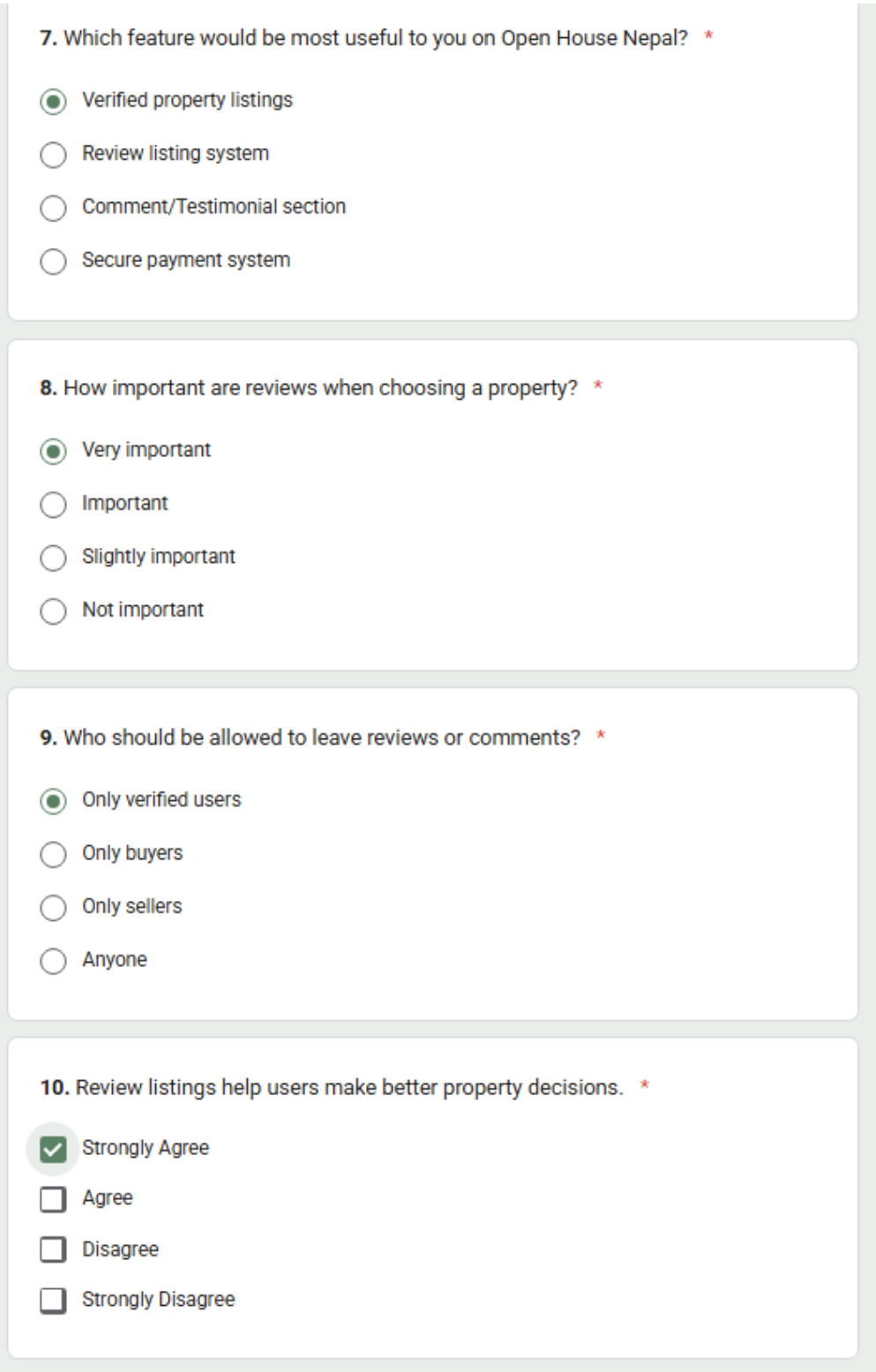


Figure 133 SAMPLE OF FILLED PRE-SURVEY FORMS

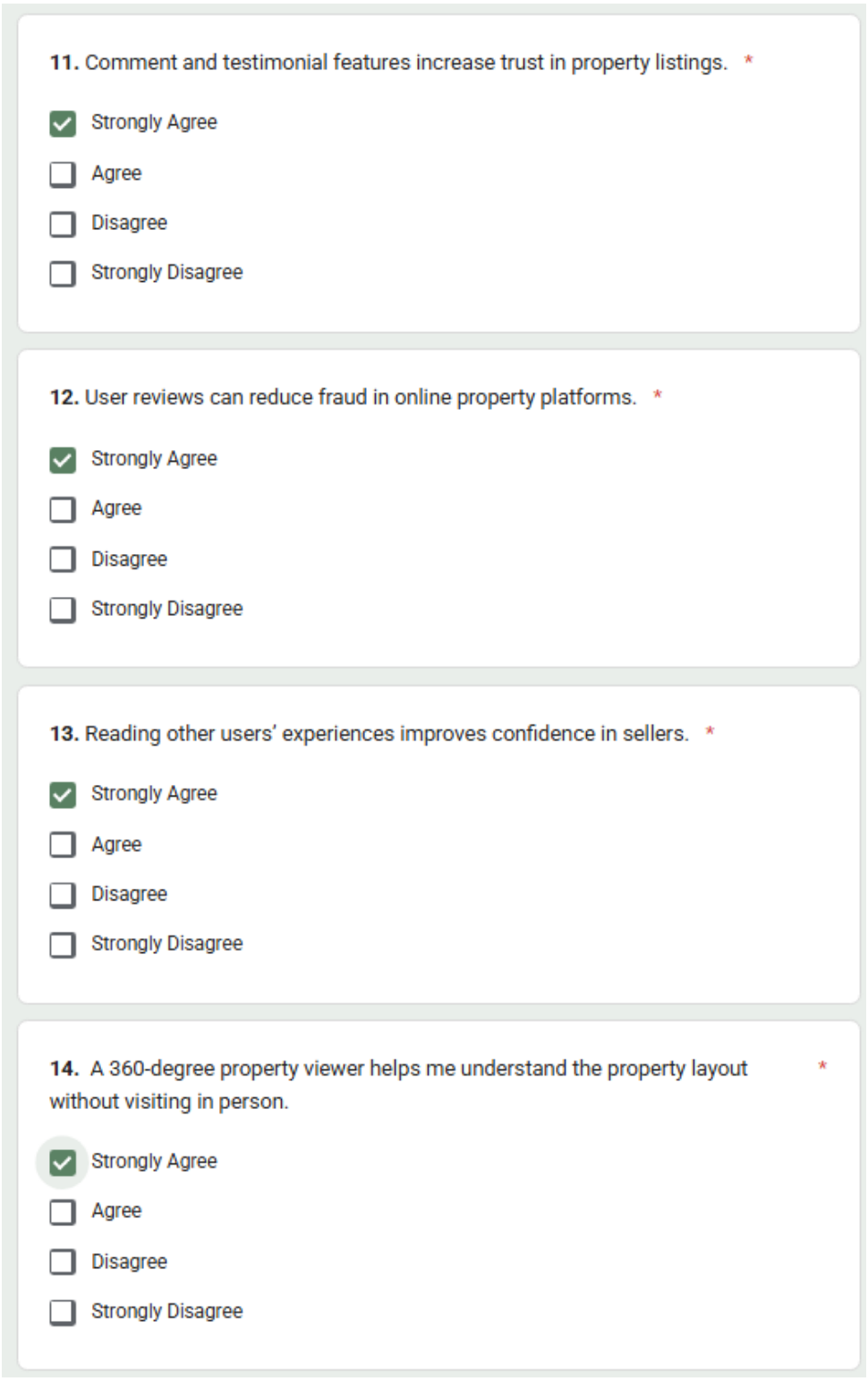


Figure 134 SAMPLE OF FILLED PRE-SURVEY FORMS

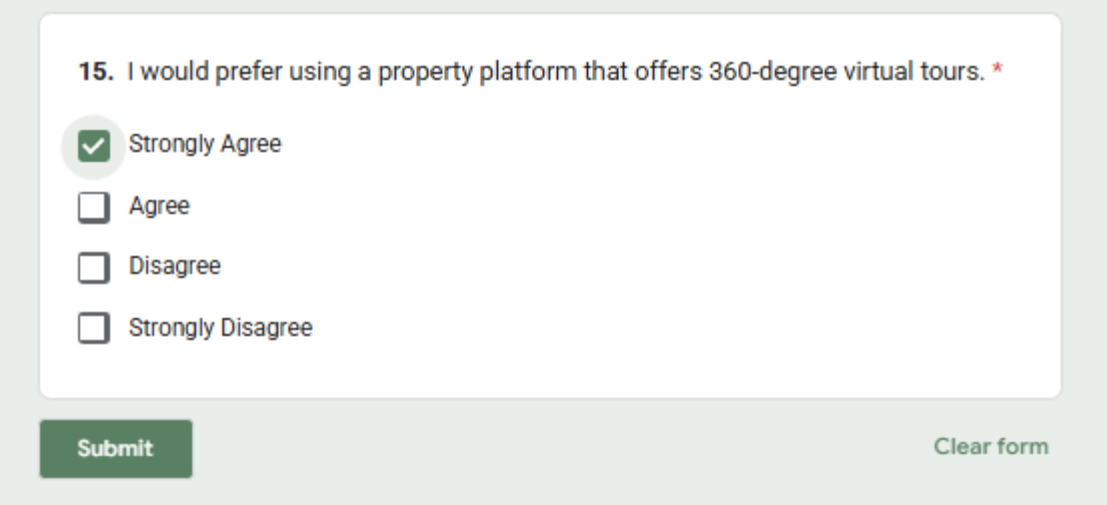


Figure 135 SAMPLE OF FILLED PRE-SURVEY FORMS

8.2.3 PRE-SURVEY RESULT

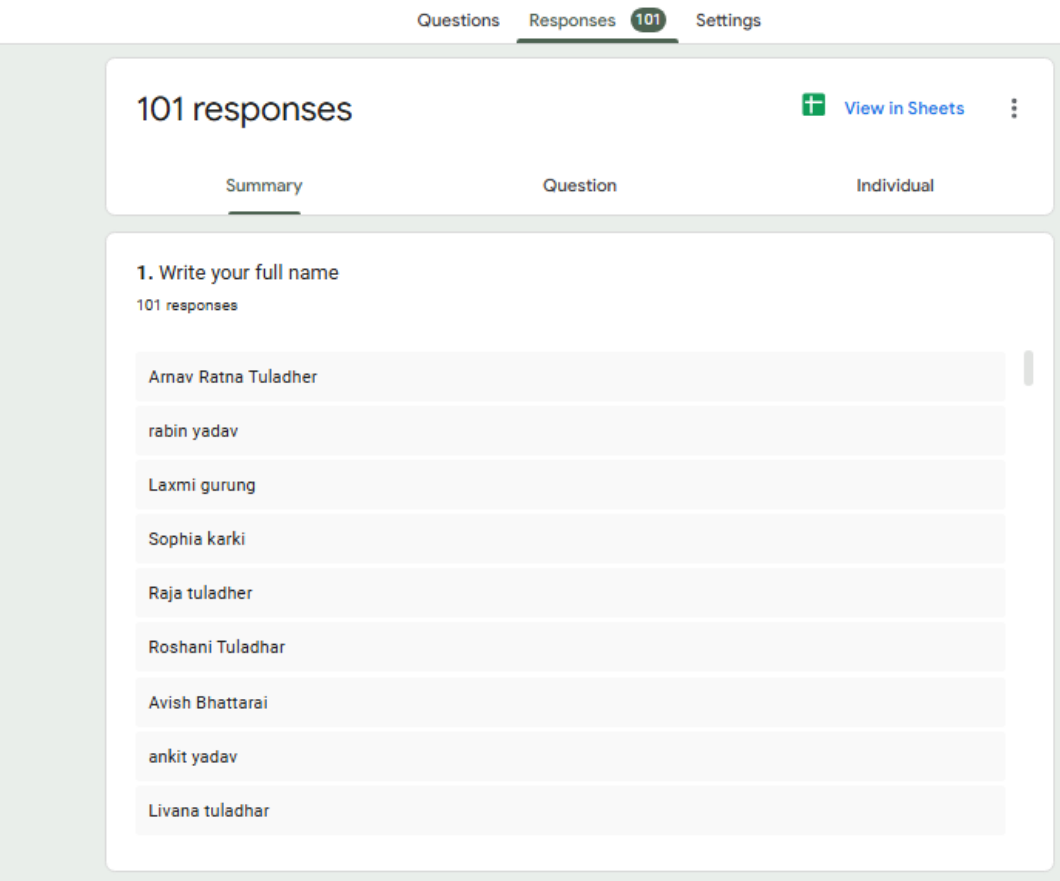


Figure 136 PRE-SURVEY RESULT

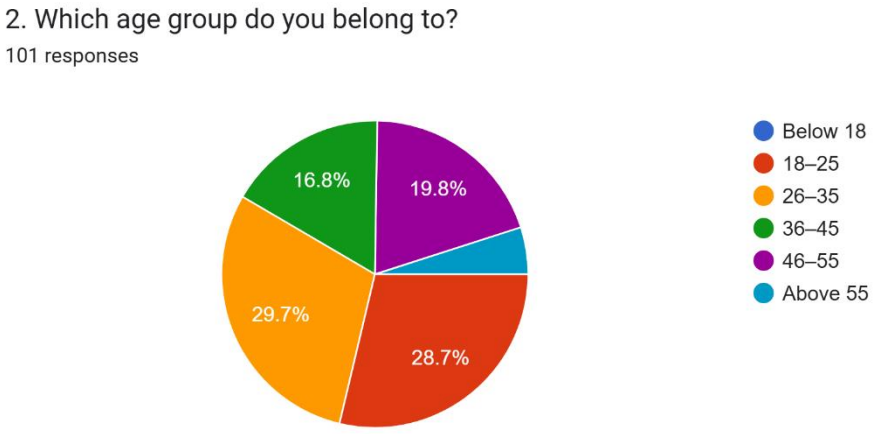


Figure 137 PRE-SURVEY RESULT

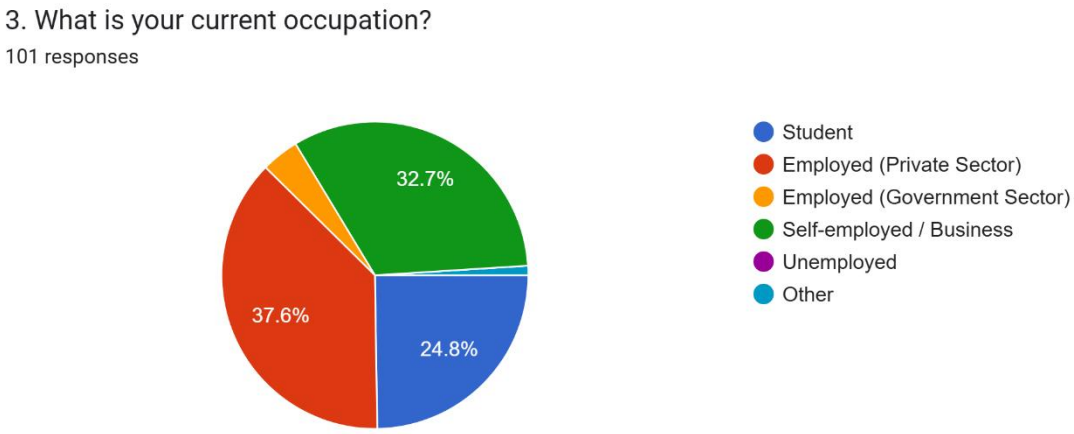


Figure 138 PRE-SURVEY RESULT

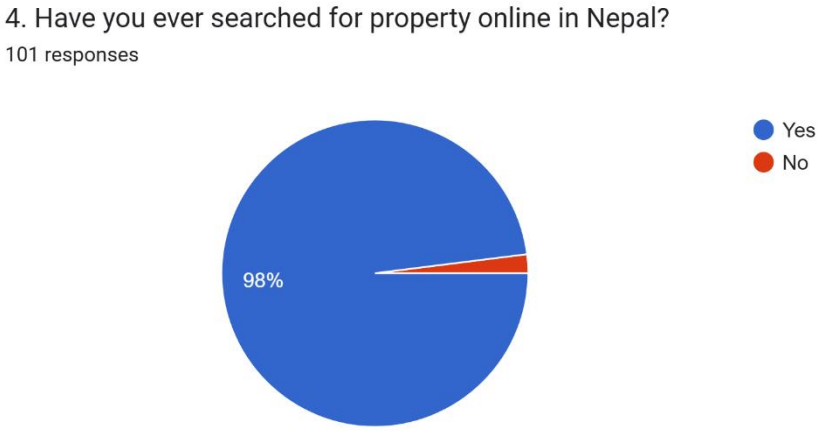


Figure 139 PRE-SURVEY RESULT

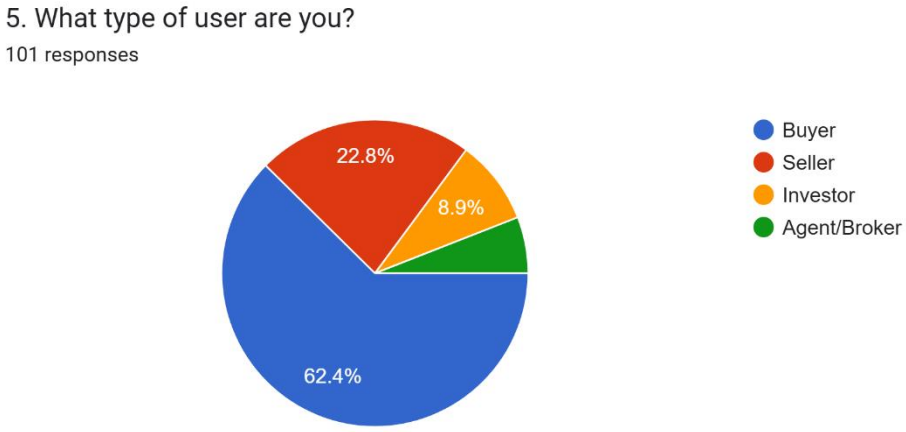


Figure 140 PRE-SURVEY RESULT

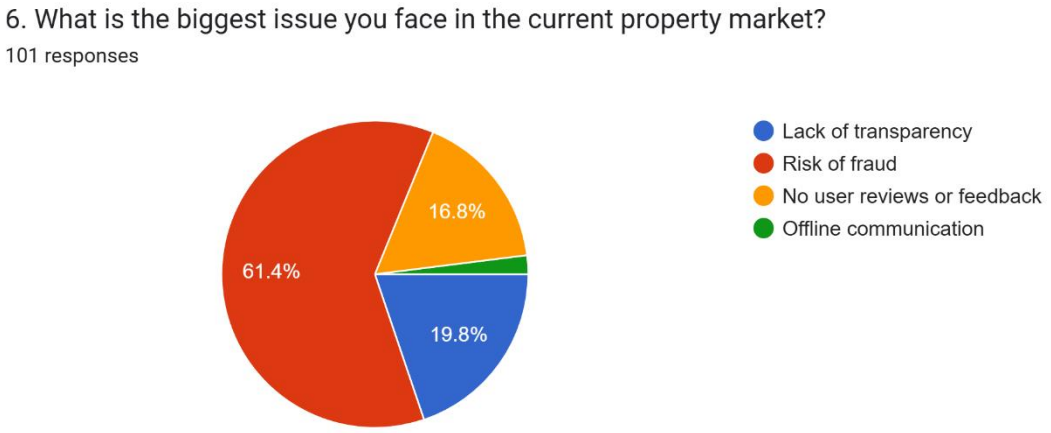


Figure 141 PRE-SURVEY RESULT

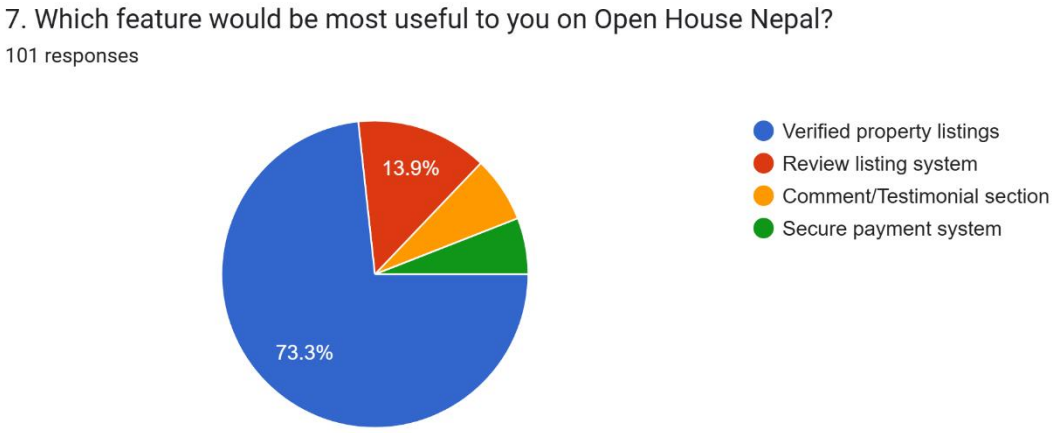


Figure 142 PRE-SURVEY RESULT

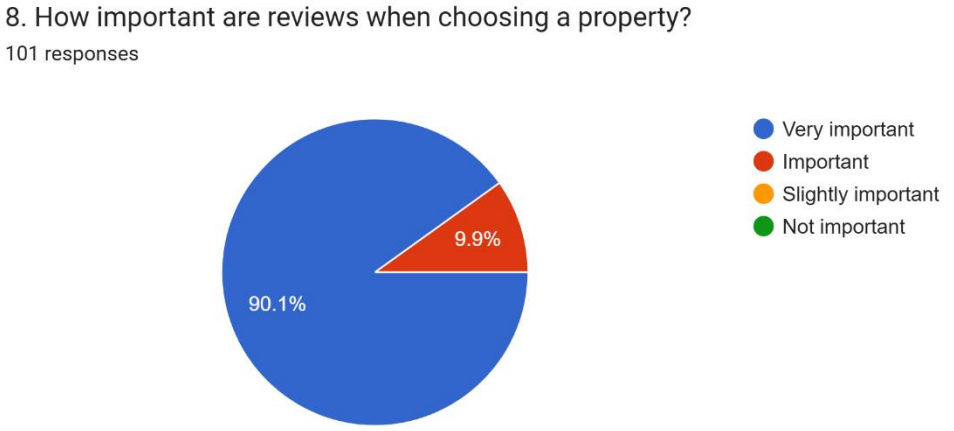


Figure 143 PRE-SURVEY RESULT

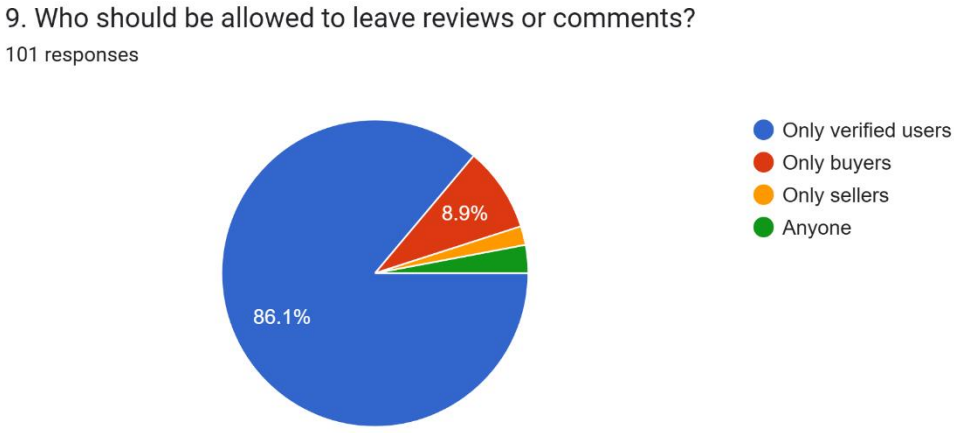


Figure 144 PRE-SURVEY RESULT

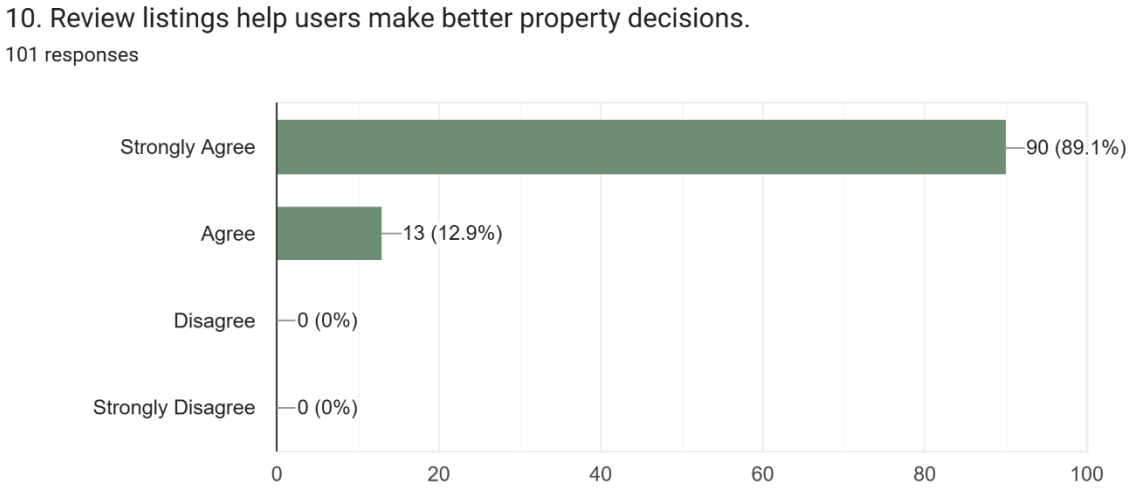


Figure 145 PRE-SURVEY RESULT

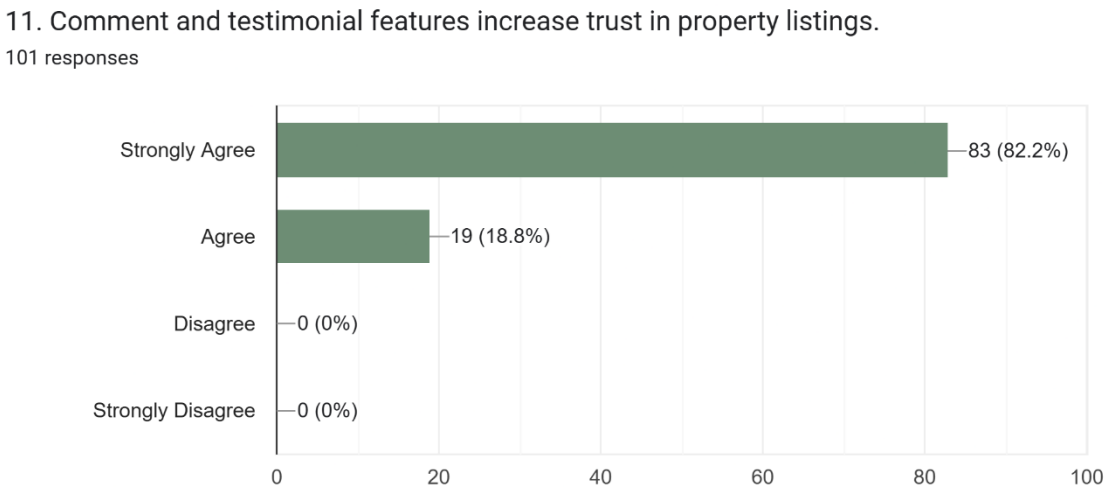


Figure 146 PRE-SURVEY RESULT

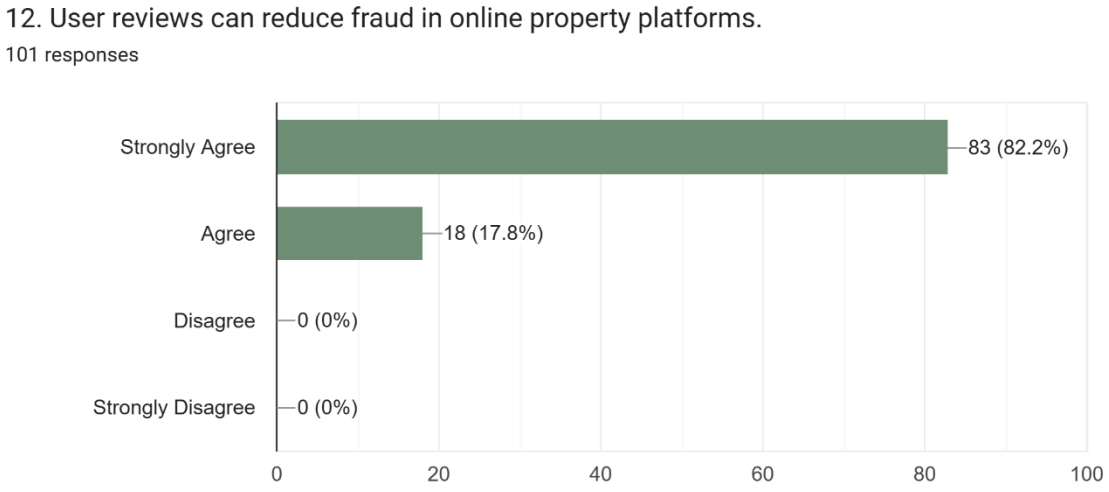


Figure 147 PRE-SURVEY RESULT

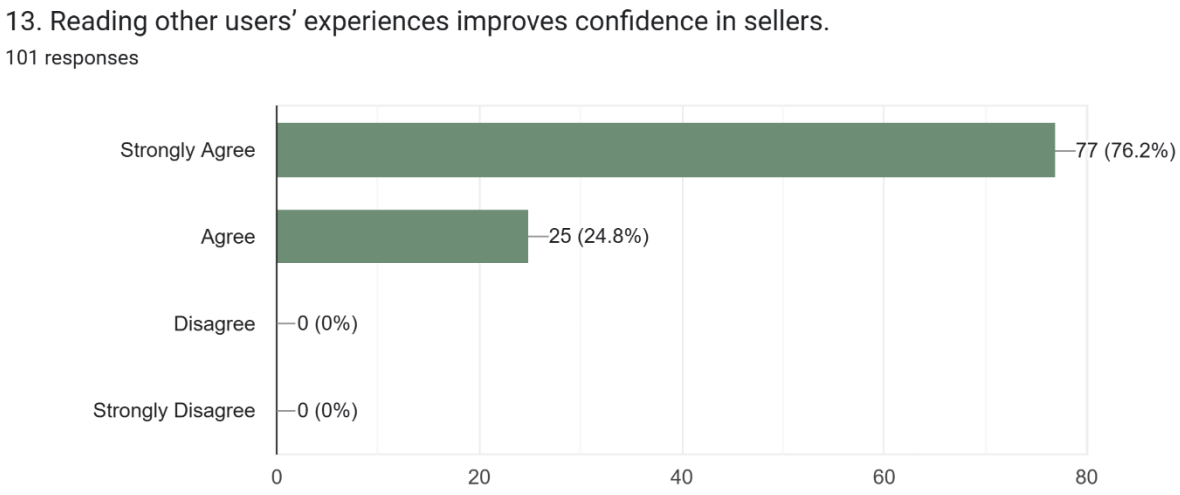


Figure 148 PRE-SURVEY RESULT

14. A 360-degree property viewer helps me understand the property layout without visiting in person.

101 responses

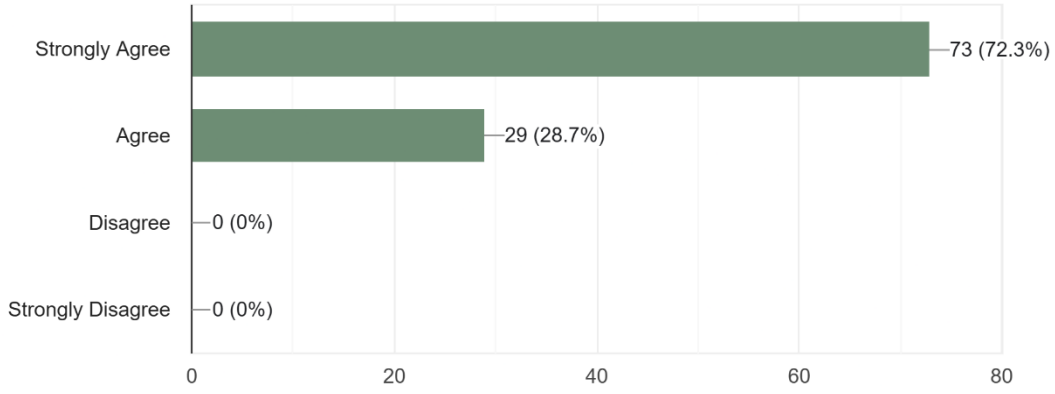


Figure 149 PRE-SURVEY RESULT

15. I would prefer using a property platform that offers 360-degree virtual tours.

101 responses

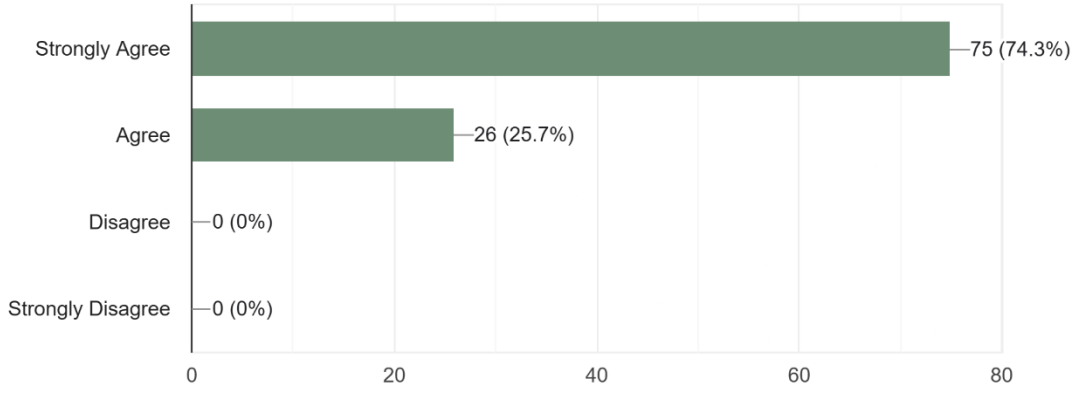


Figure 150 PRE-SURVEY RESULT

Go back to pre-survey

8.2 APPENDIX B: POST-SURVEY

8.2.1 POST-SURVEY FORM

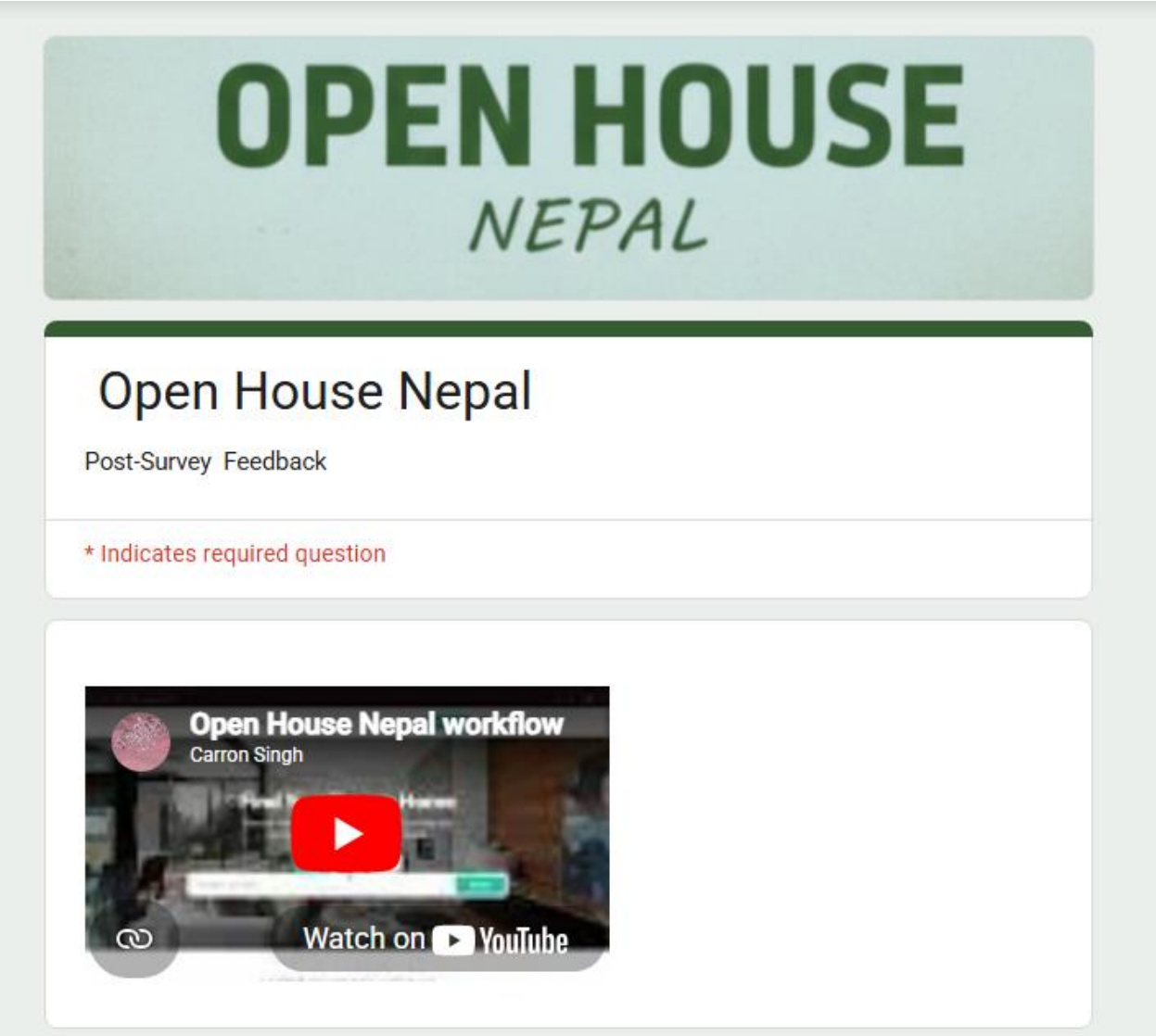


Figure 151 POST-SURVEY FORM

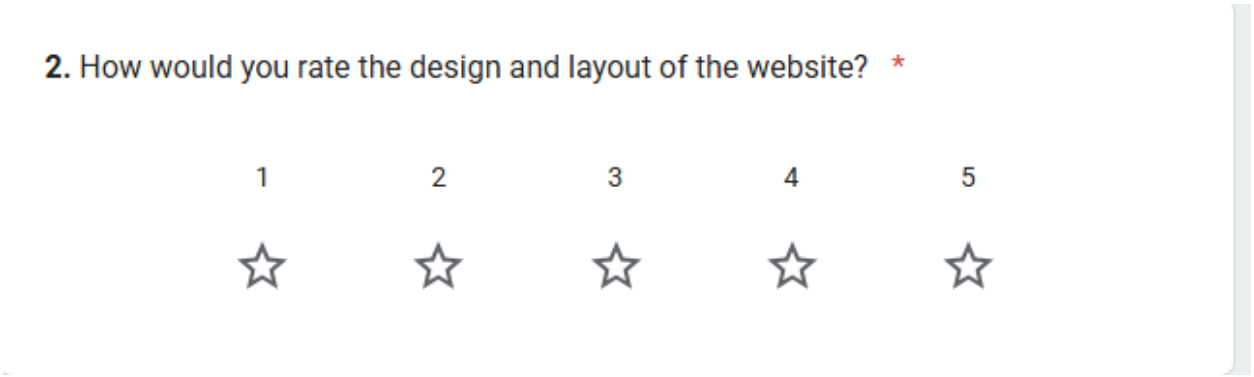


Figure 152 POST-SURVEY FORM

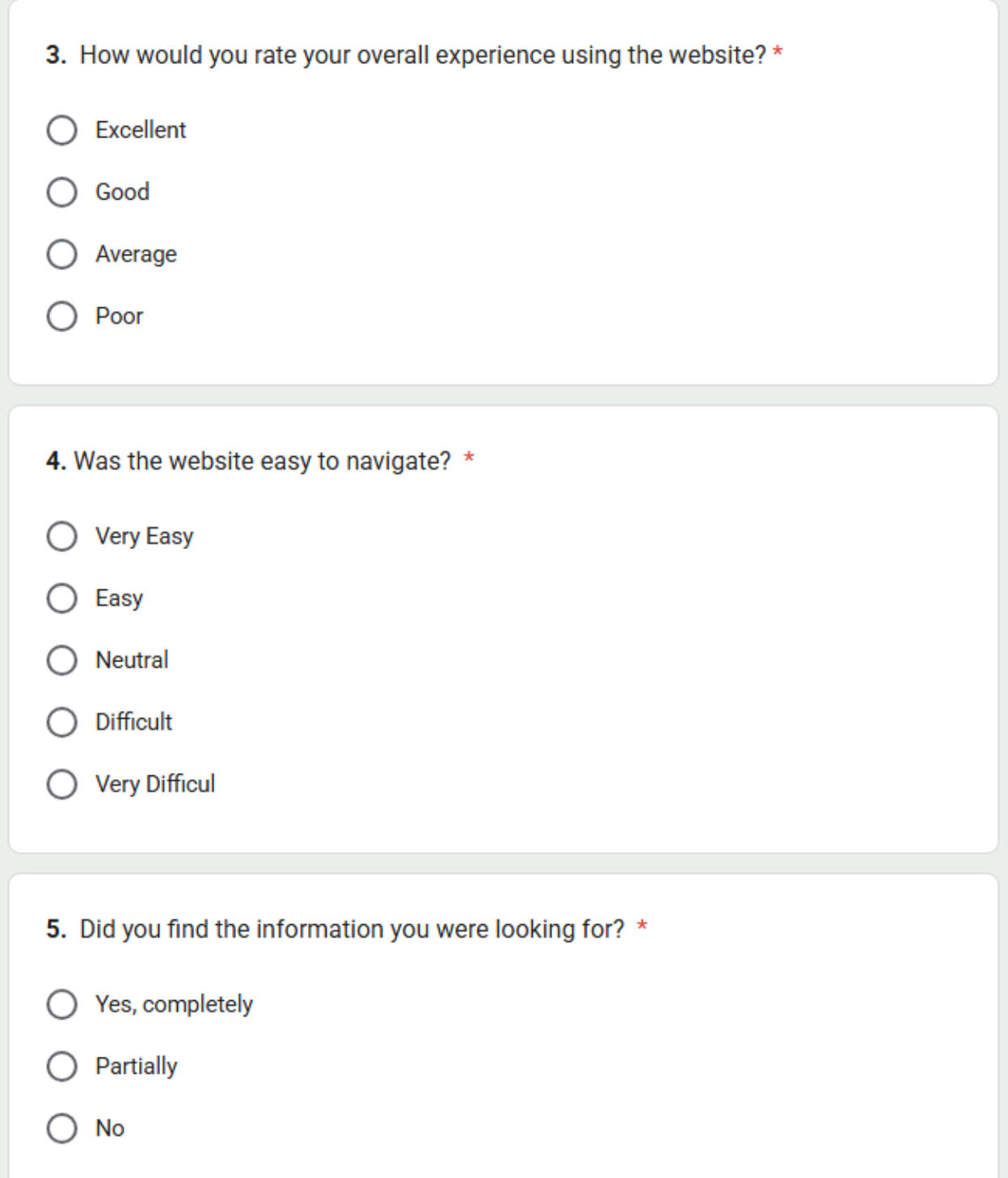


Figure 153 POST-SURVEY FORM

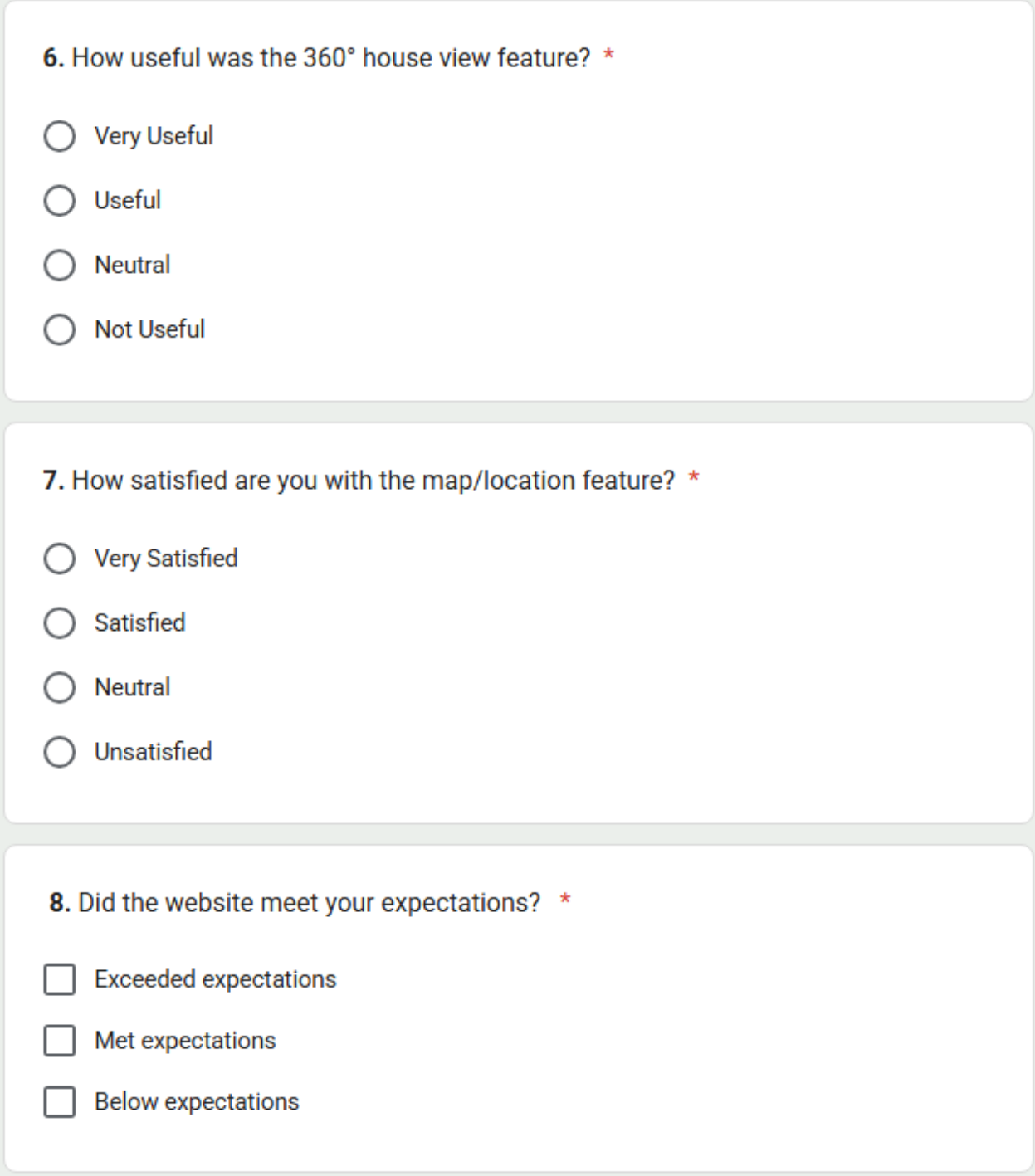


Figure 154 POST-SURVEY FORM

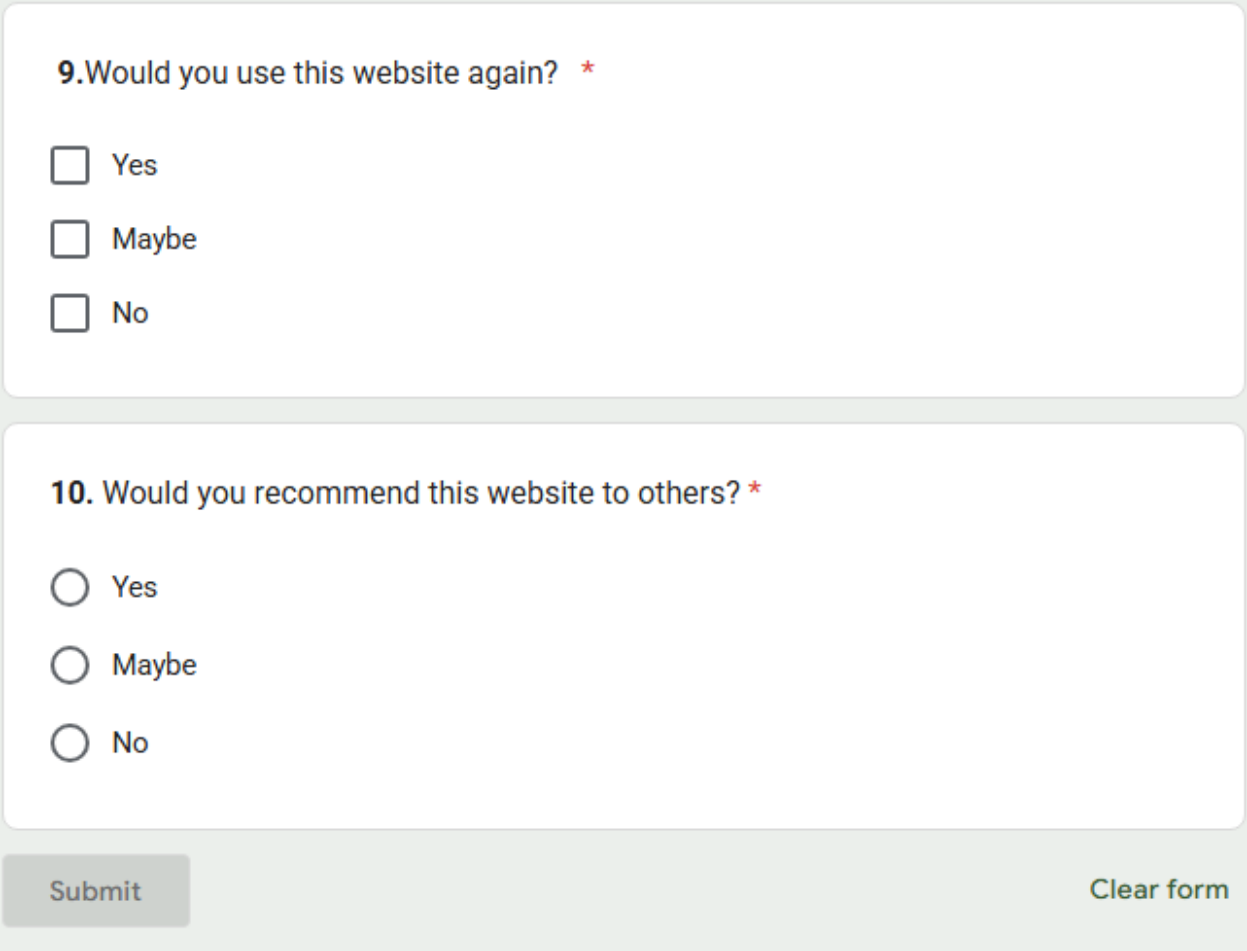


Figure 155 POST-SURVEY FORM

8.2.2 SAMPLE OF FILLED POST-SURVEY FORMS

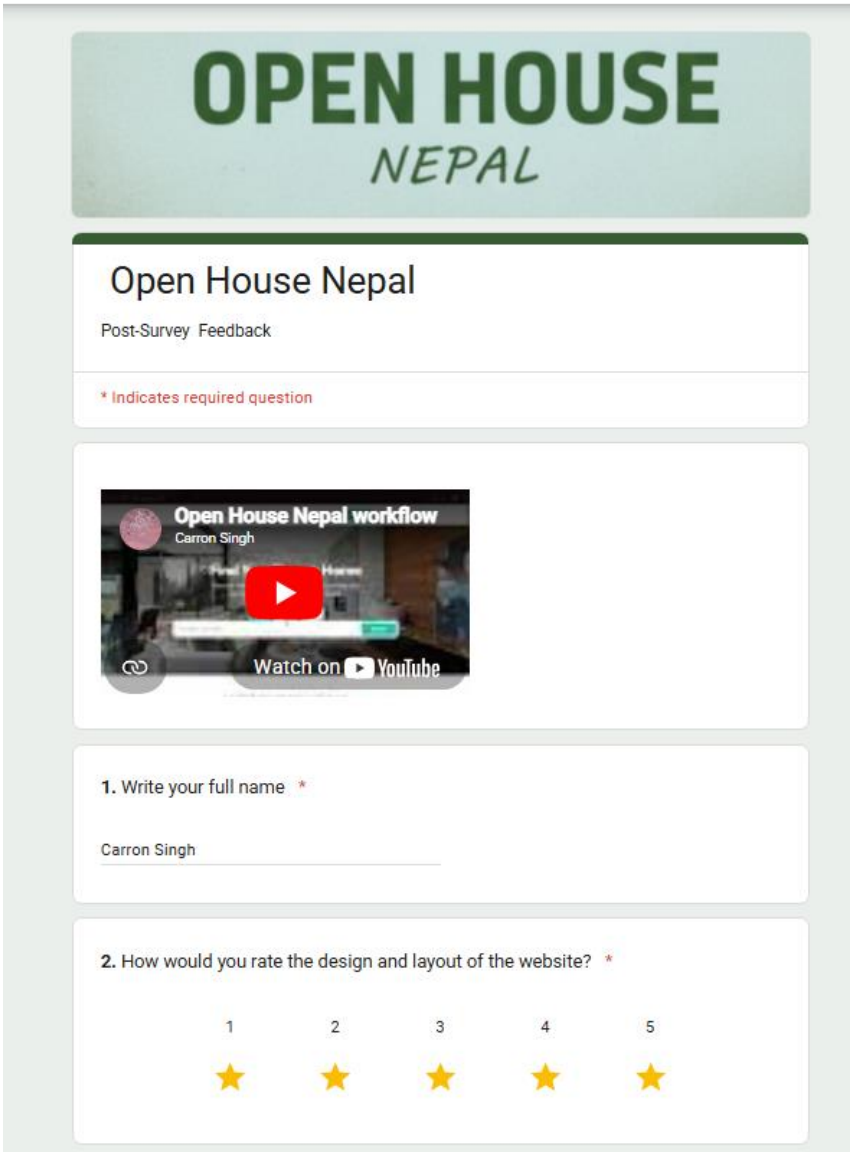


Figure 156 SAMPLE OF FILLED POST-SURVEY FORMS

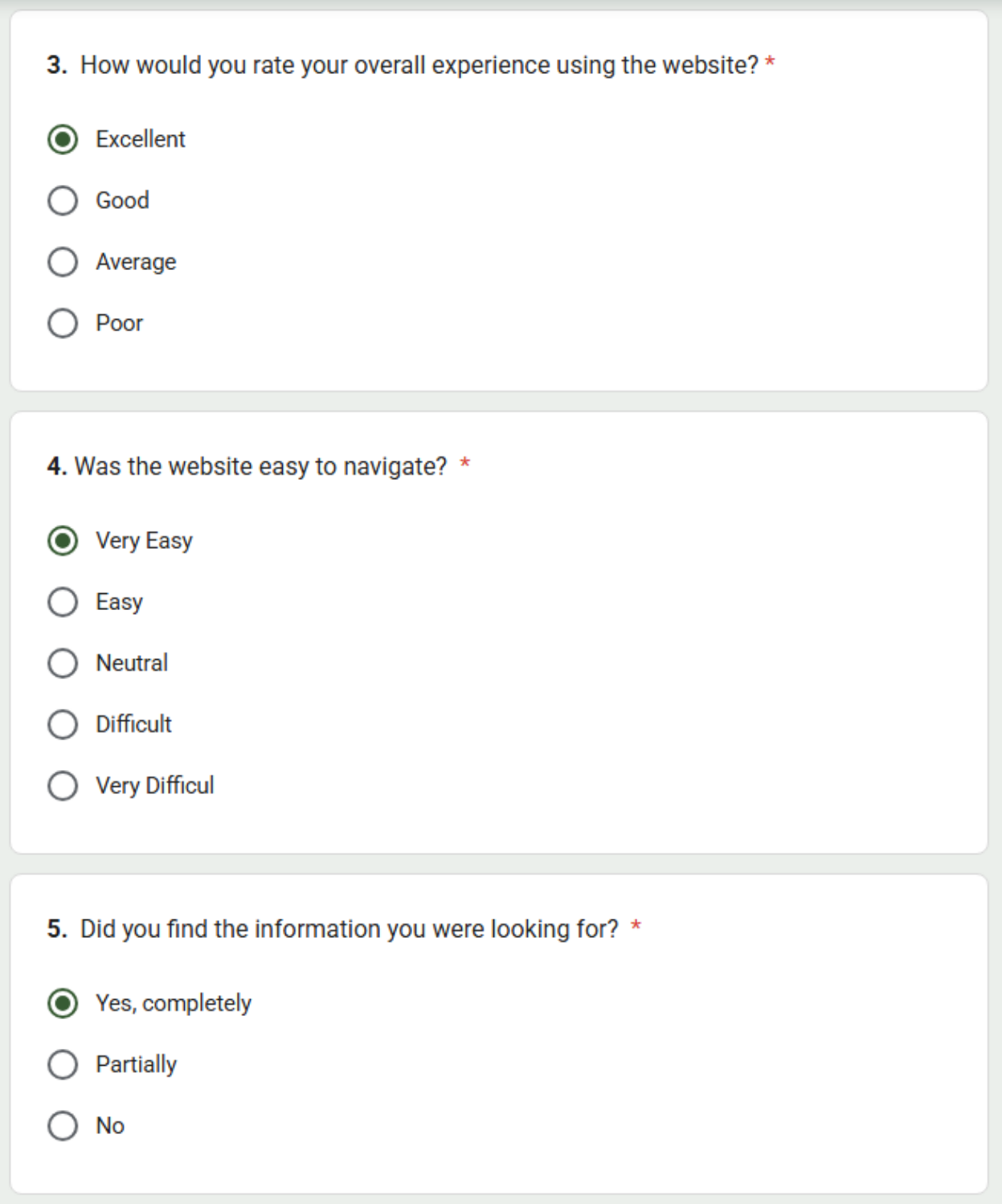


Figure 157 SAMPLE OF FILLED POST-SURVEY FORMS

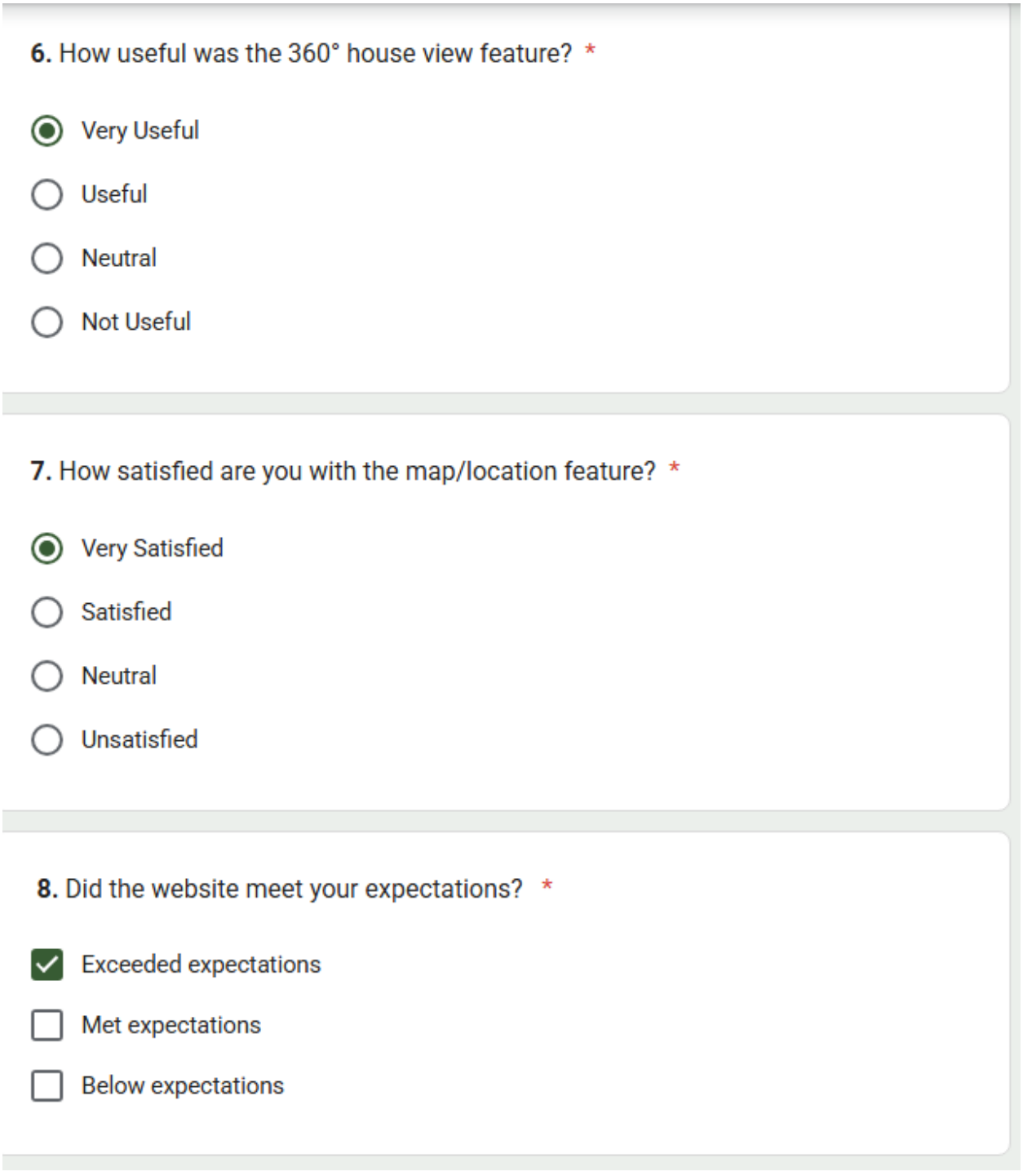


Figure 158 SAMPLE OF FILLED POST-SURVEY FORMS

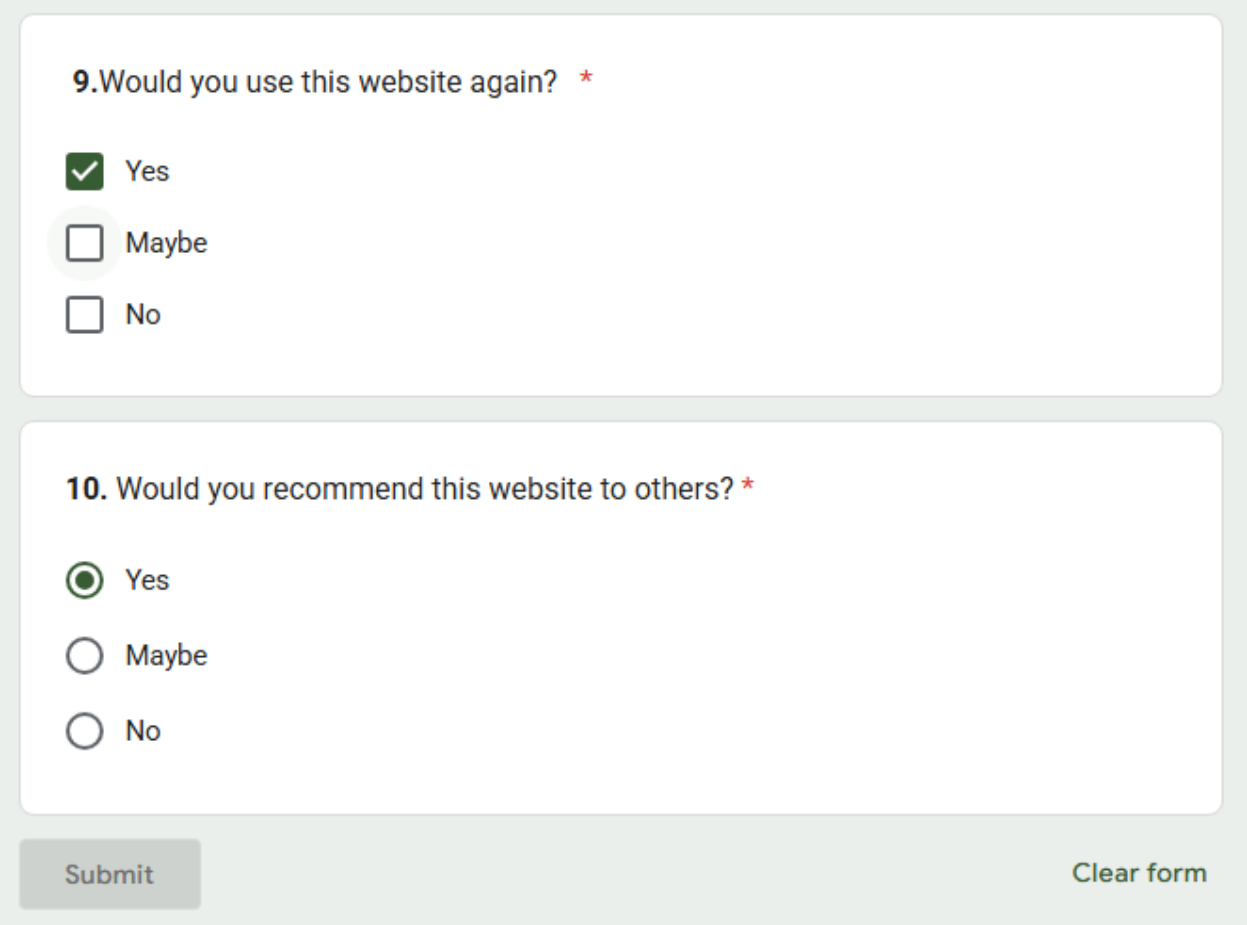


Figure 159 SAMPLE OF FILLED POST-SURVEY FORMS

8.2.3 POST-SURVEY RESULT

1. Write your full name

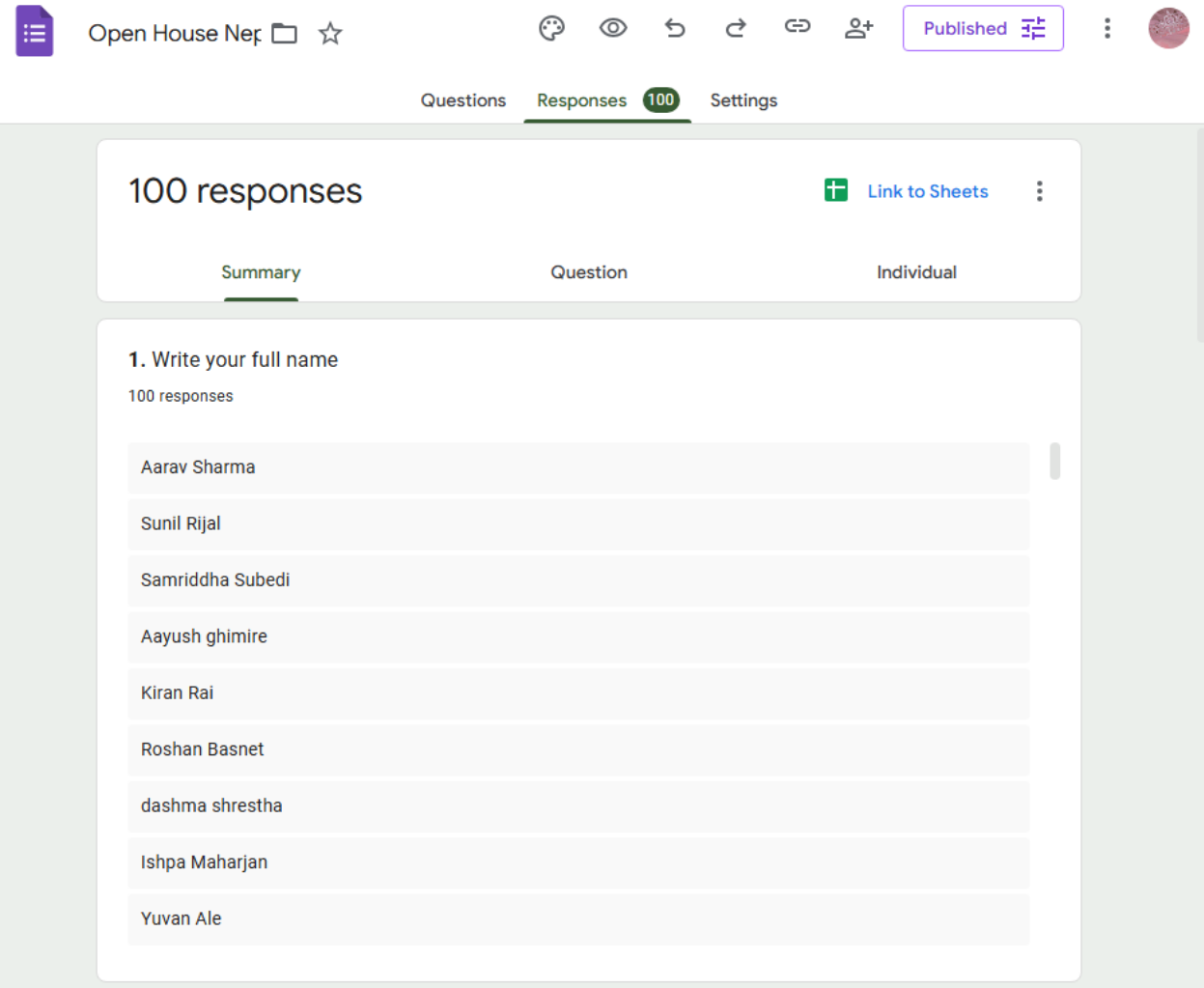


Figure 160 POST-SURVEY RESULT

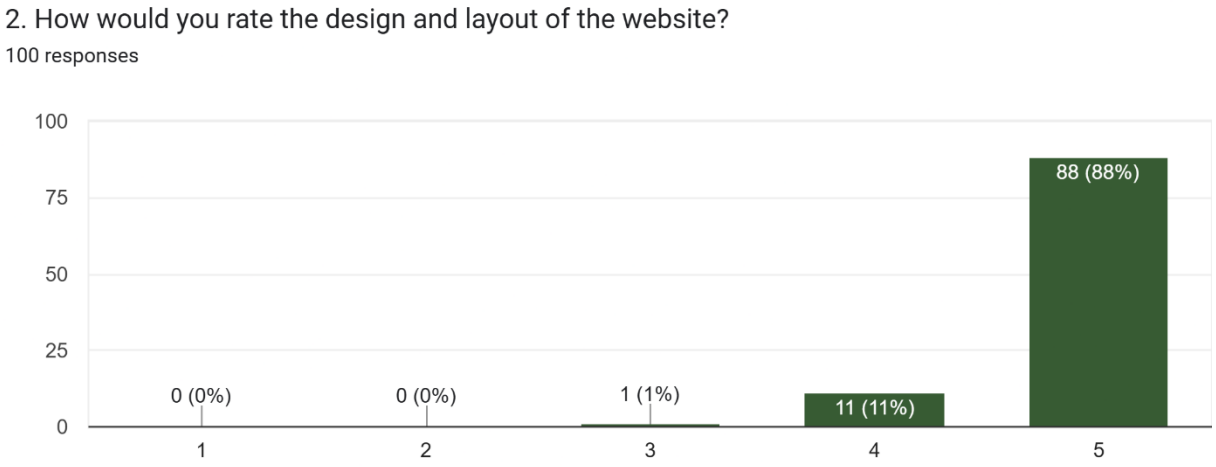


Figure 161 POST-SURVEY RESULT

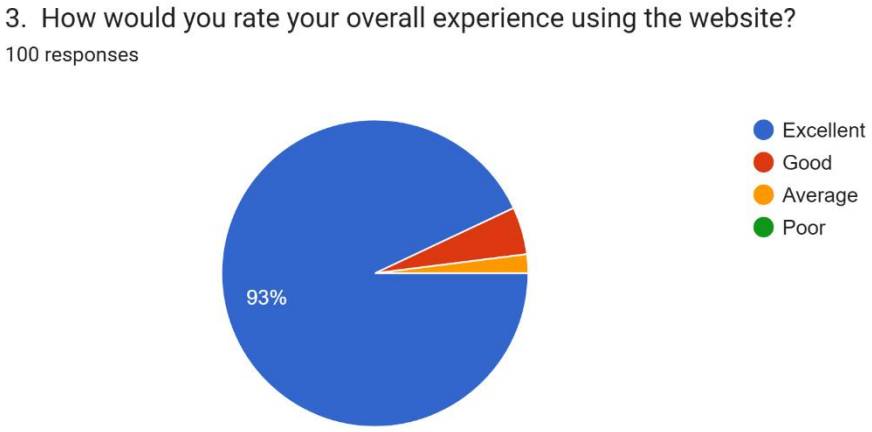


Figure 162 POST-SURVEY RESULT

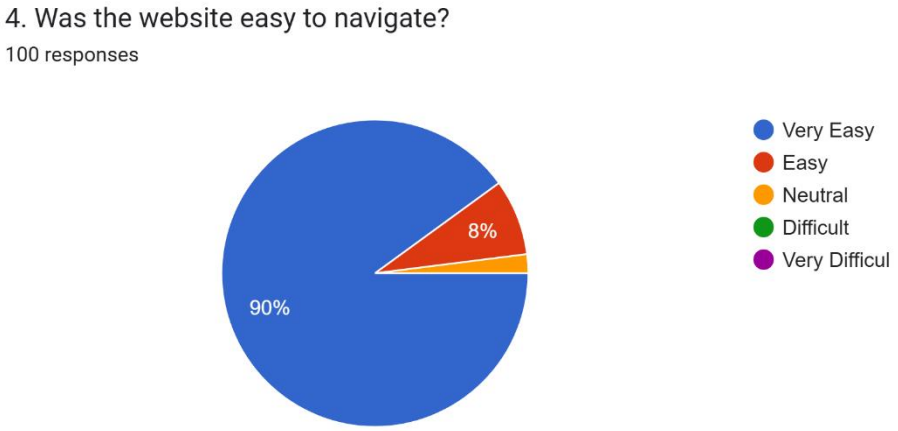


Figure 163 POST-SURVEY RESULT

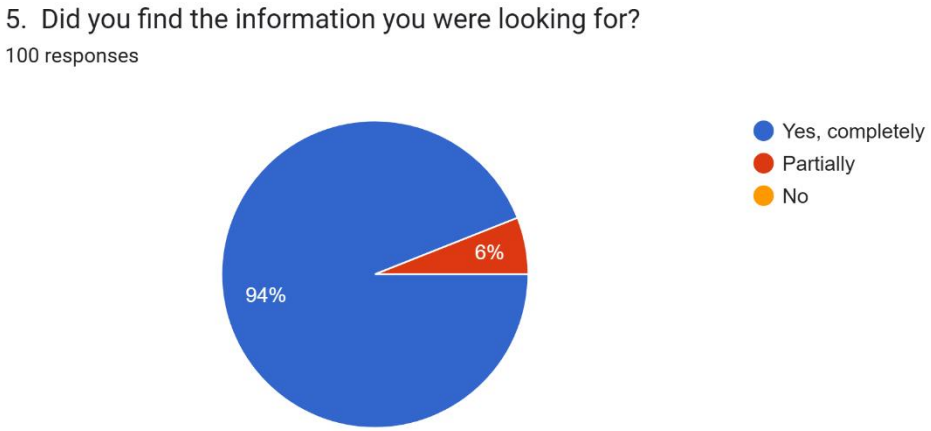


Figure 164 POST-SURVEY RESULT

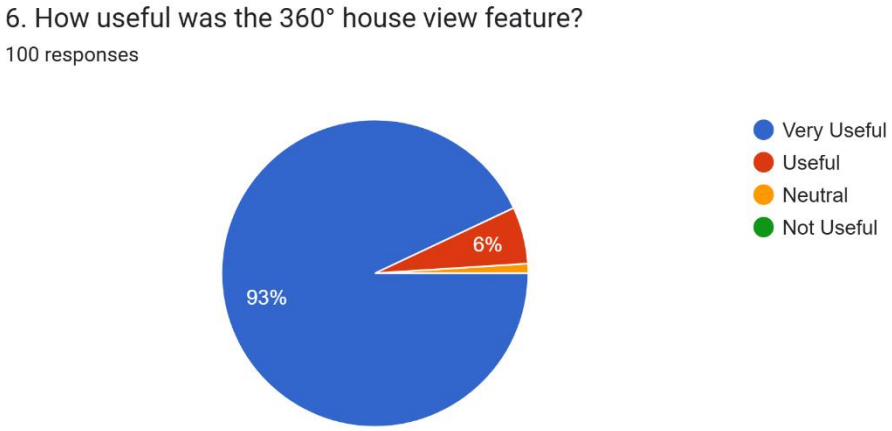


Figure 165 POST-SURVEY RESULT

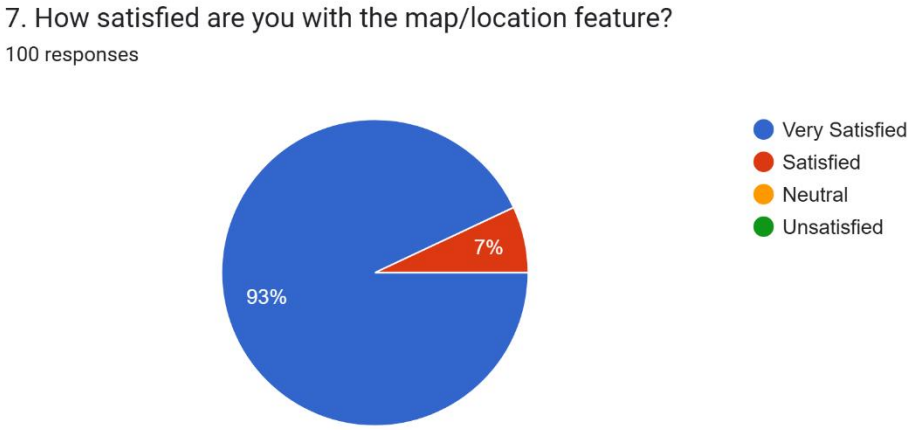


Figure 166 POST-SURVEY RESULT

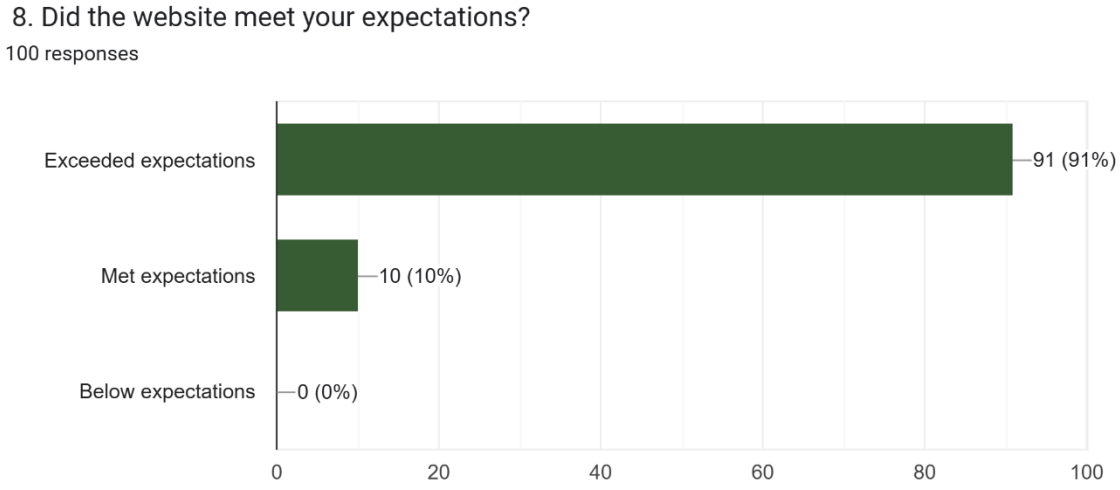


Figure 167 POST-SURVEY RESULT

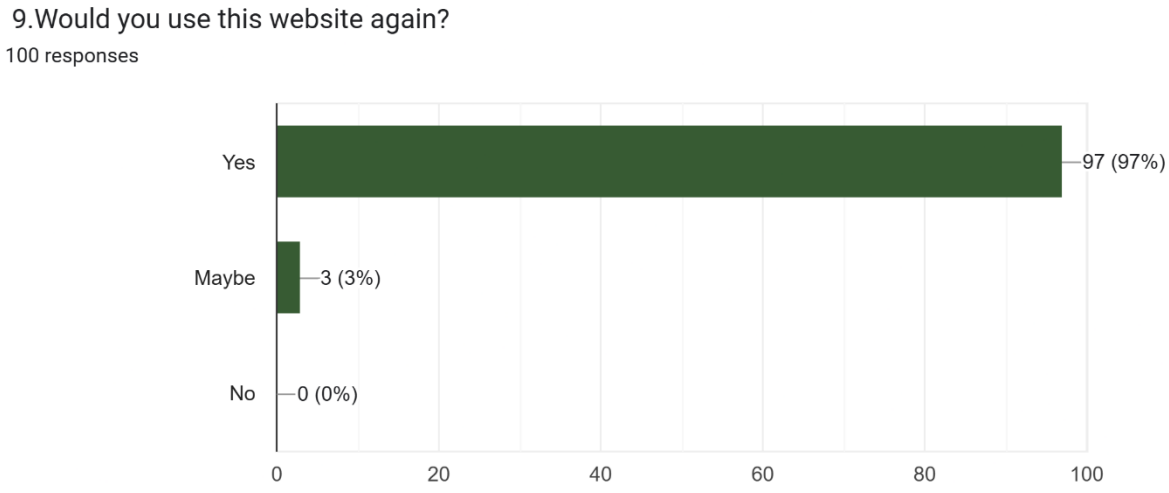


Figure 168 POST-SURVEY RESULT

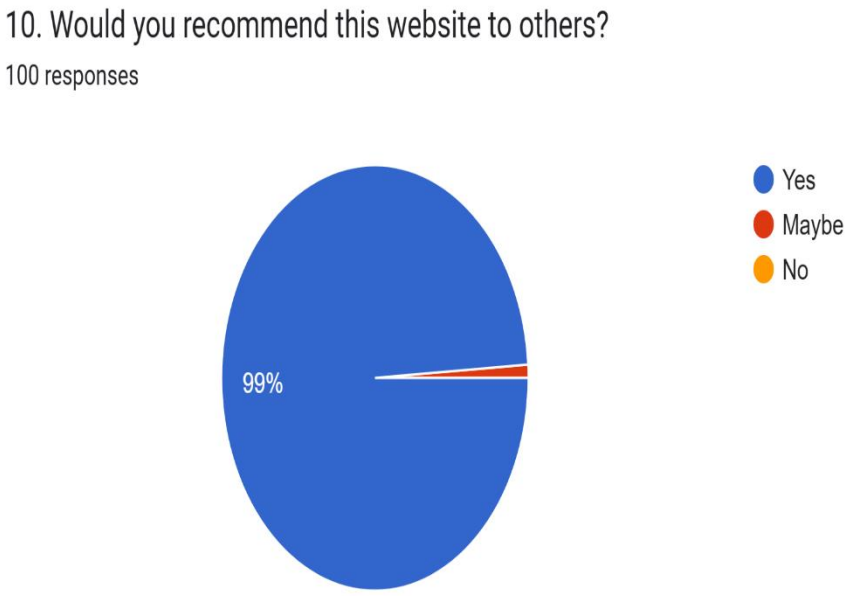


Figure 169 POST-SURVEY RESULT

Go back to post-survey

8.3 Appendix C: Sample Code

8.3.1 SAMPLE CODE OF THE UI

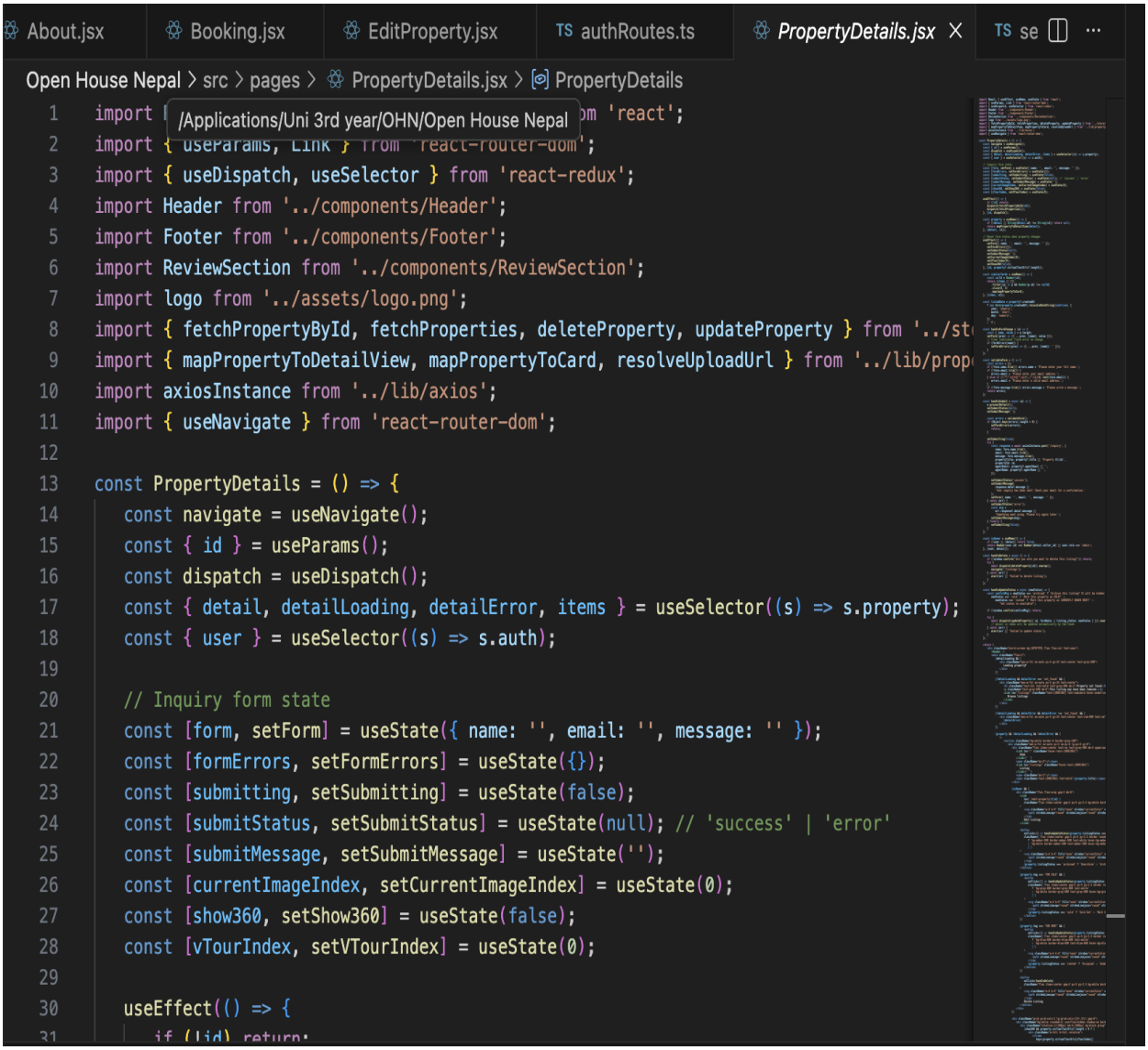


Figure 170 SAMPLE CODE OF THE UI

8.3.2 SAMPLE CODE FOR THE AUTOMATION SCRIPT

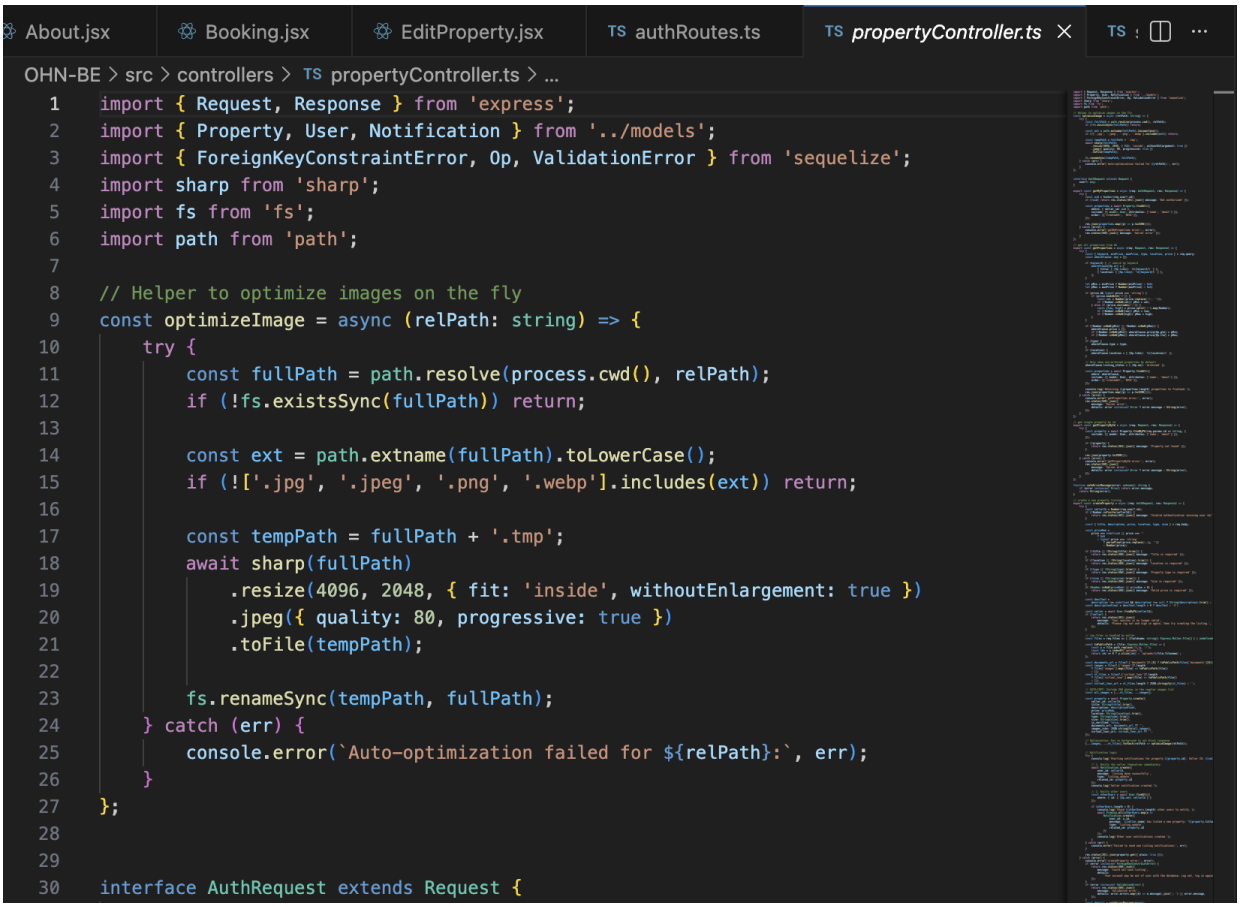


Figure 171 SAMPLE CODE FOR THE AUTOMATION SCRIPT

8.4 APPENDIX D: DESIGNS

8.4.1 WORK BREAKDOWN STRUCTURE

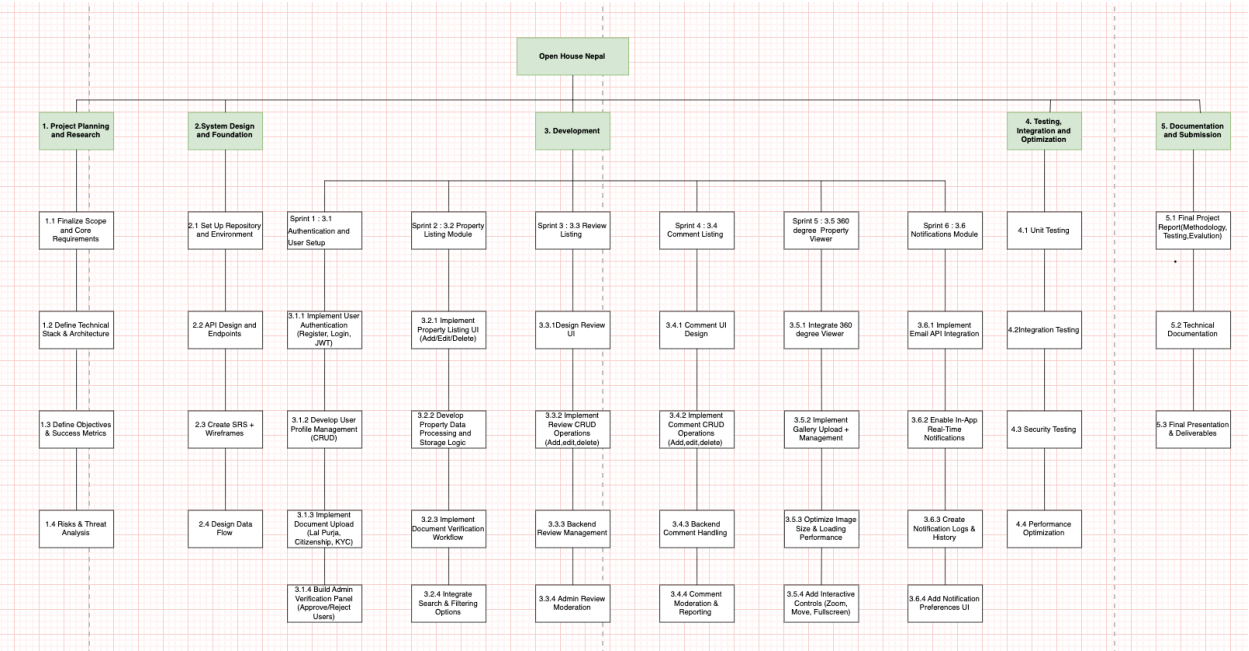


Figure 172 WORK BREAKDOWN STRUCTURE

Go back to WORK BREAKDOWN STRUCTURE

8.4.2 Gantt Chart

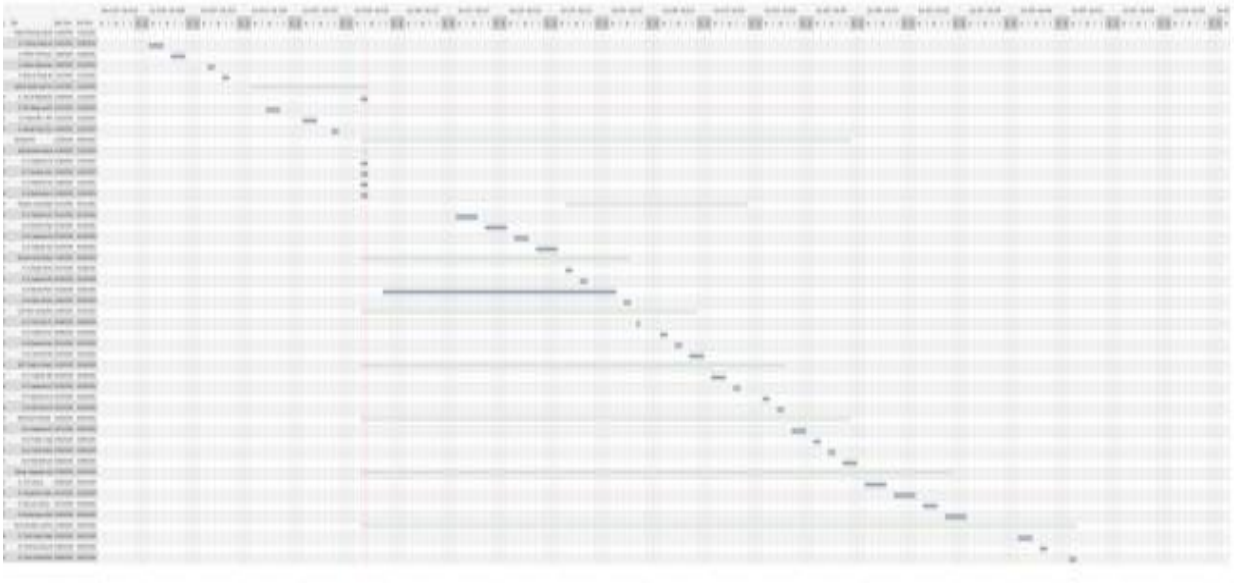


Figure 173 Gantt Chart

Go back to Gantt Chart

8.4.3 Use Case

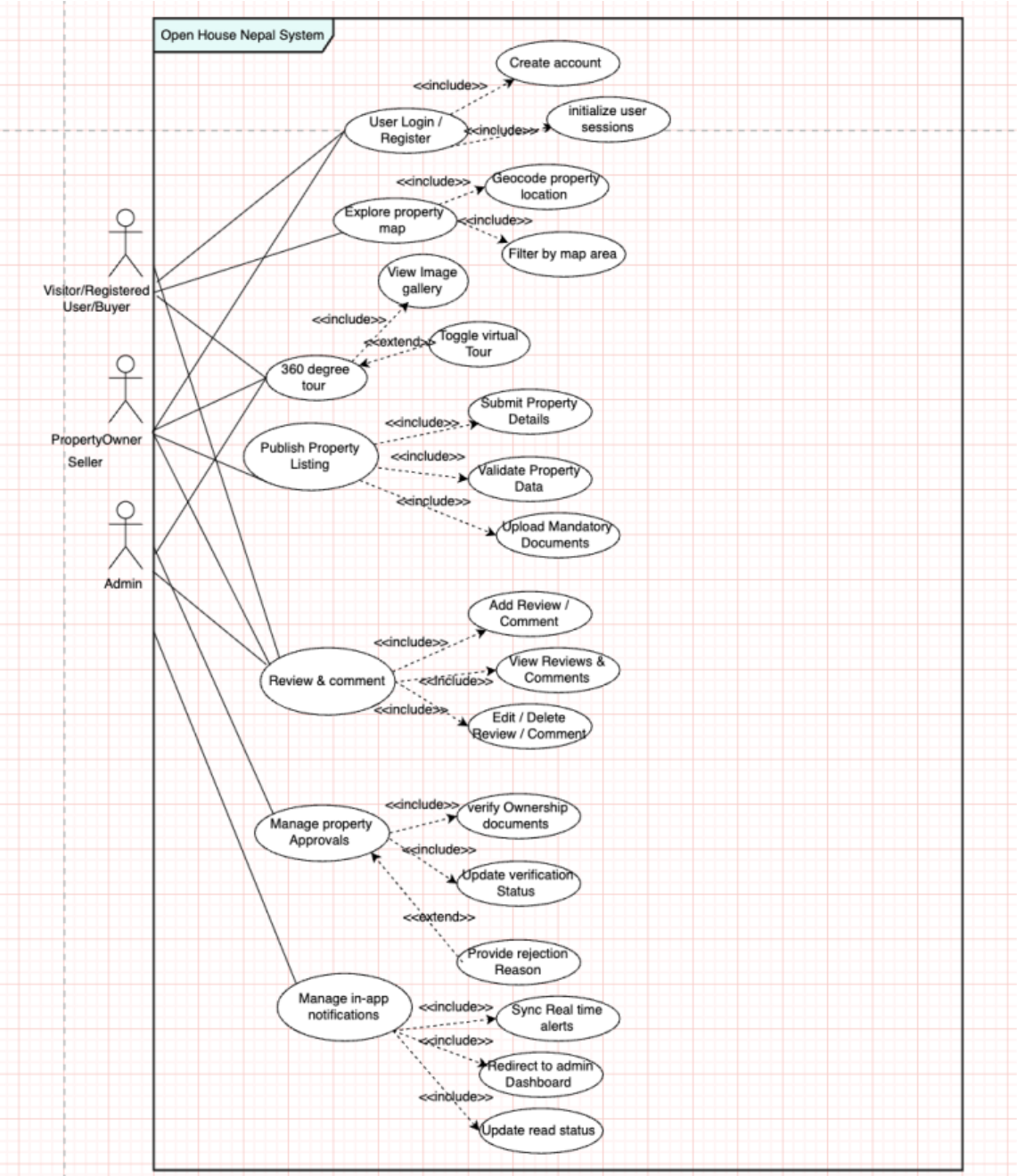


Figure 174 Use Case

Go Back to Use Case Diagram

8.4.4 ERD

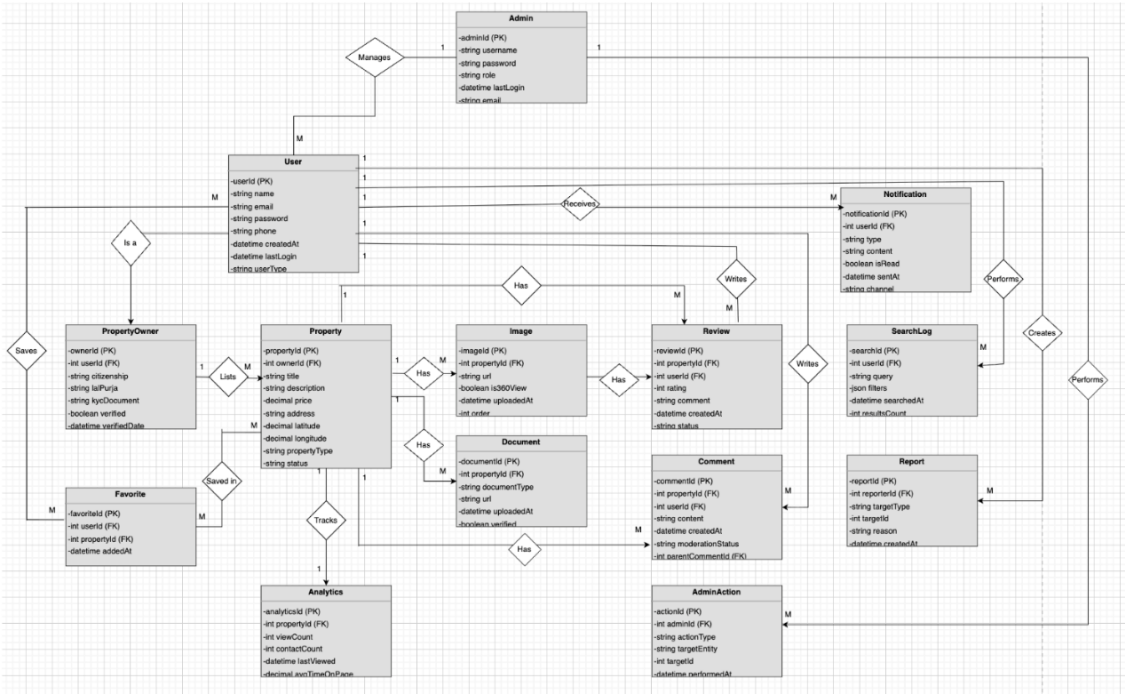


Figure 175 ERD

GO back to Entity Relationship Diagram (ERD)

8.4.5 DATA FLOW DIAGRAMS

8.4.5.1 DFD Level 0

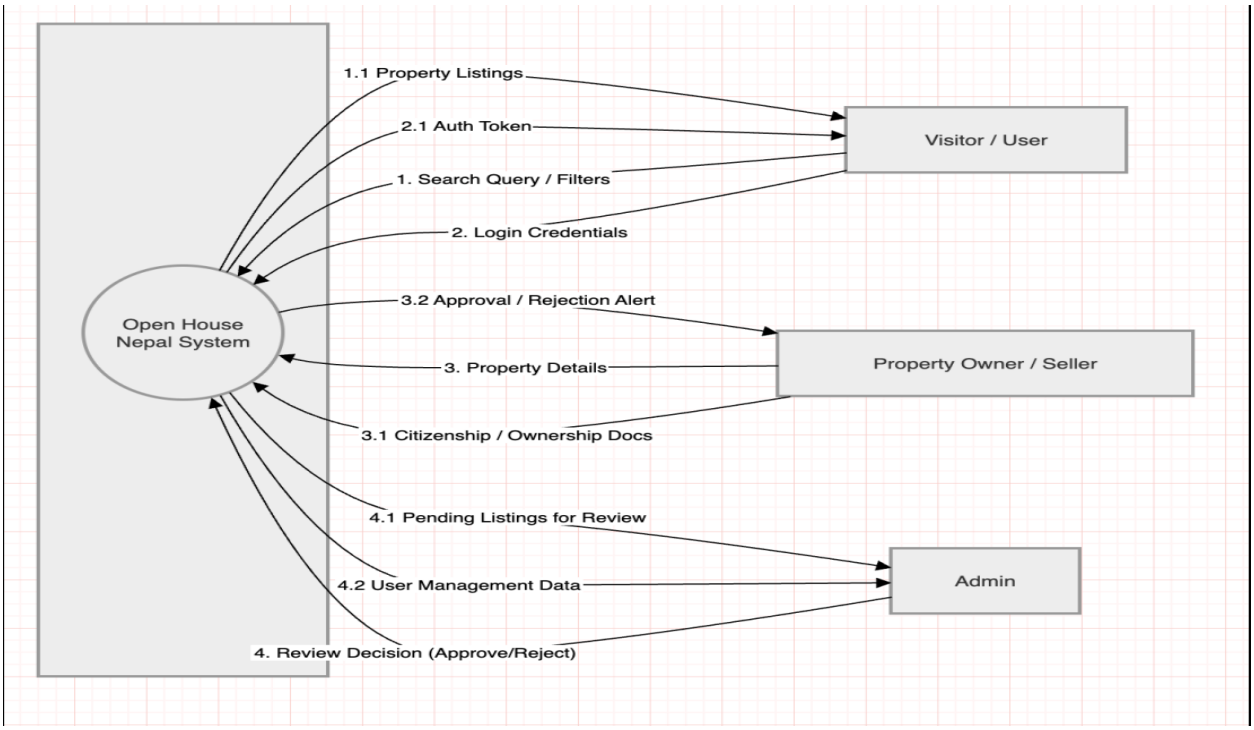


Figure 176 DFD Level 0

8.4.5.1 DFD Level 1

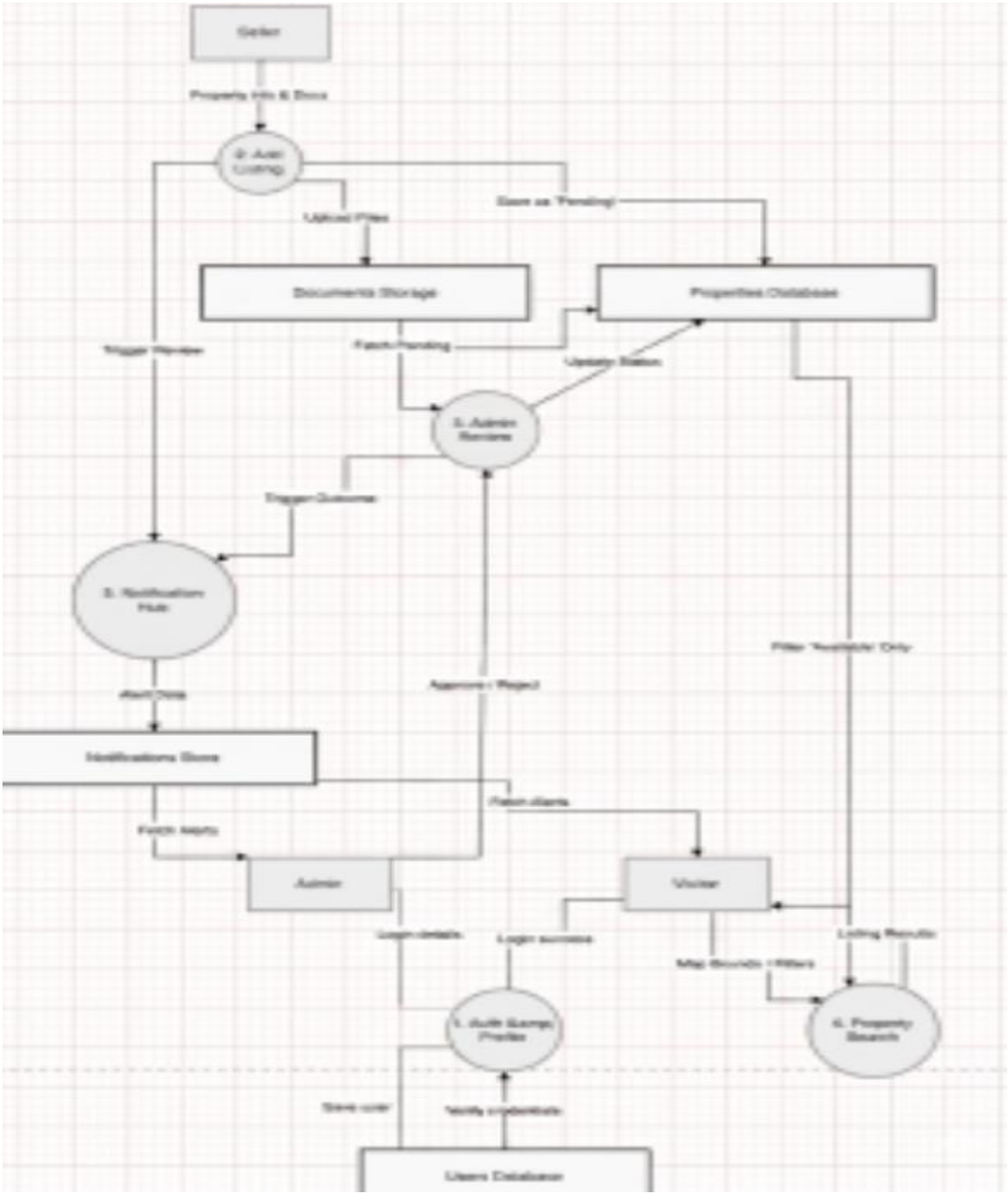


Figure 177 DFD Level 1

8.4.5.1 Final DFD

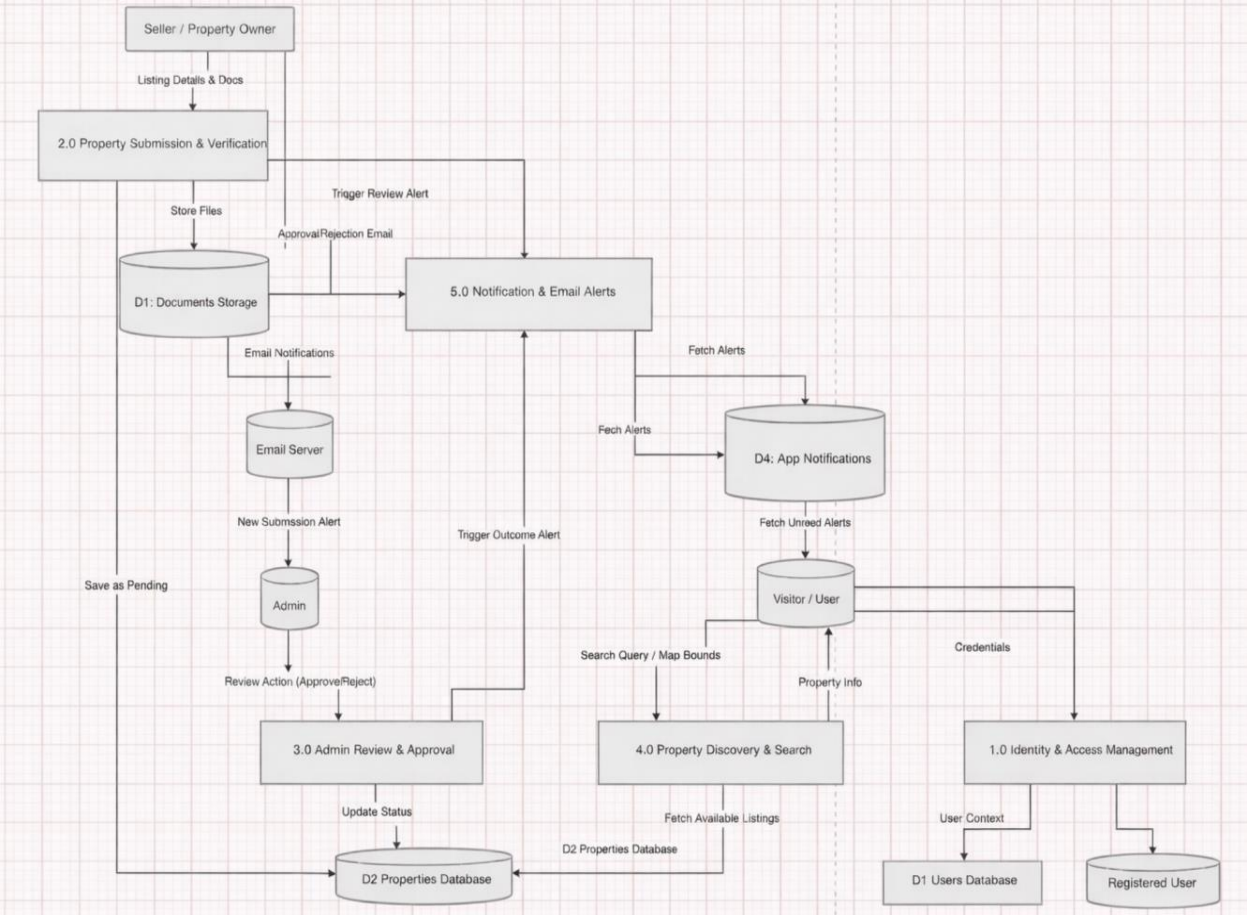


Figure 178 Final DFD

Go back to Data Flow Diagram (DFD)

8.4.6 FLOWCHART

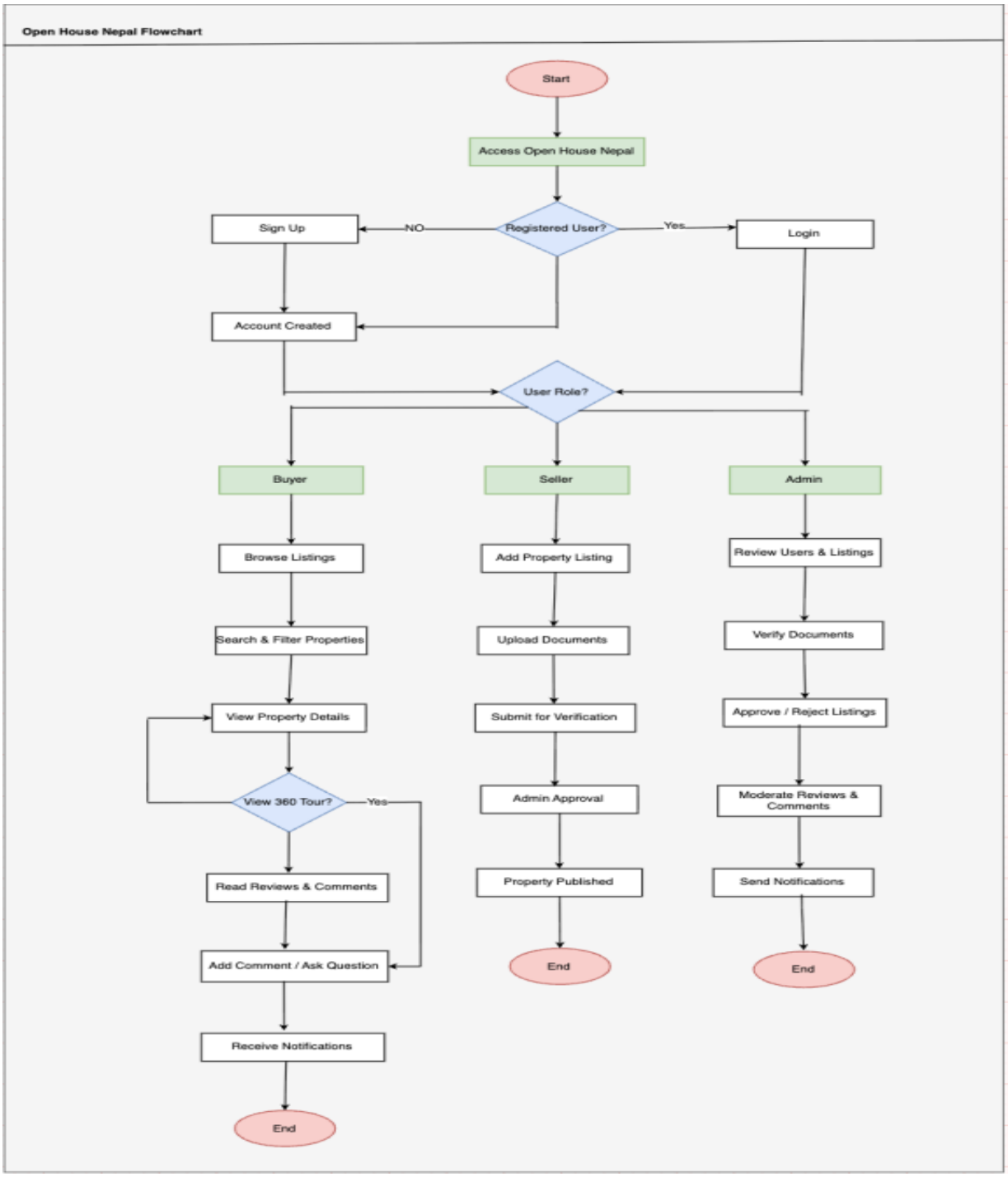


Figure 179 FLOWCHART

Go back to Flowchart

8.4.7 System Architecture

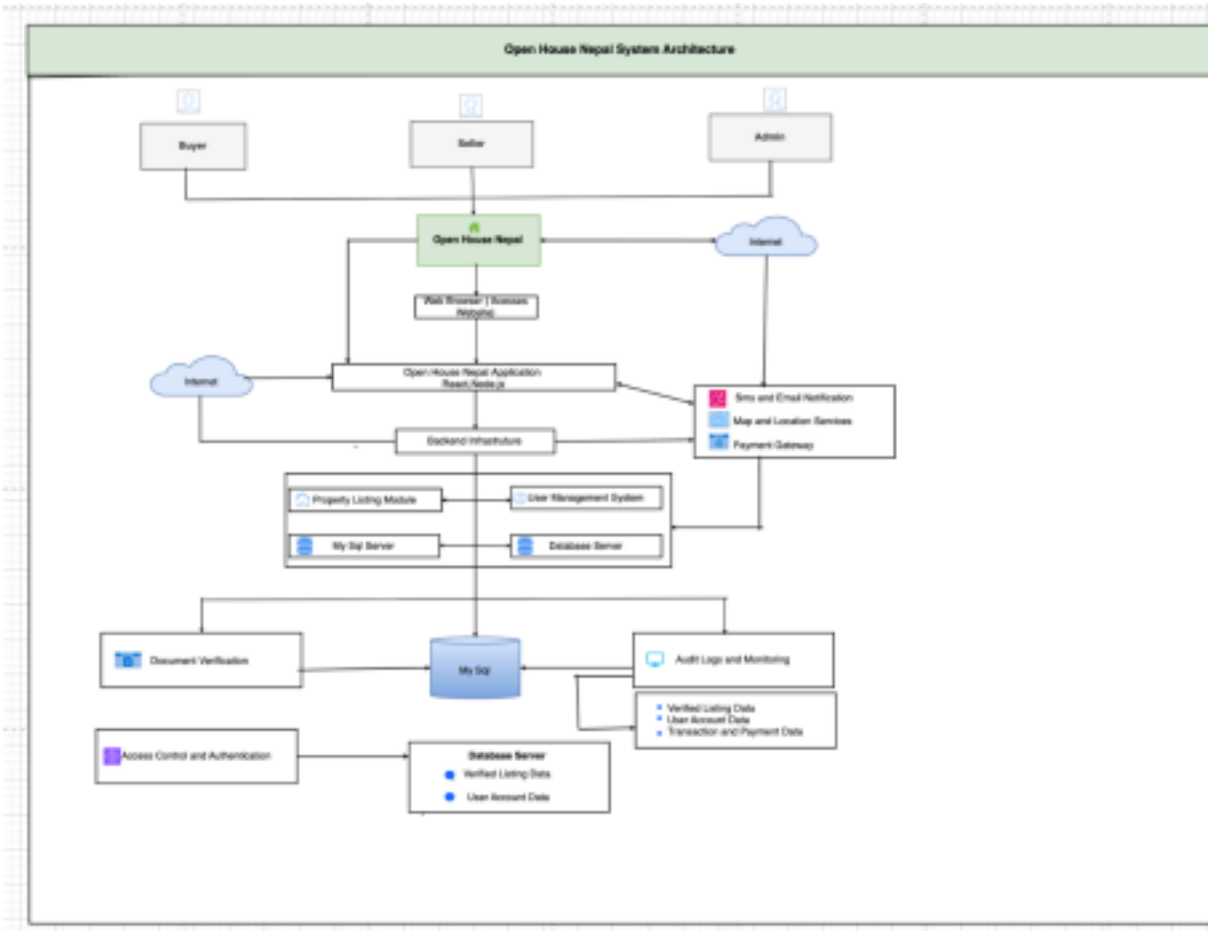


Figure 180 System Architecture

Go back to System Architecture

8.4.8 UI

8.4.8.1 Login in page



Figure 181 Login in page

8.4.8.2 Sign Up Page

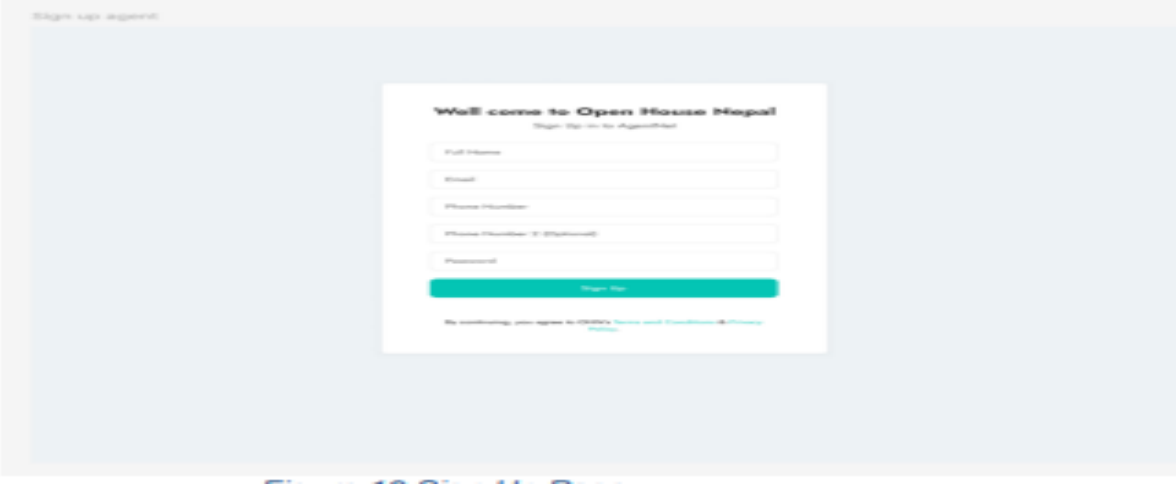


Figure 182 Sign Up Page

8.4.8.3 Forgot Password

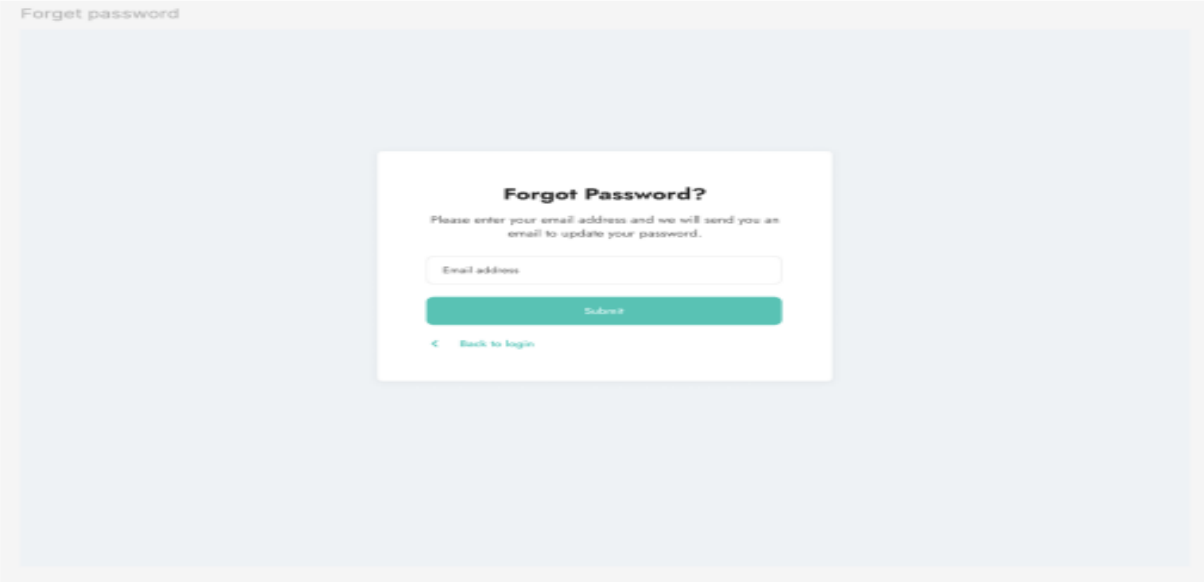


Figure 183 Forgot Password

8.4.8.4 Home Page

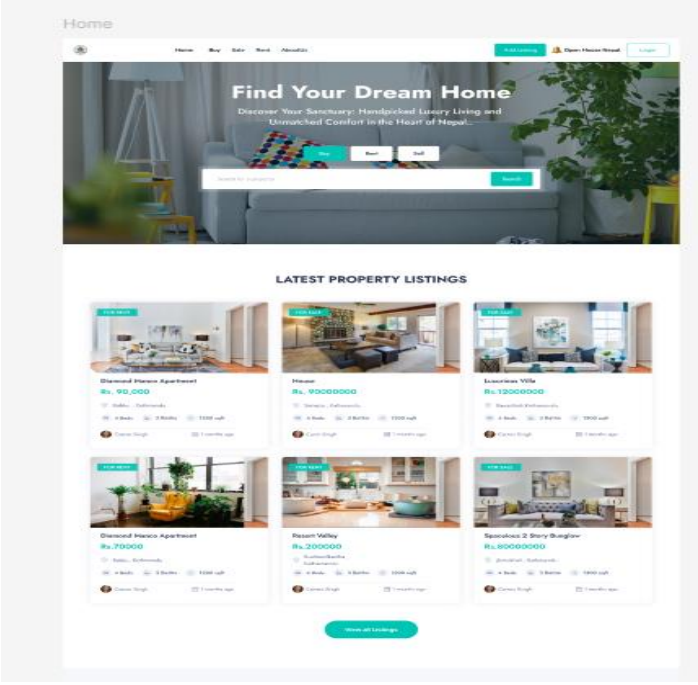


Figure 184 Home Page

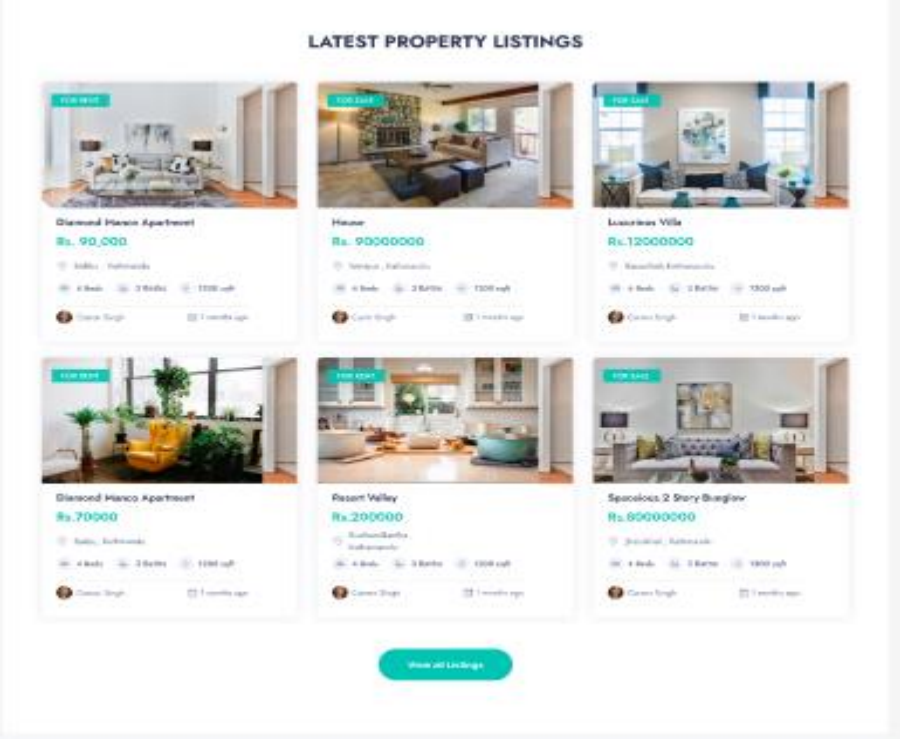


Figure 185 Home Page

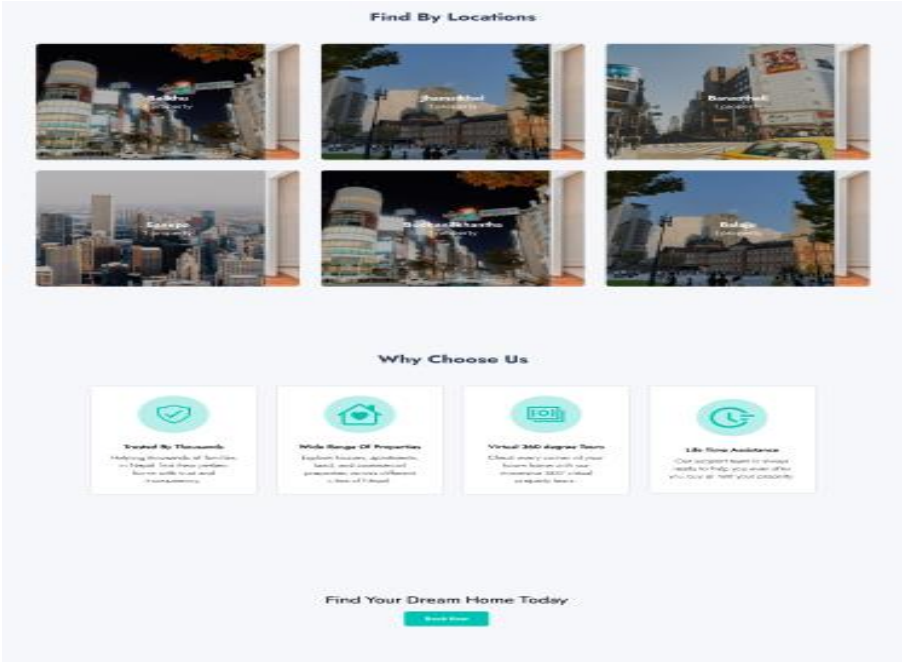


Figure 186 Home Page

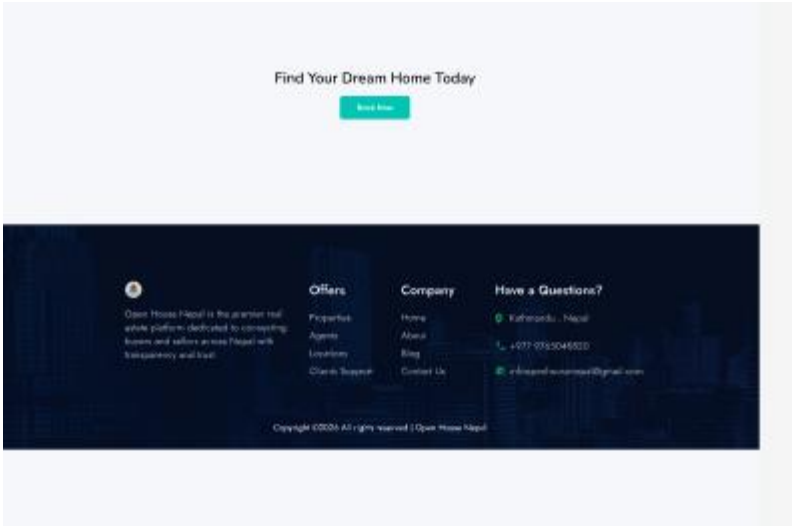


Figure 187 Home Page

8.4.8 .5 About Us



Figure 188 About Us

8.4.8.6 Listing Page

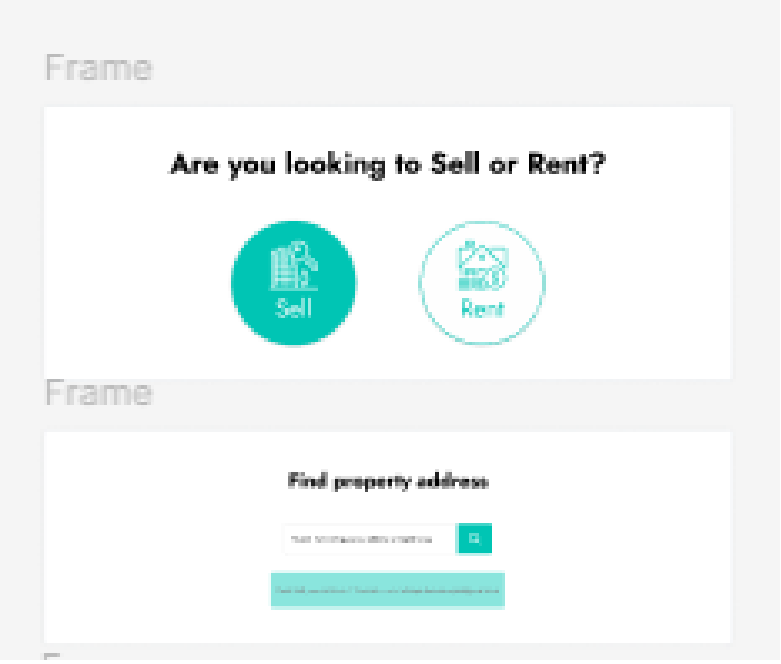


Figure 189 Listing Page

8.4.8.7 Property Details

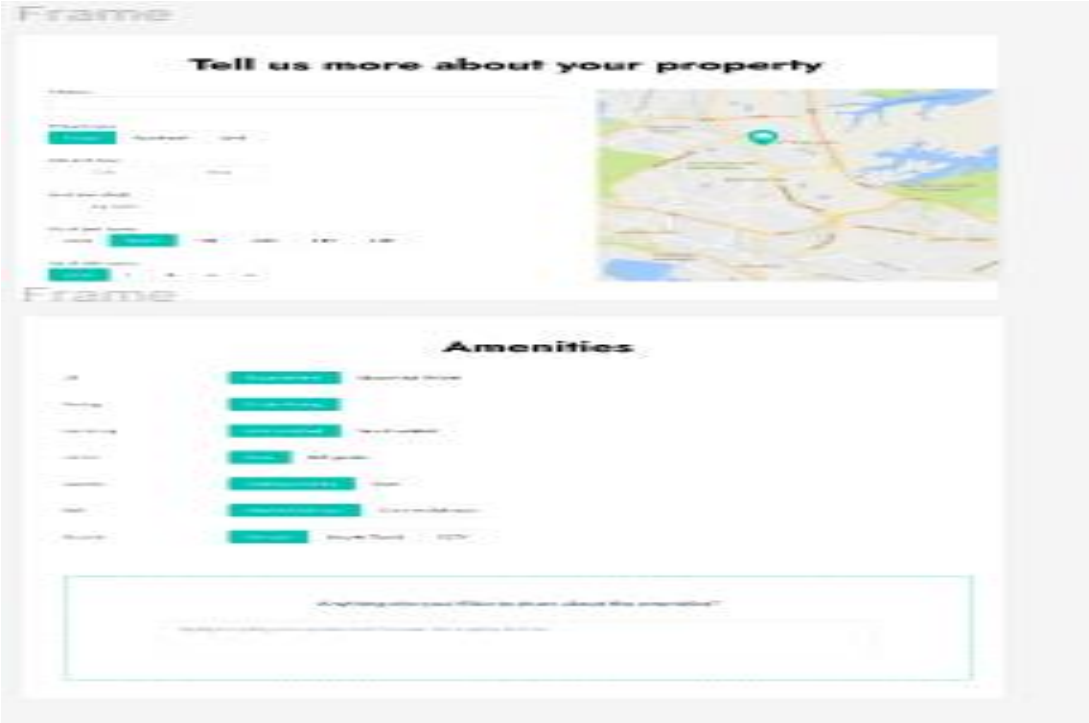


Figure 190 Property Details

Move Next page

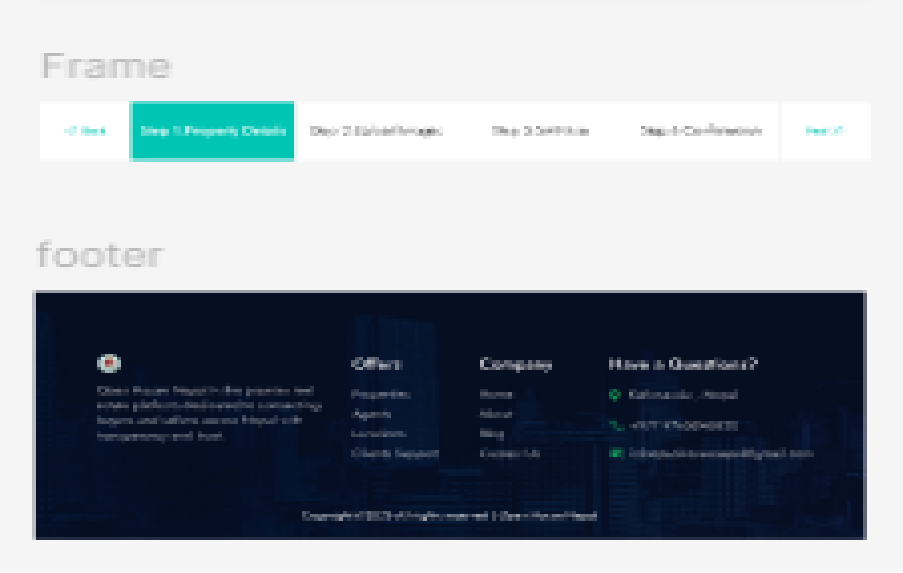


Figure 191 Move Next page

8.4.8.8 Media and Documents

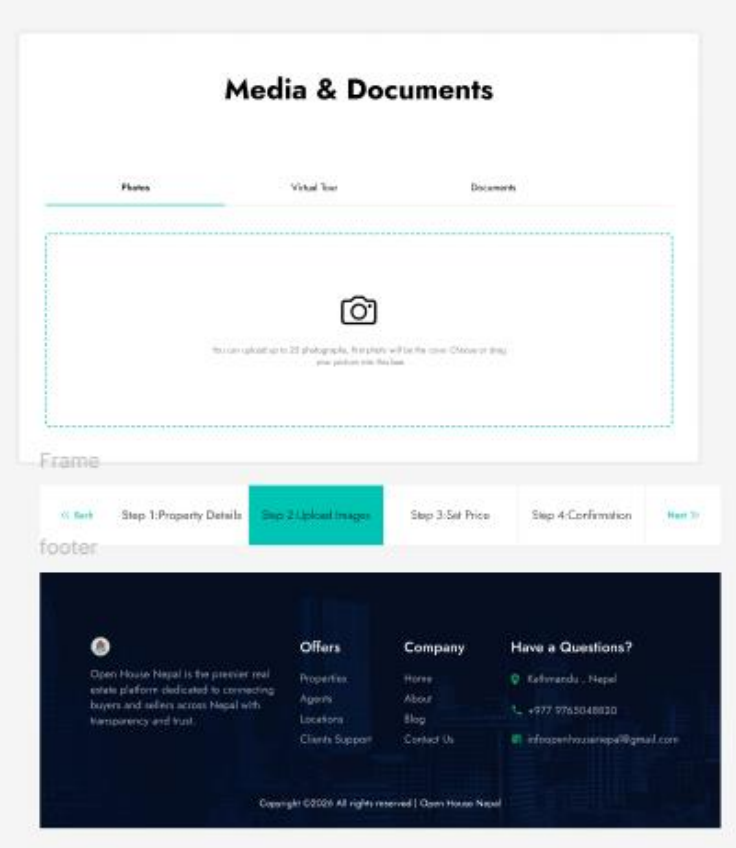


Figure 192 Media and Documents

8.4.8.9 360° photo upload

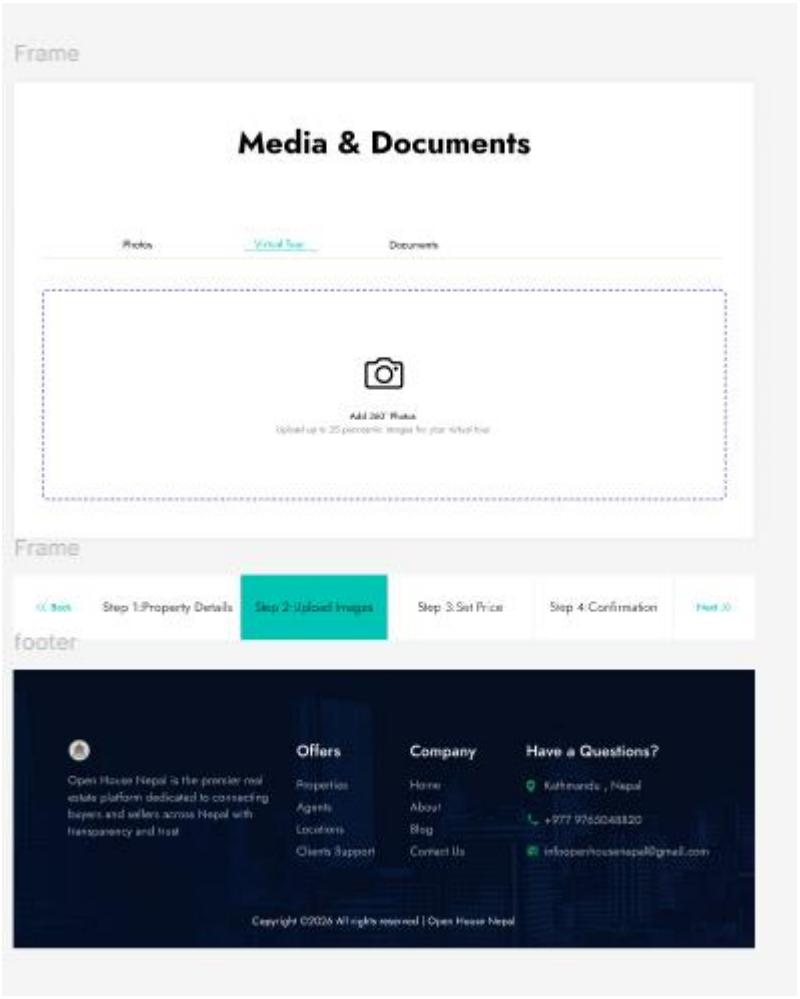


Figure 193 360° photo upload

8.4.8.10 Citizenship and Property Details

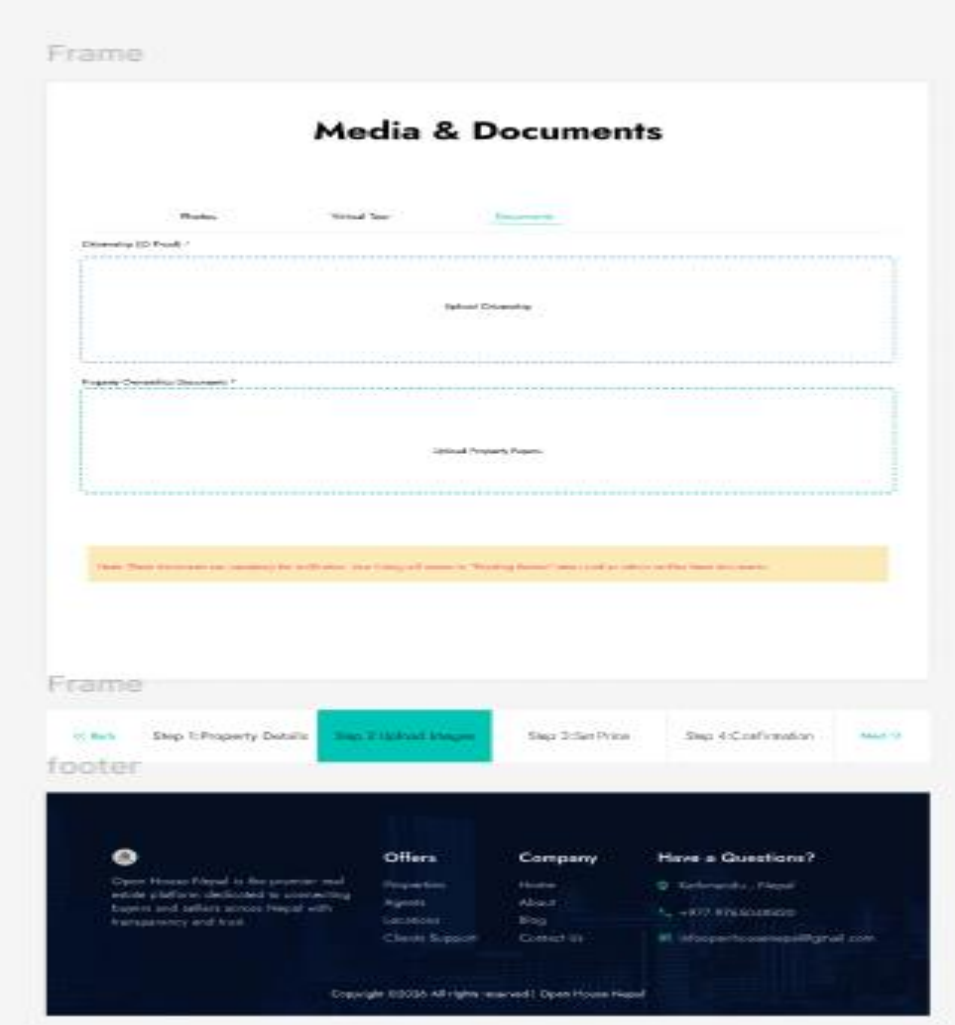


Figure 194 Citizenship and Property Details

8.4.8.11 Set your price

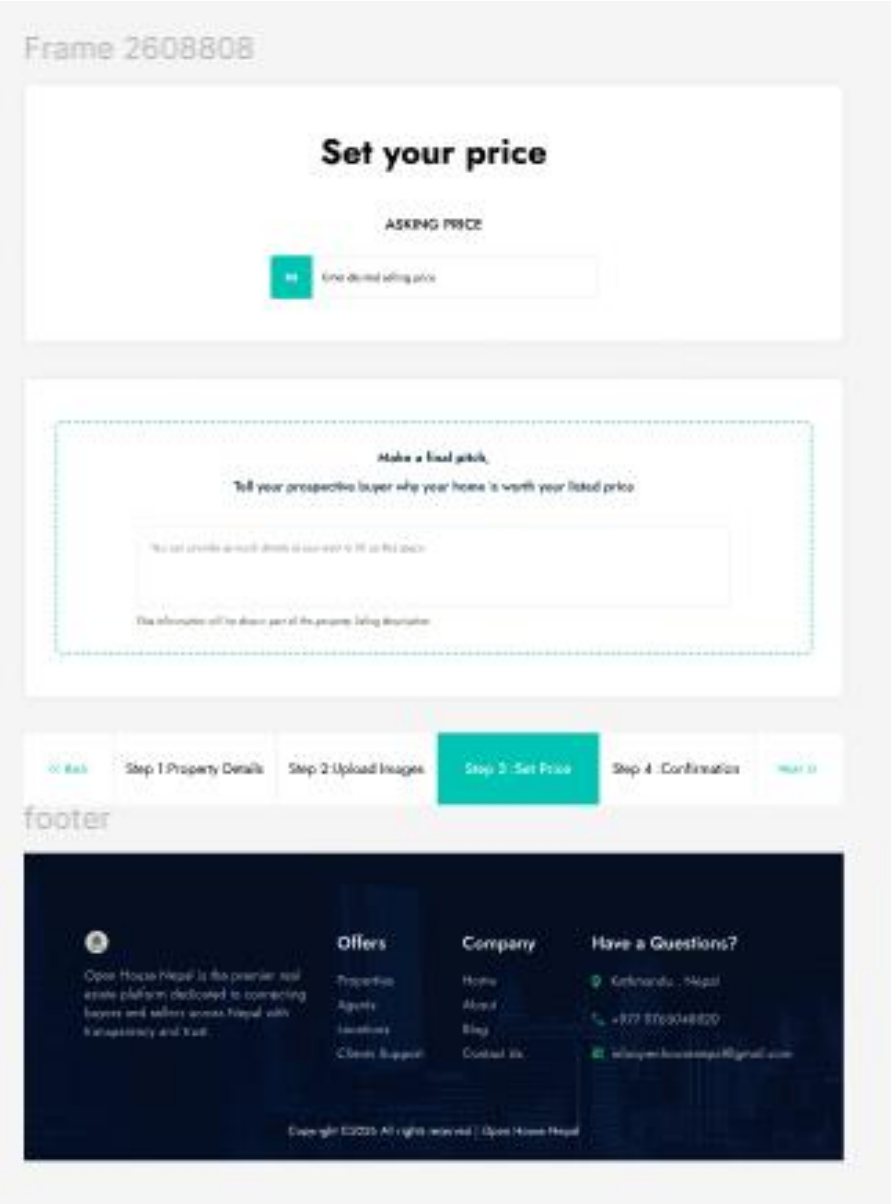


Figure 195 Set your price

8.4.8.12 Conformation

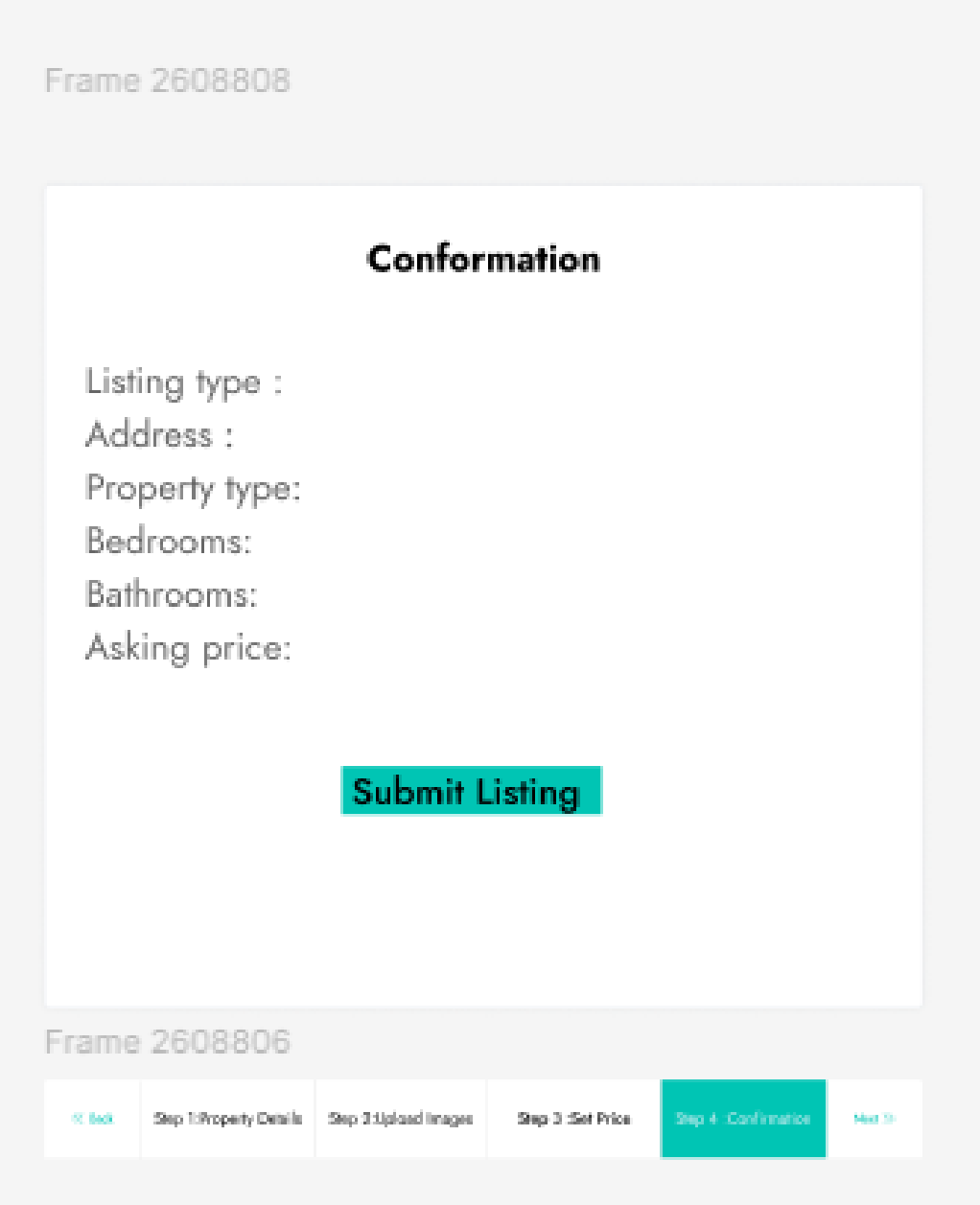


Figure 196 Conformation

8.4.8.13 Wating for Admin approval

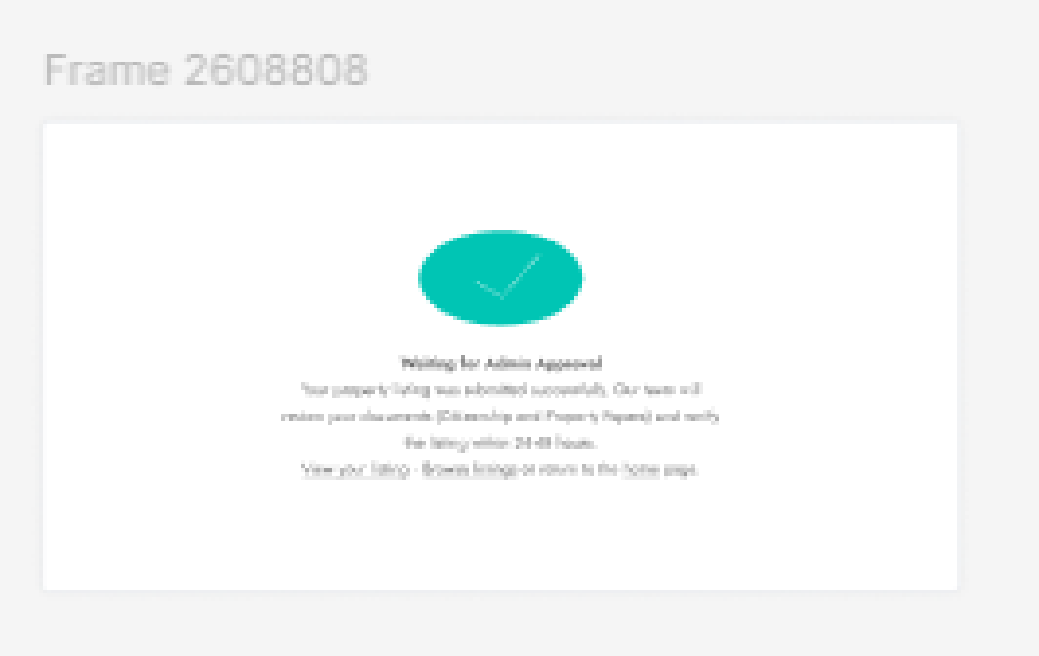


Figure 197 Wating for Admin approval

8.4.8.14 Seller Listing

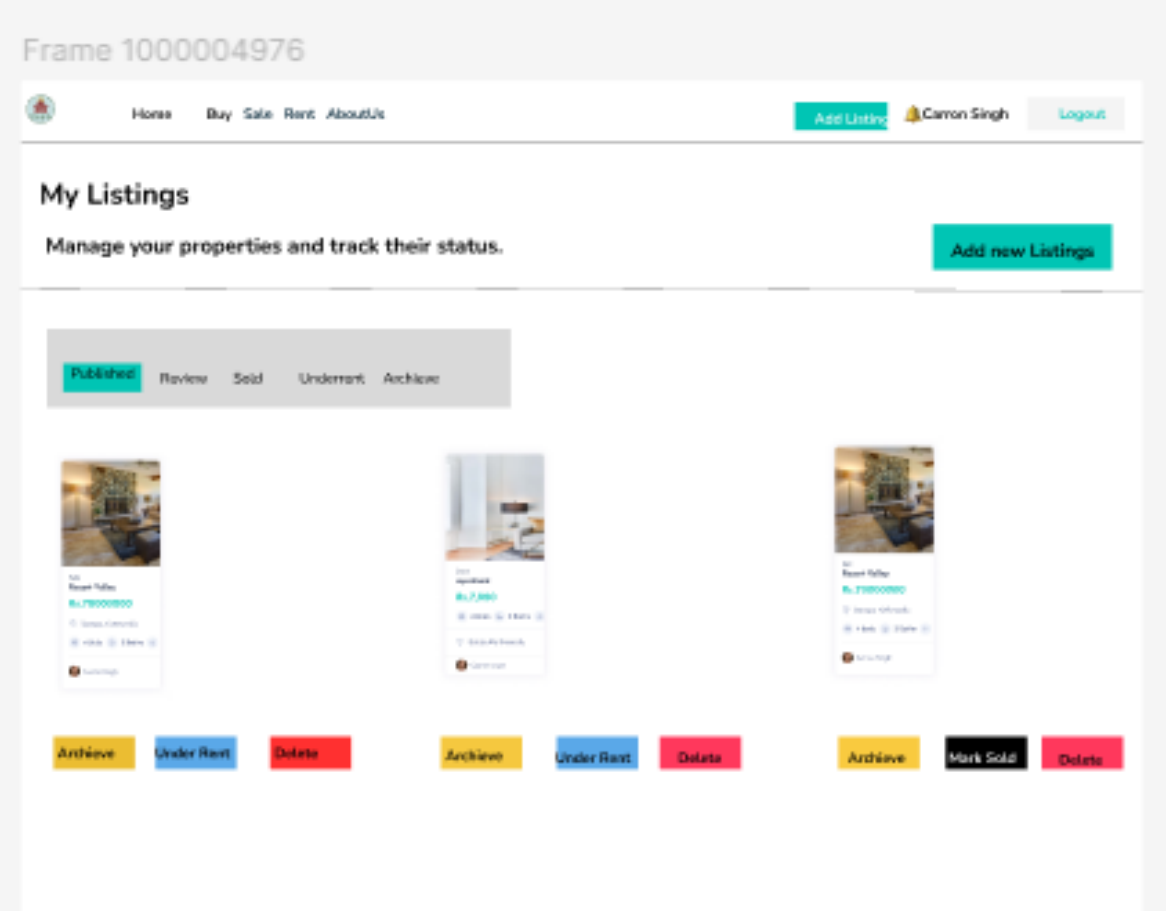


Figure 198 Seller Listing

8.4.8.15 Advance Search

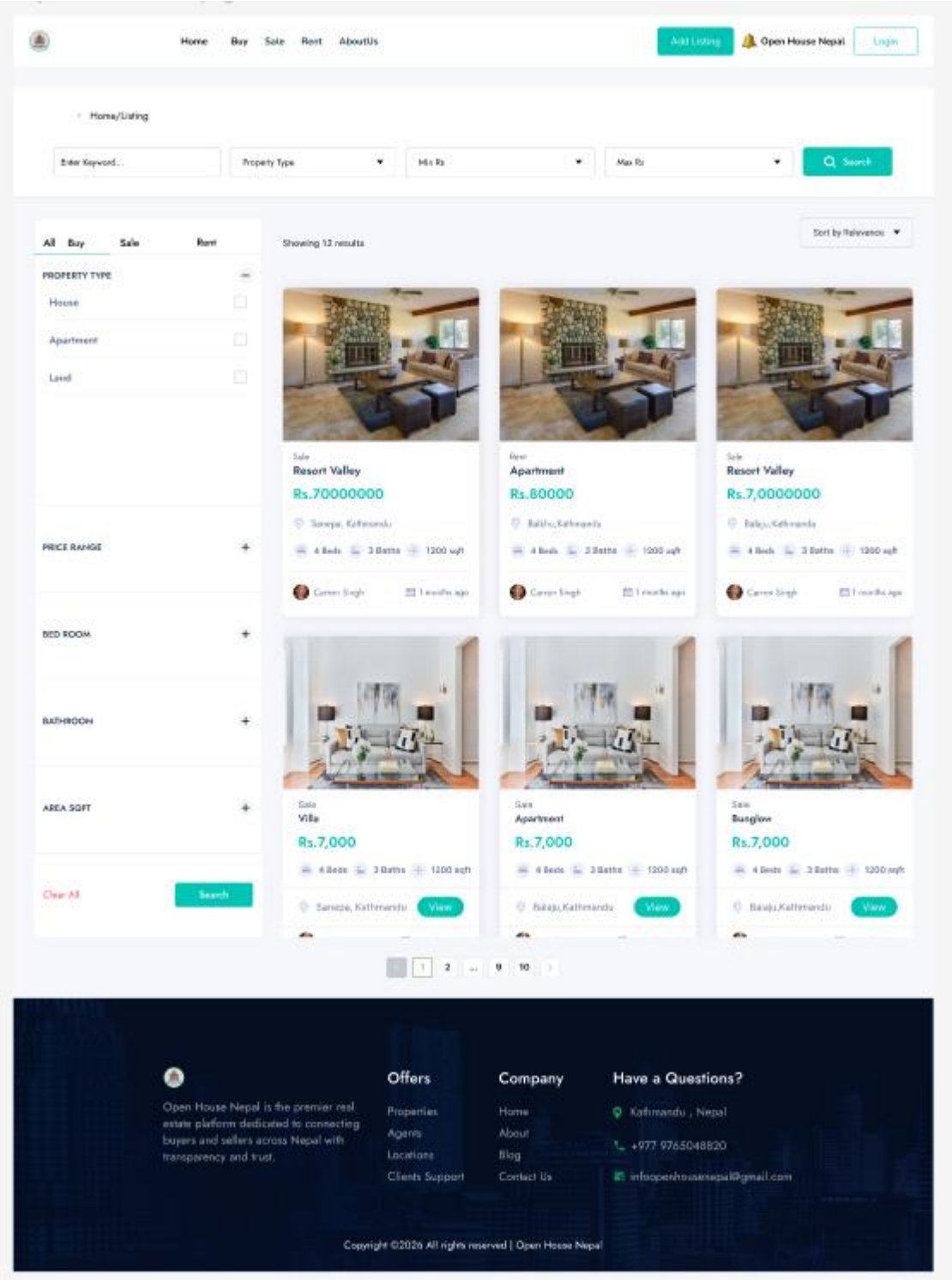


Figure 199 Advance Search

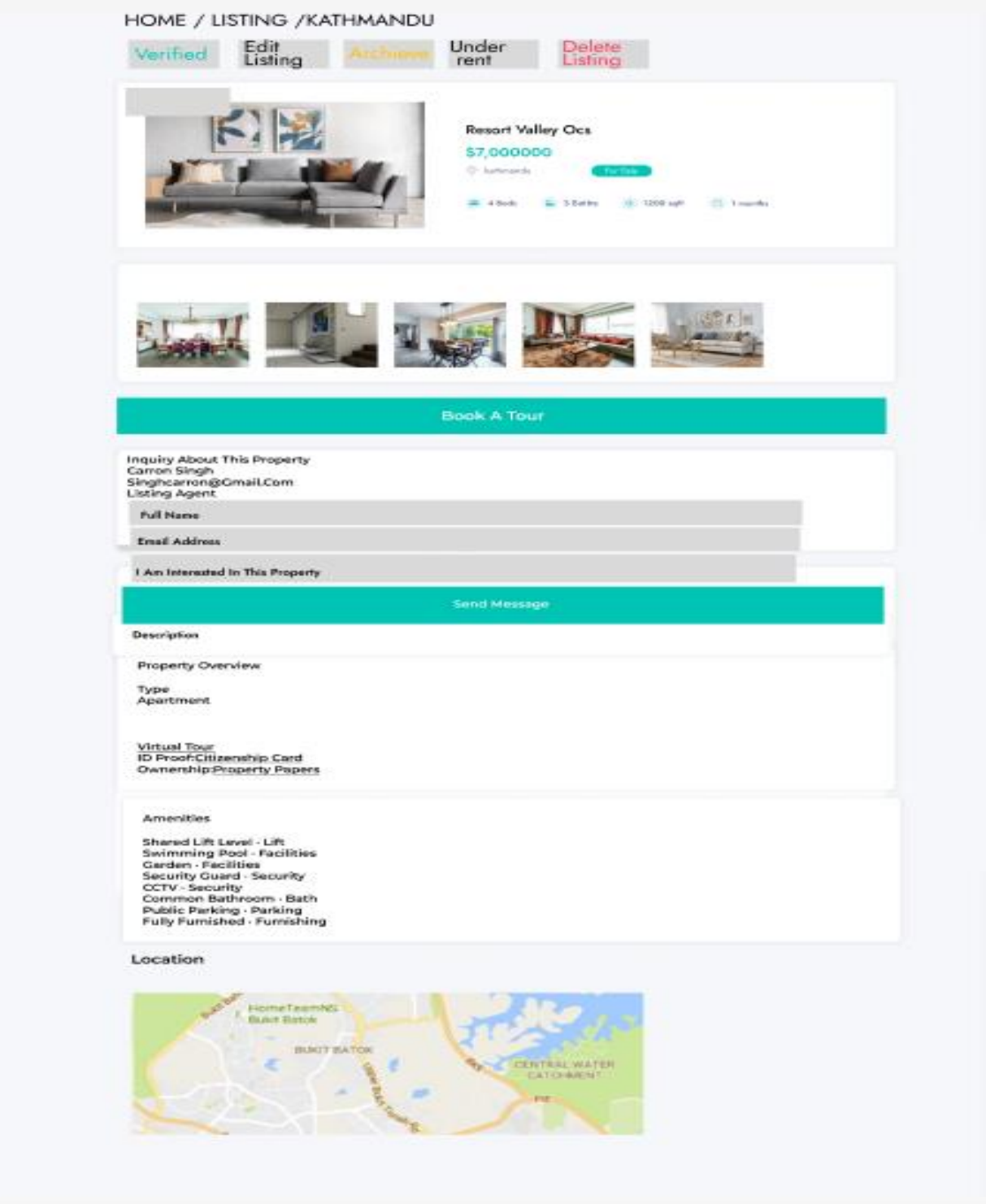


Figure 200 Advance Search

8.4.8.16 Booking

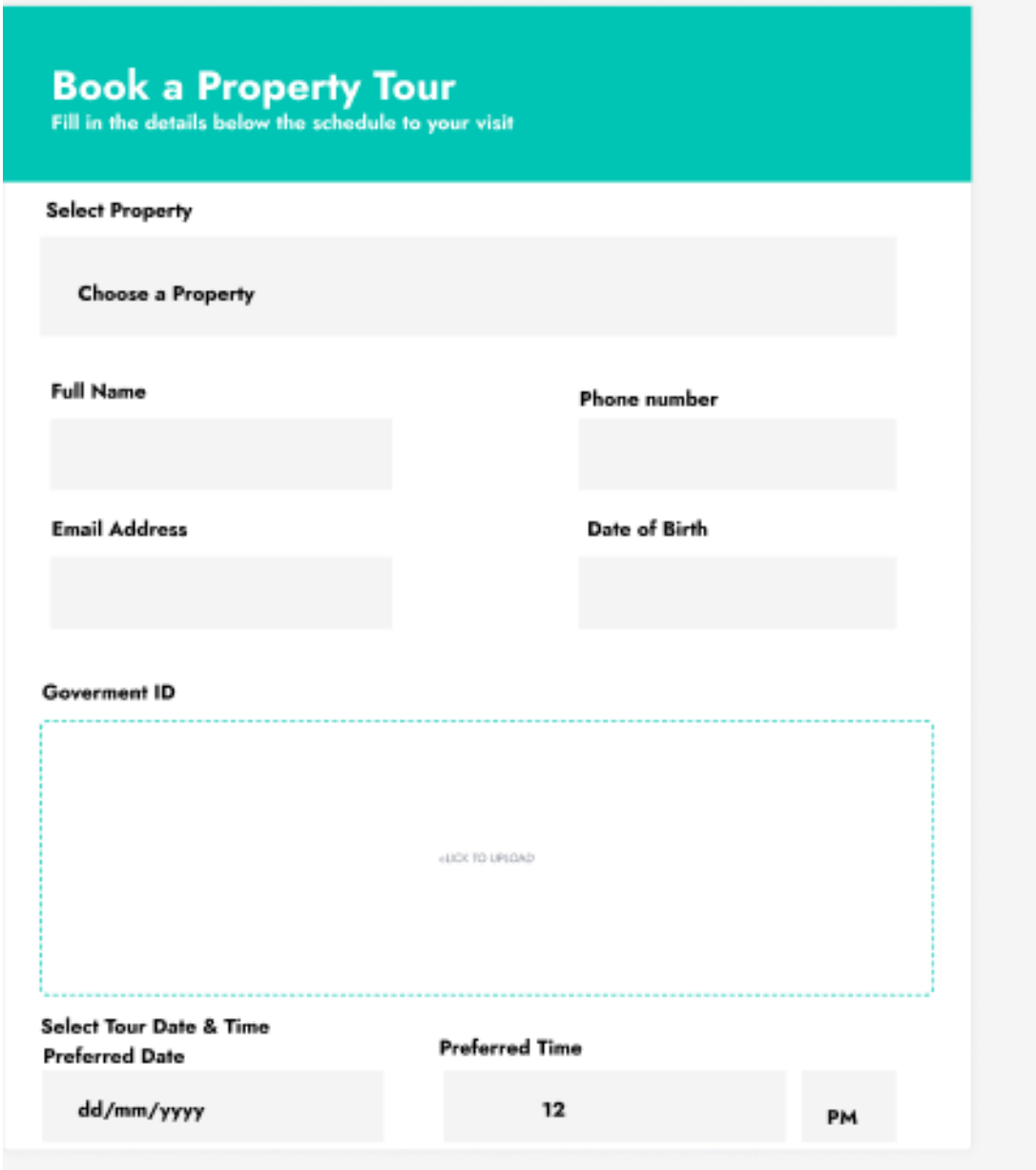


Figure 201 Booking

8.4.8.17 Property Review

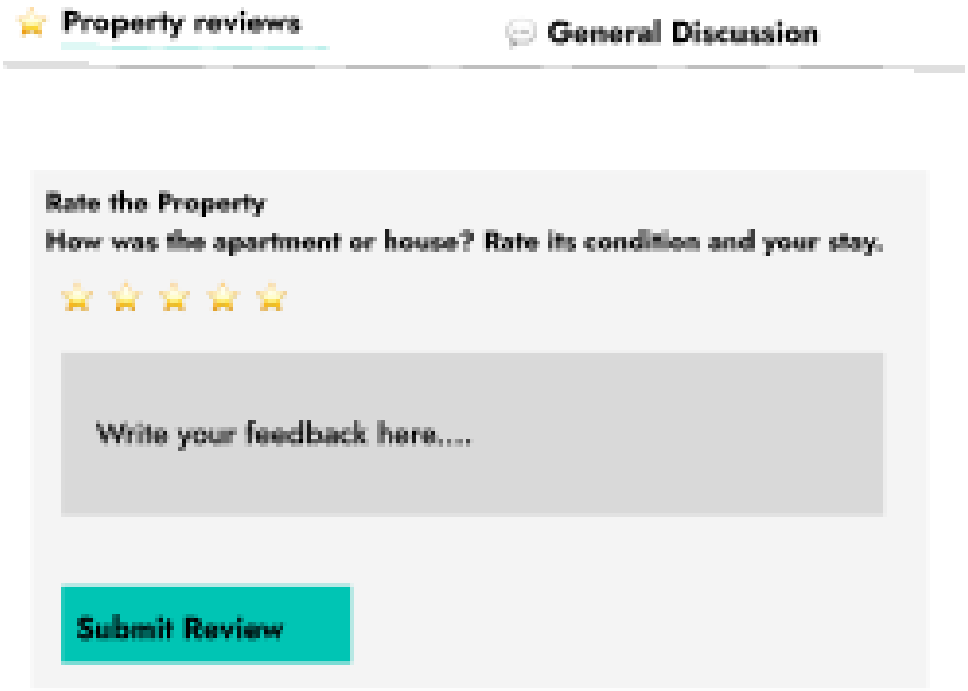


Figure 202 Property Review

8.4.8.18 Comment

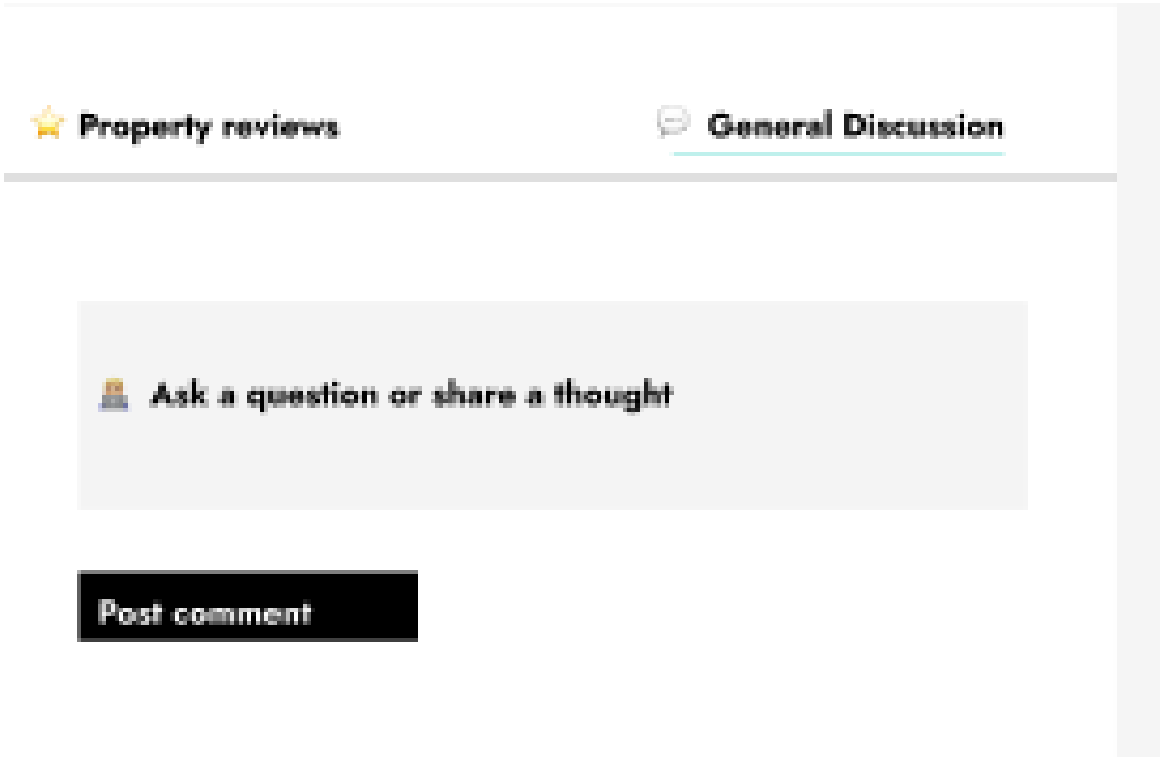


Figure 203 Comment

8.4.8.19 Admin Dashboard

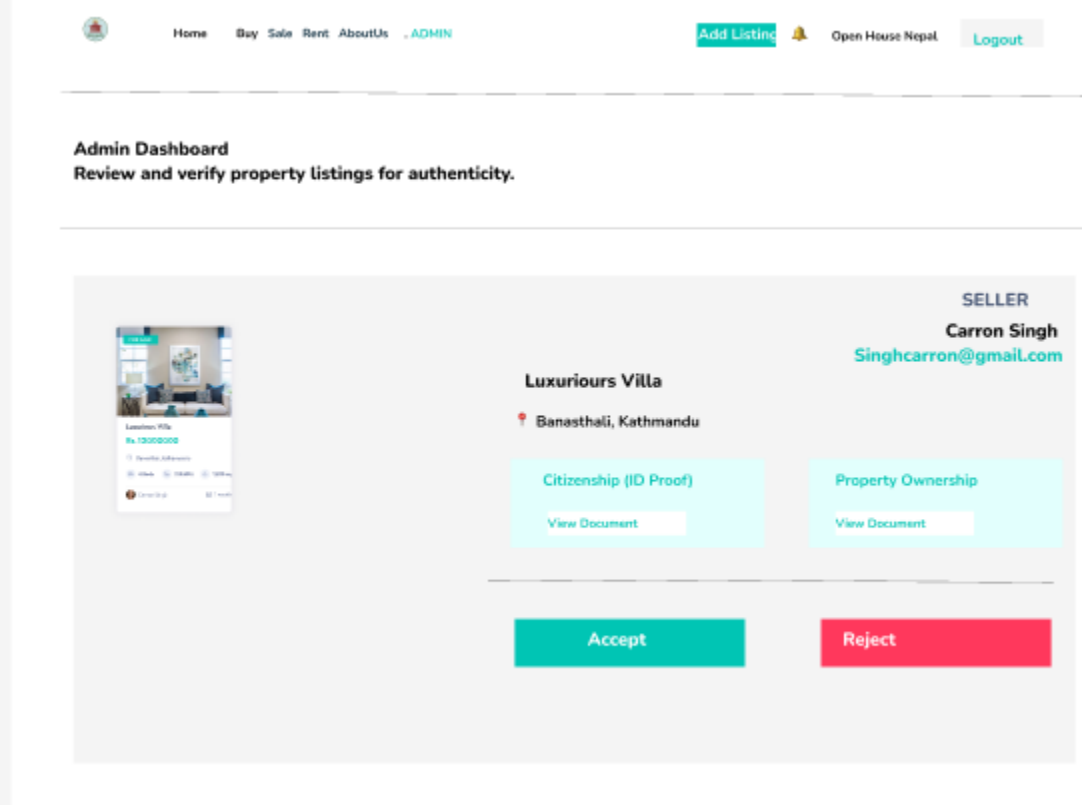


Figure 204 Admin Dashboard

Go back to UI

8.5 APPENDIX E: SCREENSHOTS OF THE SYSTEM

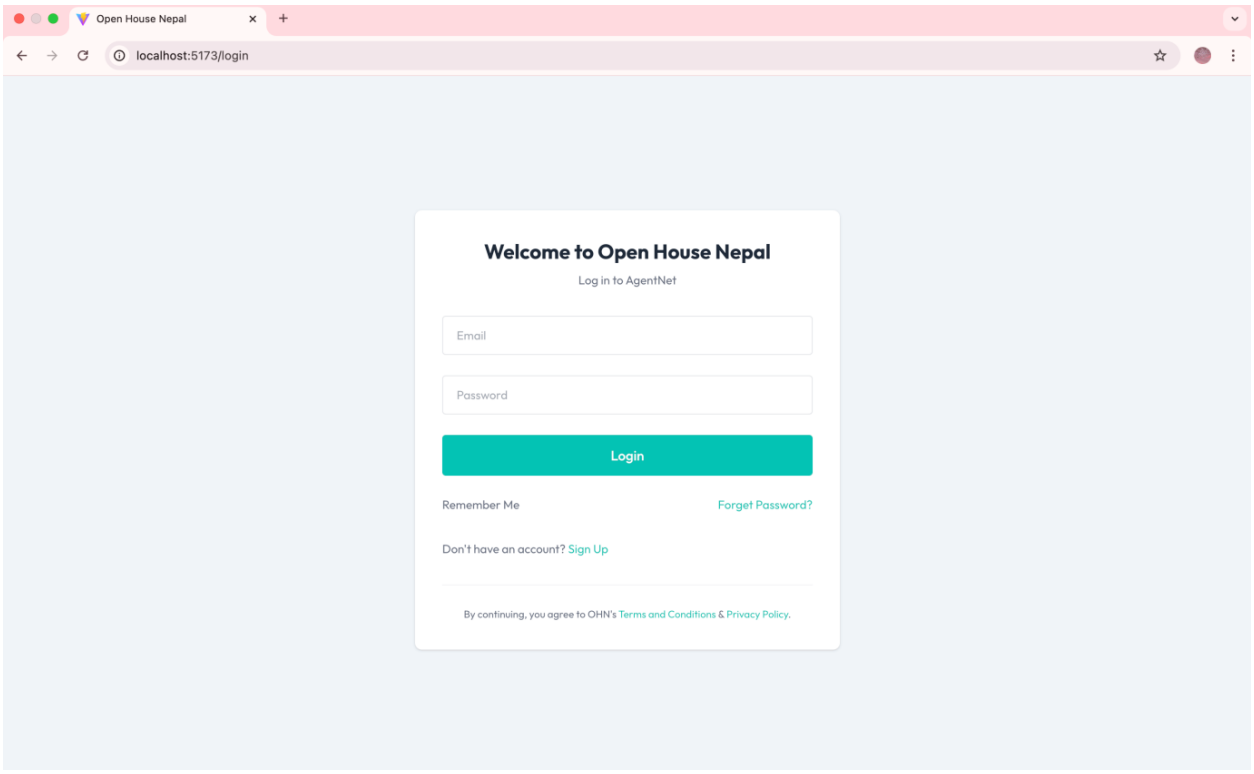


Figure 205 Login

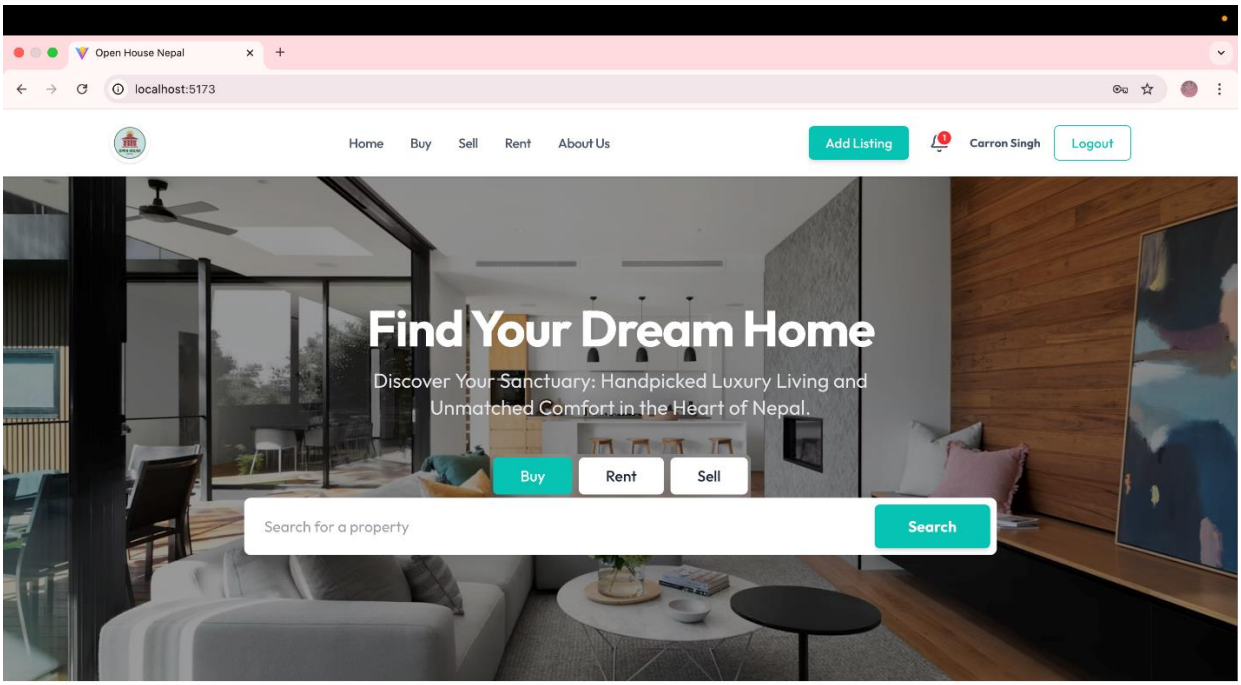


Figure 206 Homepage

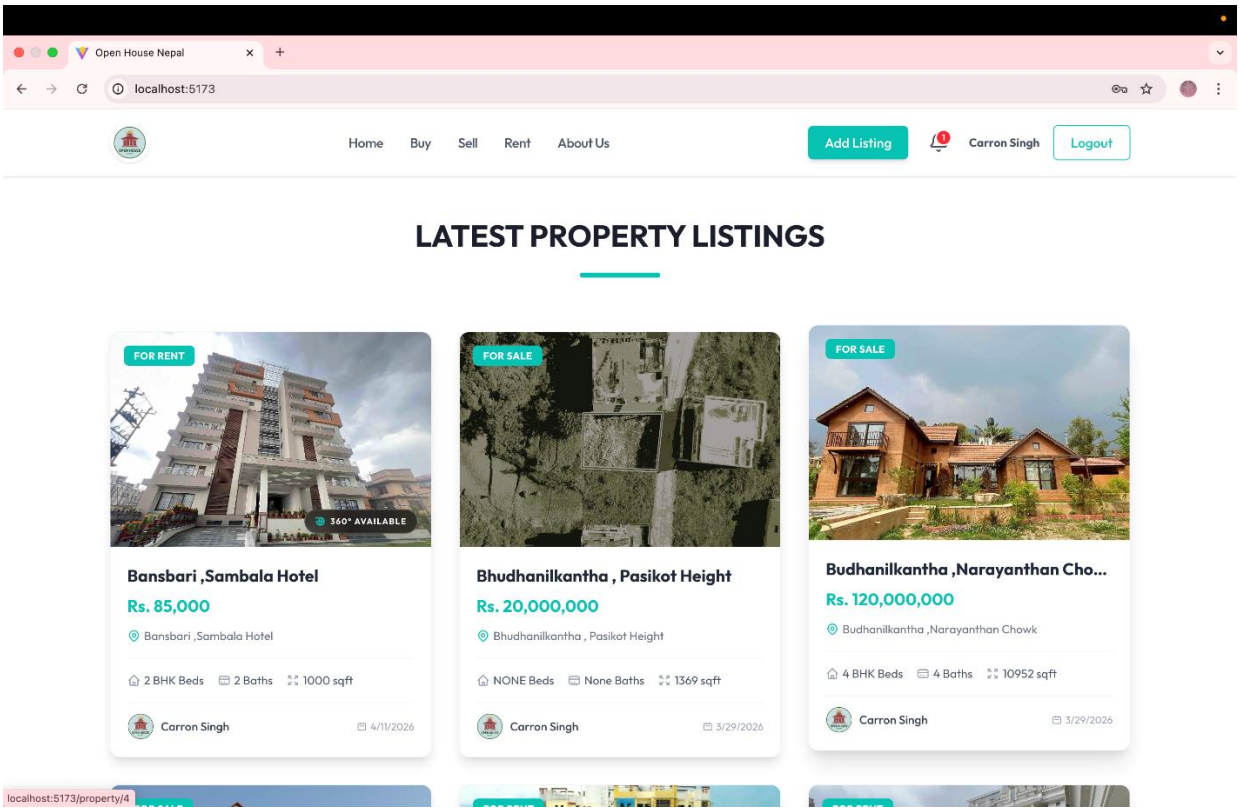


Figure 207 Latest Property Listings

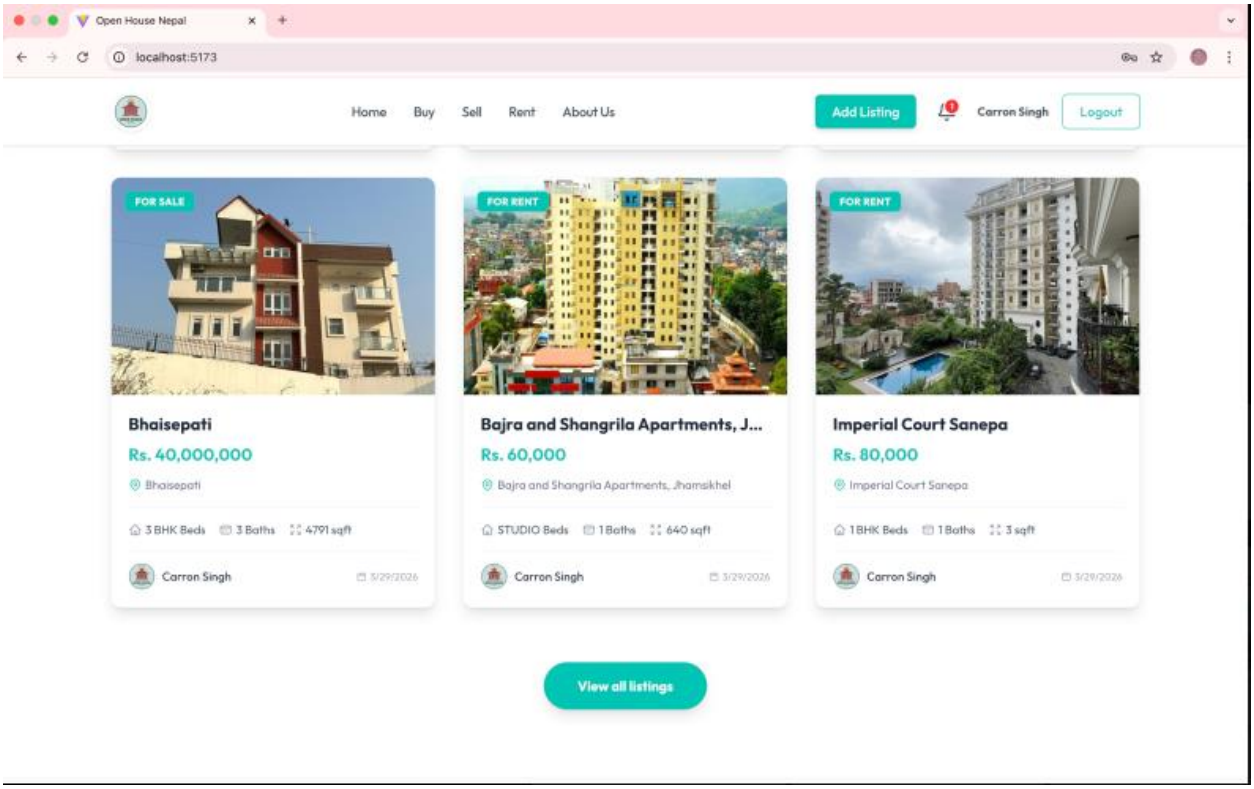


Figure 208 view listing

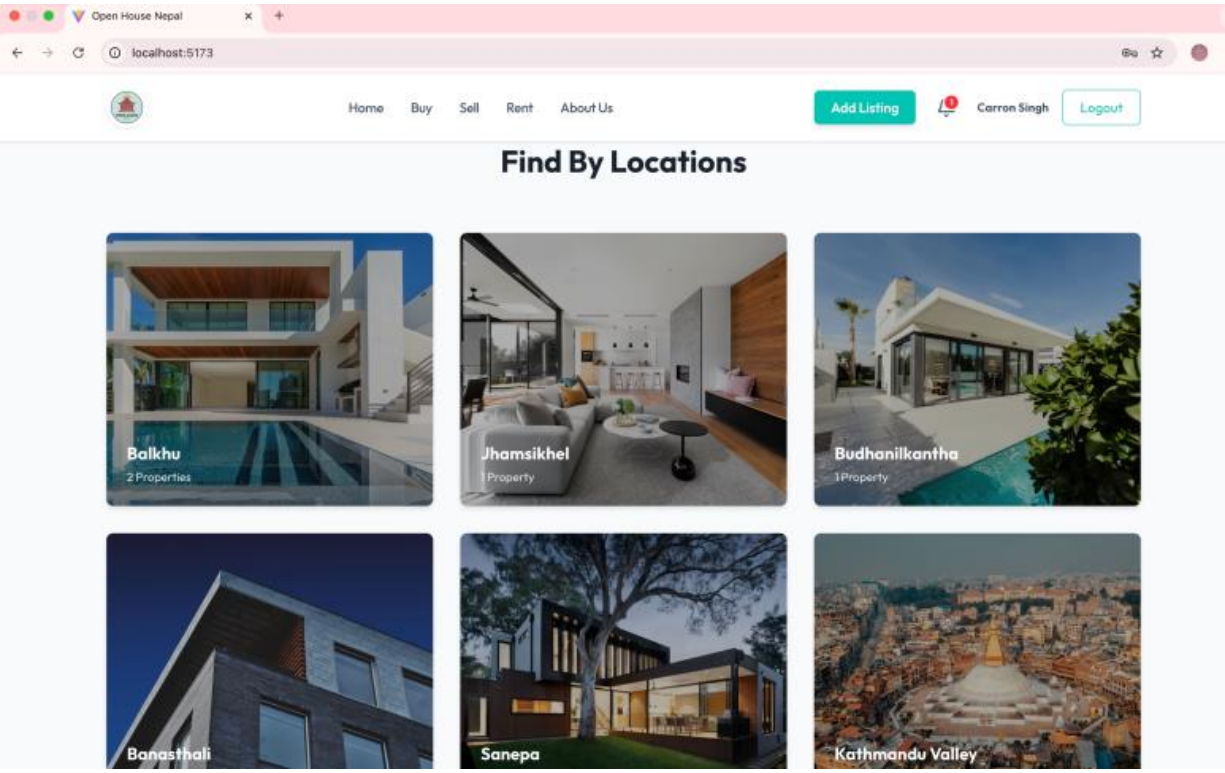


Figure 209 Find by Locations

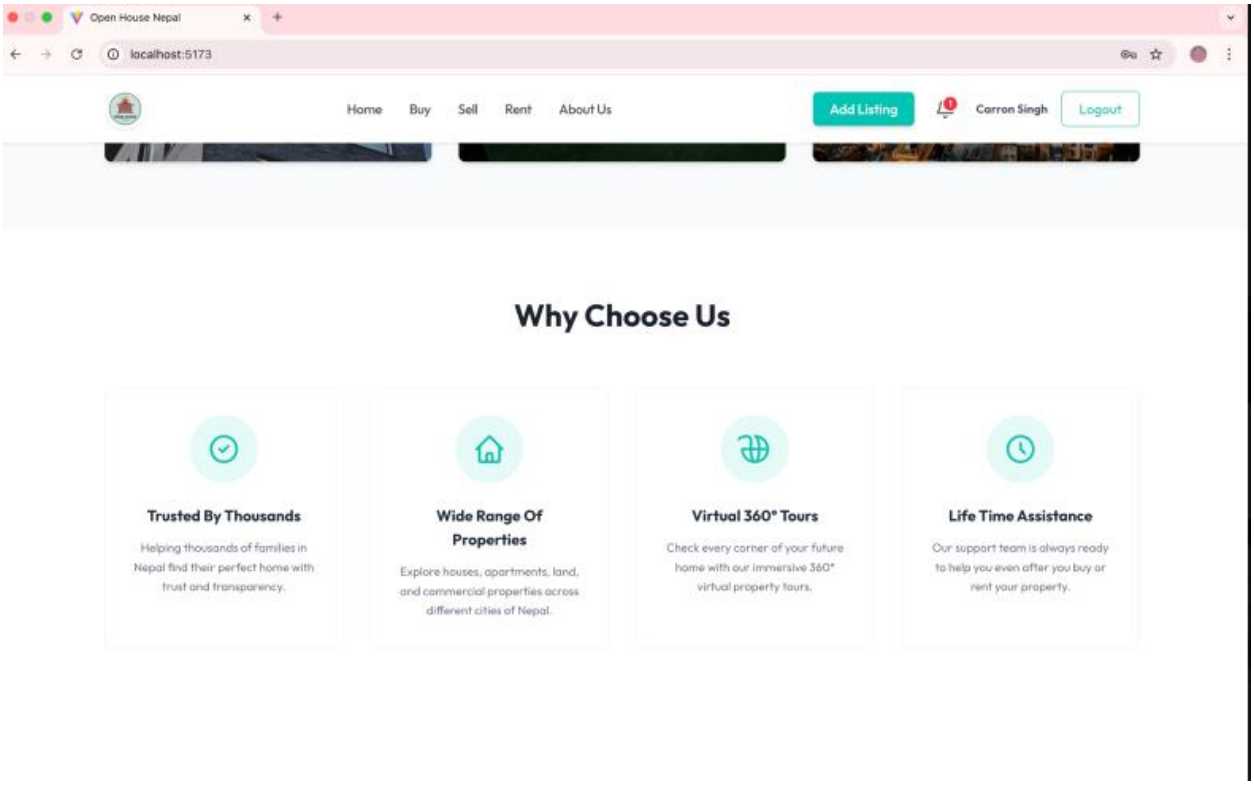


Figure 210 Why Choose Us

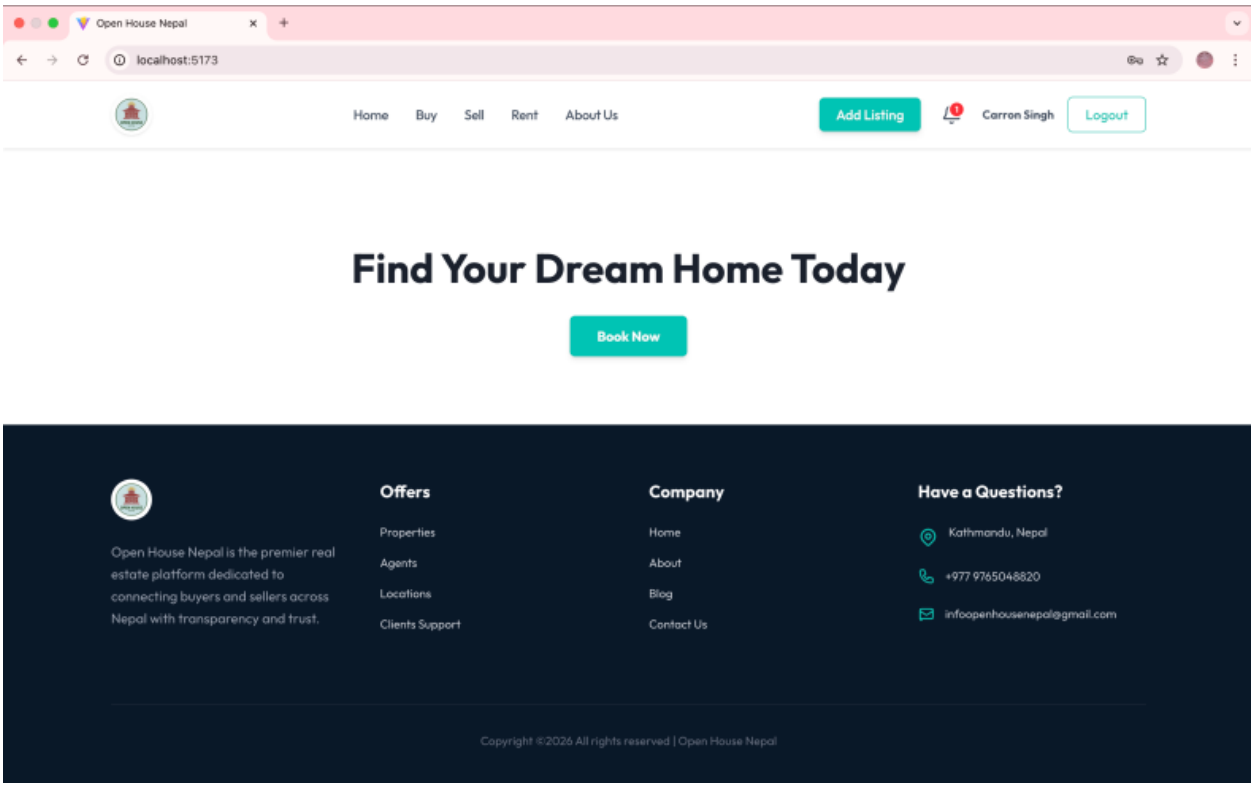


Figure 211 Book now

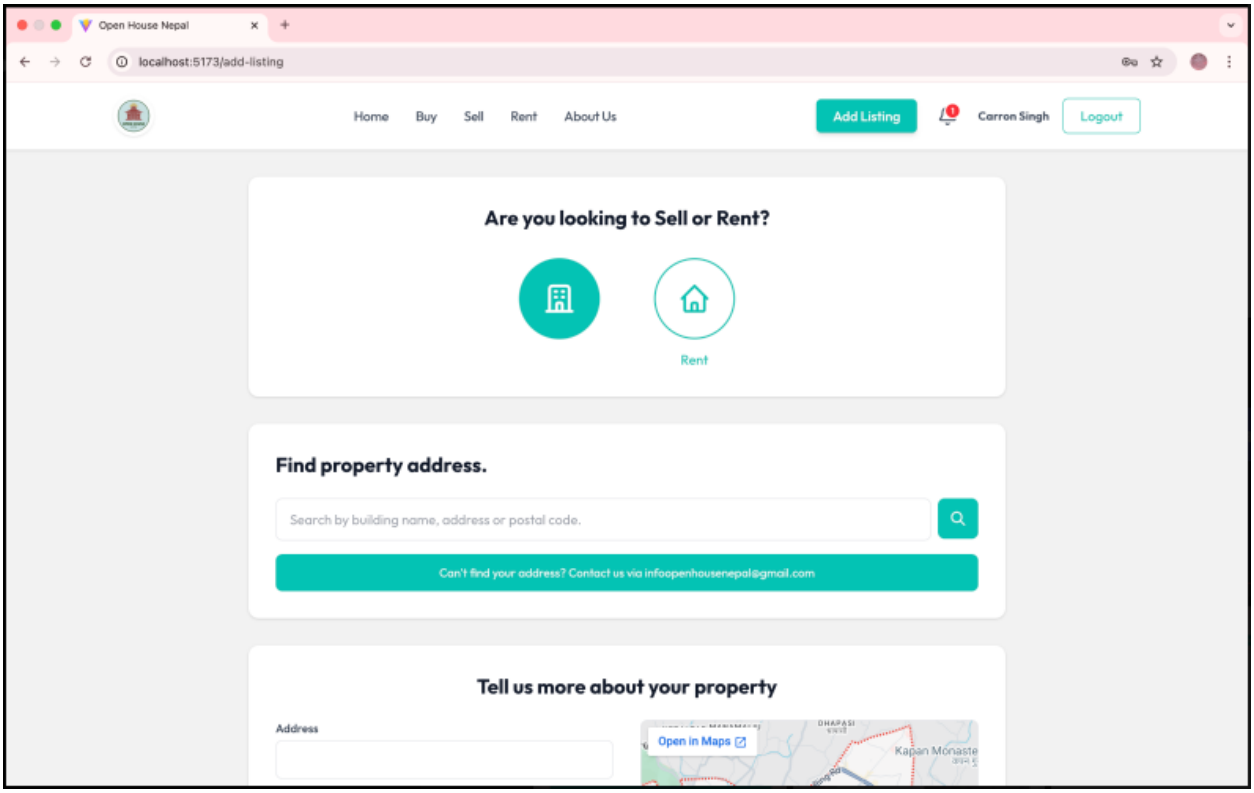


Figure 212 add listing

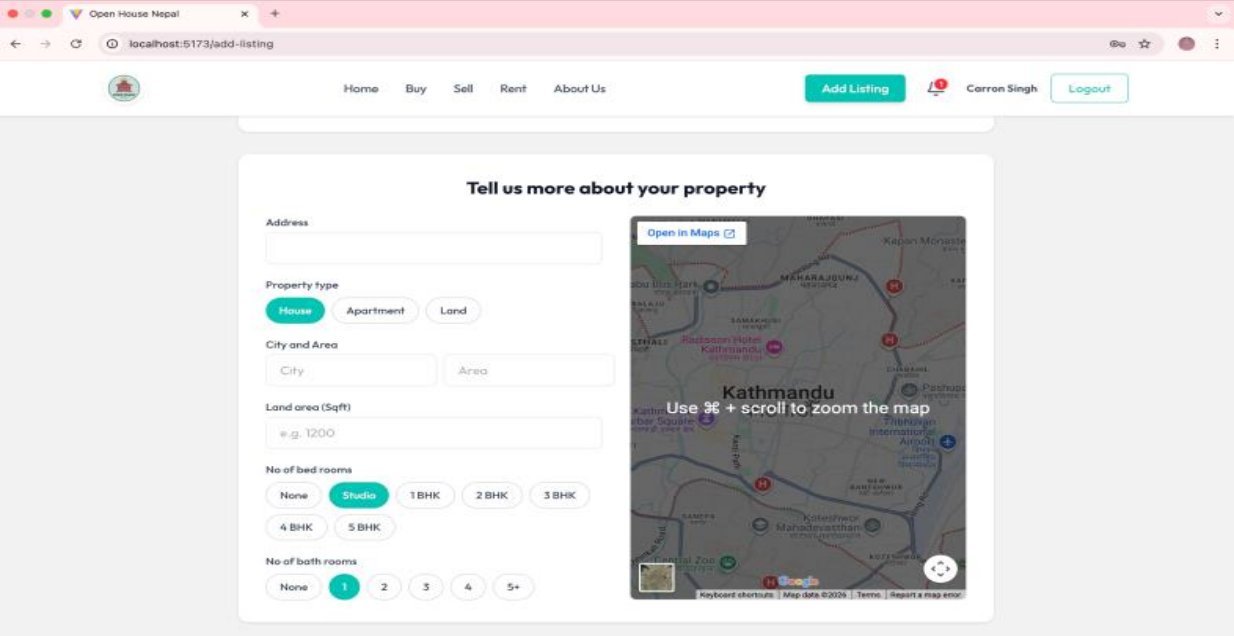


Figure 213 property details and map

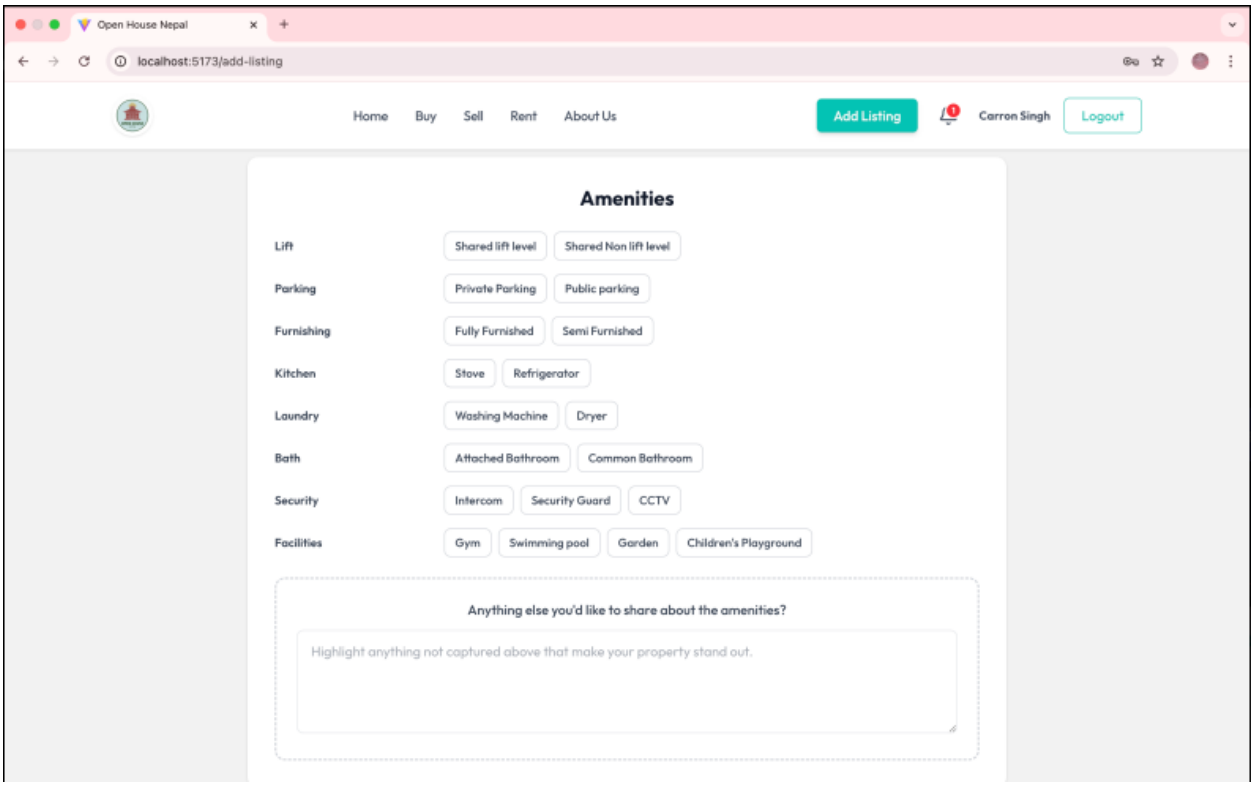


Figure 214 Amenities

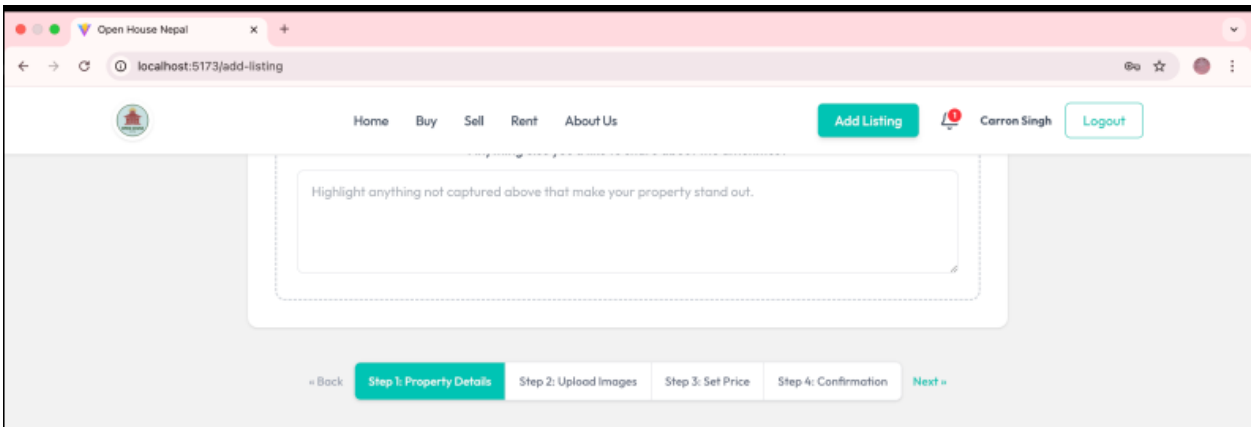


Figure 215 Property details

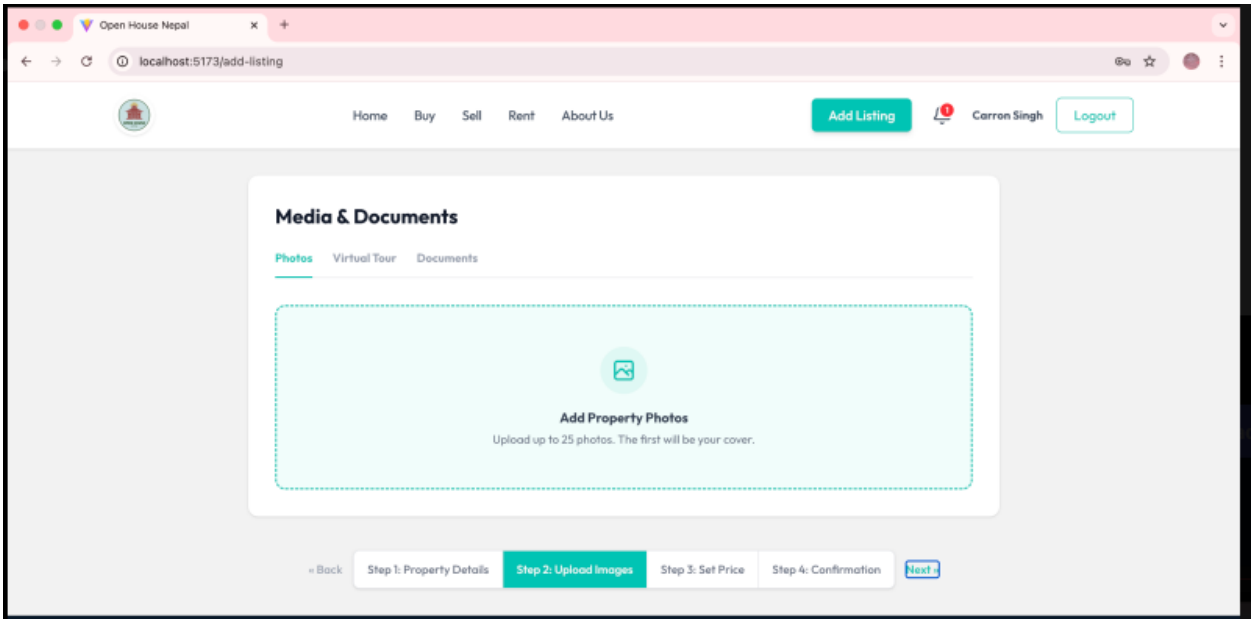


Figure 216 Property Photos

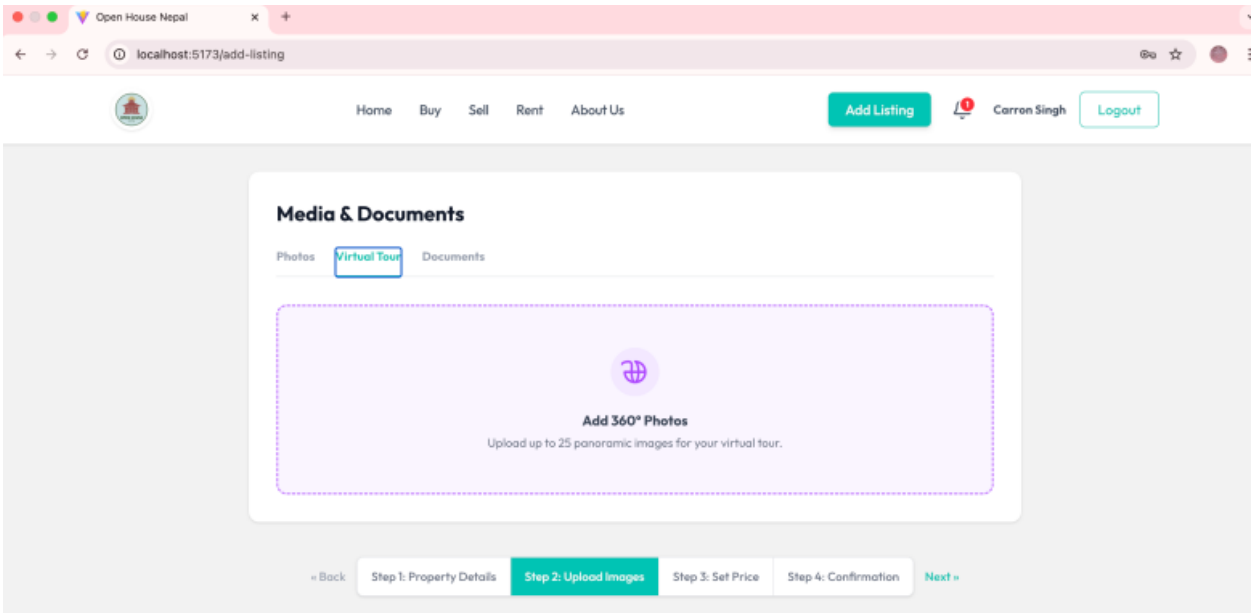


Figure 217 360°photo upload

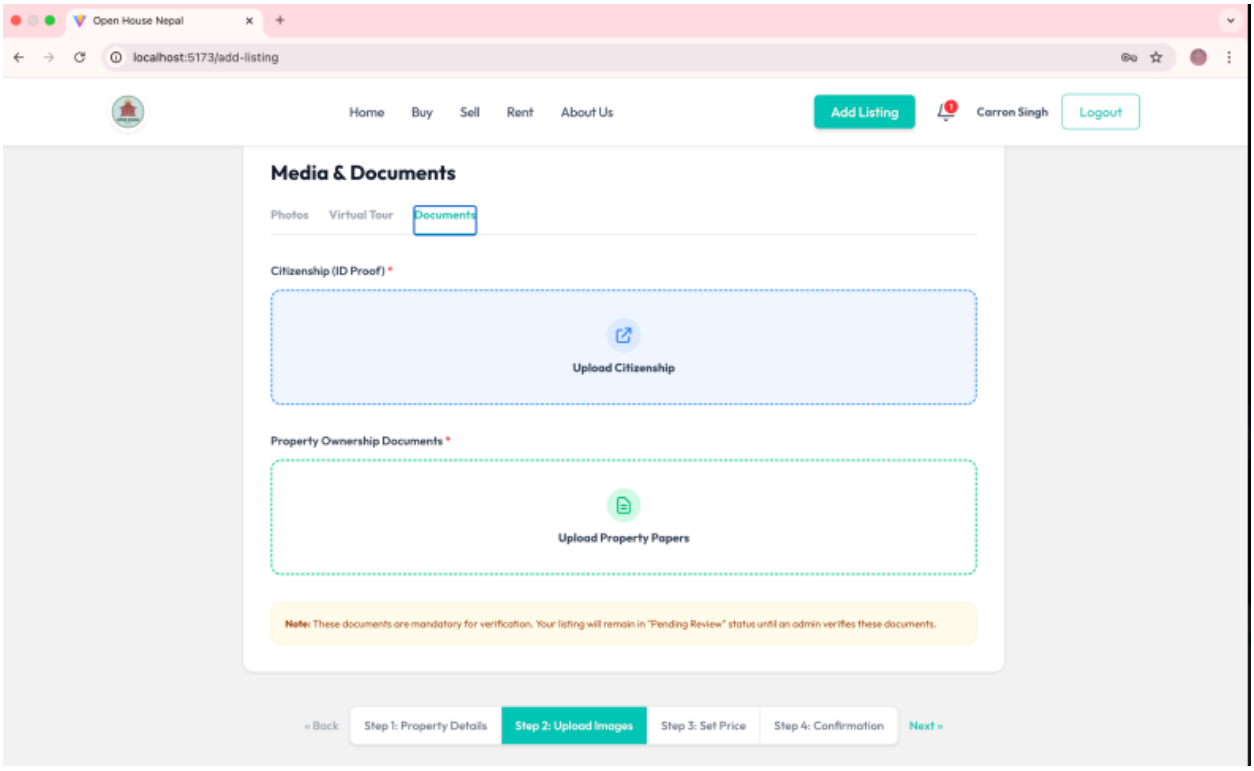


Figure 218 Documentations

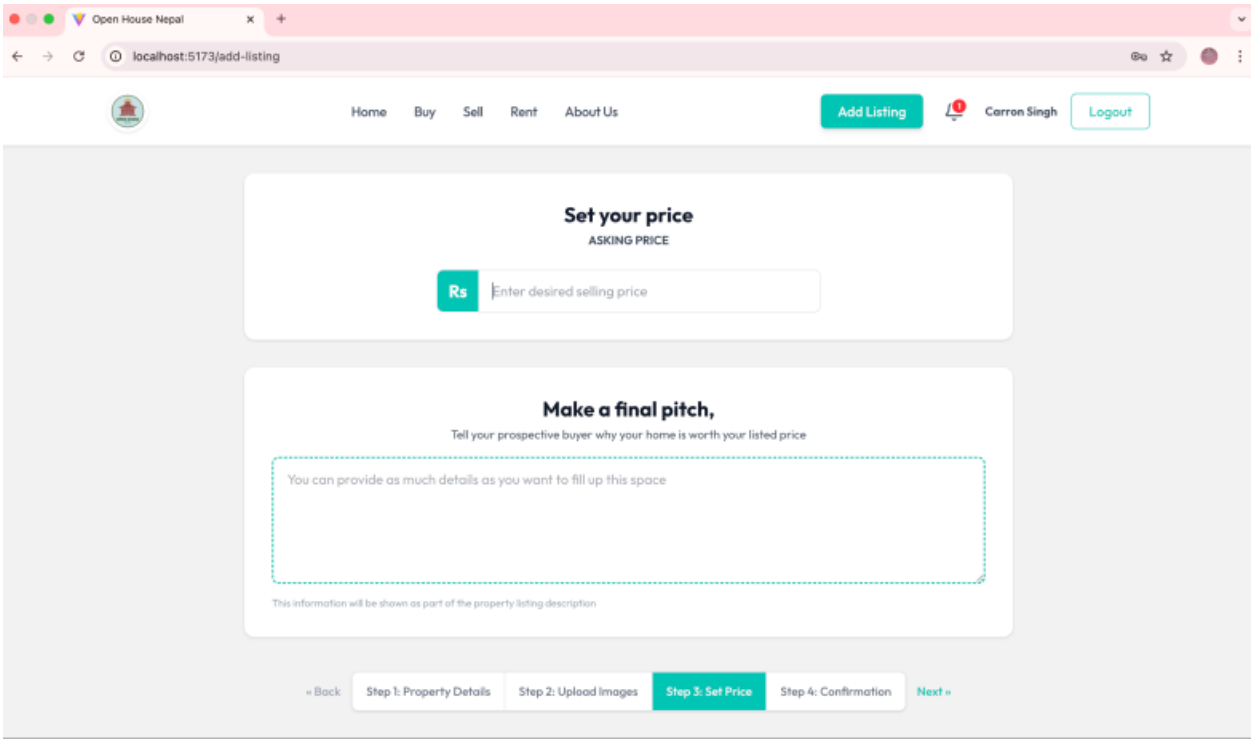


Figure 219 Set price and final pitch

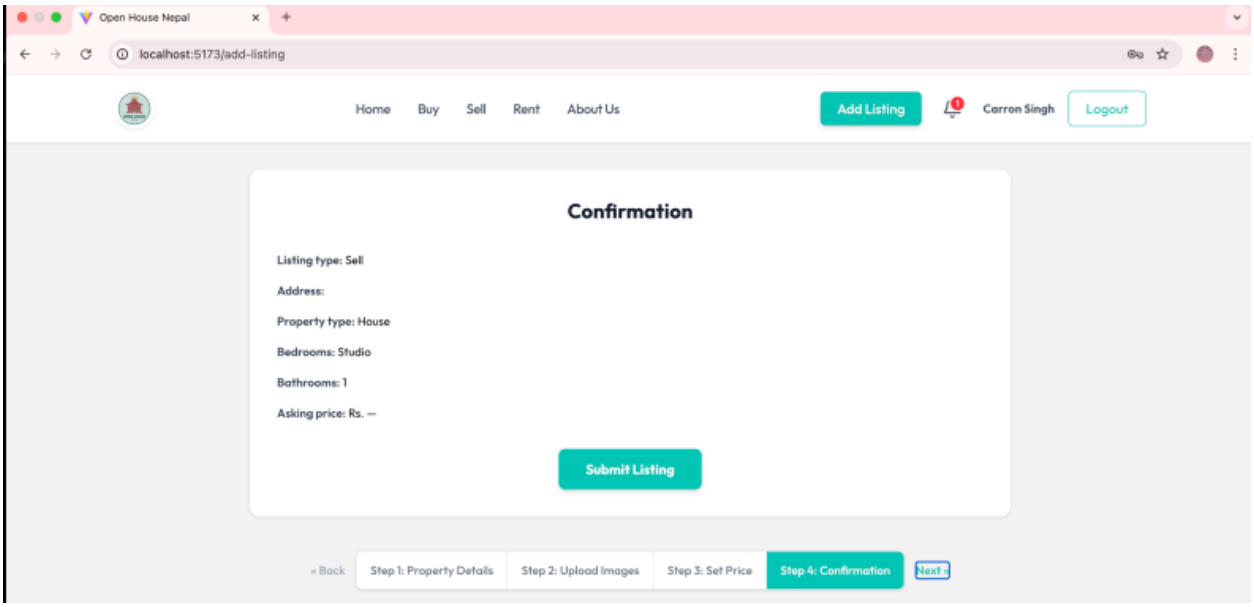


Figure 220 Confirmation

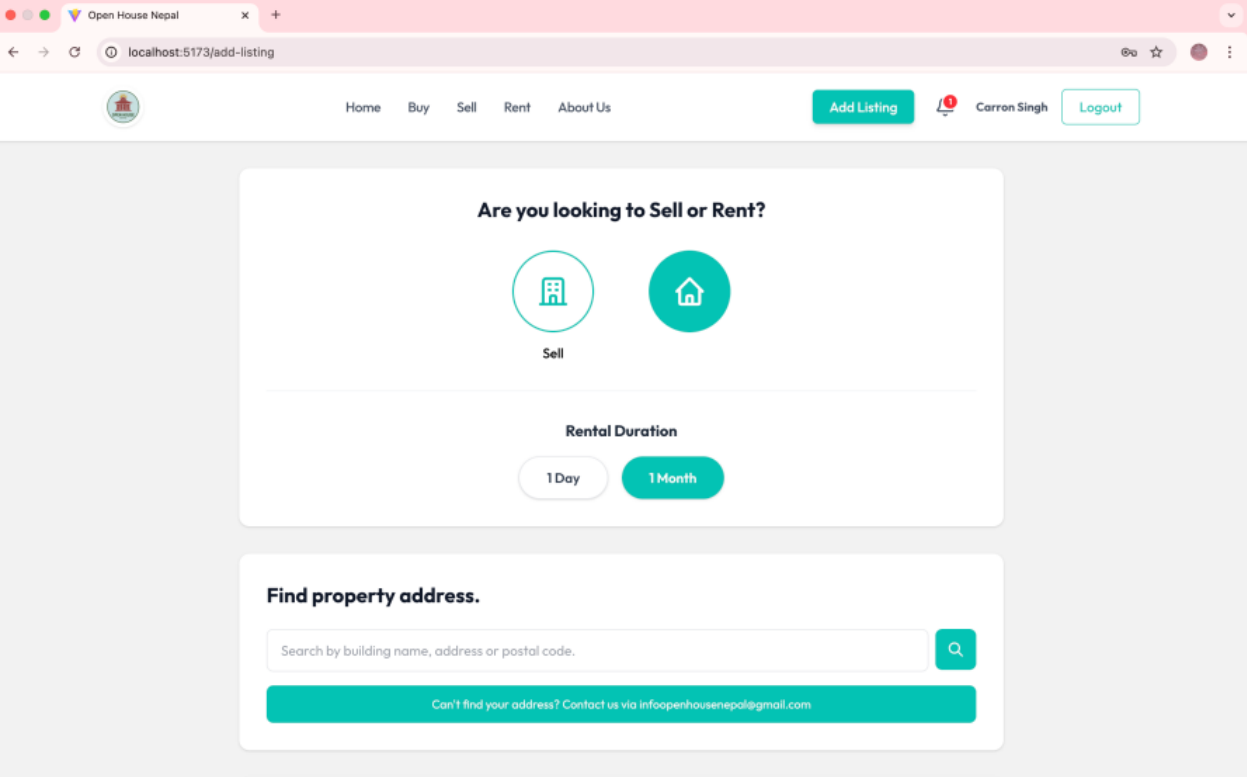


Figure 221 Rent property and rental duration

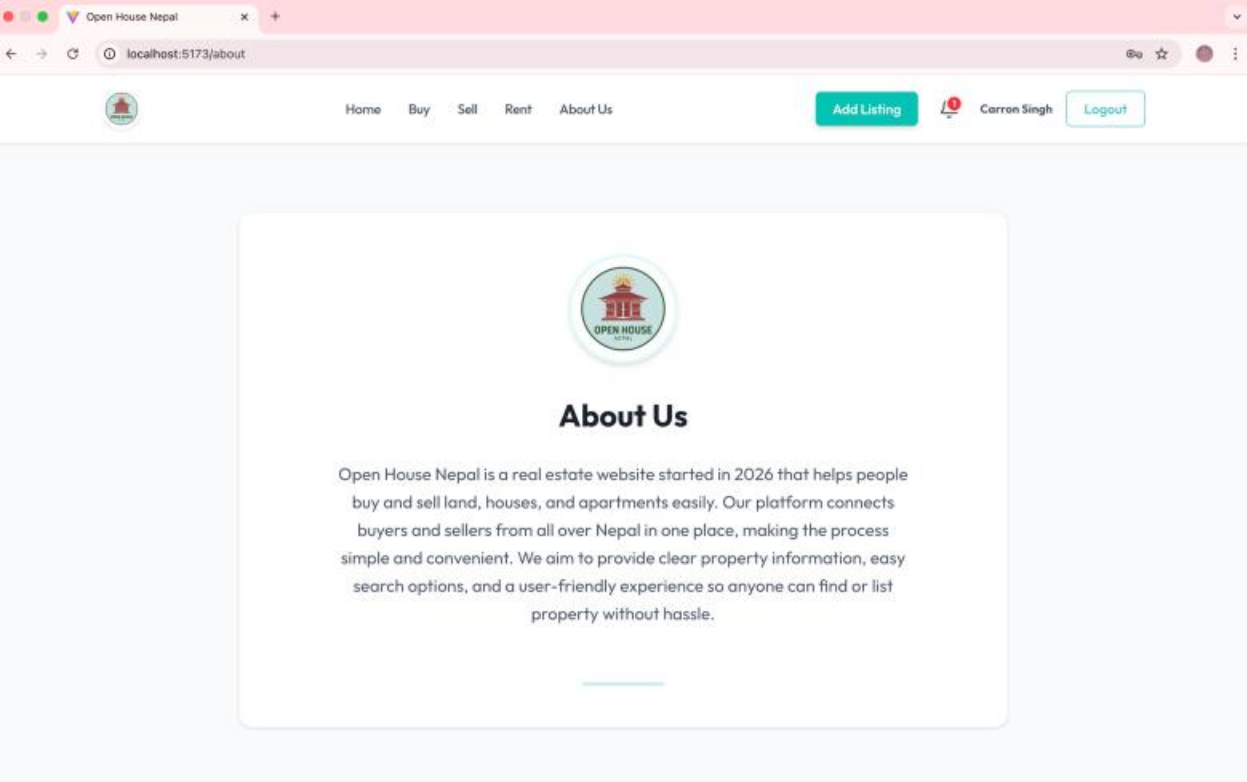


Figure 222 About Us

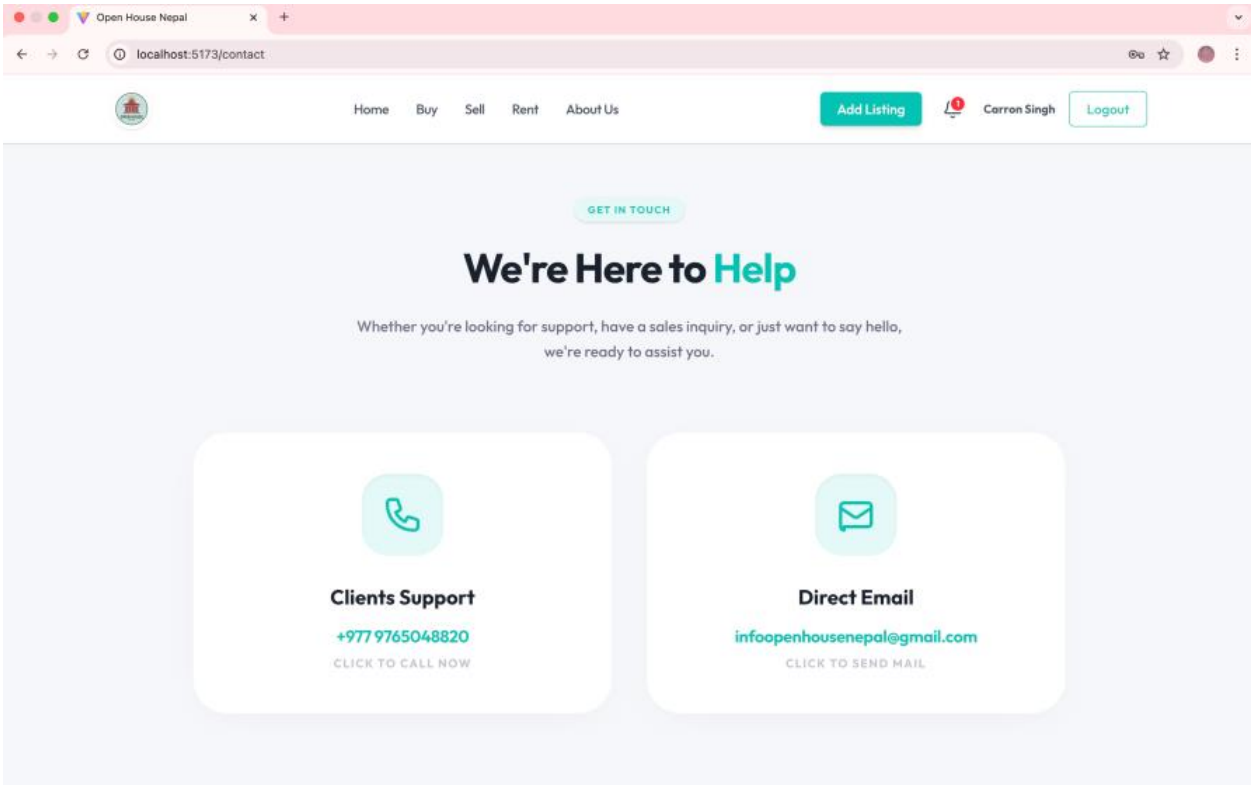


Figure 223 Support

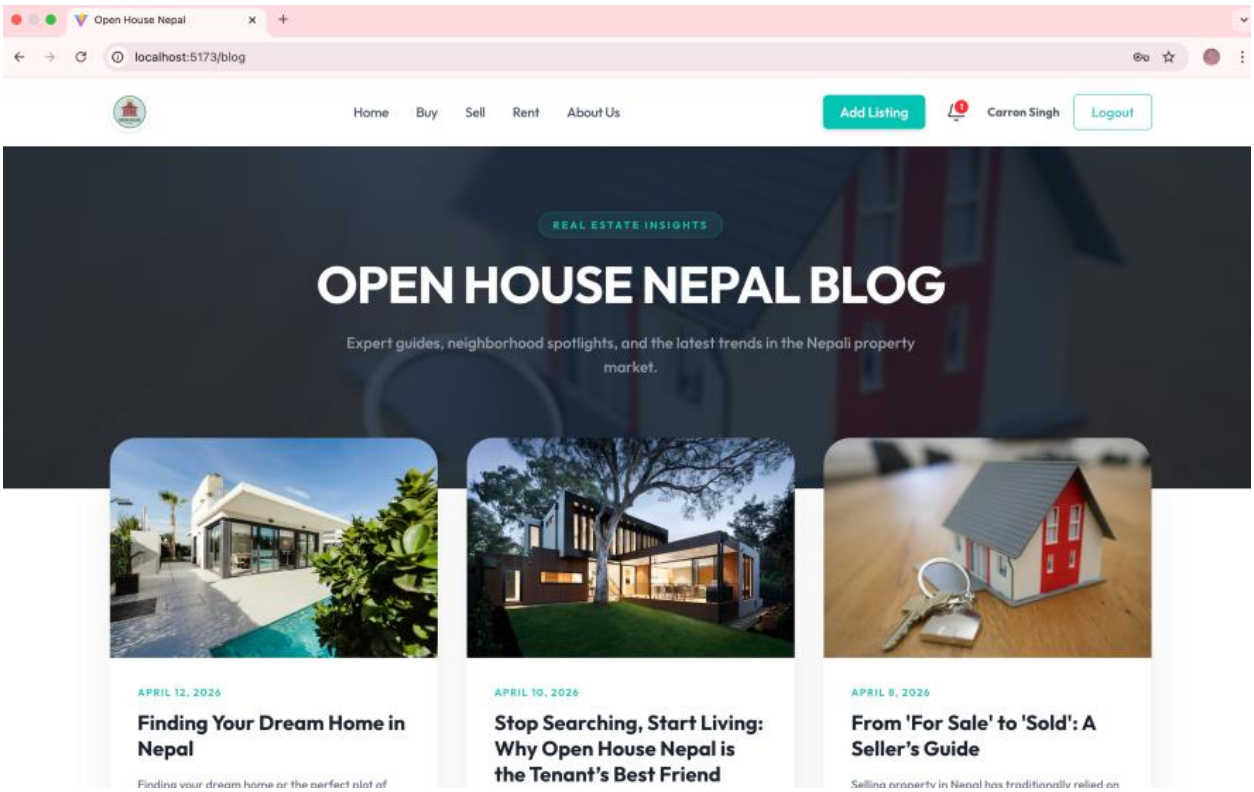


Figure 224 Blog page

8.6: APPENDIX F: USER FEEDBACK

8.6.1 USER FEEDBACK FORM

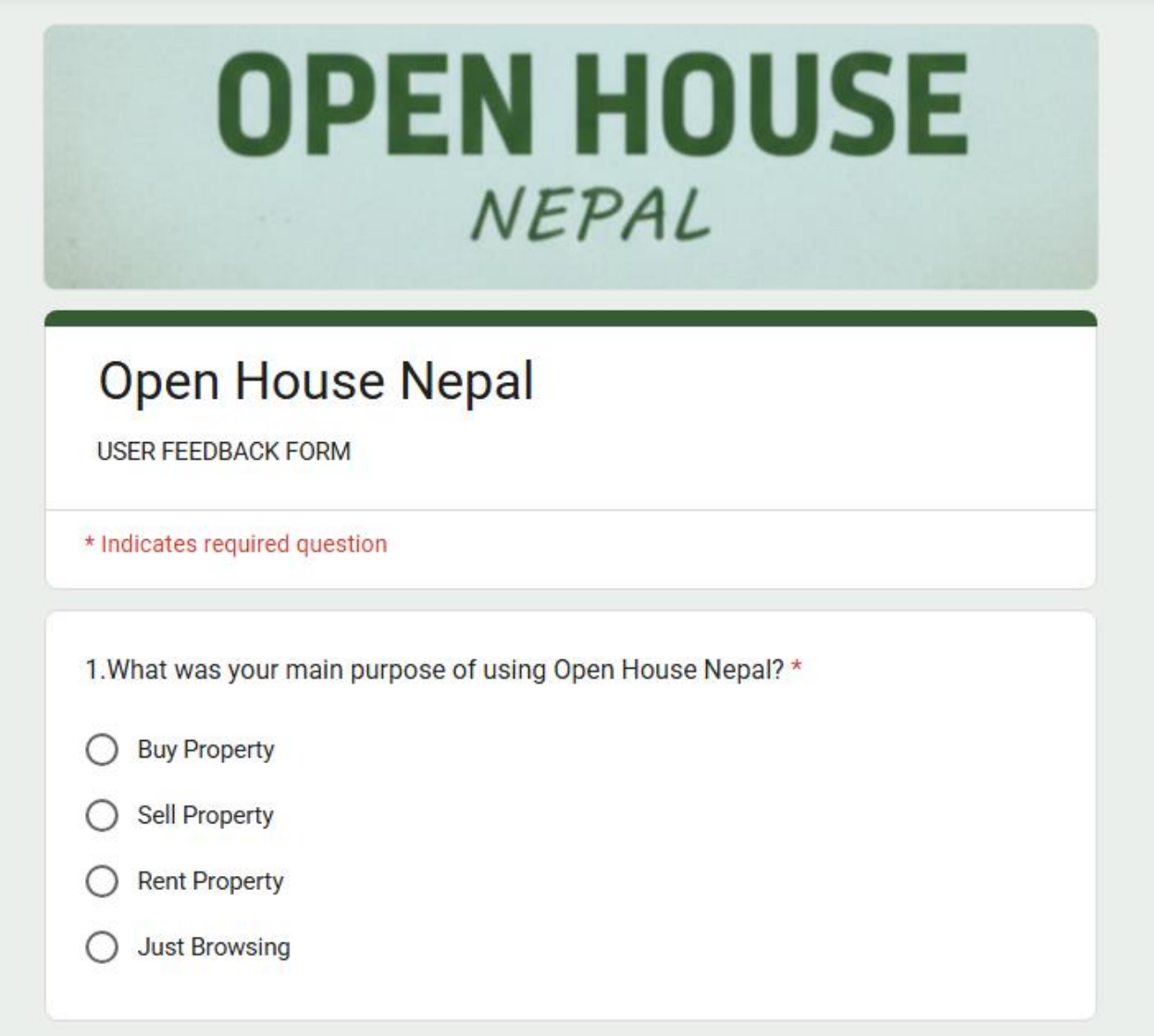


Figure 225 USER FEEDBACK FORM

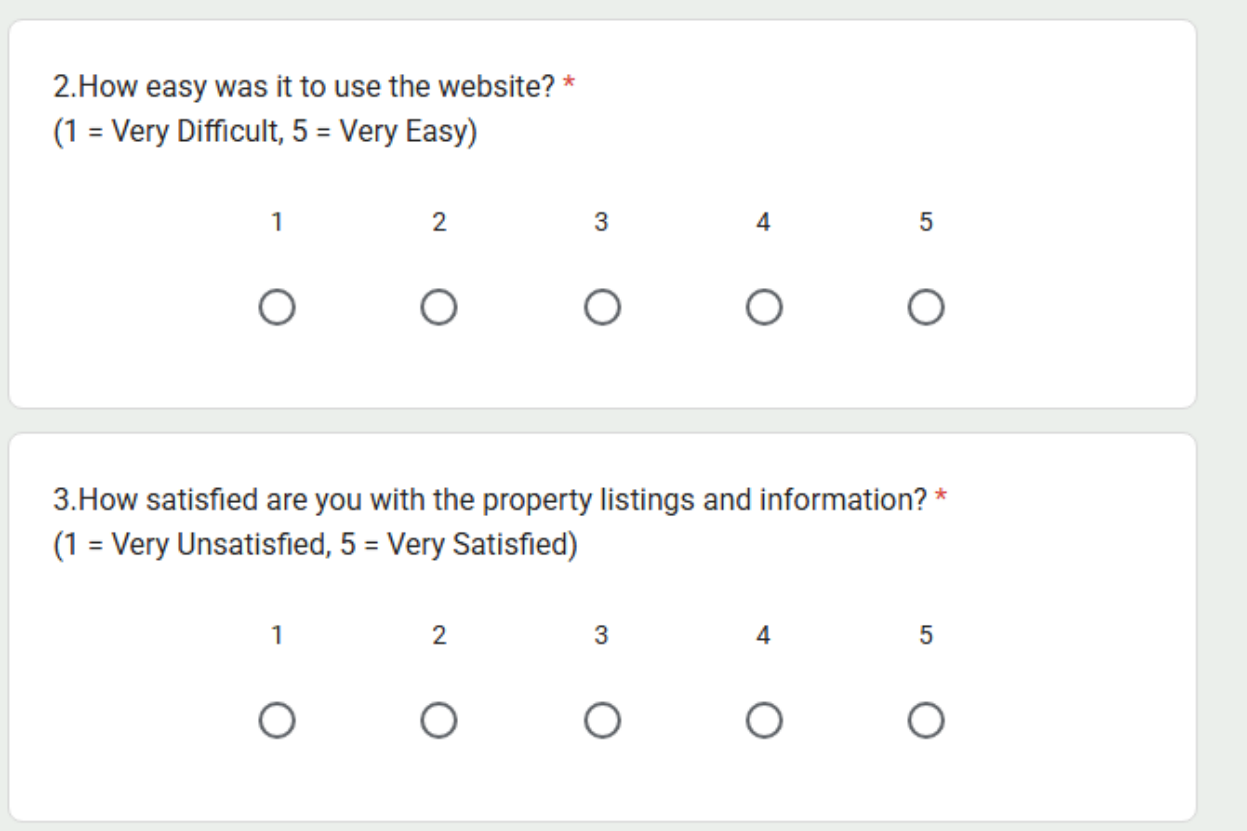


Figure 226 USER FEEDBACK FORM

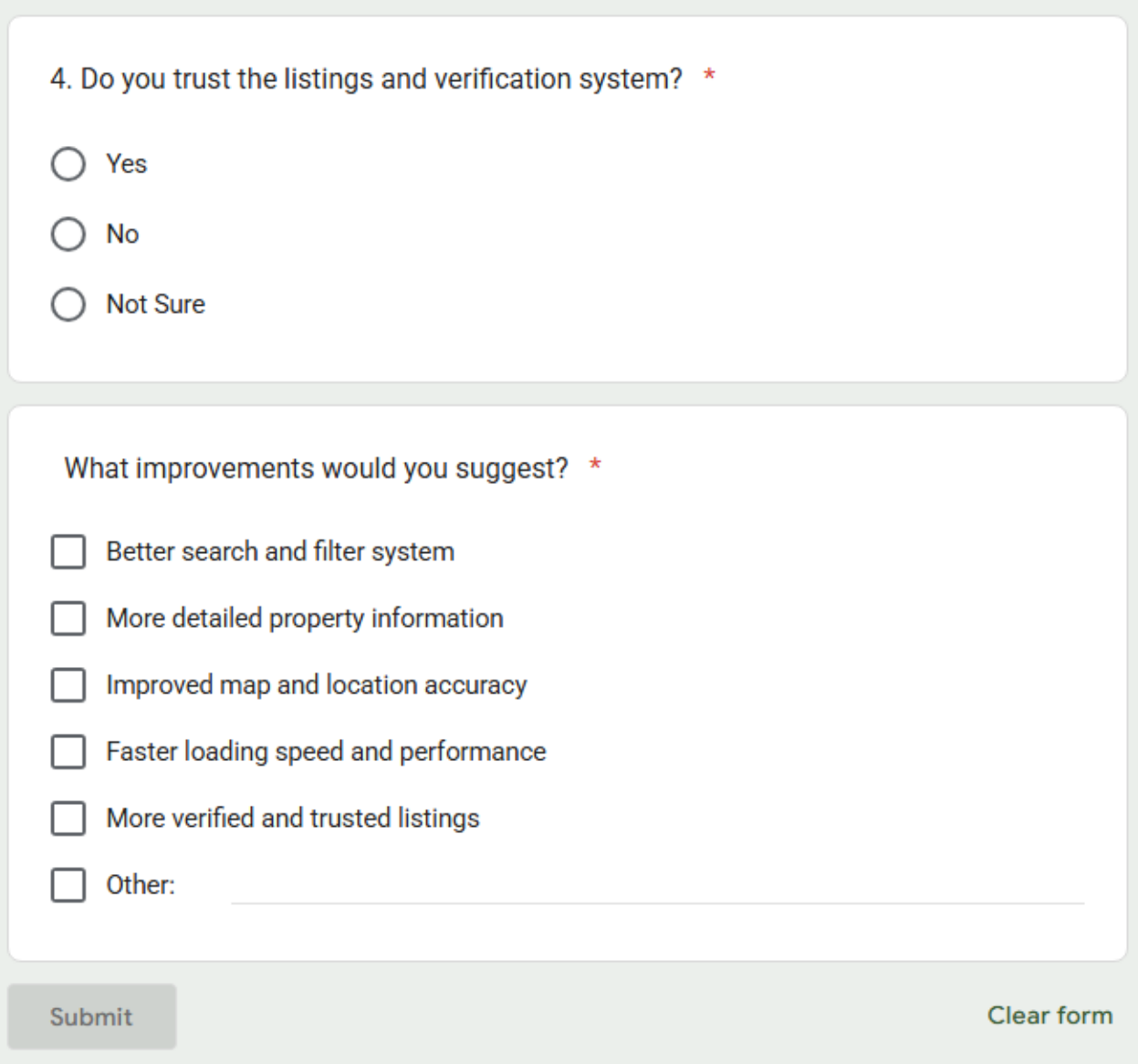


Figure 227 USER FEEDBACK FORM

8.6.2 SAMPLE OF FILLED USER FEEDBACK FORMS

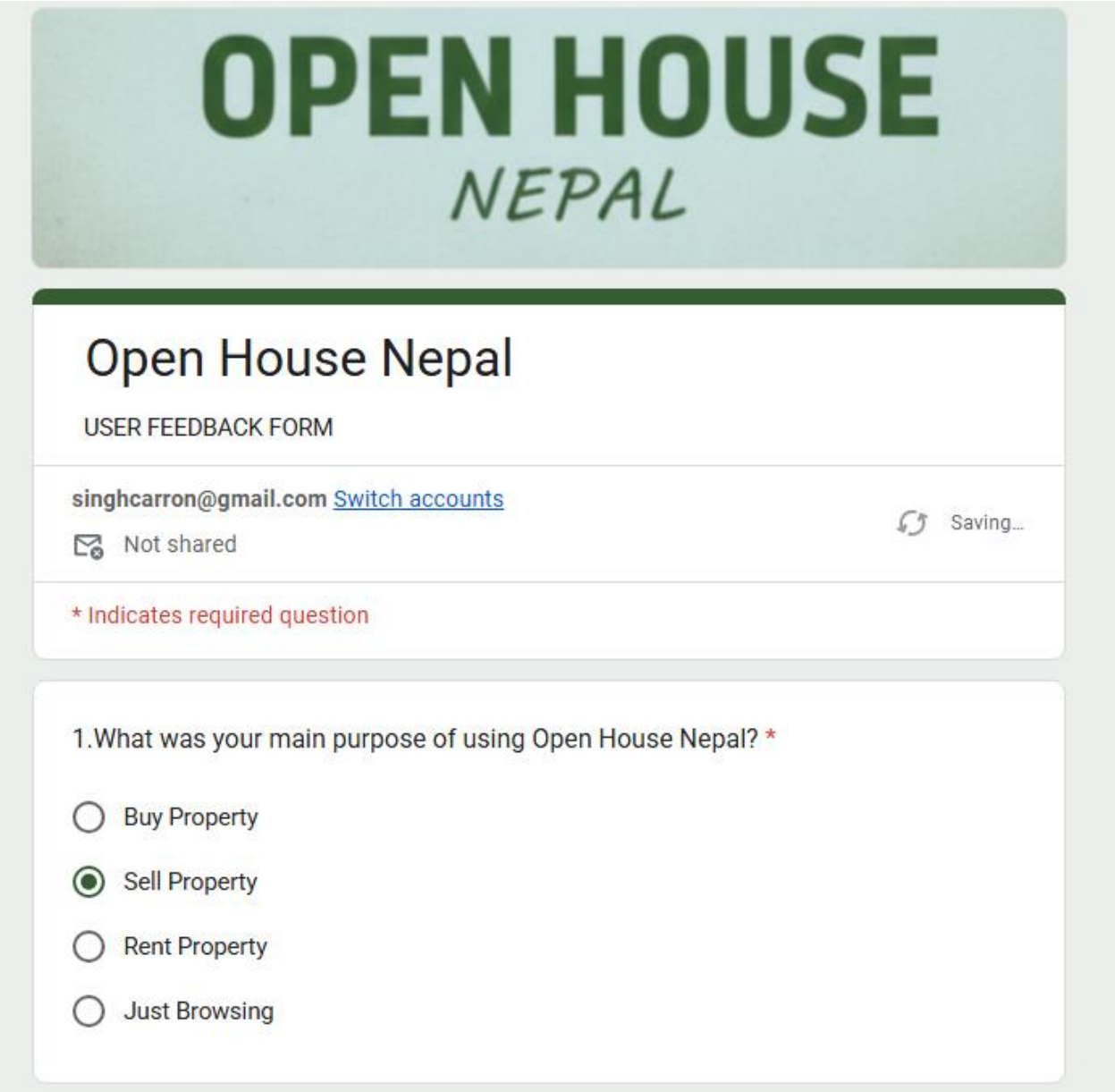


Figure 228 SAMPLE OF FILLED USER FEEDBACK FORMS

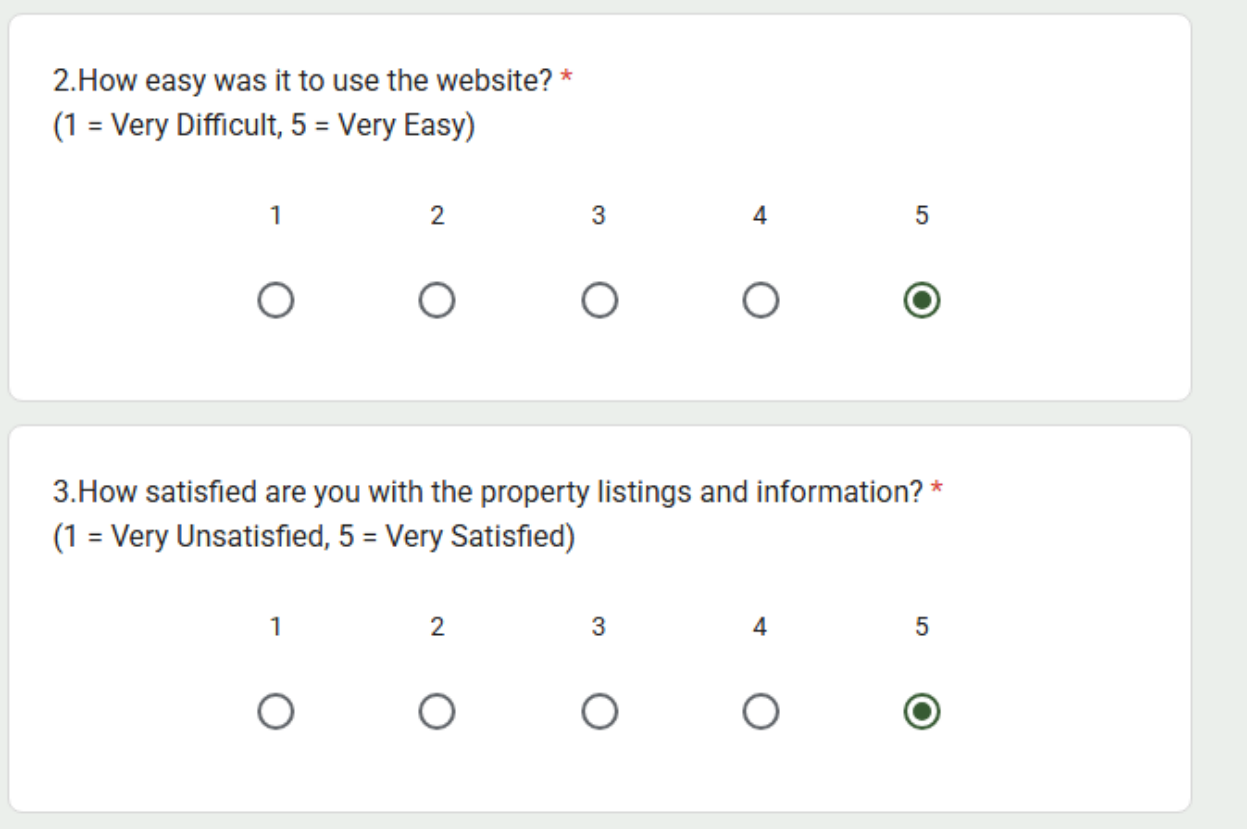


Figure 229 SAMPLE OF FILLED USER FEEDBACK FORMS

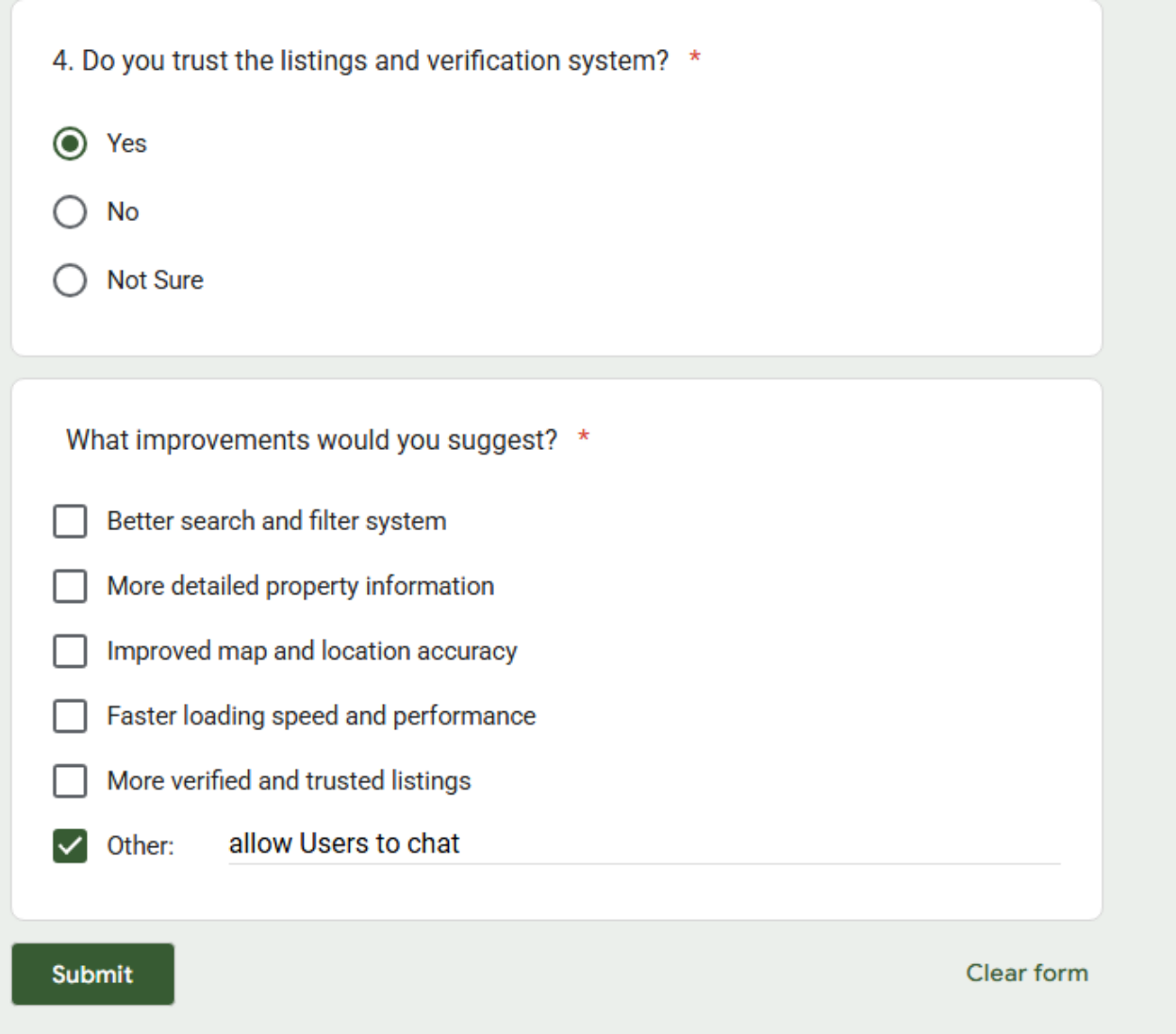


Figure 230 SAMPLE OF FILLED USER FEEDBACK FORMS

8.7 APPENDIX G: GitHub Commits

8.7.1 Open House Nepal front end commit

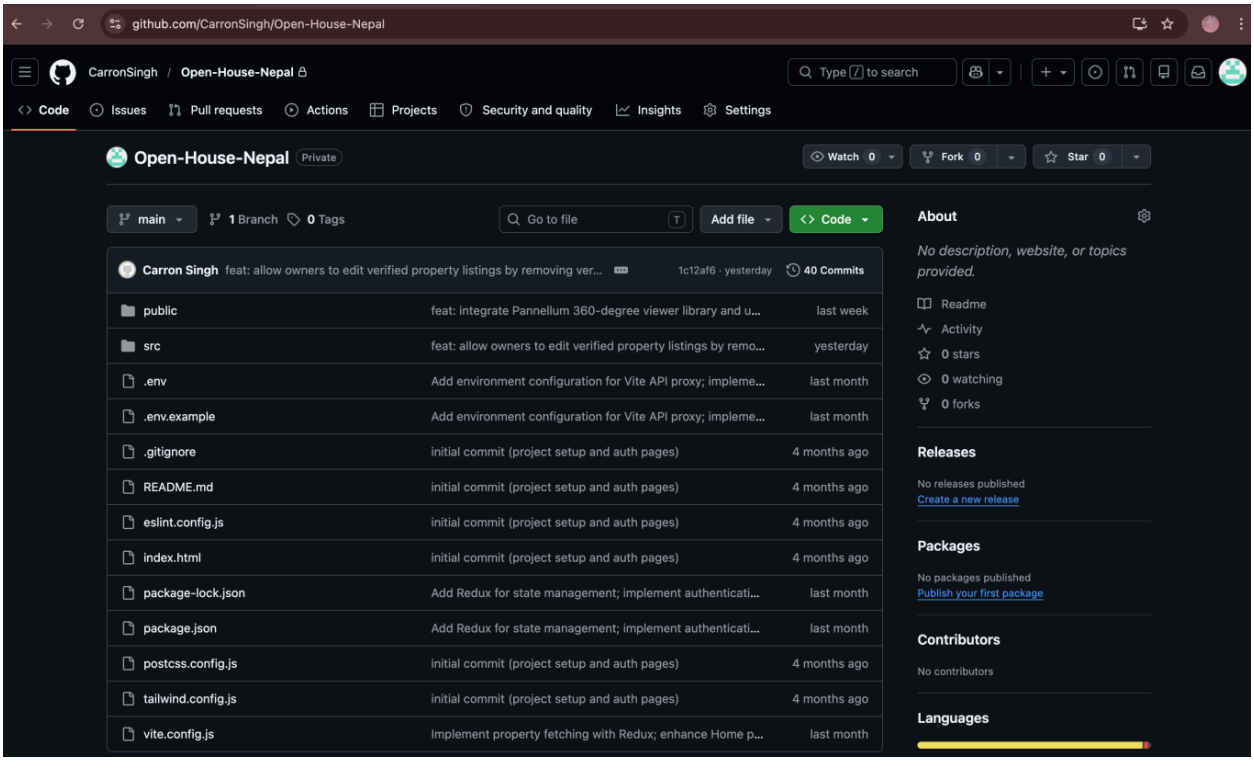


Figure 231 Open House Nepal front end commit

8.7.2 Open House Nepal Backend commit

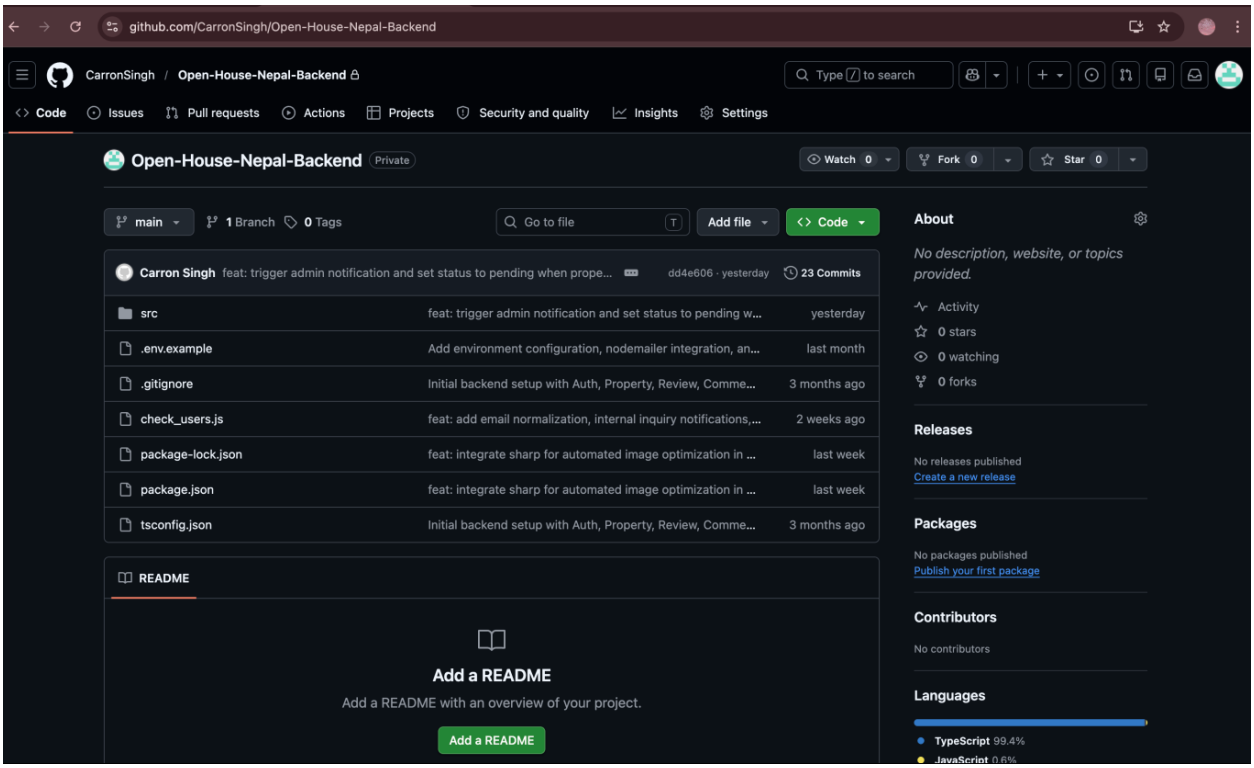


Figure 232 Open House Nepal Backend commit

8.8 APPENDIX H: FUTURE WORK

8.8.1 Future Work Readings

- **Mobile App Development**

Introduce an Android and iOS app to enhance access and user-friendliness.

- **Location-Based Features**

Include local facility data like restaurants, schools, hospitals, parks and shopping malls to facilitate improved decision-making.

- **Automated Property Valuation**

Implement a system of estimating property values according to the position, size and market trends.

- **AI-Based Property Recommendations**

Recommend properties using machine learning by considering user preferences, search history and budget.

- **User Interaction and Transactions**

Implement online reservation and chat system to enable the direct interaction between buyers and sellers.

- **System Intelligence and Checking.**

Add such features as document verification enhancement, administration analytics dashboard (user activity, most viewed properties) and property evaluation according to the market trends.